

UNIVERSITY OF PISA

PHYSICS DEPARTMENT

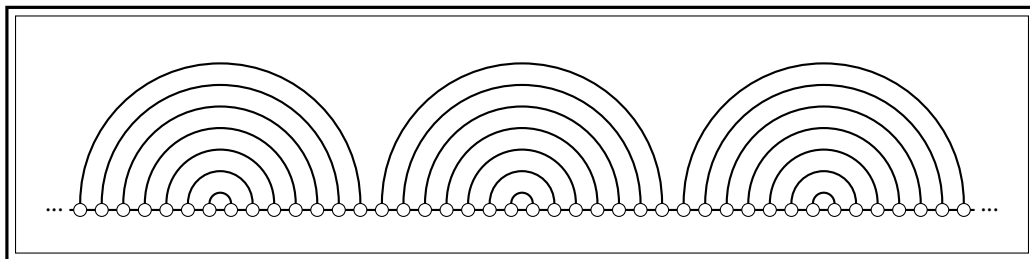
MASTER'S DEGREE IN PHYSICS - *Curriculum of Theoretical Physics*



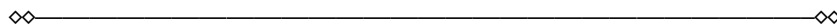
QUANTUM QUENCH DYNAMICS
◊ FROM INCREASINGLY ENTANGLED INITIAL STATES ◊
IN THE X_{XZ} MODEL

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LIST OF ABBREVIATIONS

ABA:	algebraic Bethe ansatz
ABC:	antiperiodic boundary conditions
AFM:	antiferromagnetic
BGT:	Bethe-Gaudin-Takahashi
CFT:	conformal field theory
DMRG:	density matrix renormalization group
dof:	degree of freedom
ETH:	eigenstate thermalization hypothesis
FM:	ferromagnetic
GE:	Gibbs ensemble
GGE:	generalized Gibbs ensemble
GHD:	generalized hydrodynamics
JW:	Jordan-Wigner
LSS:	local stationary state
LQSS:	local quasi-stationary state
MG:	Majumdar-Ghosh
MPO:	matrix product operator
MPS:	matrix product state
PBC:	periodic boundary conditions
QISM:	quantum inverse scattering method
SVD:	singular value decomposition
TBA:	thermodynamic Bethe ansatz
tDMRG:	time-dependent density matrix renormalization group
YBE:	Yang-Baxter equation

INTRODUCTION

The *out-of-equilibrium dynamics* in many-body 1D quantum systems has been a very active field of study in the last decades. In our work, we consider the simplest out-of-equilibrium protocol, namely the *quantum quench*: an initial state is suddenly left to evolve under a Hamiltonian of which it is not an eigenstate.

Remarkably, out-of-equilibrium dynamics is one of the main avenues for investigating one of the most fundamental conundrums of Statistical Physics itself, that is

how do statistical ensembles arise from the unitary evolution of isolated quantum systems prepared in pure states?

This is known as the problem of the *thermalization towards a Gibbs ensemble* in generic systems, where energy is almost the only local conservation law. Instead, there is numerical, experimental, and analytical evidence that *integrable systems*, which display an extensive number of (quasi)local integrals of motion, *relax to a generalized Gibbs ensemble (GGE)* [114] with one Lagrange multiplier for every conserved charge in a suitable "complete" set.

The pivotal idea in the study of this phenomenon is to focus on local observables rather than on the state of the entire system. Indeed, if an isolated quantum system is prepared in a pure state, then its state remains pure at all times, while statistical ensembles are represented by mixed states. By contrast, from the viewpoint of local observables in infinite systems, stationarity can emerge due to dephasing. The pure state of the system can become *locally indistinguishable* from many possible stationary states, which can be pure or mixed. One of these locally equivalent states is expected to be the corresponding (generalized) Gibbs ensemble [40, 51].

Simultaneously, a very convenient characterization of the stationary state is the *microcanonical description*, where a locally equivalent eigenstate of the Hamiltonian is used to compute local observables. This is called a *representative eigenstate*.

In this work, we consider quenches where the dynamics is driven by the *XXZ anisotropic Heisenberg model*, which is one of the most paradigmatic examples of integrable models.

The omnipresent theoretical framework when dealing with the XXZ model is the *Bethe ansatz* [19] and related techniques. The crucial intuition behind the Bethe ansatz is that in integrable systems, due to the presence of an extensive number of (quasi)local conserved charges, the Hamiltonian spectrum and eigenstates can still be effectively understood in terms of *stable quasiparticles*, similarly to free-fermion models. In particular, we will intensively employ both the *thermodynamic Bethe ansatz* [135] and the *algebraic Bethe ansatz* [53, 122, 123].

The former enables the characterization of eigenstates in the thermodynamic limit by means of the so-called *m-string root densities*. These can be thought of as the momentum occupation functions in free models. The main difference is that in genuinely interacting integrable models the quantization of quasimomenta is not state-independent and it admits solutions called *m-strings*, corresponding to bound states of *m* quasiparticles.

Instead, the algebraic Bethe ansatz allows us to find the complete set of quasilocal charges [71, 72] whose expectation values on the initial state of the quench completely fix the corresponding generalized Gibbs ensemble at stationarity. The root densities of the representative eigenstate can also be computed from these values by the *string-charge duality* [73].

After this general introduction, we turn to the specific topic of our work.

Indeed, their time evolution can be treated both analytically within the *quasiparticle picture* [26] or *generalized hydrodynamics (GHD)* [15], and numerically within a *matrix product states (MPS)* and *density matrix renormalization group (DMRG)* framework [118]. Moreover, their experimental realization is possible in cold-atom experiments.

However, highly entangled initial states are now attracting growing interest, as their experimental realization is becoming feasible. Some pioneering results have also been obtained from a theoretical perspective [33, 34].

Thus, we define a family of states, parametrized by an integer $n \in \mathbb{N}^{>0}$, with increasing entanglement while preserving macroscopic homogeneity, although translational invariance is microscopically broken. They are inspired by the rainbow state [111] and we call them *2n-rainbow states*, given their $2n$ -site periodicity. Every unit cell corresponds to a "rainbow" of n arcs, where a rainbow arc represents a Bell pair.

Macroscopic homogeneity is fundamental to expect relaxation to a generalized Gibbs ensemble and enables the study of entanglement dynamics and profiles of local observables respectively by generalized hydrodynamics and the quasiparticle picture. Moreover, the periodic structure of these states will allow us to perform an efficient computation of the algebraic Bethe ansatz quasilocal charges. Finally, small values of n are accessible to short-time numerical simulations by *time-dependent density matrix renormalization group (tDMRG)* [41, 118, 139].

In [99] it was conjectured that, in the XXX model, states periodically built from unit cells of length n which are linear superpositions of random states exhibit LSSs approaching the infinite-temperature thermal state as $n \rightarrow +\infty$. Since the XXZ model is a generalization of the XXX model and given the similarity between such "long random states" and the above-defined family, the first main question addressed in our work is the following:

does the 2n-rainbow state, as $n \rightarrow +\infty$, thermalize to infinite temperature in the XXZ model?

This phenomenon would be quite interesting for the following reasons.

- While local relaxation to a GGE is expected, *thermalization* is a much stronger and highly nontrivial statement in integrable systems.
- The $2n$ -rainbow state is pure and displays an elaborated, ordered, and regular Bell-pair entanglement structure. By contrast, the infinite temperature thermal state is maximally mixed and its density matrix is strikingly simple and featureless $\hat{\rho} \propto \hat{\mathbb{I}}$.
- Highly entangled initial states such as the crosscap state are locally equivalent to the infinite-temperature thermal state *from the beginning*. Instead, in the $2n$ -rainbow state, every macroscopic subsystem $\mathbb{A}$ is assumed to contain an almost infinite number of unit cells, thus it is *only slightly entangled with the environment*. This was even more evident in the states of [99] whose unit cells are random spin sequences with no entanglement.

As a first step in investigating this conjecture, we consider a simpler limit, in which many exact analytical results are accessible: the *noninteracting XX model*. The $2n$ -rainbow states are Gaussian and we exactly compute the form factor for generic n . This allows us to *analytically prove that in the XX model the 2n-rainbow state thermalizes to infinite temperature as $n \rightarrow +\infty$* .

Then, we turn to the *genuinely interacting XXZ model*. We prove that the XXZ complete set of quasilocal charges can be easily numerically evaluated on the infinite-temperature thermal state, similarly to the case of periodic product pure states. Thus, we can compute exactly both the $2n$ -rainbow state and infinite temperature root densities and study the convergence upon increasing n .

Moving beyond homogeneous quenches, we consider the XXZ time evolution of an initial state built from the junction of the (pure) $2n$ -rainbow state to the (mixed) infinite-temperature thermal state. This kind of inhomogeneous quench is called *bipartitioning protocol*.

This setting breaks translational invariance, preventing relaxation. However, GHD yields analytic predictions in a suitable scaling limit asymptotically late in time. As a complementary analysis, we simulate the finite-time entanglement dynamics by a tDMRG. It has the additional beautiful feature to automatically return the entanglement dynamics. In this procedure, we implement *purification* only on the thermal half of our initial state, allowing us to simulate the dynamics of a simple pure state. Assume that the LSS of the $2n$ –rainbow state is thermal with infinite temperature as $n \rightarrow +\infty$. Then,

according to GHD, as $n \rightarrow +\infty$ in the above bipartitioning protocol conserved charges must display exactly flat density profiles and no ballistic transport. Here, we ask what happens to ballistic transport at finite n .

In particular:

- The boundary values of conserved charges, following from the boundary conditions of the GHD equations, are expected to converge towards one another upon increasing n .
- Apart from this expected shrinking of the vertical window in which a nontrivial profile of conserved-charge densities can develop, *our main interest lies in whether ballistic transport exhibit an additional and stronger suppression, namely if density profiles approach a step function at finite n .*

This second effect would be interesting given the very different nature of the two initial halves of the bipartitioning protocol, namely the pure $2n$ –rainbow state, with its highly structured microscopic entanglement pattern, and the mixed, completely uncorrelated infinite-temperature thermal state. Remarkably, it turns out that, while in the *noninteracting XX model* ($\Delta = 0$) the energy density profile undergoes a smooth transition upon varying ζ between the two boundary values, in the *genuinely interacting XXZ antiferromagnet* ($\Delta > 1$) a sharp vertical jump at $\zeta \approx 0$ develops.

Finally, we focus on the *entanglement dynamics* in all the previous settings. First, we show that in the *XX model* it can be described within a known generalization [9, 18] of the original *quasi-particle picture* [4, 26], involving entangled multiplets of $2n$ species of quasiparticles. We give a general implicit expression to compute the entanglement evolution for the bipartitioning protocol between $2n$ –rainbow states with different n . Nicely, the two halves of the system produce multiplets with a different number of quasiparticles $2n_L \neq 2n_R$. We compute the explicit result in the bipartitioning protocol between the $2n_L = 2$ and $2n_R = 4$ rainbow states, finding agreement with the corresponding tDMRG simulation.

Turning to the *XXZ model and our half-pure half-mixed initial state*, the concept of entanglement dynamics becomes more controversial and it is correct only when referring to the purified system. The bipartite entanglement computed on the half-purified state coincides with the von Neumann entropy of the reduced density matrix of that half of the system which was initially pure. This twofold interpretation of the same von Neumann entropy allows us to ansatz a quasiparticle picture description of this dynamics in which *the initially-pure subsystem contributes with entangled pairs, while the initially-thermal subsystem with single particles carrying thermodynamic entropy*. We also give a complementary picture in the purified setting justifying the sum of these two contributions of different nature.

Remarkably, despite the fact that $2n$ –rainbow states are not *integrable initial states* [103] for $n > 1$, and that in the *XX model* they generate entangled multiplets of $2n$ quasiparticles, this prediction, based on the entangled-pairs assumption, seems to well describe the entropy slope in all the considered settings, at least within the time-interval accessible to our tDMRG.

Chapter 1

THE XXZ MODEL AND THE $2n$ -RAINBOW STATES

1.1 DEFINITION OF THE MODEL AND CONVENTIONS

The integrable model considered in this work is the *anisotropic Heisenberg model*, or *XXZ model* [135]. It is a widely studied generalization of the *isotropic Heisenberg (XXX) model*, corresponding to the introduction of an anisotropy parameter $\Delta \in \mathbb{R}$. The XXX model, recovered as $\Delta = 1$, has historically been the first model to be solved by the Bethe ansatz [19].

The XXZ model is defined on a $\frac{1}{2}$ -spin chain of L sites. It is translationally invariant and only involves nearest-neighbour interactions. This makes it natural to express the Hamiltonian as a sum over a local density $\hat{h}_{j,j+1}$ shifted along the entire lattice.

$$\begin{aligned} \hat{H}_{XXZ} &:= J \sum_{j=1}^L \hat{h}_{j,j+1} & \text{PBC: } \hat{h}_{L,L+1} &:= \hat{h}_{L,1} \\ \hat{h}_{j,j+1} &:= \left(\hat{S}_j^x \hat{S}_{j+1}^x + \hat{S}_j^y \hat{S}_{j+1}^y + \Delta \hat{S}_j^z \hat{S}_{j+1}^z \right) = \left[\frac{(\hat{S}_j^+ \hat{S}_{j+1}^- + \hat{S}_j^- \hat{S}_{j+1}^+)}{2} + \Delta \hat{S}_j^z \hat{S}_{j+1}^z \right] \\ \hat{S}^\pm &= \hat{S}^x \pm i \hat{S}^y & \hat{S}^x &= \frac{\hat{S}^+ + \hat{S}^-}{2} & \hat{S}^y &= \frac{\hat{S}^+ - \hat{S}^-}{2i} & \text{PBC: } \hat{S}_{L+1}^a &= \hat{S}_1^a \end{aligned} \quad (1.1)$$

If $\Delta = 0$ we obtain the *XX model*: in §5 we will recall how it is mappable onto a noninteracting fermionic model, immediately diagonalizable by Fourier transform. This will allow us to easily compute many exact analytical results. Instead, if $|\Delta| \rightarrow +\infty$ it reduces to the 1D *Ising model*.

Symmetries: We have already mentioned the one-step translational invariance. The unitary operator implementing this symmetry can be expressed in terms of the momentum operator $\hat{K}$ as $\hat{U} = e^{i\hat{K}}$. Unlike the generator of continuous translations, $\langle \hat{K} \rangle$ is defined modulo 2π , due to the unit lattice spacing.

Since $\hat{\sigma}^z \hat{S}^\pm = \pm \hat{S}^\pm$ and $\hat{S}^\pm \hat{\sigma}^z = \mp \hat{S}^\pm$, $\hat{H}_{XXZ}$ is also invariant under $\hat{U}(\theta) = \exp[i\theta \sum_j \hat{\sigma}_j^z]$, i.e. it has a *global $U(1)$ symmetry*, leading to the conservation of the total magnetization along z :

$$\hat{M}^z = \sum_j \hat{S}_j^z = \frac{1}{2} \sum_j \hat{\sigma}_j^z \quad [\hat{M}^z, \hat{H}_{XXZ}] = 0$$

Moreover, it exhibits $\mathbb{Z}_2$ symmetries associated to independent global spin-flips along the three directions in spin space. In particular, this means that *every common eigenspace of $\hat{M}^z$ and $\hat{H}_{XXZ}$ is mapped under a global spin-flip along the z -axis onto a degenerate eigenspace with opposite magnetization*.

Useful Notations: Once the model is defined, we introduce some convenient notations:

- The anisotropy parameter Δ will be replaced by the following variable η :

$$\begin{cases} \eta := \operatorname{arccosh}(\Delta) > 0 & \text{if } |\Delta| \geq 1 \quad (\text{gapped AFM or FM regime}) \\ \eta := \arccos(\Delta) \in [0, \pi] & \text{if } |\Delta| < 1 \quad (\text{Luttinger liquid}) \end{cases} \quad (1.2)$$

However, apart from the special $\Delta = 0$ case, we will focus exclusively on the gapped phase.

- Since the imaginary $ik(\eta/2)$ -translation of the rapidities will often appear, we define

$$f^{[k]}(\lambda) := f\left(\lambda + k\frac{i\eta}{2}\right) \quad \text{where, moreover} \quad [\pm] := [\pm 1] \quad (1.3)$$

Jordan-Wigner (JW) Transform: Note that the operators $\{\hat{S}_j^\pm\}$ satisfy the relations

$$[\hat{S}_i^+, \hat{S}_j^-] = 2\delta_{ij}\hat{S}_j^z \quad \{\hat{S}_j^+, \hat{S}_j^-\} = \hat{\mathbb{I}}_j \quad \hat{S}_j^+\hat{S}_j^+ = \hat{S}_j^-\hat{S}_j^- = 0$$

while there is no relevant relation involving $\{\hat{S}_i^+, \hat{S}_j^-\}$ when $i \neq j$.

Thus, the $\{\hat{S}_j^\pm\}$ operators are neither bosonic nor fermionic, since they satisfy the canonical anticommutation relations only when evaluated on the same site. This leads to a Pauli-like principle. By contrast, they commute as bosonic operators when acting on different sites.

Owing to this Pauli-like property, it is possible to define a JW transform which maps every $\frac{1}{2}$ -spin system onto a fermionic spinless system. The main idea is to establish a correspondence between up/down spins and full/empty sites. Thus $\hat{\sigma}^z$ is mapped onto $(2\hat{n}_j - \hat{\mathbb{I}}_j)$ and $(\hat{S}_j^+, \hat{S}_j^-)$ into $(\hat{c}_j^\dagger, \hat{c}_j)$. While the first mapping is direct, the second involves a JW string $\hat{K}_j$ to convert anticommutation into commutation relations when operators are evaluated on different sites.

$$\hat{K}_j := \prod_{i < j} (-\hat{\sigma}_i^z) \equiv \prod_{i < j} (\hat{\mathbb{I}}_i - 2\hat{n}_i) = (-1)^{\sum_{i < j} \hat{n}_i} \quad \text{JW string}$$

$$\hat{K}_j^\dagger = \hat{K}_j = \hat{K}_j^{-1} \quad [\hat{K}_i, \hat{S}_j^\pm] = [\hat{K}_i, \hat{c}_j^\dagger] = 0 \quad \forall i \leq j$$

The mapping is defined as

$$\hat{S}_j^+ \leftrightarrow \hat{K}_j \hat{c}_j^\dagger = \hat{c}_j^\dagger \hat{K}_j \quad \hat{S}_j^- \leftrightarrow \hat{K}_j \hat{c}_j = \hat{c}_j \hat{K}_j \quad \hat{S}_j^z = \hat{S}_j^+ \hat{S}_j^- - \frac{1}{2} \hat{\mathbb{I}}_j \leftrightarrow \hat{n}_j - \frac{1}{2} \hat{\mathbb{I}}_j \quad (1.4)$$

It is easily inverted observing that $\hat{K}_j^{-1} = \hat{K}_j$ and using the above two definitions of $\hat{K}_j$ in terms of spin and fermionic operators.

Fermionic Representation of the XXZ Hamiltonian: The JW transform maps strings of spin (fermion) operators with even number of terms and acting on the same site j onto their fermion (spin) counterpart directly, consistently with the mapping of $\hat{S}_j^z$. Instead, in presence of odd strings or operators acting on different sites, JW strings only partially annihilate. For example:

$$\hat{S}_j^+ \hat{S}_{j+1}^- \leftrightarrow \hat{c}_j^\dagger \hat{K}_j \hat{K}_{j+1} \hat{c}_{j+1} = \hat{c}_j^\dagger (\hat{\mathbb{I}} - 2\hat{n}_j) \hat{c}_{j+1} = \hat{c}_j^\dagger \hat{c}_{j+1}$$

$$\hat{S}_j^- \hat{S}_{j+1}^+ \leftrightarrow \hat{c}_j \hat{K}_j \hat{K}_{j+1} \hat{c}_{j+1}^\dagger = \hat{c}_j (\hat{\mathbb{I}} - 2\hat{n}_j) \hat{c}_{j+1}^\dagger = -\hat{c}_j \hat{c}_{j+1}^\dagger = \hat{c}_{j+1}^\dagger \hat{c}_j$$

Moreover, in transforming $\hat{H}_{XXZ}$, PCB together with the non-locality of the JW transform lead to an interaction term depending on the total number of fermions:

$$\hat{S}_L^+ \hat{S}_1^- + \hat{S}_L^- \hat{S}_1^+ \leftrightarrow \hat{K}_L (\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L \hat{c}_1^\dagger) = (-1)^{\hat{N}_{tot}+1} (\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L \hat{c}_1^\dagger)$$

Finally, the anisotropy term, despite its trivial mapping, leads to genuine (i.e. non quadratic) interactions. These are due to $\hat{n}_j \hat{n}_{j+1}$, which involves four c -operators.

$$\hat{S}_j^z \hat{S}_{j+1}^z \leftrightarrow \hat{n}_j \hat{n}_{j+1} - \frac{\hat{n}_j + \hat{n}_{j+1}}{2} + \frac{\hat{\mathbb{I}}}{4}$$

In this new fermionic picture, the $U(1)$ symmetry leads to the *conservation of the number of fermions*, since $\hat{M} \leftrightarrow \left(\sum_j \hat{n}_j\right) - \frac{L}{2} \hat{\mathbb{I}} := \hat{N}_{tot} - \frac{L}{2} \hat{\mathbb{I}}$.

Instead, a global spin flip along the z -axis here replaces empty with occupied sites and vice versa. In particular, the number conservation implies that, if we consider the dynamics of a state which is an eigenstate of $\hat{N}_{tot}$, then the effect of $(-1)^{\hat{N}_{tot}+1}$ is equivalent to have *ABC/PBC if the eigenvalue N_{tot} is even/odd*, respectively.

Putting everything together, the final XXZ fermionic Hamiltonian is

$$\frac{\hat{H}_{XXZ}}{J} \leftrightarrow \left[\sum_{j=1}^{L-1} \frac{(\hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_{j+1}^\dagger \hat{c}_j)}{2} \right] + (-1)^{\hat{N}_{tot}+1} \frac{(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L \hat{c}_1^\dagger)}{2} + \Delta \left[\frac{\hat{\mathbb{I}}L}{4} - \hat{N}_{tot} + \sum_{j=1}^L \hat{n}_j \hat{n}_{j+1} \right] \quad (1.5)$$

However, we will never use this more complicated fermionic representation, except for the $\Delta = 0$ case in §5. Instead, later in this chapter we will use the inverse JW transform to map the $2n$ -rainbow state, defined in a fermionic framework, onto an $\frac{1}{2}$ -spin setting.

1.2 OUT-OF-EQUILIBRIUM DYNAMICS FROM HIGHLY ENTANGLED INITIAL STATES

In this section we briefly outline the state-of-the-art regarding the dynamics from highly entangled initial states. Many concepts, methods and procedures mentioned here will be defined and discussed in more detail later in this and other chapters.

Until recently, in the study of the out-of-equilibrium dynamics in 1D integrable systems, low-entangled initial states have been the primary choice. Indeed, their time evolution can be treated both analytically within the *quasiparticle picture* [26] or *generalized hydrodynamics* [15], and numerically within a *matrix product states* and *density matrix renormalization group* framework [118]. Moreover, their experimental realization is possible in cold-atoms experiments.

The main limitation of the aforementioned numerical methods for the dynamics simulation is that they generally fail as the entanglement content of the evolved state increases.

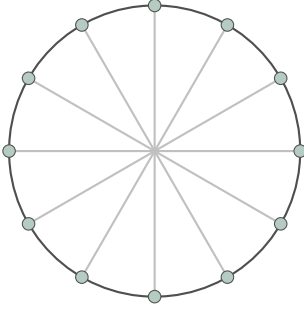
However, highly entangled initial states are now attracting growing interest, as their experimental realization is becoming feasible, for example through trapped-ion systems [61, 95].

Some pioneering results have been achieved from the theoretical viewpoint as well. For example, it has been shown that quenches from *crosscap states* [8, 33, 35, 50] can be efficiently studied with both analytical and numerical techniques. In particular, in integrable systems the post-quench entanglement dynamics can be described within a generalized quasiparticle picture. In an L -site lattice with PBC, which can be viewed as a circle, crosscap states are defined by entangling in a Bell pair antipodal sites. This structure is exactly translationally invariant.

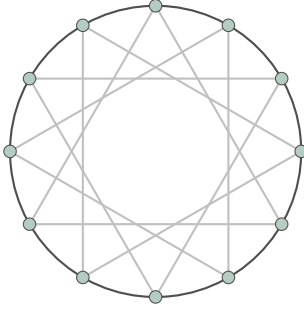
Moreover, it leads to volume-law scaling of entanglement. Indeed, for every subsystem A given by a segment of length $\leq L/2$, the reduced density matrix is $\hat{\rho}_A \propto \hat{\mathbb{I}}_A$. This also implies that the crosscap state is locally indistinguishable from the infinite-temperature state.

In another recent article [34] a generalization of crosscap states was introduced, i.e. the family of the *entangled multipodal states*. Different states of this family are labelled by integers $n \in \mathbb{N}^{\geq 2}$, where $n = 2$ corresponds to crosscap states.

Crosscap state
($n = 2$ multipodal state)



Triangle state
($n = 3$ multipodal state)



$L = 12 \quad (1.6)$

$i \text{ --- } j \leftrightarrow \frac{\hat{\mathbb{I}} + \hat{c}_i^\dagger \hat{c}_j^\dagger}{\sqrt{2}}$

$\begin{matrix} j \\ \diagup \quad \diagdown \\ i \end{matrix} \text{ --- } k \leftrightarrow \frac{1}{\sqrt{2}} \left(\hat{\mathbb{I}} + \frac{\hat{c}_i^\dagger \hat{c}_j^\dagger + \hat{c}_i^\dagger \hat{c}_k^\dagger + \hat{c}_j^\dagger \hat{c}_k^\dagger}{\sqrt{3}} \right)$

For $n > 2$, these states are still translationally invariant, but entangled sites sit on the vertices of n -sided polygons. For example, the $n = 3$ state is also called triangle state. As n increases, the entanglement content decreases, and area-law scaling is gradually recovered as $n \approx L$.

The fermionic multipodal states are Gaussian. This allows their dynamics in free models to be studied exactly in terms of the two-point correlation matrix.

Moreover, it has been shown that the corresponding time evolution of the entanglement entropy associated to a subsystem $\mathbb{A}$ which is a segment of length smaller than L/n can be described within a modified quasiparticle picture.

1.3 THE $2n$ -RAINBOW STATES

Motivations for the Choice of the Initial States: In this work, we are not going to consider states with extensive-range ($\propto L$) entanglement, such as multipodal states. We define a family of states, parametrized by an integer $n \in \mathbb{N}^{>0}$ with increasing entanglement while preserving macroscopic homogeneity, despite translational invariance is microscopically broken. They are inspired by the rainbow state [111] and we call them $2n$ -rainbow states, given their $2n$ -site periodicity.

Macroscopic homogeneity is fundamental to expect relaxation to a generalized Gibbs ensemble (see §3) and enables the employment of generalized hydrodynamics (see §4) and the quasiparticle picture (see §5). Moreover, the periodic structure of these states will allow us to perform an efficient computation of the complete set of quasilocal charges for the GGE (see §7).

Finally, while small values of n are accessible to short-time numerical simulations by time-dependent density matrix renormalization group, *their similarity to the "long random states" of [99] suggests that they thermalize to infinite temperature as $n \rightarrow +\infty$* . This is the central question addressed in the present work. We prove the conjecture analytically in the noninteracting XX model, and we provide numerical evidence supporting it in the XXZ model.

We now formally define our family of increasingly entangled initial states.

The Rainbow State: The original rainbow state was first introduced in [143] as *concentric singlet phase* and then gained its more familiar name in [110]. It was further studied in [111, 112, 115, 116].

In terms of fermionic creation and annihilation operators it is defined as

$$|R\rangle := \lim_{L \rightarrow +\infty} \left(\hat{b}_{-\frac{L-1}{2}, \frac{L-1}{2}}^{[(-1)^{L/2+1}] \dagger} \dots \left(\hat{b}_{-\frac{5}{2}, \frac{5}{2}}^{[+] \dagger} \right) \left(\hat{b}_{-\frac{3}{2}, \frac{3}{2}}^{[-] \dagger} \right) \left(\hat{b}_{-\frac{1}{2}, \frac{1}{2}}^{[+] \dagger} \right) \right) |\Omega\rangle$$

where $|\Omega\rangle$ is the vacuum (no fermion present) and $(\hat{b}_{ij}^{[\pm]})^\dagger$ are the following entangling operators:

$$(\hat{b}_{ij}^{[\pm]})^\dagger := \frac{\hat{c}_i^\dagger \pm \hat{c}_j^\dagger}{\sqrt{2}} \quad (1.7)$$

This construction gives rise to the rainbow-like structure which justifies its name: sites at the same distance from the centre of the lattice form a Bell pair. This entanglement link is usually represented in literature as a rainbow arc.

The rainbow state clearly satisfies the volume-law of entanglement scaling under arbitrary bipartitions obtained by cutting the system in two halves. Instead, this in general fails when considering a subsystem $\mathbb{A}$ which is a segment including the centre of the lattice. For example, symmetric segments are in a product state with the rest of the system, since this bipartition does not cut any rainbow arc.

Finally, this state is known to be the ground state of the following inhomogeneous free-fermion Hamiltonian [111, 143], where the hopping amplitudes between adjacent sites decay exponentially outwards from the centre of the chain, in the strong inhomogeneity regime $\alpha \rightarrow 0^+$.

$$\hat{H}_R := -\frac{1}{2} \hat{c}_{\frac{1}{2}}^\dagger \hat{c}_{-\frac{1}{2}} - \sum_{n=\frac{1}{2}}^{L-3/2} \frac{\alpha^{2n}}{2} [\hat{c}_n^\dagger \hat{c}_{n+1} + \hat{c}_{-n}^\dagger \hat{c}_{-(n+1)}] + h.c. \quad \alpha \in [0, 1]$$

By tuning α , the ground state of the system evolves smoothly from a logarithmic entanglement scaling towards the volume-law associated to the rainbow state.

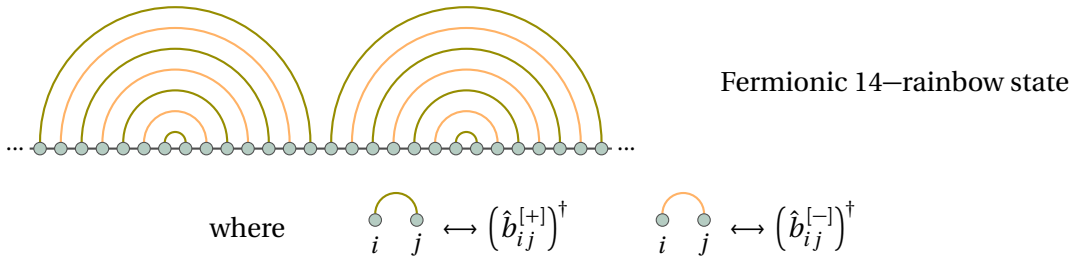
Definition of the $2n$ –Rainbow State: We define the $2n$ –rainbow state by modifying the proper rainbow state. In particular, we assume a finite rainbow support of size $2n$ (i.e. only n arcs) and use the obtained segment as unit cell to fill a lattice of $L \rightarrow +\infty$ sites.

This construction implies that, even when we consider infinite-sized rainbows $n \rightarrow +\infty$, the number $N := L/2n$ of unit cells in the system is also required to be infinite.

Thus *macroscopic homogeneity* is granted.

$$|R_n\rangle := \lim_{N \rightarrow +\infty} \prod_{\ell=0}^{\leftarrow(N-1)} \left[\prod_{j=1}^{\leftarrow n} \left(\hat{b}_{-(j-\frac{1}{2})+2n\ell, (j-\frac{1}{2})+2n\ell}^{[(-1)^{j+1}]} \right)^\dagger \right] |\Omega\rangle \quad (1.8)$$

where $\leftarrow$ denotes that terms inside the tensor product have indices running over the specified set from left to right. Pictorially:



Note that the staggered green/yellow colour refers to the alternation of $(\hat{b}_{ij}^{[\pm]})^\dagger$ in the definition (1.8). Instead, for example, with the convention $|\{j_1, \dots, j_N\}\rangle = \hat{c}_{j_N} \dots \hat{c}_{j_1} |\Omega\rangle$, where $|\{j_1, \dots, j_N\}\rangle$ is the state in which sites $j_1 < \dots < j_N$ have occupation 1 and the others are empty, all rainbow arcs correspond to triplet Bell pairs $2^{-1/2}(|10\rangle + |01\rangle)$. However, we never employ this explicit representation. Staggered signs are recovered at the level of the correlation matrix $\langle \hat{c}_i^\dagger \hat{c}_j \rangle$ (see §5).

To study the time evolution of this state under the spin Hamiltonian (1.1), we need to map it onto its $1/2$ –spin counterpart through a JW transform.

We can factorize on the right the common part of the JW string. This way inside $[]$ there are only operators acting on sites between $-(j - \frac{1}{2}) + 2n\ell$ and $(j - \frac{1}{2}) + 2n\ell$:

$$\left(\hat{b}_{-(j-\frac{1}{2})+2n\ell, (j-\frac{1}{2})+2n\ell}^{[(-1)^{j+1}]} \right)^\dagger \leftrightarrow \frac{1}{\sqrt{2}} \left[\hat{S}_{-(j-\frac{1}{2})+2n\ell}^+ + (-1)^{j-1} \hat{S}_{(j-\frac{1}{2})+2n\ell}^+ \prod_{-(j-\frac{1}{2})+2n\ell}^{(i-\frac{1}{2})+2n\ell-1} (-\hat{\sigma}_i^z) \right] \hat{K}_{-(j-\frac{1}{2})+2n\ell}$$

where the lattice sites run from $-(n - \frac{1}{2})$ to $(n - \frac{1}{2}) + 2n(N-1) = L - (n + \frac{1}{2})$.

We aim to simplify the blue and green terms using that the whole string of the b -operators is acting on $|\downarrow\downarrow\downarrow\dots\rangle$, where $-\hat{\sigma}_j^z \equiv \hat{\mathbb{I}}$.

The key observation is that, when an (ℓ, j) b -operator is acting, sites between $-(n - \frac{1}{2})$ and $(n - \frac{1}{2}) + 2n(\ell - 1)$ are filled with $\ell - 1$ complete $2n$ -rainbows, due to the action of the previous b -operators. Thus, the action of $\hat{K}_{-(j-\frac{1}{2})+2n\ell}$ only yields a global $(-1)^{n(\ell-1)}$ sign, which can be neglected since it does not affect any observable value. Instead, sites between $-(n - \frac{1}{2}) + 2n\ell$ and $-(j - \frac{1}{2}) + 2n\ell$ are still untouched, i.e. have spin $\downarrow$ and the JW string acts trivially.

The blue term is therefore simplified to

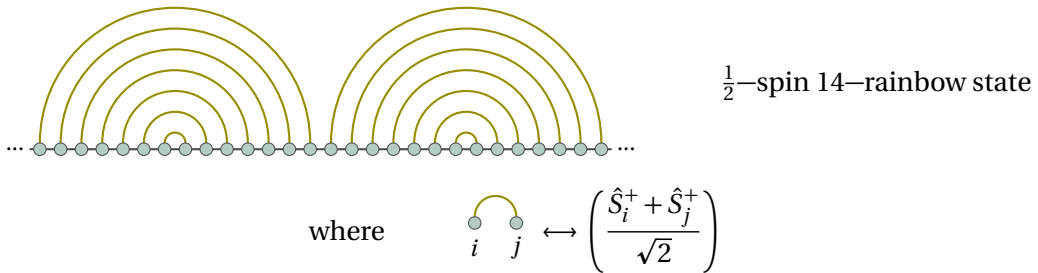
$$\hat{K}_{-(j-\frac{1}{2})+2n\ell} \equiv (-1)^{n(\ell-1)} \hat{\mathbb{I}} \equiv \hat{\mathbb{I}}$$

Analogously, when the green term acts, the site $-(j - \frac{1}{2}) + 2n\ell$ is untouched and $-\hat{\sigma}^z \equiv \hat{\mathbb{I}}$. Instead, sites between $-(j - \frac{1}{2}) + 2n\ell + 1$ and $(j - \frac{1}{2}) + 2n\ell - 1$ are filled with a complete $2(j-1)$ -rainbow, which counts $j-1$ up spins.

$$\prod_{-(j-\frac{1}{2})+2n\ell}^{(i-\frac{1}{2})+2n\ell-1} (-\hat{\sigma}_i^z) \equiv (-1)^{j-1} \hat{\mathbb{I}}$$

Finally, the $1/2$ -spin $2n$ -rainbow state turns out to be defined as

$$\begin{aligned} |R_n\rangle &\leftrightarrow \lim_{N \rightarrow +\infty} \prod_{\ell=0}^{N-1} \left[\prod_{j=1}^n \left(\frac{\hat{S}_{-(j-\frac{1}{2})+2n\ell}^+ + \hat{S}_{(j-\frac{1}{2})+2n\ell}^+}{\sqrt{2}} \right) \right] |\Omega\rangle = \\ &= \lim_{N \rightarrow +\infty} \bigotimes_{\ell=0}^{N-1} \bigotimes_{j=1}^n \left(\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}} \right)_{-(j-\frac{1}{2})+2n\ell, (j-\frac{1}{2})+2n\ell} \end{aligned}$$



Fermionic alternate $(\hat{b}_{ij}^{[\pm]})^\dagger$ operators have been replaced by $\frac{1}{2}$ -spin triplet Bell-pair creators.

Gaussianity: A quantum state is said to be Gaussian if all N -point functions $\langle \hat{c}_{i_1}^\dagger \dots \hat{c}_{i_N}^\dagger \hat{c}_{j_1} \dots \hat{c}_{j_N} \rangle$ factorize via Wick's theorem as sums of products of two-point functions, fully encoded in the correlation matrix $C_{ij} = \langle \hat{c}_i^\dagger \hat{c}_j \rangle$. In fermionic Gaussian states the 4-point function is e.g.

$$\langle \hat{c}_{i_1}^\dagger \hat{c}_{i_2}^\dagger \hat{c}_{j_1} \hat{c}_{j_2} \rangle = C_{i_1 j_1} C_{i_2 j_2} - C_{i_1 j_2} C_{i_2 j_1}$$

Moreover, using the anticommutation rules it is possible to prove that Slater determinant states

$$|j_1, \dots, j_N\rangle = \prod_{k=1}^{\leftarrow N} \hat{c}_k^\dagger |\Omega\rangle \quad \text{are Gaussian.}$$

To prove that the $2n$ –rainbow state is Gaussian, we need to do some preliminary observations. Consider a Fock space whose basis is constructed by L creation operators $\{\hat{c}_1^\dagger, \dots, \hat{c}_L^\dagger\}$, which we assume to be fermionic. Let $|\psi\rangle$ be a state in this space given by

$$|\psi\rangle = \prod_{j=1}^{\leftarrow N} \hat{d}_j^\dagger |\Omega\rangle \quad N \leq L \quad (1.9)$$

where $\{\hat{d}_1, \dots, \hat{d}_N\}$ is a set of operators which can be completed to another set $\{\hat{d}_1, \dots, \hat{d}_L\}$ such that a unitary matrix $U \in \text{Mat}_{L \times L} \mathbb{C}$ exists that maps this new set onto the original annihilators of the Fock space.

$$\hat{d}_i = U_{ij} \hat{c}_j \quad \rightarrow \quad \hat{d}_i^\dagger = U_{ij}^* \hat{c}_j^\dagger = \hat{c}_j^\dagger (U^\dagger)_{ji} \quad U^\dagger U = U U^\dagger = \mathbb{I}$$

This guarantees that fermionic anticommutation rules are preserved.

$$\{\hat{d}_i, \hat{d}_j\} = \{\hat{d}_i^\dagger, \hat{d}_j^\dagger\} = 0 \quad \{\hat{d}_i^\dagger, \hat{d}_j\} = U_{ja} (U^\dagger)_{bi} \{\hat{c}_b^\dagger, \hat{c}_a\} = U_{ja} (U^\dagger)_{bi} \delta_{ba} = \delta_{ij}$$

Two-point functions in terms of d –operators are fully determined by that of c –operators:

$$\langle \hat{d}_i^\dagger \hat{d}_j \rangle = U_{ja} (U^\dagger)_{bi} C_{ba}$$

Thus, Wick's theorem can be equivalently proved at the level of d –operators. However, from the viewpoint of d –operators (1.9) is simply a Slater determinant, thus Wick's theorem is satisfied. This means that states which can be written as $|\psi\rangle$ are Gaussian.

At this point, it only remains to prove that the fermionic $2n$ –rainbow state can be expressed as (1.9). However, this is transparent from (1.8), where $\hat{b}$ operators have the role of $\hat{d}$ operators in $|\psi\rangle$. To complete their set we can add the complementary $[\pm]$ –creators. The completed set is connected to the original $\{\hat{c}_j^\dagger\}$ by the unitary transformation described in (1.7).

Therefore, the $2n$ –rainbow state is Gaussian.

Energy Density: Owing to the periodicity of the state and the Hamiltonian, it is sufficient to compute $\langle \hat{h}_{j,j+1} \rangle$ on $2n$ sites only and then average the results.

For a more compact and transparent derivation we employ the total angular momentum $|l, m\rangle_{a,b}$ basis for the space of two $1/2$ –spins corresponding to sites a and b . When $\hat{h}_{j,j+1}$ acts on two sites belonging to two different rainbow arcs, it maps $|R_n\rangle$ into an orthogonal state, since

$$\hat{h}_{b,c} |1, 0\rangle_{a,b} \otimes |1, 0\rangle_{c,d} = \frac{|1, 1\rangle_{a,b} \otimes |1, -1\rangle_{c,d} + |1, -1\rangle_{a,b} \otimes |1, 1\rangle_{c,d}}{2} - \frac{\Delta}{4} |0, 0\rangle_{a,b} \otimes |0, 0\rangle_{c,d}$$

Therefore, the only nontrivial energetic contribution comes from the $\hat{h}_{j,j+1}$ acting on the two central sites of every $2n$ –rainbow, which belongs to the same rainbow arc.

$$\hat{h}_{b,c} |1, 0\rangle_{a,b} = \frac{2-\Delta}{4}$$

This means that energy density for the $2n$ –rainbow state is given by (independently of j_0)

$$\frac{E}{L} = \frac{J}{2n} \sum_{j=j_0}^{j_0+2n-1} \langle \hat{h}_{j,j+1} \rangle = \frac{J(2-\Delta)}{8n} = O\left(\frac{1}{n}\right) \quad (1.10)$$

In particular, $\forall \Delta$ the energy density is zero when $n \rightarrow +\infty$, while it is always zero for $\Delta = 2$.

Chapter 2

THERMODYNAMIC BETHE ANSATZ

2.1 COORDINATE BETHE ANSATZ

Heuristic Intuition Behind the Bethe Ansatz: The name "Bethe ansatz" refers to a family of diagonalization methods for genuinely interacting (i.e. non quadratic) integrable systems.

Note that the word "integrable" has not still a universally accepted formal definition [32], as we will discuss in §3 and §7. Throughout this chapter, we adopt the existence of an extensive set of (quasi)local conserved charges as our working definition of *integrability*.

Typically, these systems display a $U(1)$ symmetry which can be interpreted as the conservation of the number N of some kind of *stable quasiparticles*. The Hamiltonian can then be diagonalized separately on every subspace with fixed N . Thus, one may wonder whether, similarly to Slater determinants in free systems, a basis of eigenstates for every eigenspace with fixed N could be found considering N -body wavefunctions which are linear combination of $N!$ plane waves, each corresponding to a permutation of N selected quasimomenta. This fails for generic interacting systems with an $U(1)$ symmetry. From a physical viewpoint, scattering events are not elastic and the Hamiltonian connects states with different sets of quasimomenta.

However, heuristically, the presence of an extensive number of (quasi)local conserved charges can constrain these scattering events so that all quasimomenta are individually conserved as in free models. The effect of interactions is reduced to a dephasing between the different terms of the eigenstate wavefunction corresponding to permutations of the quasimomenta [62, 135].

The original Bethe ansatz, i.e. the *coordinate Bethe ansatz*, was introduced in 1931 by Hans Bethe (during his postdoctoral stay in Rome at the laboratory of Enrico Fermi) in his famous work [19], to solve the XXX isotropic Heisenberg model.

The first generalization of the method from the XXX to the XXZ model was performed by Orbach [96] and Walker [146] almost thirty years later.

In this chapter we will discuss this original formulation and its recasting in the thermodynamic limit, i.e. the *thermodynamic Bethe ansatz*. Then, in §7 we will introduce a more abstract and elegant formulation of the method, namely the *algebraic Bethe ansatz*, deeply connected to the very mathematical nucleus of integrability.

The Bethe Wavefunction for Spin/Fermionic Eigenstates: The $U(1)$ symmetry of the XXZ model in the spin picture is given by the conservation of the total magnetization $\hat{M}^z$ along the z -axis. The Hamiltonian and $\hat{M}^z$ are simultaneously diagonalizable. Thus, we can look for eigenstates with fixed number $N = \langle \hat{M} \rangle + L/2$ of up spins, for every $N \in \{0, \dots, L\}$ separately. This coincides with the number of fermions after the JW-mapping.

Such an eigenstate is completely specified by a wavefunction ψ depending on N not-ordered site coordinates. These must be all different, given the Pauli-like property of the $\hat{S}^\pm$ operators. Not-ordered N -uples are equivalent to ordered N -uples inside a given *simplex* Q_N , i.e. a domain

in which one and only one permutation of every set of positions $\{x_1, \dots, x_N\}$ appears.

$$Q_N := \{ \{x_1, \dots, x_N\} \in \{1, \dots, L\}^N \mid x_1 < \dots < x_N \}$$

$$|\psi\rangle := \sum_{\{x_1, \dots, x_N\} \in Q_N} \psi(x_1, \dots, x_N | Q_N) \hat{S}_{x_N}^+ \dots \hat{S}_{x_1}^+ |\downarrow\downarrow\downarrow \dots\rangle \quad (2.1)$$

We could have done the same in the fermionic framework after JW mapping, which leads to the equivalent description

$$|\psi\rangle \leftrightarrow (-1)^{N-1} \sum_{\{x_1, \dots, x_N\} \in Q_N} \psi(x_1, \dots, x_N | Q_N) \hat{c}_{x_N}^\dagger \dots \hat{c}_{x_1}^\dagger |\Omega\rangle$$

where it is remarkable that *this fermionic wavefunction is the same as the spin picture*, since the suitable symmetry/antisymmetry property is encoded inside the corresponding creation and annihilation operators. Owing to the restriction to the single simplex Q_N , Bethe's ansatz to find $\psi(x_1, \dots, x_N | Q_N)$ is identical.

Note that the phase $(-1)^{N-1}$ arising from the JW transform can be arbitrarily removed from the above definition of the fermionic Bethe eigenstates.

As parenthesis, we mention that *this still holds for properly bosonic systems*, where the only difference is that in general equal site-indices are admitted in $\{x_1, \dots, x_N\}$ in absence of hard-core constraints, thus the simplex is rather defined by $x_1 \leq \dots \leq x_N$.

As we motivated in the previous paragraph, Bethe's proposal was to look for N -body eigenstates assuming their wavefunctions to be linear superpositions of $N!$ plane waves, each corresponding to a permutation $\sigma \in \mathcal{P}_N$ of the N selected quasimomenta $\{k_1, \dots, k_N\}$.

$$\psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N) := \sum_{\sigma \in \mathcal{P}_N} A_\sigma(\{k_1, \dots, k_N\}) e^{i \sum_{q=1}^N k_{\sigma(q)} x_q} \quad \begin{array}{l} \text{Bethe-ansatz} \\ \text{wavefunction} \end{array} \quad (2.2)$$

We recall that, from the Pauli-like principle of the operators $\hat{S}^+$ and $\hat{c}^\dagger$, the wavefunction must vanish when $x_n = x_{n+1}$. This holds for both spin and fermionic systems.

Total Momentum, Energy and $\{A_\sigma\}$ of a Bethe State with Fixed Quasimomenta: Up to now, we have not yet defined any quantization condition for quasimomenta. However, we temporarily shelve this issue and assume that a suitable set of quasimomenta $\{k_1, \dots, k_N\}$ has been chosen. We aim to determine the corresponding energy E and coefficients A_σ [62, 135].

In these calculations we assume an infinite lattice, i.e. we accept in $\psi(x_1, \dots, x_N | Q_N)$ arguments spanning all $\mathbb{Z}$. In particular, what we use is that $x_j \pm 1$ is always a well defined position to plug into (2.2) without any complication. In the next chapter we explicitly require the domain of $\psi(x_1, \dots, x_N | Q_N)$ to be extended out of $x \in \{1, \dots, L\}$ by periodicity.

However, the extension of ψ outside the physical interval is merely a convenient mathematical device used to ensure that the results derived here remain valid on a finite lattice. Indeed, in (2.2) we are never required to evaluate ψ out of $x \in \{1, \dots, L\}$. This requirement leads to a quantization condition for the quasimomenta, namely the Bethe equations.

By consistency, in this paragraph we mean Kronecker deltas as periodically extended:

$$\delta_{i,j} := \delta_{\text{mod}(i-1, L)+1, \text{mod}(j-1, L)+1}$$

Let us now come back to total momentum. The total momentum operator is defined so that the *one-step shift operator* is given by $e^{i\hat{K}}$. The latter acts on a generic wavefunction as

$$e^{i\hat{K}} \phi(x_1, \dots, x_N) := \phi(x_1 + 1, \dots, x_N + 1)$$

Remarkably, *the Bethe wavefunction is an eigenstate of $e^{i\hat{K}}$* , since

$$\psi^{\{k_1, \dots, k_N\}}(x_1 + 1, \dots, x_N + 1) = \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_{j=1}^N k_{\sigma(j)}(x_j + 1)} = \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_{j=1}^N k_{\sigma(j)} x_j} e^{i \sum_{j=1}^N k_{\sigma(j)}}$$

where $e^{i \sum_{j=1}^N k_{\sigma(j)}} = e^{i \sum_{j=1}^N k_j}$ independently of $\sigma \in \mathcal{P}_N$

This exponential can be factorized out of $\sum_{\sigma}$ and corresponds to the eigenvalue of $e^{i\hat{K}}$.

$$\rightarrow \langle \hat{K} \rangle = \sum_{j=1}^N k_j \mod(2\pi) \quad \text{Total momentum of a Bethe eigenstate}$$

As in §1, we stress that, by defining the one-step shift operator as $e^{i\hat{K}}$, we are implicitly assuming that the spatial coordinate has unit lattice spacing. Thus, two total momentum spaced by $\Delta K \in 2\pi\mathbb{Z}$ are physically equivalent and we can (arbitrarily) restrict $\langle \hat{K} \rangle$ values inside $[0, 2\pi]$ by applying $\mod(2\pi)$. Note that, despite the quasimomenta $\{k_j\}$ can have complex values (see §2.3), $\hat{K}$ is Hermitian, thus their sum $\langle \hat{K} \rangle$ has to be real. We recall that quasimomenta are non-observable quantities, without any direct physical meaning, thus it does not matter if they are complex.

Computing total energy is a bit more cumbersome, since it does not simply depend on the wavefunction structure, but also on the action of the Hamiltonian on it.

$$E |k_1, \dots, k_N\rangle = \hat{H}_{XXZ} |k_1, \dots, k_N\rangle \quad (2.3)$$

By expliciting the decomposition (2.1) in the computational basis and then projecting on each of its vectors with N up spins, one obtains $\binom{N}{L}$ independent equations.

In order to explicitly write them, it is useful to split $\hat{H}_{XXZ}$ into an hopping term $\hat{H}_{hop}$, which exchanges spins in adjacent sites, and the Ising term $\hat{H}_I$, which is diagonal in the computational basis. The component along $S_{x_N}^+ \dots S_{x_1}^+ |\downarrow\downarrow\downarrow \dots\rangle$ of $\hat{S}_j^z \hat{S}_{j+1}^z |k_1, \dots, k_N\rangle$ is given by:

$$\begin{aligned} & \frac{1}{4} \left[2 \left(\sum_{k=1}^N \delta_{j, x_k} \right) - 1 \right] \left[2 \left(\sum_{q=1}^N \delta_{j+1, x_q} \right) - 1 \right] \psi(x_1, \dots, x_N | Q_N) = \\ & = \left\{ \sum_{k=1}^N \left[\delta_{j, x_k} \delta_{x_{k+1}, x_k+1} - \frac{1}{2} (\delta_{j, x_k} + \delta_{j+1, x_k}) \right] + \frac{L}{4} \right\} \psi(x_1, \dots, x_N | Q_N) \end{aligned}$$

The corresponding component of the entire $\hat{H}_I |k_1, \dots, k_N\rangle = J\Delta \sum_j \hat{S}_j^z \hat{S}_{j+1}^z |k_1, \dots, k_N\rangle$ is

$$\langle \downarrow\downarrow\downarrow \dots | S_{x_1}^- \dots S_{x_N}^- \hat{H}_I |k_1, \dots, k_N\rangle = J\Delta \left\{ \sum_{k=1}^N [\delta_{x_{k+1}, x_k+1} - 1] + \frac{L}{4} \right\} \psi(x_1, \dots, x_N | Q_N)$$

Instead, $\hat{H}_{hop}$ is nondiagonal in the computational basis.

The component $\psi(x'_1, \dots, x'_N | Q_N)$ of $|k_1, \dots, k_N\rangle$ can contribute to the projection along the vector $S_{x_N}^+ \dots S_{x_1}^+ |\downarrow\downarrow\downarrow \dots\rangle$ of $\hat{H}_{hop} |k_1, \dots, k_N\rangle$ if and only if $\{x'_1, \dots, x'_N\}$ and $\{x_1, \dots, x_N\} \in \{0, \dots, L\}^N$ differ for exactly one site-index, but the former becomes the latter after a suitable one-step hopping.

$$\begin{aligned} & \langle \downarrow\downarrow\downarrow \dots | S_{x_1}^- \dots S_{x_N}^- \hat{H}_{hop} |k_1, \dots, k_N\rangle = \\ & = \frac{J}{2} \sum_{q=1}^N [1 - \delta_{x_q+1, x_{q+1}}] [\psi(x_1, \dots, x_q, x_{q+1}-1, \dots, x_N | Q_N) + \psi(x_1, \dots, x_q+1, x_{q+1}, \dots, x_N | Q_N)] \end{aligned}$$

By plugging these results into (2.3) one finally obtains

$$\begin{aligned} 0 & = \left[J\Delta \sum_{q=1}^N (\delta_{x_{q+1}, x_q+1} - 1) - \left(E - \frac{J\Delta L}{4} \right) \right] \psi(x_1, \dots, x_N | Q_N) + \\ & + \frac{J}{2} \sum_{q=1}^N [1 - \delta_{x_q+1, x_{q+1}}] [\psi(x_1, \dots, x_q, x_{q+1}-1, \dots, x_N | Q_N) + \psi(x_1, \dots, x_q+1, x_{q+1}, \dots, x_N | Q_N)] \end{aligned} \quad (2.4)$$

At this point we are able to compute the energy by considering an element of Q_N in which there are no adjacent $\uparrow\uparrow$. This is always possible if $N \leq L/2$, but even when $N \leq L/2$ we can always map our eigenstate onto a degenerate eigenstate with $N \leq L/2$ by global spin-flip.

In this case $\delta_{x_{q+1}, x_{q+1}} = 0 \ \forall k$ and we can exploit the following property, for $\alpha = \pm 1$:

$$\begin{aligned} \sum_q \psi(x_1, \dots, x_q + \alpha, \dots, x_N + 1 | Q_N) &= \sum_q \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_{j=1}^N k_{\sigma(j)} x_j} e^{i \alpha k_{\sigma(q)}} = \\ &= \left(\sum_q e^{i \alpha k_{\tilde{\sigma}(j)}} \right) \psi(x_1, \dots, x_N + 1 | Q_N) \end{aligned}$$

where $\tilde{\sigma} \in \mathcal{P}_N$ is an arbitrarily chosen permutation. Finally we obtain

$$E = E_0 + J \sum_{i=1}^N (\cos(k_i) - \Delta) \quad E_0 = \frac{J\Delta L}{4} \quad \text{Total energy} \quad (2.5)$$

where we denoted by E_0 the energy of the vacuum state $|\downarrow\downarrow\downarrow \dots\rangle$ which is the same as that of the other fully polarized state $|\uparrow\uparrow\uparrow \dots\rangle$. We recall that, despite the name, in general this is not the ground state of $\hat{H}_{XXZ}$: only when $J\Delta \rightarrow -\infty$ (i.e. ferromagnetic Ising model) the fundamental eigenspace is $\text{span}\{|\downarrow\downarrow\downarrow \dots\rangle, |\uparrow\uparrow\uparrow \dots\rangle\}$.

Note that the energy E of a Bethe eigenstate $|k_1, \dots, k_N\rangle$ only depends on the set of quasimomenta and not on their order. This implies that the space of wavefunctions corresponding to a given set $\{k_j\}$ can be decomposed as a direct sum of some irreducible representations of the permutation group $\mathcal{P}_N$. In the next paragraph we will prove the stronger statement that this space coincides with a single irreducible representation.

We have not yet fully exploited (2.4), since we can still consider the situations in which there exist configurations $\uparrow\uparrow$. Plugging the above value of E into (2.4) leads to

$$\begin{aligned} 0 &= \left[J\Delta \sum_{q=1}^N \delta_{x_{q+1}, x_{q+1}} \right] \psi(x_1, \dots, x_N | Q_N) - \\ &- \frac{J}{2} \sum_{q=1}^N \delta_{x_{q+1}, x_{q+1}} \left[\psi(x_1, \dots, x_q, x_{q+1} - 1, \dots, x_N | Q_N) + \psi(x_1, \dots, x_q + 1, x_{q+1}, \dots, x_N | Q_N) \right] \end{aligned}$$

$$\text{By expanding } \psi \text{ with (2.1):} \quad 0 = \frac{J}{2} \sum_{q=1}^N \delta_{x_{q+1}, x_{q+1}} \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_{j \notin \{q, q+1\}} k_{\sigma(j)} x_j}.$$

$$\cdot \left[2\Delta e^{i[k_{\sigma(q)} x_q + k_{\sigma(q+1)}(x_{q+1})]} - e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]x_q} - e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}](x_{q+1})} \right] =$$

$$0 = \sum_{q=1}^N \delta_{x_{q+1}, x_{q+1}} \sum_{\sigma \in \mathcal{P}_N} e^{i \sum_{j \notin \{q, q+1\}} k_{\sigma(j)} x_j} e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]x_q} A_\sigma \left[2\Delta e^{i k_{\sigma(q+1)}} - 1 - e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]} \right]$$

The first two exponentials are fully symmetric under a permutation $\pi_{q, q+1}$ applied before σ . Thus we can equivalently recast the above equation in the following way:

$$\begin{aligned} 0 &= \sum_{q=1}^N \delta_{x_{q+1}, x_{q+1}} \sum_{\sigma \in \mathcal{P}_N} e^{i \sum_{j \notin \{q, q+1\}} k_{\sigma(j)} x_j} e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]x_q} \cdot \\ &\cdot \frac{1}{2} \left[A_\sigma (1 + e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]}) - 2\Delta e^{i k_{\sigma(q+1)}} + A_{\sigma \pi_{q, q+1}} (1 + e^{i[k_{\sigma(q+1)} + k_{\sigma(q)}]}) - 2\Delta e^{i k_{\sigma(q)}} \right] \end{aligned}$$

This is clearly satisfied if

$$\frac{A_\sigma}{A_{\sigma\pi_{q,q+1}}} = -\frac{1 + e^{i[k_{\sigma(q+1)} + k_{\sigma(q)}]} - 2\Delta e^{ik_{\sigma(q)}}}{1 + e^{i[k_{\sigma(q)} + k_{\sigma(q+1)}]} - 2\Delta e^{ik_{\sigma(q+1)}}} \quad \forall \sigma \in \mathcal{P}_N \quad (2.6)$$

Then every possible $A_{\sigma'}$ with $\sigma' = \sigma\tau$ can be obtained from A_σ by a chain rule, since every possible permutation τ can be decomposed as a sequence of adjacent exchanges $\pi_{q,q+1}$:

$$\frac{A_\sigma}{A_{\sigma\pi_{q,q+1}\pi_{k,k+1}}} = \frac{A_\sigma}{A_{\sigma\pi_{q,q+1}}} \cdot \frac{A_{\sigma\pi_{q,q+1}}}{A_{\sigma\pi_{q,q+1}\pi_{k,k+1}}} \quad (2.7)$$

This way, all the $\{A_\sigma\}$ result to be fixed up to a global normalization constant $\mathcal{N}$ [135]:

$$A_\sigma = \mathcal{N} \cdot \text{sgn}(\sigma) \prod_{j=1}^N \prod_{0 < q < j} (1 + e^{i[k_{\sigma(j)} + k_{\sigma(q)}]} - 2\Delta e^{ik_{\sigma(q)}}) \quad \{A_\sigma\}\text{-Coefficients} \quad (2.8)$$

Pauli-Like Property for the Bethe-Wavefunction Momenta: Given $\pi(k_i, k_j)$ the operator exchanging the variables $k_i \leftrightarrow k_j$ of a generic function $f(k_1, \dots, k_N)$, the coefficients (2.8) enjoy

$$\pi(k_i, k_j) A_\sigma = -A_{\sigma\pi_{ij}}$$

$$\text{This implies that } \pi(k_i, k_j) \psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N) = -\psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N)$$

i.e. the wavefunctions $\psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N)$ transform like an irreducible representation of the permutations acting on momenta and in particular are antisymmetric under pair exchange. This means that all these wavefunctions correspond to the same eigenstate $|k_1, \dots, k_N\rangle$ up to a non-physical phase. This is consistent with our previous observation that the energy of an eigenstate does not depend on the order of quasimomenta.

We emphasize that this is not linked to the antisymmetry in coordinates of fermionic systems. Indeed, we have already observed that the required symmetry properties in (2.1) are already encoded in the commutation or anticommutation rules of the creation operators. Moreover, the found $\{A_\sigma\}$ return the correct eigenstates for both the fermionic and spin picture, where the second is nor bosonic nor fermionic. The properly bosonic Lieb-Lininger model has a Bethe wavefunction which is manifestly anti-symmetric in quasimomenta as well [135].

In all these cases, sets $\{k_1, \dots, k_N\}$ with two equal quasimomenta do not correspond to meaningful eigenstates. This is sometimes called *Pauli-like property of Bethe eigenstates*.

However, once we restrict to sets $\{k_1, \dots, k_N\}$ with all distinct momenta, then the antisymmetry property has no physical meaning. More formally, quasimomenta are not observables, but just quantum numbers which identify eigenstates. One can define an operator $\hat{\pi}(k_i, k_j)$ acting on $\hat{H}$, but its definition is tautologically given by its action on the states $|k_1, \dots, k_N\rangle$. With our convention $\hat{\pi}(k_i, k_j) |k_1, \dots, k_N\rangle = -|k_1, \dots, k_N\rangle$. Nevertheless, we can arbitrarily modify (2.2) changing a global sign depending on the permutation of momenta such that $\psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N)$ is symmetric. With this new convention $\hat{\pi}(k_i, k_j) |k_1, \dots, k_N\rangle = |k_1, \dots, k_N\rangle$.

Henceforth we can assume every Bethe eigenstate to be one-to-one defined by the not-ordered set of quasimomenta $\{k_j\}$, being the Bethe wavefunction (2.2) built from it following some re-ordering prescription. This way our discussion becomes explicitly independent of the chosen convention. For example, when in §7 we will mention the second-quantization-like construction of Bethe eigenstates, creation operators will turn out to commute.

2.2 S-MATRIX, BETHE EQUATIONS AND RAPIDITIES

S-Matrix, Factorization and Order-Independence in "Integrable Scattering": Summarizing, in the previous chapter we observed that starting from Bethe's ansatz for eigenstate wavefunctions (2.2) and requiring $\hat{H} |k_1, \dots, k_N\rangle = E |k_1, \dots, k_N\rangle$, then one finds the relations (2.6) between

the A_σ -coefficients. These have the form of ratios $A_\sigma/A_{\sigma\pi}$ only depending on the two exchanged quasimomenta. Since these ratios play a very pivotal role in all developments of the Bethe-ansatz theory, it is convenient to work directly with them rather than the A_σ -coefficients.

To this end we define the S_2 -matrix in momentum space. It is often recast in exponential form by introducing the corresponding *phase shift* θ_2 . We also draw a pictorial representation, which will turn out convenient later, to be read from right to left.

$$S_2(k, k') := -\frac{e^{i(k+k')} + 1 - 2\Delta e^{ik'}}{e^{i(k+k')} + 1 - 2\Delta e^{ik}} := -e^{-i\theta_2(k, k')} \quad \frac{1}{S_2(k, k')} := \begin{array}{c} k \leftarrow \text{---} \text{---} k' \\ \text{---} \text{---} k' \leftarrow k \end{array}$$

$$\frac{A_\sigma}{A_{\sigma\pi_{q,q+1}}} = \frac{1}{S_2(k_{\sigma(q)}, k_{\sigma(q+1)})} := \begin{array}{c} k_{\sigma(q)} \leftarrow \text{---} \text{---} k_{\sigma(q+1)} \\ \text{---} \text{---} k_{\sigma(q+1)} \leftarrow k_{\sigma(q)} \end{array} \quad (2.9)$$

The S_2 -matrix, or equivalently the phase shift θ_2 , is the elementary building block of every Bethe-ansatz-based theory (coordinate, thermodynamic, algebraic...), *encoding all the relevant information about the peculiar integrable model considered*. A Bethe-ansatz solvable model is usually called "(genuinely) interacting" if its S_2 -matrix is momentum-dependent.

Notice the following inversion property:

$$\frac{1}{S_2(k_{\sigma(q)}, k_{\sigma(q+1)})} = \frac{A_\sigma}{A_{\sigma\pi_{q,q+1}}} = \frac{A_{\sigma\pi_{q,q+1}\pi_{q,q+1}}}{A_{\sigma\pi_{q,q+1}}} = S_2(k_{\sigma(q+1)}, k_{\sigma(q)})$$

$$\rightarrow \begin{cases} S_2(k, k) = -1 \\ S_2(k, k') S_2(k', k) = 1 \end{cases} \quad \begin{array}{c} k' \leftarrow \text{---} \text{---} k' \\ k \leftarrow \text{---} \text{---} k \end{array} = \begin{array}{c} k' \leftarrow k' \\ k \leftarrow k \end{array}$$

In principle $\theta_2(k, k')$ is defined modulo 2π . Throughout this work we choose the convention

$$\theta_2(k, k') = i \log(-S_2(k, k'))$$

where $\log$ denotes the standard branch of the complex logarithm, with the cut along the negative real axis. With this choice, the phase shift turns out to be antisymmetric.

$$\rightarrow \begin{cases} \theta_2(k, k) = 0 \\ \theta_2(k, k') = -\theta_2(k', k) \end{cases} \quad (2.10)$$

As discussed in the previous chapter, the ratios $A_\sigma/A_{\sigma\pi}$ satisfy the chain rule (2.7) which enable the computation of $A_\sigma/A_{\sigma'}$ with σ and σ' connected by an arbitrary permutation τ , i.e. $\sigma' = \tau\sigma$. From (2.9), it follows that for every exchange of adjacent quasimomenta, the corresponding $1/S_2(k, k')$ factor appears. For example, in deriving the Bethe equations we will deal with the cyclic permutation $\tau_{1 \rightarrow N} := 1 \rightarrow N \rightarrow (N-1 \rightarrow \dots \rightarrow 2 \rightarrow 1)$. The simplest way to decompose it with adjacent exchanges is $\pi_{1N} \dots \pi_{13} \pi_{12}$, hence

$$\frac{A_\sigma}{A_{\sigma\tau_{1 \rightarrow N}}} = \begin{array}{c} k_{\sigma(1)} \leftarrow \text{---} \text{---} k_{\sigma(N)} \\ k_{\sigma(N)} \leftarrow \text{---} \text{---} k_{\sigma(N-1)} \\ \dots \leftarrow \text{---} \text{---} \dots \\ k_{\sigma(3)} \leftarrow \text{---} \text{---} k_{\sigma(2)} \\ k_{\sigma(2)} \leftarrow \text{---} \text{---} k_{\sigma(1)} \end{array} = \prod_{q=2}^N \frac{1}{S_2(k_{\sigma(1)}, k_{\sigma(q)})} \quad (2.11)$$

A remarkable fact is that, *in general, there are infinite possible sequences of exchanges leading from σ to σ' and they must be all equivalent, i.e. the ratio $A_\sigma/A_{\sigma'}$ must be independent of the particular chosen path*. Trivially, applying two times in sequence the same exchange have to be equal to apply no exchange, consistently with the inversion property (2.10).

A more interesting relation is obtained if we look for equivalence rules for permutations of three objects, e.g.

$$(2.12)$$

consistently with $S_2(k_b, k_c) S_2(k_a, k_c) S_2(k_a, k_b) = S_2(k_a, k_b) S_2(k_a, k_c) S_2(k_b, k_c)$

The above picture expresses an equivalence among two permutation in which the same three exchanges are performed in a different order. Remarkably, it coincides with the pictorial representation of the Yang-Baxter equation in (7.2), which is the fundamental equation of the algebraic Bethe ansatz (see §7). Indeed, the exchange operator is probably the simplest solution of the Yang-Baxter equation after identity.

Coming back to the coordinate Bethe ansatz, when rewritten in terms of S_2 -matrices this result is trivial, since they are scalars and commute with each other. However, this triviality arose from the very special fact that in Bethe-ansatz integrable models the ratio $A_\sigma / A_{\sigma\pi_{q,q+1}}$ only depends on the two exchanged quasimomenta through a fixed function $S_2(k, k')$.

Physically, if we view the terms included in the Bethe wavefunction as states connected by scattering events (as in our original heuristic justification of the ansatz), we can think of Yang-Baxter integrability as the property that every N -quasiparticle scattering is factorizable into a sequence of 2-particle elastic scatterings, where the order of this decomposition does not matter.

Bethe Equations: In the previous calculations we admitted ψ to have arguments out of $\{0, \dots, L\}$, by assuming it extended by periodicity out of this domain. This way x_q and $x_q + (\alpha L + 1)$ were meant to lead to a $\uparrow\uparrow$ configuration. By consistency, Kronecker deltas were extended this way as well. However, we have not explicitly investigated the consequences of this requirement on the wavefunction, yet. This is the starting point of this paragraph. As in the elementary problem of the free particle in a box, it will lead to a quantization condition for quasimomenta.

Formally, PBC in the simplex Q_N correspond to

$$\psi^{\{k_1, \dots, k_N\}}(x_2, \dots, x_N, x_1 + L | Q_N) = \psi^{\{k_1, \dots, k_N\}}(x_1, \dots, x_N | Q_N)$$

By plugging this requirement into (2.2), one obtains

$$\begin{aligned} \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_j k_{\sigma(j)} x_j} &= \sum_{\sigma \in \mathcal{P}_N} A_\sigma e^{i \sum_j k_{\sigma(j)} x_{\tau_{1 \rightarrow L}(j)}} e^{i k_{\sigma(L)} L} = \\ &= \sum_{\sigma \in \mathcal{P}_N} A_{\sigma \tau_{1 \rightarrow L}} e^{i \sum_j k_{\sigma \tau_{1 \rightarrow L}(j)} x_{\tau_{1 \rightarrow L}(j)}} e^{i k_{\sigma \tau_{1 \rightarrow L}(L)} L} = \quad \text{where} \quad \left[\sum_j k_{\sigma \tau_{1 \rightarrow L}(j)} x_{\tau_{1 \rightarrow L}(j)} = \sum_j k_{\sigma(j)} x_j \right] \\ &= \sum_{\sigma \in \mathcal{P}_N} A_{\sigma \tau_{1 \rightarrow L}} e^{i \sum_j k_{\sigma(j)} x_j} e^{i k_{\sigma(1)} L} \end{aligned}$$

Equalling terms with the same $e^{i \sum_j k_{\sigma(j)} x_j}$ and recalling (2.11),

$$e^{i k_{\sigma(1)} L} = \frac{A_\sigma}{A_{\sigma \tau_{1 \rightarrow L}}} = \prod_{q=2}^N \frac{1}{S_2(k_{\sigma(1)}, k_{\sigma(q)})} \quad \forall \sigma \in \mathcal{P}_N$$

To have $e^{i k_j L}$ on the left, we can choose $\sigma = \pi_{1,j}$. This finally leads to the so-called *Bethe equations, representing quantization conditions for quasimomenta*.

$$e^{i k_j L} \prod_{q \neq j}^N S_2(k_j, k_q) = 1 \quad \forall j \in \{1, 2, \dots, N\}$$

Since $S_2(k, k) = -1$, it is possible to reintroduce in the product $k = j$ by a simple sign change.

Rewriting in Terms of Rapidities: Since in general (and here, too) the fundamental object S_2 (and θ_2) has the unpleasant property of not being invariant under a global shift of momenta, it is convenient to re-parametrize eigenstates in terms of *rapidities* $\{\lambda_j\}$ specifically defined to grant translational invariance. For the XXZ model the conversion relation is [62, 134, 135]

$$e^{ik(\lambda)} = \frac{\sin^{[+]}(\lambda)}{\sin^{[-]}(\lambda)} \quad \text{s.t.} \quad \text{Re}(\lambda) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right) \quad \text{momentums} \leftrightarrow \text{rapidities} \quad (2.13)$$

where $[-\pi/2, \pi/2]$ has been arbitrarily chosen, but the restriction to a π -width interval of $\text{Re}[\lambda]$ comes from the π -periodicity of this specific conversion relation (i.e. $p(\lambda + \pi) = p(\lambda)$).

$$-e^{ik(\lambda_j)L} \prod_{k=1}^N S_2(\lambda_j - \lambda_k) = 1 \quad \forall j \in \{1, 2, \dots, N\} \quad \text{Bethe equations} \quad (2.14)$$

This way, the Bethe equations become primarily a quantization condition for rapidities.

Note that *in principle Bethe equations admit solutions with coincident rapidities (i.e. coincident quasimomenta). However, their derivation from Bethe's ansatz for the wavefunction required rapidities to be all different. Solutions of (2.14) containing equal rapidities do not lead to meaningful eigenstates.* Thus, henceforth we restrict a priori to solutions which are consistent with the Pauli-like property of the Bethe eigenstates: we require rapidities to be all different.

In the XXZ model, by introducing the following notation [134]

$$S_n(\lambda) := \frac{\sin^{[-n]}(\lambda)}{\sin^{[+n]}(\lambda)} = \frac{e^{i\lambda+n\eta} - e^{-(i\lambda+n\eta)}}{e^{i\lambda-n\eta} - e^{-(i\lambda-n\eta)}} := -e^{-i\theta_n(\lambda)} \quad (2.15)$$

$$\text{it follows that:} \quad e^{ik(\lambda)} = \frac{1}{S_1(\lambda)} = -e^{i\theta_1(\lambda)} \quad S_2[k(\lambda), k(\mu)] = S_2(\lambda - \mu) = -e^{-i\theta_2(\lambda)}$$

The explicit form of θ_n from the above definition is

$$\theta_n(\lambda) = 2 \arctan\left(\frac{\tan(\lambda)}{\tanh(n\eta/2)}\right) \quad \theta_n(\lambda) \in [-\pi, \pi] \quad \text{if } \lambda \in \mathbb{R} \quad (2.16)$$

$$\theta_n\left(-\frac{\pi}{2}\right) = -\pi \quad \theta_n\left(\frac{\pi}{2}\right) = \pi \quad \forall n \in \mathbb{N}^{>0}$$

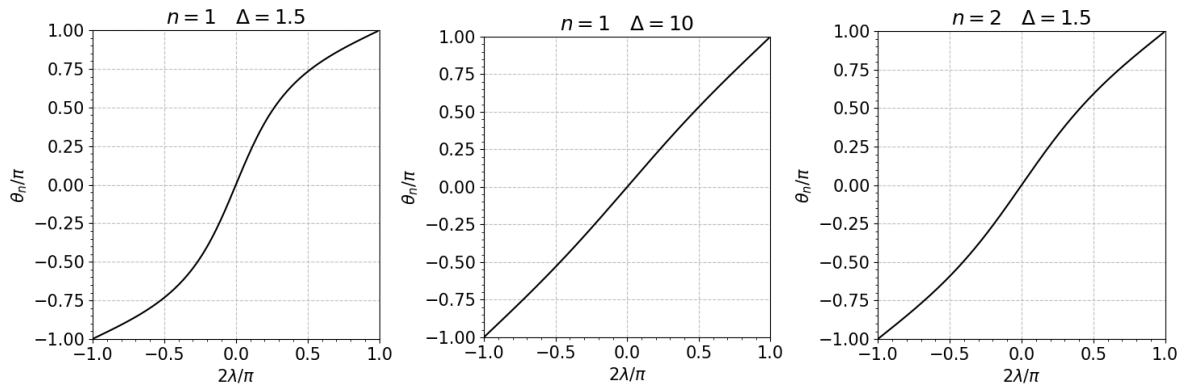


FIGURE 2.1: The $\theta_n(\lambda)$ function for various choices of Δ and n . For large anisotropies (e.g. $\Delta = 10$) $k(\lambda) = \theta_1(\lambda) + \pi \approx 2\lambda + \pi$, since the XXZ model is nearly the Ising model.

In Fig.2.1 we plotted the above defined θ_n for some choices of Δ and n . This definition maps the $[-\pi/2, \pi/2)$ rapidity interval onto $\theta_n \in [-\pi, \pi)$ in a monotonic and continuous way.

Since quasimomenta are defined up to a 2π -translation, if we choose as *first Brillouin zone* $k \in [0, 2\pi)$ and restrict ourselves to quasimomenta inside it, then we can rewrite (2.13) as

$$k(\lambda) := \theta_1(\lambda) + \pi$$

Before proceeding, an instructive observation is that, from both the above formula and the pictures, $k(\lambda) = \theta_1(\lambda) + \pi = 2\lambda + \pi$ as $\Delta \rightarrow +\infty$, i.e. $\tanh(n\eta/2) = 1$. This is consistent with the fact that for infinite anisotropy the XXZ becomes the Ising model, which is mapped by JW-transforming onto the free-fermion Kitaev model, whose eigenstates are Slater determinants, without any need of the Bethe ansatz and rapidities.

Taking the logarithm of (2.14), there must exist an integer $\alpha_j \in \mathbb{Z}$ s.t.

$$\theta_1(\lambda_j) = k(\lambda_j) - \pi = \frac{2\pi}{L} \left(\alpha_j - \frac{(L-N)}{2} \right) + \frac{1}{L} \sum_{k=1}^N \theta_2(\lambda_j - \lambda_k)$$

$$\theta_1(\lambda_j) = k(\lambda_j) - \pi = 2\pi \frac{\tilde{J}_j}{L} + \frac{1}{L} \sum_{k=1}^N \theta_2(\lambda_j - \lambda_k) \quad \tilde{J}_j \begin{cases} \text{integer} & \leftrightarrow L-N \text{ even} \\ \text{semi-integer} & \leftrightarrow L-N \text{ odd} \end{cases} \quad (2.17)$$

A set of N rapidities $\{\lambda_j \in [-\pi/2, \pi/2]\}$ is a solution of the Bethe equations (i.e. corresponds to a wavefunction with quasimomenta $\{k(\lambda_j)\}$ satisfying PBC on Q_N) if and only if there exists a set of N (semi-)integers $\{\tilde{J}_j\}$ satisfying (2.17).

Bethe Numbers: We mentioned above that, in order for the wavefunction not to vanish identically, all rapidities (thus all quasimomenta) must be pairwise distinct. If we assume that, for fixed $\{\tilde{J}_j\}$, the solution of the logarithmic Bethe equations, whenever it exists, is unique, then *the integers $\{\tilde{J}_j\}$ have to be all different* as well.

Moreover, if a set $\{\tilde{J}_j\}$ corresponding to a solution $\{\lambda_j\}$ is transformed by shifting one of its elements by L , namely $\tilde{J}_q \rightarrow \tilde{J}_q + L$, then this is equivalent to a 2π -shift of $k(\lambda_q)$, leading to a physically identical solution.

This means that the truly independent sets leading to possible solutions are only $\binom{L}{N}$.

Therefore, in order to have a more convenient classification of eigenstates, we can associate to every set $\{\tilde{J}_j\}$ admitting a solution a new set $\{\tilde{I}_j\}$, whose elements are called *Bethe numbers* of the state and where $\tilde{I}_j \in \{0, \dots, L-1\}$ coincides with $\tilde{J}_j$ modulo L .

If we assume that every set $\{\tilde{I}_j\} \in \{0, \dots, L-1\}^N$ corresponds to one and only one solution, then we have correctly found a complete basis of eigenstates, since $\binom{L}{N}$ is exactly the dimension of the Hilbert space with N quasiparticles, i.e. our method returns a complete set of eigenstates.

Summarizing, *Bethe equations are a set of N coupled nonlinear equations to be solved for fixed $\{\tilde{I}_j\}$ finding the corresponding set of N rapidities*: if we assume that every possible choice of Bethe numbers leads to one and only one solution for rapidities $\{\lambda_j\}$, then they fully identify a complete basis of eigenstates and can be regarded as its quantum numbers.

All these results have as assumption the existence and uniqueness of the Bethe-equations solution for fixed $\{\tilde{I}_j\}$. For the Lieb-Lininger model, mainly owing to the fact that rapidities can be required to be purely real, this is proven by introducing the so called Yang-Yang functional $B(k_1, \dots, k_N)$ [135, 136], whose extremal points are solutions of the Bethe equations: since the Hessian matrix $\partial^2 B / \partial k_i \partial k_j$ is positive definite, B has one and only one minimum.

By contrast, in XXX and XXZ models rapidities are in general complex and this method fails. Existence and uniqueness are taken as assumptions and an a priori investigation of solutions is highly nontrivial. In the next chapter we will introduce an additional hypothesis about the structure of solutions. This will allow us to recast the problem in a more accessible way.

2.3 STRING HYPOTHESIS FOR THE THERMODYNAMIC LIMIT

In few particularly simple models, e.g. the Bose gas with repulsive δ -potential, which was first solved by Lieb and Liniger by the Bethe ansatz in [88, 89], quasimomenta are all real. When considering an attractive potential, complex quasimomenta appear here as well.

In his original article about the solution of the XXX model [19], Bethe already noticed that real rapidities do not yield sufficiently many solutions to identify a complete basis of eigenstates. Thus it was necessary to admit pairs of rapidities with complex conjugate values.

Emergence of Strings in a "Pseudo-Thermodynamic" Limit: In the XXZ model (here we always assume $\Delta > 1$, since for $|\Delta| < 1$ the string condition is more complicated, see e.g. [135]), for finite (L, N) the Bethe equations (2.14) also admit solutions with complex rapidities. This seems problematic [62, 136] when taking $L \rightarrow +\infty$ while keeping finite the number of particles N . Note that this is not the proper thermodynamic limit, which requires $N/L = \text{constant}$.

$$|e^{ik(\lambda_j)L}| \rightarrow \begin{cases} 0 & \text{if } \text{Im}(k) > 0 \\ +\infty & \text{if } \text{Im}(k) < 0 \end{cases}$$

Let us consider first the $\text{Im}[k(\lambda_j)] > 0$ case: in order for (2.14) to be valid, at least one of the $S_2(\lambda_j - \lambda_k)$ must have a pole, i.e., looking at (2.16),

$$\exists \lambda_k \quad \text{s.t.} \quad i(\lambda_k - \lambda_j) = \eta \rightarrow \lambda_k = \lambda_j - i\eta$$

On the opposite, if $\text{Im}[k(\lambda_j)] < 0$, then at least one of the $S_2(\lambda_j - \lambda_k)$ must be zero, i.e.

$$\exists \lambda_k \quad \text{s.t.} \quad i(\lambda_k - \lambda_j) = -\eta \rightarrow \lambda_k = \lambda_j + i\eta$$

To proceed, we need to make two observations:

- For the XXZ model, the Bethe equations are invariant under $\{\lambda_j\} \rightarrow \{\lambda_j^*\}$ [144].
- Consider (2.16). The explicit expression of the $\arctan(y)$ function is

$$y = \tan(x) = \frac{1}{i} \left(\frac{e^{2ix} - 1}{e^{2ix} + 1} \right) \rightarrow x = \arctan(y) = \frac{\log(1 + iy) - \log(1 - iy)}{2i}$$

$$\text{Im}[k(\lambda)] = \text{Im}[\theta_1(\lambda)] = \text{Re} \left\{ \log \left[1 + \frac{(e^{2i\lambda} - 1)}{(e^{2i\lambda} + 1) \tanh(\eta/2)} \right] - \log \left[1 - \frac{(e^{2i\lambda} - 1)}{(e^{2i\lambda} + 1) \tanh(\eta/2)} \right] \right\}$$

As above, we consider the standard branch of the complex logarithm. Finally, it is possible to prove that $\text{Im}(\lambda)$ and $\text{Im}[k(\lambda)] = \text{Im}[\theta_1(\lambda)]$ have the same sign.

This implies that, as $L \rightarrow +\infty$, Bethe roots align themselves in configurations of n elements with same real part and symmetrically displaced along the imaginary direction with spacing η .

$$\{\lambda_j^{n,\alpha}\} := \left\{ \lambda^{n,\alpha} + (n+1-2j) \frac{i\eta}{2} \mid j = 1, 2, \dots, n \right\} \quad \text{where} \quad \lambda^{n,\alpha} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right) \quad \forall n, \alpha \quad (2.18)$$

This set of rapidities is said to be an n -string with central value $\lambda^{n,\alpha}$, belonging to the string only if n is odd. In the above definition α is an additional label distinguishing strings with same length but different central rapidity. We recall that, given the Pauli-like property of the wavefunction, there cannot be strings with both equal n and α .

Summarizing, these alignments are so that the products of the corresponding n Bethe equations have on the left $\exp \left[i \sum_{j=1}^n k(\lambda_j^{n,\alpha}) \right]$, with a purely real result for the sum. This way the exponential does not become zero or divergent for $L \rightarrow +\infty$ and the initial problem is solved.

Physically, using for simplicity the Lieb-Lininger model, it is possible to show [62, 135] that an n -string in the Bethe wavefunction can be viewed as signalling a *bound state of n quasiparticles*.

String Hypothesis and Bethe-Gaudin-Takahashi Equations: *The above proof fails within the proper thermodynamic limit, where N/L stays constant while $L \rightarrow +\infty$. Indeed, an infinite product of S_2 -matrices can have a pole/zero without this being a pole/zero of any individual S_2 .*

Thus, the presence of strings can only be regarded as an hypothesis: the *string hypothesis*.

To perform the proper thermodynamic limit of Bethe equations, we need to come back to a finite lattice of length $L < +\infty$ with PBC and only after take $L, N \rightarrow +\infty$ s.t. $N/L = \text{constant}$.

It is numerically observed that for configurations interpreted as bound states of n quasiparticles deviations from the string structure typically decay exponentially with L . In some extreme limits counter-examples are known. However, in most physically relevant cases the equations obtained under the string hypothesis reproduce the correct thermodynamics [135].

Let us henceforth assume the string hypothesis (2.18) to hold. The entire set of rapidities $\{\lambda_j\}$ is fully determined by the central rapidities $\{\lambda^{n,\alpha}\}$. It is thus convenient to write a specific formulation of (2.14) for these new independent variables.

Let N_n be the number of n -strings in an eigenstate. The equations corresponding to the n rapidities of a same string are not more independent and can be replaced by a single equation given by their multiplication. In doing this, one notices a very nice property:

$$\frac{1}{S_1(\lambda)S_1(\lambda+i\eta)} = \frac{\sin[\lambda+i\eta/2]}{\sin[\lambda-i\eta/2]} \frac{\sin[\lambda+i3\eta/2]}{\sin[\lambda+i\eta/2]} = \frac{\sin[\lambda+i\eta/2]}{\sin[\lambda-i\eta/2]}$$

By proceeding with similar chain cancellations, if we consider an n -string with central (real) rapidity $\bar{\lambda} \in [-\pi/2, \pi/2]$, it follows that

$$\lambda_0 := \bar{\lambda} - i \frac{(n-1)}{2} \eta \rightarrow \frac{1}{\prod_{b=0}^{n-1} S_1(\lambda_0 + i b \eta)} = \frac{\sin[\lambda_0 + i(n-\frac{1}{2})\eta]}{\sin[\lambda_0 - i\frac{\eta}{2}]} = \frac{\sin^{[+n]}(\bar{\lambda})}{\sin^{[-n]}(\bar{\lambda})} = \frac{1}{S_n(\bar{\lambda})} \quad (2.19)$$

This allows one to rewrite a particularly simple expression for the total momentum of an n -string, only depending on its central (real) rapidity.

$$e^{ik_{tot}^{n,\alpha}} = \frac{1}{\prod_{j=1}^n S_1(\lambda_j^{n,\alpha})} = \frac{1}{S_n(\lambda^{n,\alpha})} = -e^{i\theta_n(\lambda^{n,\alpha})}$$

Note that the $k = j$ case has been reintroduced by a -1 global multiplication. Simultaneously, recasting the product in the Bethe equations (2.14) in terms of strings

$$-\left(\frac{1}{S_1(\lambda_j^{n,\alpha})}\right)^L = \prod_{m=1}^{+\infty} \prod_{\beta=1}^{N_m} \frac{1}{\prod_{l=1}^m S_2(\lambda_j^{n,\alpha} - \lambda_l^{m,\beta})}$$

and multiplying these expressions for each rapidity in the (n, α) string:

$$(-1)^n \left(\frac{1}{S_n(\lambda^{n,\alpha})}\right)^L = \prod_{m=1}^{+\infty} \prod_{\beta=1}^{N_m} \frac{1}{\prod_{j=1}^n \prod_{l=1}^m S_2(\lambda_j^{n,\alpha} - \lambda_l^{m,\beta})} \quad (2.20)$$

It can be proven that the denominator can be rewritten as a function of the two n - and m -string central rapidities only.

$$\prod_{j=1}^n \prod_{l=1}^m S_2(\lambda_j^{n,\alpha} - \lambda_l^{m,\beta}) = S_{nm}(\lambda^{n,\alpha} - \lambda^{m,\beta}) \quad \text{where}$$

$$\begin{aligned} S_{nm}(\lambda) &:= S_{|n-m|}(\lambda) S_{|n-m|+2}^2(\lambda) S_{|n-m|+4}^2(\lambda) \dots S_{n+m-2}^2(\lambda) S_{n+m}(\lambda) := e^{-i\theta_{nm}(\lambda)} \quad [S_0(\lambda) = 1] \\ \rightarrow \quad \theta_{nm}(\lambda) &= \theta_{|n-m|}(\lambda) + 2 \left(\sum_{k=1}^{\min[m,n]-1} \theta_{|m-n|+2k}(\lambda) \right) + \theta_{n+m}(\lambda) \quad [\theta_0(\lambda) = \pi] \end{aligned} \quad (2.21)$$

By plugging this result into (2.20) one obtains the string recasting of Bethe equations

$$\begin{aligned}
 (-1)^n \left(\frac{1}{S_n(\lambda^{n,\alpha})} \right)^L &= \prod_{m=1}^{+\infty} \prod_{\beta=1}^{N_m} \frac{1}{S_{nm}(\lambda^{n,\alpha} - \lambda^{m,\beta})} && \text{Bethe-Gaudin-Takahashi (BGT) equations} \\
 \iff \quad \exists J^{n,\alpha} \quad \text{s.t.} \quad \theta_n(\lambda^{n,\alpha}) &= 2\pi \frac{J^{n,\alpha}}{L} + \frac{1}{L} \sum_{m=1}^{+\infty} \sum_{\beta=1}^{N_m} \theta_{nm}(\lambda^{n,\alpha} - \lambda^{m,\beta}) && (2.22) \\
 \text{where} \quad J^{n,\alpha} &\begin{cases} \text{integer} & \iff L+n \text{ even} \\ \text{semi-integer} & \iff L+n \text{ odd} \end{cases}
 \end{aligned}$$

Note that $\theta_{nm}(0) = \pi \delta_{nm}$, since $\theta_n(0) = \pi \delta_{n0}$.

Analogously to the $\{\tilde{J}^{n,\alpha}\}$ in (2.17):

- By the Pauli-like property of the Bethe wavefunction $J^{n,\alpha} \neq J^{m,\beta}$ if n and m have the same parity, to avoid the presence of two equal rapidities. By contrast, we cannot exclude that $J^{n,\alpha} = J^{m,\beta}$ if n and m are one odd and one even.
- The $J^{n,\alpha}$ leading to physically different solutions for fixed n are defined modulo L . This makes convenient to label the Bethe eigenstates with quantum numbers $I^{n,\alpha} \in [0, L)$, each of them coinciding with $J^{n,\alpha}$ up to a $\mathbb{Z}L$ -shift.

The quantum numbers $\{I^{n,\alpha}\}$ are called *Bethe-Takahashi (BT) numbers*.

In the infinite lattice limit $L \rightarrow +\infty$, the possible values for

$$Y^{n,\alpha} := \frac{I^{n,\alpha}}{L} \quad \text{Rescaled BT numbers} \quad (2.23)$$

become a dense set in the unit interval.

2.4 PARTICLES/HOLES DENSITIES

Densities of Particles and Holes: In *noninteracting systems*, e.g. the XX model ($\Delta = 0$), the momentum quantization obtained by requiring PBC is state-independent. From a Bethe-ansatz viewpoint (which is unnecessary), the S_2 -matrix is independent of λ and Bethe equations decouple enabling a separate solution for each rapidity. It follows that every rapidity λ_j is in one-to-one correspondence with its Bethe number I_j in (2.17), while does not depend on the other BT-numbers/rapidities of the eigenstate.

In this framework the set of all possible Bethe numbers induces a fixed set of "available rapidities". Each of these rapidities can be independently labelled as present or not in an eigenstate. Thus, in the thermodynamic limit, where the available $\{\lambda_j\}$ become a dense set in an interval, it appears natural to introduce a state-independent density of available rapidities $\rho_n^{tot}(\lambda)$.

The particle/hole densities $\rho_n(\lambda)$ and $\rho_n^h(\lambda)$ of an eigenstate, satisfying $\rho^{tot}(\lambda) = \rho(\lambda) + \rho^h(\lambda)$, are analogously easily defined.

In *interacting models* this formulation with densities is still possible and very useful. Nevertheless, its definition is less automatic (and not only for the presence of strings). BGT equations (2.22) are coupled, and it has no sense to talk about an individual "available rapidity" $\lambda^{n,\alpha}$ without specifying the other elements of the solution $\{\lambda^{n,\alpha}\}$.

What really remains independently eligible to be present or not are the BT numbers $\{I^{n,\alpha}\}$, while only their total chosen set collectively fixes the rapidities. It is thus possible to introduce the particles/holes concepts starting from BT numbers.

Let $\{Y\}_a$ be the set of L distinct available rescaled BT numbers (defined in (2.23)) and $\{Y^{n,\alpha}\}_p$ the N chosen ones, which one-to-one determine a set of rapidities $\{\lambda^{n,\alpha}\}_p$ according to (2.22).

Inversely, given the set $\{\lambda^{n,\alpha}\}$, the $\{Y^{n,\alpha}\}_p$ numbers are defined by satisfying

$$Y^{n,\alpha} = c_n(\lambda^{n,\alpha}) \pmod{1} \leftrightarrow (2.22)$$

where $c_n(\lambda)$ is the n -string *counting function* [136]

$$c_n(\lambda) = \frac{\theta_n(\lambda)}{2\pi} - \frac{1}{L} \sum_{m=1}^{+\infty} \sum_{\beta=1}^{N_m} \frac{\theta_{nm}(\lambda - \lambda^{m,\beta})}{2\pi}$$

which clearly depends itself from all the rapidities of the given set $\{\lambda^{n,\alpha}\}$.

Here we assume monotonicity and continuity of $c_n(\lambda) \pmod{1}$ for $\lambda \in [-\pi/2, \pi/2]$. This guarantees that it is invertible. While e.g. for the Lieb-Lininger model monotonicity and continuity can be explicitly proven, here we have to regard them as assumptions.

In this picture, the concept of "particle" is defined as follows: if $Y \in \{Y^{n,\alpha}\}_p$, then the state contains an n -string with rapidity $\lambda = c_n^{-1}(Y) \in \{\lambda^{n,\alpha}\}_p$. Instead, every $Y \in \{Y\}_a$ is said to identify an "*available rapidity*" $\lambda = c_n^{-1}(Y)$. Rapidities corresponding to BT numbers in $\{Y\}_h := \{Y\}_a \setminus \{Y\}_p$ are said to represent "*holes*". The value of every available rapidity $\lambda(Y)$ depends on the entire present rapidity set $\{\lambda^{n,\alpha}\}_p$.

The *root densities* $\{\rho_n(\lambda), \rho_n^{tot}(\lambda), \rho_n^h(\lambda)\}$ for n -strings, i.e. the densities of "Bethe roots" λ , are defined such that, in the $L \rightarrow +\infty$ limit,

$$\begin{aligned} L\rho_n(\lambda)d\lambda &:= \#\{\{\lambda\}_p \cap [\lambda, \lambda + d\lambda]\} := dN_n^p \\ L\rho_n^{tot}(\lambda)d\lambda &:= \#\{\{\lambda\}_a \cap [\lambda, \lambda + d\lambda]\} := dN_n^{tot} \rightarrow \rho_n^{tot}(\lambda) = \rho_n(\lambda) + \rho_n^h(\lambda) \\ L\rho_n^h(\lambda)d\lambda &:= \#\{\{\lambda\}_h \cap [\lambda, \lambda + d\lambda]\} := dN_n^h \end{aligned} \quad (2.24)$$

The main difference with noninteracting models is that $\rho_n^{tot}(\lambda)$ becomes state-dependent.

For every given set $\{\lambda\}_p$, $\rho_n(\lambda)$ is automatically known. Now we aim to write $\rho_n^{tot}(\lambda)$ explicitly, so that $\rho_n^h(\lambda)$ is determined as well. We exploit that the distribution of available BT-numbers $\rho_Y^{tot}(Y)$ in the chosen unit interval is trivial.

$$L\rho_Y^{tot}(Y)dY := \#\{\{Y\}_a \cap [Y, Y + dY]\} := dN_Y^{tot} = LdY \rightarrow \rho_Y^{tot}(Y) = 1$$

For corresponding intervals dY and $d\lambda$, $dN_Y^{tot} = dN_n^{tot}$ and the counting function $c_n(\lambda)$ has the role of the Jacobian for this change of variables in densities, according to $Y = c_n(\lambda)$:

$$\begin{aligned} \rho_n^{tot}(\lambda) &= \frac{1}{L} \frac{dN_n^{tot}}{d\lambda} = \frac{1}{L} \frac{dN_Y^{tot}}{dY} \frac{dY}{d\lambda} = \rho_Y^{tot}(c_n^{-1}(\lambda)) \frac{dc_n(\lambda)}{d\lambda} = \frac{dc_n(\lambda)}{d\lambda} \\ \rightarrow \rho_n(\lambda) + \rho_n^h(\lambda) &= \rho_n^{tot}(\lambda) = \frac{dc_n(\lambda)}{d\lambda} = a_n(\lambda) - \frac{1}{L} \sum_{m=1}^{+\infty} \sum_{\beta=1}^{N_m} a_{nm}(\lambda - \lambda^{m,\beta}) \end{aligned} \quad (2.25)$$

where by convenience we replaced (2.16) and (2.21) by

$$\begin{aligned} a_n(\lambda) &:= \frac{\partial_\lambda \theta_n(\lambda)}{2\pi} = \frac{1}{\pi} \frac{d}{d\lambda} \arctan\left(\frac{\tan(\lambda)}{\tanh(n\eta/2)}\right) = \frac{1}{\pi} \left(\frac{\sinh(n\eta)}{\cosh(n\eta) - \cos(2\lambda)} \right) \\ a_{nm}(\lambda) &:= \frac{\partial_\lambda \theta_{nm}(\lambda)}{2\pi} = a_{|n-m|}(\lambda) + 2 \left(\sum_{k=1}^{\min[m,n]-1} a_{|m-n|+2k}(\lambda) \right) + a_{n+m}(\lambda) \quad [a_0(\lambda) = 0] \end{aligned} \quad (2.26)$$

Notations for the Root-Density Formalism: In the thermodynamic limit, sums over present rapidities are expected to be well approximated by integrals:

$$\sum_{\alpha=1}^{N_n} f(\lambda^{n,\alpha}) \rightarrow L \int_{-\pi/2}^{\pi/2} \rho_n(\lambda) f(\lambda) d\lambda \quad (2.27)$$

The same holds for $\rho_n^{tot}(\lambda)$ and $\rho_n^h(\lambda)$, in case of sums over available rapidities or holes, respectively. Another useful definition is

$$(f * g)(\lambda) := \int_{-\pi/2}^{\pi/2} f(\lambda - \mu) g(\mu) d\mu \quad \text{Convolution in rapidity space} \quad (2.28)$$

Densities Formulation of BGT Equations in the Thermodynamic Limit: The main problem of the (2.25) form of the BGT equations is that it still contains explicitly the infinitely many discrete rapidities $\{\lambda^{m,\beta}\}_p$. However, in the thermodynamic limit it is possible to replace the last sum with an integral using (2.27). Then, recognising a convolution as defined in (2.28),

$$\rho_n^{tot}(\lambda) = a_n(\lambda) - \sum_{m=1}^{+\infty} (a_{nm} * \rho_m)_\lambda \quad \text{BGT eq. with Densities} \quad (2.29)$$

a_n and a_{nm} are defined in (2.26). To summarize:

- The main advantage of this rewriting is the replacement of equations in terms of infinitely many quantized rapidities by new equations in terms of the continuous and analytic distributions $\rho_n(\lambda)$ and $\rho_n^h(\lambda)$.
- Given that $\rho_n^{tot}(\lambda) = \rho_n(\lambda) + \rho_n^h(\lambda)$, for every fixed n only two of the three root densities are independent. Moreover, if the phase shifts $\theta_n(\lambda)$ of the model are known, then a single set of root densities suffices. Owing to (2.29), the density of particle rapidities, together with the S_2 -matrix encoding all the relevant information about interactions, completely determines the density of hole rapidities.

2.5 YANG-YANG ENTROPY AND THERMODYNAMIC BETHE ANSATZ (TBA) EQUATIONS

The thermodynamic Bethe equations for the XXX model were first derived by Takahashi [133] in 1971. The same year this formulation was extended by Gaudin [64] to the XXZ model.

Given (2.5) and the vacuum state $|\downarrow\downarrow\downarrow\dots\rangle$ with energy $E_0 = J\Delta L/4$, we can define the energy of an excitation/quasiparticle of quasimomentum k as

$$\epsilon_1(k) = J(\cos(k) - \Delta)$$

To rewrite $\epsilon(k)$ as a function of the rapidity λ , we have simply to use (2.13):

$$\begin{aligned} \cos(k) &= \frac{e^{ik} + e^{-ik}}{2} = \frac{1}{2} \left(\frac{\sin^{[+]}(\lambda)}{\sin^{[-]}(\lambda)} + \frac{\sin^{[-]}(\lambda)}{\sin^{[+]}(\lambda)} \right) = \frac{\cos(2\lambda) \cosh(\eta) - 1}{\cos(2\lambda) - \cosh(\eta)} \\ -\Delta &= -\cosh(\eta) = \frac{\cosh^2(\eta) - \cosh(\eta) \cos(2\lambda)}{\cos(2\lambda) - \cosh(\eta)} = \frac{\sinh^2(\eta) + 1 - \cosh(\eta) \cos(2\lambda)}{\cos(2\lambda) - \cosh(\eta)} \\ \rightarrow \quad \epsilon_1(k) &= \frac{J \sinh^2(\eta)}{\cos(2\lambda) - \cosh(\eta)} = -J\pi \sinh(\eta) a_1(\lambda) \end{aligned}$$

However, in presence of strings a more convenient quantity to work with is the energy of an entire string with central rapidity $\lambda \in \mathbb{R}$. It can be easily derived from the above result exploiting the property (2.19), which implies that

$$\begin{aligned} a_n(\lambda) &= \sum_{b=0}^{n-1} a_1 \left(\lambda - i \left[\frac{(n-1)}{2} + b \right] \eta \right) \\ \rightarrow \quad \epsilon_n(\lambda) &= \sum_{i=1}^n \epsilon_1(\lambda_i) = -J\pi \sinh(\eta) a_n(\lambda) = \frac{J \sinh(\eta) \sinh(n\eta)}{\cos(2\lambda) - \cosh(n\eta)} \end{aligned} \quad (2.30)$$

$$\left[\text{and in particular} \quad \epsilon_n(\lambda) \approx -\frac{J \sinh(\eta) \sinh(n\eta)}{\cosh(n\eta)} \approx -J \sinh(\eta) \quad \text{when } n\eta \gg 1 \right]$$

where $\{\lambda_i\}$ are the rapidities composing the string.

The total energy density of a Bethe eigenstate is thus, employing (2.27):

$$\frac{E}{L} := \frac{J\Delta}{4} + \sum_{n=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \epsilon_n(\lambda) \rho_n(\lambda) d\lambda \quad (2.31)$$

As all local observables, it is univocally associated root densities. However, there are many possible eigenstates corresponding to these same root densities. Indeed, these are only defined by the particle/hole rapidities in each infinitesimal interval $[\lambda, \lambda + d\lambda]$, independently of which rapidities are selected. If we view root densities as the macrostate of a system and all the compatible Bethe eigenstates as the possible equiprobable microstates, then it is natural to introduce a counterpart of the Boltzmann entropy in this framework by counting these eigenstates.

Using the definition (2.24), the number of possible rapidity choices in the interval $d\lambda$ is:

$$d\mathcal{N}(\lambda) = \prod_{n=1}^{+\infty} \frac{(dN_n^{tot})!}{(dN_n^p)!(dN_n^h)!} = \frac{(L\rho_n^{tot}(\lambda)d\lambda)!}{(L\rho_n(\lambda)d\lambda)!(L\rho_n^h(\lambda)d\lambda)!}$$

$$dS_{YY}(\lambda) := \log[d\mathcal{N}(\lambda)] = \sum_{n=1}^{+\infty} \{\log[(L\rho_n^{tot}(\lambda)d\lambda)!] - \log[(L\rho_n(\lambda)d\lambda)!] - \log[(L\rho_n^h(\lambda)d\lambda)!]\}$$

Exploiting the Stirling approximation and integrating over rapidities finally leads to the so-called *Yang-Yang entropy*, which was first introduced by Yang and Yang [150] in 1969, within their derivation of the Lieb-Lininger model thermodynamics:

$$\frac{S_{YY}[\{\rho_n\}]}{L} := \sum_{n=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \{\rho_n^{tot}(\lambda) \log[\rho_n^{tot}(\lambda)] - \rho_n(\lambda) \log[\rho_n(\lambda)] - \rho_n^h(\lambda) \log[\rho_n^h(\lambda)]\} d\lambda$$

Here the three root densities are not independent, but related by $\rho_n^{tot}(\lambda) = \rho_n(\lambda) + \rho_n^h(\lambda)$ and the BGT equations (2.29). Before proceeding with the computation of the thermal root densities, for later use within the quasiparticle picture (see §5), we mention that this formula naturally introduces the concept of the *contribution to the entropy density of an n -strings with rapidity λ* :

$$s_n(\lambda) := \rho_n^{tot}(\lambda) \log[\rho_n^{tot}(\lambda)] - \rho_n(\lambda) \log[\rho_n(\lambda)] - \rho_n^h(\lambda) \log[\rho_n^h(\lambda)] \quad (2.32)$$

In order to shift to a *canonical description at temperature T* , we compute the partition function and define the corresponding free energy functional $F[\{\rho_n\}, T]$.

Exploiting that $e^{S_{YY}[\{\rho_n\}]}$ by definition counts the number of microstates with root densities $\{\rho_n\}$, which have almost degenerate energy $E[\{\rho_n\}]$:

$$\begin{aligned} Z(T) &= \int [D\{\rho_n\}] e^{S_{YY}[\{\rho_n\}]} e^{-\beta E[\{\rho_n\}]} := \int [D\{\rho_n\}] e^{-\beta F[\{\rho_n\}, T]} \\ &\rightarrow F[\{\rho_n\}, T] = E[\{\rho_n\}] - TS_{YY}[\{\rho_n\}] \end{aligned}$$

Since F is an extensive quantity, in the thermodynamic limit this integral can be performed by *saddle-point approximation*, i.e. observables on the thermal state at temperature T can be efficiently computed by only considering the root densities $\{\rho_n(\lambda)\}$ minimizing $\{F[\{\rho_n\}, T]\}$.

By plugging $\rho_n^{tot}(\lambda) = \rho_n(\lambda) + \rho_n^h(\lambda)$ and (2.22) into $F = E - TS_{YY}$, we obtain:

$$\frac{F}{L} = \frac{J\Delta}{4} + \sum_n \int_{-\pi/2}^{\pi/2} d\lambda \left\{ [\epsilon_n + T \log(1 + \eta_n)] \rho_n + T \log\left(1 + \frac{1}{\eta_n}\right) \rho_n^h \right\}$$

$$\text{where} \quad \eta_n(\lambda) := \frac{\rho_n^h(\lambda)}{\rho_n(\lambda)}$$

Since we are looking for a minimum of F , we compute its first order variation corresponding to an infinitesimal modification $\delta\rho_n$.

$$\frac{\delta F}{L} = \sum_n \int_{-\pi/2}^{\pi/2} d\lambda \left\{ [\epsilon_n + T \log(1 + \eta_n)] \delta\rho_n + T \log\left(1 + \frac{1}{\eta_n}\right) \delta\rho_n^h \right\}$$

$\delta\rho_n$ and $\delta\rho_n^h$ are connected by the BGT equations (2.31). Hence, for every function $f_n(\lambda)$,

$$\begin{aligned} \sum_n \int_{-\pi/2}^{\pi/2} d\lambda f_n \delta\rho_n^h &= - \sum_n \int_{-\pi/2}^{\pi/2} d\lambda f_n \delta\rho_n - \sum_{n,m} \int_{-\pi/2}^{\pi/2} d\lambda d\mu f_n(\lambda) a_{nm}(\lambda - \mu) \delta\rho_n(\mu) \\ &\rightarrow \sum_n \int_{-\pi/2}^{\pi/2} d\lambda f_n \delta\rho_n^h = - \sum_n \int_{-\pi/2}^{\pi/2} d\lambda \left\{ f_n + \sum_m [a_{nm} * f_m] \right\} \delta\rho_n \end{aligned}$$

where we employed that $a_{nm}(\lambda - \mu)$ is symmetric in both $\lambda \leftrightarrow \mu$ and $n \leftrightarrow m$ to shift the convolution from $\delta\rho_n$ to f_n . Upon plugging this into the above expression of δF , the variation is fully expressed in terms of $\delta\rho_n$.

$$\frac{\delta F}{L} = \sum_n \int_{-\pi/2}^{\pi/2} d\lambda \left\{ \epsilon_n + T \log(\eta_n) - T \sum_m \left[a_{nm} * \log\left(1 + \frac{1}{\eta_m}\right) \right] \right\} \delta\rho_n$$

Requiring $\delta F = 0 \ \forall \delta\rho_n$, the content of $\{\}$ has to be zero $\forall n \in \mathbb{N}^{>0}$, $\lambda \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. [3, 134, 135]:

$$\log(\eta_n(\lambda)) = -\frac{\epsilon_n(\lambda)}{T} + \sum_{m=1}^{+\infty} \left[a_{nm} * \log\left(1 + \frac{1}{\eta_m}\right) \right]_{\lambda} \quad \text{TBA equations} \quad (2.33)$$

In this infinite set of coupled nonlinear integral equation, $\{\eta_n\}$ is an infinite set of unknown functions. In general, the solution has to be found numerically, except for very special limits, e.g. at infinite temperature, as we will show in §2.7.

2.6 QUASI-DECOUPLING OF THE BGT AND TBA EQUATIONS

Here we present a useful technique to quasi-decouple equations of the form

$$f_n(\lambda) = \xi_n(\lambda) - \sum_m (a_{nm} * g_m)_{\lambda} \quad (2.34)$$

obtaining equations involving only three adjacent string indices $(k-1, k, k+1)$. Above the convolution kernel $a_{nm}(\lambda)$ is the object defined in (2.26), while f_n , β_n and g_m are arbitrary functions.

Note that both BGT equations (2.29) and TBA equations (2.33) fall within this category. This structure will arise again in the computation of dressed velocities (4.17) in §4.3.

The key idea, introduced in [135], is the definition of a function $s(\lambda)$ satisfying

$$\begin{cases} a_1(\lambda) - [s * a_2]_{\lambda} = s(\lambda) \\ a_n(\lambda) - [s * (a_{n-1} + a_{n+1})]_{\lambda} = 0 \quad \forall n > 1 \end{cases} \quad (2.35)$$

This function will have a pivotal role again in §7.8.

Owing to (2.35), it enables an easy inversion of $(a + \delta)_{kn}(\lambda)$. Indeed

$$(a + \delta)_{kn}^{-1}(\lambda) := \delta_{kn} \delta(\lambda) - (\delta_{k-1,n} + \delta_{k+1,n}) s(\lambda) \quad (2.36)$$

$$\text{is such that} \quad \sum_n [(a + \delta)_{kn}^{-1} * (a + \delta)_{nm}]_{\lambda} = \delta_{km} \delta(\lambda)$$

The proof that the above $(a + \delta)_{kn}^{-1}(\lambda)$ satisfies the inverse definition can be done by direct check. However, it requires noticing the following property of $a_{mn}(\lambda)$, following from (2.35):

$$a_{mk}(\lambda) = [s * (a + \delta)_{(m-1)k}]_{\lambda} + [s * (a + \delta)_{(m+1)k}]_{\lambda} \quad (2.37)$$

Moreover, it is straightforward that

$$\sum_n [(a + \delta)_{kn}^{-1} * f_n]_{\lambda} = f_k(\lambda) - [s * (f_{k-1} + f_{k+1})]_{\lambda}$$

Thus, by defining $h_n := f_n - \xi_n$ and applying $(a + \delta)_{kn}^{-1}$ in convolution on the left to (2.34),

$$\begin{aligned} \sum_n [(a + \delta)_{kn}^{-1} * h_n]_{\lambda} &= h_k(\lambda) - [s * (h_{k-1} + h_{k+1})]_{\lambda} \\ \sum_{nm} [(a + \delta)_{kn}^{-1} * a_{nm} * g_m]_{\lambda} &= \sum_m \left\{ \left[\sum_n (a + \delta)_{kn}^{-1} * a_{nm} \right] * g_m \right\}_{\lambda} = \\ &= \sum_m \left\{ \left[\sum_n (a + \delta)_{kn}^{-1} * (a + \delta - \delta)_{nm} \right] * g_m \right\}_{\lambda} = \quad \text{by using (2.36)} \\ &= g_k(\lambda) - \sum_m [(a + \delta)_{km}^{-1} * g_m]_{\lambda} = [s * (g_{k-1} + g_{k+1})]_{\lambda} \\ \rightarrow f_k(\lambda) &= \xi_k(\lambda) - [s * (\xi_{k-1} + \xi_{k+1})]_{\lambda} + \{s * [(f - g)_{k-1} + (f - g)_{k+1}]\}_{\lambda} \end{aligned} \quad (2.38)$$

with the convention that $\xi_0 = (f - g)_0 = 0$.

In the following two paragraphs we plug into this result the explicit form of the functions f , ξ and g corresponding to the BGT and TBA equation. In §4.3 we will do the same for the equations of dressed velocities.

Explicit Form and Some Properties of $s(\lambda)$: An explicit solution for $s(\lambda)$ is the following real even function, periodic of period π [135]:

$$s(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} \frac{e^{-i2\lambda k}}{\cosh(k\eta)} = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{+\infty} \frac{\cos(2\lambda k)}{\cosh(k\eta)}$$

Since $\eta = \text{arccosh}(\Delta) > 0$, high- k terms are increasingly suppressed by $e^{-\eta k}$. Thus, in numerical practice it is possible to truncate this series to a finite k_{max} .

$$s'(\lambda) = -\frac{1}{\pi} \sum_{k=1}^{+\infty} \frac{2k \sin(2\lambda k)}{\cosh(k\eta)} \quad s''(\lambda) = -\frac{1}{\pi} \sum_{k=1}^{+\infty} \frac{(2k)^2 \sin(2\lambda k)}{\cosh(k\eta)}$$

$s'(\lambda)$ is null in $\lambda \in \{-\pi/2, 0, \pi/2\}$. In particular, $s''(0) < 0$, hence $\lambda = 0$ is a maximum of $s(\lambda)$.

In Fig.2.2 we plotted $s(\lambda)$ for the two main values of Δ used in this work, i.e. $\Delta = 3$ and $\Delta = 10$.

In the next chapter we will use the integral of $s(\lambda)$ as well, which only receives contribution from the constant term $k = 0$:

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} s(\mu) d\mu &= \frac{1}{2\pi \cosh(k\eta)} \sum_{k=-\infty}^{+\infty} \int_{-\pi/2}^{\pi/2} e^{-i2\mu k} d\mu = \\ &= \frac{1}{2} + \sum_{k \neq 0} \frac{1}{4k\pi \cosh(k\eta)} \int_{-k\pi}^{k\pi} e^{-i\gamma} d\mu = \frac{1}{2} \end{aligned} \quad (2.39)$$

Before proceeding, we also mention that, in view of §7.8, it is convenient to introduce *the inverse of $s*$, i.e. the operator s^{-1} defined by $s^{-1} * (s * f) = f$* .

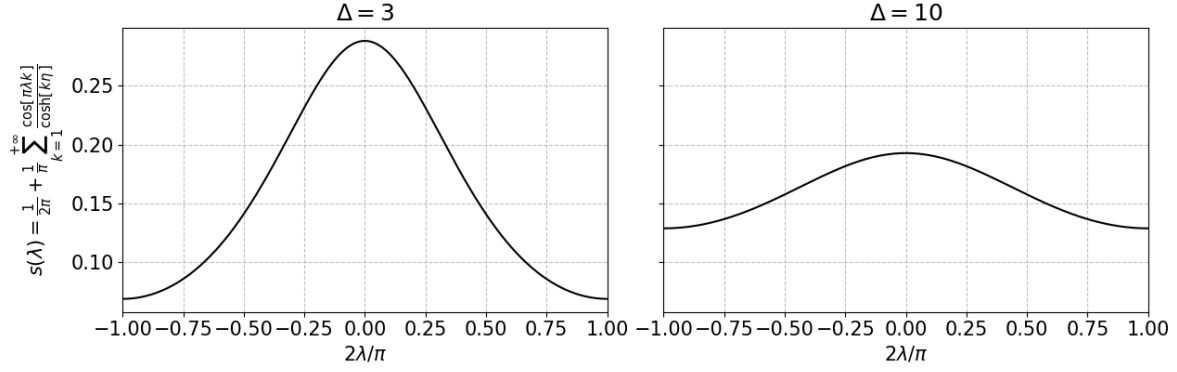


FIGURE 2.2: $s(\lambda)$ for $\Delta = 3$ and $\Delta = 10$, with rapidities rescaled by $\pi/2$.

Its action can be proven to represent a symmetrized combination of the spectral parameter shifts $[\pm]$ defined in (1.3) [73]:

$$[s^{-1} * f]_{\lambda} = \lim_{\epsilon \rightarrow 0} [f^{[+]}(\lambda - i\epsilon) + f^{[-]}(\lambda + i\epsilon)] \quad \lambda \in \mathbb{R} \quad (2.40)$$

for functions free of singularities inside the domain

$$\mathcal{P}_{\eta} := \left\{ \lambda \in \mathbb{C} \mid \text{Im}(\lambda) \in \left(-\frac{\eta}{2}, \frac{\eta}{2} \right) \right\} \quad \text{"physical strip"} \quad (2.41)$$

Quasi-Decoupling of the BGT Equations (2.29): Here $\xi_n = a_n$. Thus, when $n > 1$, the first term $\xi_k(\lambda) - [s * (\xi_{k-1} + \xi_{k+1})]_{\lambda}$ of (2.38) vanishes because of (2.35). Instead, when $n = 1$ it returns $s(\lambda)$. Moreover $f_k = \rho_k^{tot}$ and $(f - g)_k = \rho_k^{tot} - \rho_k = \rho_k^h$, hence:

$$\begin{cases} \rho_1^{tot}(\lambda) = s(\lambda) + [s * \rho_2^h]_{\lambda} \\ \rho_k^{tot}(\lambda) = [s * (\rho_{k-1}^h + \rho_{k+1}^h)]_{\lambda} \quad \forall k > 1 \end{cases}$$

If the ratio $\zeta_n = \rho_n^h / \rho_n^{tot}$ is known, e.g. from the TBA equations or string-charge duality (see §7.8),

$$\begin{cases} \rho_1^{tot}(\lambda) = s(\lambda) + [s * (\zeta_2 \rho_2^{tot})]_{\lambda} \\ \rho_k^{tot}(\lambda) = [s * (\zeta_{k-1} \rho_{k-1}^{tot} + \zeta_{k+1} \rho_{k+1}^{tot})]_{\lambda} \quad \forall k > 1 \end{cases} \quad (2.42)$$

For the XXZ model the length of string configurations has no upper limit, thus an exact solution of the BGT or TBA equations would require k spanning from 1 to infinity. However, this is clearly unfeasible in numerical implementation, where we need to approximate the maximal string length to a finite value m_{max} .

Note that there is not a clear optimal way to perform the string-length truncation. For example, two extremal methods are:

- "Standard" truncation: $\rho_{m_{max}}^{tot} = s * \rho_{m_{max}-1}^h$
- "Reflected" truncation: $\rho_{m_{max}}^{tot} = 2s * \rho_{m_{max}-1}^h$

We will discuss further these two possibilities in the next section.

Quasi-Decoupling of the TBA Equations (2.33): Again, we can employ (2.35) to annihilate the ξ term in (2.38).

$$\begin{aligned} \xi_k(\lambda) &= -\beta \epsilon_k(\lambda) = J\beta\pi \sinh(\eta) a_k(\lambda) \\ f_k(\lambda) &= \log(\eta_k(\lambda)) \quad (f - g)_k = \log(\eta_k) + \log\left(1 + \frac{1}{\eta_k}\right) = \log(\eta_k + 1) \end{aligned}$$

Since $J\pi \sinh(\eta)$ can be regarded as a harmless prefactor, calculations follows identically as above:

$$\begin{cases} \log(\eta_1(\lambda)) = J\beta\pi \sinh(\eta) s(\lambda) + [s * \log(\eta_2 + 1)]_\lambda \\ \log(\eta_k(\lambda)) = [s * (\log(\eta_{k-1} + 1) + \log(\eta_{k+1} + 1))]_\lambda \quad \forall k > 1 \\ \lim_{k \rightarrow +\infty} \log \eta_k / k = 0 \end{cases} \quad (2.43)$$

The third equation is obtained starting from (2.33) with $n = 1$.

For the proof, the reader is referred to [134, 135].

2.7 EXACT SOLUTION OF THE INFINITE-TEMPERATURE TBA EQUATIONS

Exact Solution of the TBA Equations at Infinite Temperature: As we will discuss in next chapters, the case of our interest is $\beta = 0$, where the driving term $J\beta\pi \sinh(\eta) s(\lambda)$ vanishes.

However, this is not the only simplification. Let us extend η over all $\mathbb{R}$ assuming periodicity of period π . Then the system (2.43) becomes totally independent of λ , thus we expect the $\eta_k(\lambda)$ functions to be constant.

$$\begin{cases} \log(\eta_1) = \left[\int_{-\pi/2}^{\pi/2} s(\mu) d\mu \right] \cdot \log(\eta_2 + 1) \\ \log(\eta_k) = \left[\int_{-\pi/2}^{\pi/2} s(\mu) d\mu \right] \cdot \log[(\eta_{k-1} + 1)(\eta_{k+1} + 1)] \quad \forall k > 1 \\ \lim_{k \rightarrow +\infty} \log \eta_k / k = 0 \end{cases}$$

The integral of $s(\lambda)$ has already been computed in (2.39).

$$\rightarrow \begin{cases} \eta_1 = \sqrt{\eta_2 + 1} \\ \eta_k = \sqrt{(\eta_{k-1} + 1)(\eta_{k+1} + 1)} \quad \forall k > 1 \\ \lim_{k \rightarrow +\infty} \log \eta_k / k = 0 \end{cases}$$

or equivalently: $\eta_2 = \eta_1^2 - 1 \quad \eta_k = \frac{\eta_{k-1}^2}{(\eta_{k-2} + 1)} - 1 \quad \forall k \geq 3 \quad \lim_{k \rightarrow +\infty} \frac{\log \eta_k}{k} = 0$

This recursion has the following known simple analytical solution [134]:

$$\eta_n = \frac{\rho_n^h}{\rho_n} = (n+1)^2 - 1 \quad \rightarrow \quad \theta_n = \frac{\rho_n}{\rho_n^{tot}} = \frac{1}{(n+1)^2} \quad \zeta_n = \frac{\rho_n^h}{\rho_n^{tot}} = 1 - \frac{1}{(n+1)^2} \quad (2.44)$$

In particular, when $n \rightarrow +\infty$, $\eta_n \approx n^2$ is divergent, while $\theta_n = O(1/n^2)$ and $\zeta_n = 1 + O(1/n^2)$.

Therefore $\rho_n \ll \rho_n^h \approx \rho_n^{tot}$ for long strings.

Asymptotic Decay of ρ_m^{tot} : While the TBA equations at infinite temperature can be solved exactly as just described, the corresponding root densities have to be obtained numerically from the BGT equations. However, here we give a rough estimate of ρ_m^{tot} for high m (long strings).

We start by approximating $\rho_m^{tot}(\lambda)$ as independent of λ in this regime, consistently with the observed numerical results. The decoupled BGT equations become:

$$\rho_m^{tot} \approx \frac{\zeta_{m-1} \rho_{m-1}^{tot} + \zeta_{m+1} \rho_{m+1}^{tot}}{2} = \frac{\rho_{m-1}^{tot} + \hat{\rho}_{m+1}^{tot}}{2} + O\left(\frac{1}{m^2}\right)$$

This is incompatible with an exponential decay (the characteristic length would be infinite). Thus we expect saturation to a constant or power-law decay.

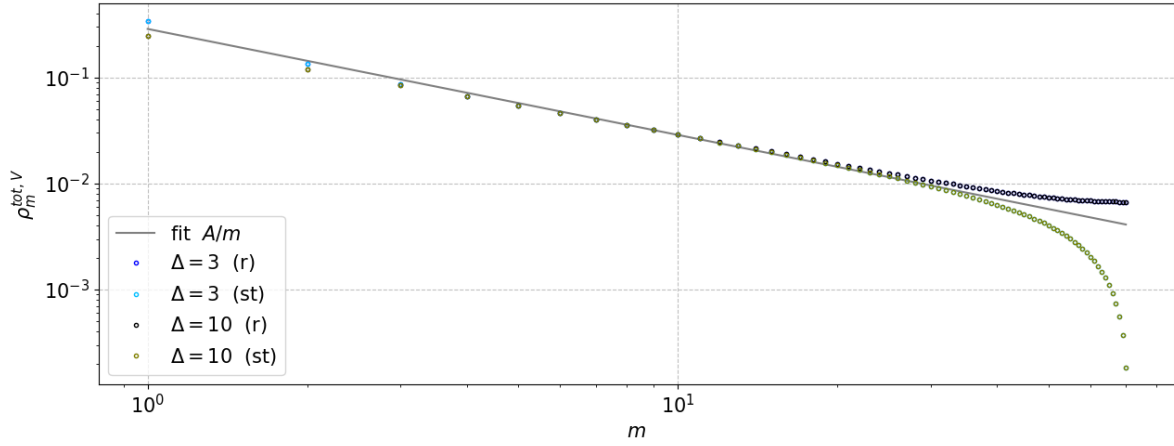


FIGURE 2.3: Empty dots denote the $\lambda = 0$ value of each $\{\hat{\rho}_m^{tot}(\lambda)\}$. The BGT equations were solved numerically with fixed $m_{max} = 70$, varying $\Delta \in \{3, 10\}$ and the truncation scheme (standard or reflected). The grey line shows the best-fit curve obtained from the ansatz $\hat{\rho}_m^{tot} = A/m$.

Nevertheless, saturation is not possible, since the thermal state has finite particle density $1/2$ (i.e. zero local magnetization).

$$\frac{1}{2} = \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} m \rho_m(\lambda) d\lambda \geq \sum_{m=\bar{m}}^{+\infty} \int_{-\pi/2}^{\pi/2} \frac{m}{(m+1)^2} \rho_m^{tot}(\lambda) d\lambda \approx \pi \int_{\bar{m}}^{+\infty} \frac{\rho_m^{tot}}{m} dm$$

We therefore conjecture $\rho_m^{tot} \approx Am^{-\alpha}$.

To obtain an estimate of α and A , we numerically solved the BGT equations with $m_{max} = 70$, while changing $\Delta \in \{3, 10\}$ and the truncation scheme, trying both the two possibilities (standard and reflected) defined in the previous section. The corresponding $\lambda = 0$ value for each $\{\rho_m^{tot}(\lambda)\}$ is shown in Fig.2.3. These data clearly display three different regimes:

- An initial transient, say $m \leq 3$, in which ρ_m^{tot} mainly depends on Δ due to the driving term $s(\lambda)$, while the dependence on the truncation scheme is negligible.
- An intermediate region in which all data collapse, meaning substantial independence of both Δ and the truncation scheme. This is the suitable region to perform a meaningful fit with power-law decay. In Fig.2.3 the grey line is the curve obtained fitting the points $m \in \{5, \dots, 20\}$ assuming $\alpha = 1$. Otherwise, if we perform the fit leaving the exponent as free parameter, we obtain $\alpha \in [0.9, 1]$ depending on the selected data interval. The constant A in $\rho_m^{tot} \approx Am^{-1}$ turns out to be $A \approx 0.3$.

We will exploit this result in §8 to ansatz the suitable fitting functions for the convergence of the infinite-temperature root densities upon increasing m_{max} .

- For $m \geq 20 - 25$ root densities become sensible to the truncation scheme. Indeed, the reflected truncation induces saturation to a constant value, while the standard truncation requires root densities to vanish at $m \approx m_{max}$.

In this work we will always compare both possibilities when the differences are relevant.

Chapter 3

RELAXATION, THERMALIZATION AND QUASILOCAL CHARGES

3.1 LOCAL STATIONARY STATES

Possibility of Relaxation: In this chapter we will introduce and summarize the concepts of *thermalization and relaxation in isolated quantum systems*. Literature is extremely vast, but for a pedagogical review near to our approach see e.g. [51].

The striking proliferation of studies about this topic is mainly due to the fact that, in the last two decades, experiments on trapped ultra-cold atomic gases (see e.g. [67, 137]), where high levels of isolation and control are possible, have suggested that, even in isolated quantum systems initially prepared in pure states, local observables generically relax to time-independent values.

The theoretical explanation of this fact deals with the fundamental issue of quantum Statistical Mechanics itself, that is *how statistical ensembles can asymptotically arise from the dynamics of many-body systems, driven by the Schrödinger equation*. Indeed, the main problem is that the time evolution $|\psi(t)\rangle = e^{i\hat{H}t} |\psi(0)\rangle$ is unitary, thus the state of the system is maintained pure at all times. By contrast, statistical ensembles are by definition represented by mixed states.

A crucial observation is that, *while the entire system remains in a pure state, what in experiments is factually observed to reach stationarity are local observables*. The convenience to focus on physical observables rather than wavefunctions or density matrices describing the entire system was already pointed out by von Neumann in 1929, while discussing the problem of quantum relaxation to a statistical ensemble [145]. Thus, rather than directly considering $|\psi(t)\rangle$, the connection between the unitary Schrödinger evolution and stationary quantum statistical ensembles has to be looked for focussing on the dynamics of local observables:

$$\langle \psi(t) | \hat{O} | \psi(t) \rangle = \sum_{\alpha, \beta} e^{-i(E_\beta - E_\alpha)t} \psi_\alpha^* \psi_\beta \langle \alpha | \hat{O} | \beta \rangle \quad \psi_\alpha := \langle \alpha | \psi(0) \rangle \quad (3.1)$$

where we denoted by $|\alpha\rangle$ the Hamiltonian eigenstates and by E_α the corresponding energies.

Before proceeding, a simple example which shows that there always exist nonlocal observables which do not reach stationarity is that

$$\hat{O}_{\alpha\beta} := |\alpha\rangle \langle \beta| + |\beta\rangle \langle \alpha| \quad \rightarrow \quad \langle \psi(t) | \hat{O} | \psi(t) \rangle = e^{-i(E_\beta - E_\alpha)t} \psi_\alpha^* \psi_\beta \langle \alpha | \hat{O} | \beta \rangle + c.c.$$

Both terms oscillate periodically with $\omega = E_\beta - E_\alpha$.

We thus restrict to local observables. If $\hat{O}$ has support inside a finite subregion $\mathbb{A}$, its expectation value is fully determined by the reduced density matrix $\hat{\rho}_{\mathbb{A}}(t)$ of $|\psi(t)\rangle$ with respect to $\mathbb{A}$. In general, $\hat{\rho}_{\mathbb{A}}(t)$ corresponds to a mixed states.

Moreover, heuristically, *this way we can think of relaxation as due to the entire system acting as a heat bath with respect to the subregion $\mathbb{A}$ of size $\ell \ll L$.*

Even with this breakthrough, a second immediate problem is that unitary evolution is periodic or quasiperiodic. After a sufficiently long time, any finite system returns to its initial state or arbitrarily close to it (*Poincaré recurrence* [21, 75, 104]). The periodic behaviour is inherited at a local level, preventing local observables to be described by a stationary statistical ensemble.

We stress that *here relaxation is meant in a strong sense*: at sufficiently late times the instantaneous values of observables approach an equilibrium value and remain close to it at almost all subsequent times. Indeed, this is the relaxation observed in cold-atom experiments. By contrast, some theoretical explanations involve relaxation in a weak sense, dealing with long-time averages of observables instead of their value at long times. We will never consider this meaning.

This makes periodicity highly problematic. However, the period of quantum recurrences typically diverges with the system size, so that *if the $t \rightarrow +\infty$ limit is applied after the thermodynamic limit $L \rightarrow +\infty$, it is no longer impossible for local observables to reach stationarity*.

Indeed, once the period diverges, while at short times the oscillations in the summed terms of (3.1) lead to a nontrivial evolution, we should expect that asymptotically *dephasing* leads to relaxation to an equilibrium value.

Later we will discuss that, in order for dephasing to lead to relaxation, some other assumptions are required at the level of Hamiltonian spectra (the so-called "*eigenstate thermalization*").

Formal Definition of Local Relaxation to a Stationary State: As just discussed, an isolated quantum system prepared in a pure state cannot relax in its entirety. However, it can relax locally in space, i.e. local observables can assume stationary values.

Expectation values of all local observables supported on an arbitrary finite compact subsystem $\mathbb{A}$ are fully determined by the reduced density matrix

$$\hat{\rho}_{\mathbb{A}}(t) = \text{Tr}_{\bar{\mathbb{A}}} |\psi(t)\rangle \langle \psi(t)| \rightarrow \langle \hat{O}(t) \rangle = \text{Tr}[\hat{O}_{\mathbb{A}} \hat{\rho}_{\mathbb{A}}(t)]$$

Thus, if in an *infinite* ($L \rightarrow +\infty$) system, for every *finite* compact subsystem $\mathbb{A}$ of length $\ell \ll L$

$$\text{the limit } \hat{\rho}_{s,\mathbb{A}} := \lim_{t \rightarrow +\infty} \hat{\rho}_{\mathbb{A}}(t) \text{ exists,} \quad (3.2)$$

we can state that the system relaxes to a *local stationary state (LSS)* $\hat{\rho}_s$ defined by the property

$$\text{Tr}_{\bar{\mathbb{A}}}[\hat{\rho}_s] = \hat{\rho}_{s,\mathbb{A}} = \lim_{t \rightarrow +\infty} \hat{\rho}_{\mathbb{A}}(t) \quad \forall \mathbb{A} \quad [\hat{H}, \hat{\rho}_s] = 0 \quad (3.3)$$

In general (3.3) does not admit a unique solution and in particular can be fulfilled by both pure and mixed states, albeit $\hat{\rho}(t) = |\psi(t)\rangle \langle \psi(t)|$ describes a pure state.

We emphasize that, while $\hat{\rho}_{\mathbb{A}}(t)$ is a state of the subsystem $\mathbb{A}$ only, *both $\hat{\rho}_s$ (stationary mixed state) and $\hat{\rho}(t) = |\psi(t)\rangle \langle \psi(t)|$ (time-dependent pure state) are states of the entire system*. However, from the factual viewpoint of local observables, they are indistinguishable and it makes no sense to ask which of them is the "true" state of the system. The same holds for the above-mentioned different possible candidates for $\hat{\rho}_s$.

3.2 GENERALIZED GIBBS ENSEMBLE (GGE)

Thermalization and Relaxation to a GGE: For generic (i.e. nonintegrable) systems, where energy is almost the only conserved charge, (3.3) is expected to admit a thermal state as solution [40, 113], since heuristically the rest of the system can be viewed as a heat bath, leading to *thermalization*. More formally, a thermal state for the whole system is defined as

$$\hat{\rho}_{GE} := \frac{e^{-\beta \hat{H}}}{Z} \quad \text{where} \quad Z := \text{Tr}[e^{-\beta \hat{H}}] \quad \text{Gibbs ensemble}$$

If a value of β exists s.t. $\hat{\rho}_s = \hat{\rho}_{GE}$ is a solution of (3.3), we can state that the system as $t \rightarrow +\infty$ becomes locally thermal with $T = 1/\beta$.

However, as mentioned, the Gibbs ensemble is based on the assumption that the (almost) only local conservation law is that of energy. This definitely fails in *quantum integrable systems*, which by definition (we will discuss this point further in the next paragraph), display an extensive number of extensive (quasi)local conserved charges $\{\hat{H}_n\}$ commuting with each other and with the Hamiltonian¹, which can itself be included in their number.

The formal definition of extensive (quasi)local operator is introduced in §3.4. For now the reader can imagine $\hat{H}_n$ as given by sums of densities which are local operators. In this sense, the Hamiltonian $\hat{H}_{XXZ} = J \sum_j \hat{h}_{j,j+1}$ is clearly such a kind of operator.

Let us assume to know a suitable "complete set" $\{\hat{H}_n\}$ of extensive (quasi)local conserved charges and that an equilibrium state exists. As originally proposed in [114], requiring that the von Neumann entropy $S = -\text{Tr}[\hat{\rho}_s \log \hat{\rho}_s]$ is maximized under the constraint that the conserved charges are fixed by their values in the initial state, $\hat{\rho}_s$ turns out to have the same structure as $\hat{\rho}_{GGE}$, but with a Lagrange multiplier for each $\hat{H}_n$ [58, 65, 70, 93, 126, 142].

$$\hat{\rho}_{GGE} := \frac{e^{-\sum_n \beta_n \hat{H}_n}}{Z} \quad \text{where} \quad Z := \text{Tr}[e^{-\sum_n \beta_n \hat{H}_n}] \quad \begin{array}{l} \text{Generalized} \\ \text{Gibbs ensemble (GGE)} \end{array} \quad (3.4)$$

All the information about the initial state which is necessary to fix $\hat{\rho}_{GGE}$ is encoded in the Lagrange multipliers $\{\beta_n\}$, which are computed by requiring that

$$\text{Tr}[\hat{\rho}_{GGE} \hat{H}_n] = \langle \psi(0) | \hat{H}_n | \psi(0) \rangle \quad \forall n$$

However, this fundamentally simple conjecture contains a subtle ambiguity, namely how to choose in practice the set $\{\hat{H}_n\}$ that correctly fixes ρ_{GGE} and how to state and check its *completeness*. This nontrivial issue is strongly connected to the very difficulty of a formal definition of integrability in Quantum Mechanics.

A Glimpse into the Problem of Defining Quantum Integrability: In Classical Mechanics, the requirement of existence of a set of integrals of motion (functionally independent of the Hamiltonian and each other) equal in number to the degrees of freedom is a sufficient definition of integrability. Their presence confines time evolution to a lower-dimensional hypersurface of the corresponding microcanonical shell of the phase space, preventing thermalization.

By contrast, in quantum systems it is not clear under what conditions conserved quantities provide independent constraints on the relaxation dynamics. To begin with, in quantum systems it is linear independence and not functional independence that matters.

Moreover, every $\hat{H}$ has $\dim(\mathcal{H})$ integrals of motion, i.e. the projectors $\hat{P}_\alpha = |\alpha\rangle \langle \alpha|$ on its eigenstates. The GGE built with all these operators is by definition equivalent to the so-called diagonal ensemble², which describes the long-time average (i.e. weak relaxation) of arbitrary observables.

$$\lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T \langle \psi(t) | \hat{O} | \psi(t) \rangle dt = \sum_\alpha |\psi_\alpha|^2 \langle \alpha | \hat{O} | \alpha \rangle = \text{Tr} \left[\hat{O} \sum_\alpha |\psi_\alpha|^2 |\alpha\rangle \langle \alpha| \right]$$

However, this fact does not imply exact solvability of the model.

Trivially, a $\hat{\rho}_{GGE}$ built using the $\{\hat{P}_\alpha\}$ encodes almost all information about the initial pure state (i.e. all the parameters $|\psi_\alpha|^2$ up to their relative phases), rather than only a minimal set of integrals of motion which, together with the maximal entropy principle, allows one to find the

¹While commutation with $\hat{H}$ is required by the very definition of conserved charge, pairwise commutation of the $\{\hat{H}_n\}$ is not strictly necessary [27, 54, 125]. However, here we always assume conserved charges to be in involution.

²...assuming for simplicity that the energy spectrum is non-degenerate. When there are degeneracies, it is possible that one should include additional integrals of motion in order to correctly capture the stationary behaviour. However, in this work we are not going to discuss the diagonal ensemble, the eigenstate thermalization hypothesis and the dephasing mechanism for relaxation in more detail than what is strictly needed for our purposes. For a more formal and complete discussion the reader is referred to [40].

stationary state. Explicitly, if we consider a lattice system with local Hilbert space dimension d , then the number of projectors $\hat{P}_\alpha$ is $\dim(\mathcal{H}) = d^L$. By contrast, we expect the number of GGE Lagrange multipliers to be extensive ($\propto L$), in agreement with the high compression of needed information implicit in the concept of relaxation to a statistical ensemble.

However, an important fact is that the conservation laws associated to $\{\hat{P}_\alpha\}$ are nonlocal, while *only conservation laws associated to (quasi)local extensive operators $\{H_n\}$, turn out to have important consequences for both solvability and local relaxation.*

Thus, as we mentioned in the previous paragraph, the very distinction between this kind of GGE and that of an integrable system is that, in the latter, the charges can be chosen to have some notion of *locality*. We will give the suitable formal definitions in §3.4.

A discussion and comparison of some common definitions of integrability and why none of them can be considered fully general can be found in [32].

GGE in Terms of Momenta Occupation Numbers in Noninteracting Models: In §5 we deal with the noninteracting XX model, namely the anisotropic Heisenberg model with $\Delta = 0$.

The corresponding fermionic Hamiltonian is easily diagonalized by Fourier-transforming. The L obtained mode-occupation operators $\{\hat{n}_k\}$ are conserved and their number is extensive. Therefore, one may wonder whether, in noninteracting models, these operators may be a suitable choice of conserved charges for the GGE, despite they are nonlocal. The answer is in most cases positive and it is equivalently possible to define a set of L conserved charges, linearly related to the $\{\hat{n}_k\}$, which can be expressed as sums of local densities [3, 51, 57, 107]. For the XX model:

$$\hat{H}_n = \sum_k h_n(k) \hat{n}_k \quad \begin{array}{ll} h_n(k) = \epsilon(k) \cos[nk/2] & \text{if } n \text{ even} \\ h_n(k) = J \cos[(n+1)k/2] & \text{if } n \text{ odd} \end{array} \quad n \in \{0, \dots, L-1\} \quad (3.5)$$

where $\hat{H}_0$ consistently coincides with the Hamiltonian (we have clearly some arbitrariness in defining the charges). Both sets of conserved operators are valid choices for the GGE.

We just mention that in integrable quantum field theories an analogous correspondence exists between local conserved charges and the occupation operators for Faddeev-Zamolodchikov quasiparticles [52, 59, 127, 152]. However, even in free models there are cases in which local conservation laws exist that cannot be expressed in terms of mode occupation operators only [51].

We do not pursue this further, since in interacting lattice models it is in general more convenient to directly work with local charges. We will only use mode occupation operators for the XX model in §5, where the $\{\hat{n}_k\}$ represent a suitable complete set.

The Role of the Initial State: Before proceeding, above we said that relaxation to a statistical ensemble (and in particular a suitable $\hat{\rho}_{GGE}$) is "expected", since it has been numerically and analytically checked in many settings and, heuristically, we can think of the complementary infinite subsystem $\bar{\mathbb{A}}$ as a heat bath.

However, a general answer to this question is complicated, even after the restriction to one-dimensional systems. Indeed, it depends both on the driving Hamiltonian and the initial state.

For example, let us consider as initial state the superposition of a finite number of eigenstates of the Hamiltonian (but more than one). This implies that (3.1) receives contributions only from a finite set of nonzero frequencies $\Delta E_{\alpha\beta}$. Hence, the time evolution of every observable is periodic with finite period (or quasi-periodic), even if $L \rightarrow +\infty$.

Moreover, for noninteracting Hamiltonians, where the GGE can be expressed in terms of suitable mode occupation operators $\hat{n}_k$, a sufficient requirement on the initial state, weaker than Gaussianity, has been proven to be the so-called *cluster-decomposition property* [126], which states that any two infinitely-distant subsets of points must be uncorrelated.

$$\lim_{r \rightarrow +\infty} (\text{Tr}[\hat{\rho}(0) \hat{O}(x) \hat{O}(x+r)] - \text{Tr}[\hat{\rho}(0) \hat{O}(x)] \text{Tr}[\hat{\rho}(0) \hat{O}(x+r)]) = 1$$

The $2n$ -rainbow state displays correlations with range up to $2n$ and, even when we assume $n \rightarrow +\infty$, in order to preserve macroscopic homogeneity we still require $L/2n \rightarrow +\infty$. This means that local observables whose supports are separated by more than $2n$ sites exist and are uncorrelated, i.e. our state satisfies the cluster-decomposition property.

However, in §1 we more specifically proved that it is Gaussian.

3.3 STATIONARY STATES AND TBA: THE GENERALIZED MICROCANONICAL DESCRIPTION

Up to now, in this and the previous chapter, it may seem that we have introduced methods for the solution of two different and disjoint problems:

- In §2 the TBA has provided an efficient description of the eigenstates of $\hat{H}$ in terms of root densities. Note that this is primarily a diagonalization method, with no direct connection to dynamics.
- When considering an initial state $|\psi(0)\rangle$ whose evolution is driven by an integrable Hamiltonian $\hat{H}$, local observables are expected to reach stationary values. Asymptotically, the state of the system becomes indistinguishable from a GGE whose stationary Lagrange multipliers are fully determined by the expectation values on the initial state of a suitable complete set of (quasi)local conserved charges.

Here we aim to highlight their connection.

$\hat{\rho}_{GGE}$ provides a complete description of the local physics at stationarity.

Nevertheless, being the GGE density matrix impractical to handle directly, it is convenient to exploit once more the concept of *local equivalence of states* and wonder whether, analogously to the replacement of the time-dependent $|\psi(t)\rangle \langle \psi(t)|$ by the stationary $\hat{\rho}_{GGE}$, we can also replace the mixed state $\hat{\rho}_{GGE}$ with a single eigenstate $|\alpha\rangle \langle \alpha|$, called *representative eigenstate*.

$$\text{Tr}_{\bar{A}}[|\alpha\rangle \langle \alpha|] = \text{Tr}_{\bar{A}}[\hat{\rho}_s] = \lim_{t \rightarrow +\infty} \text{Tr}_{\bar{A}}[|\psi(t)\rangle \langle \psi(t)|] \quad \forall \text{A finite compact subsystem in an infinite system} \quad (3.6)$$

This is the so-called *generalized microcanonical description* of the local stationary state³.

It would be a striking simplification, since even if $|\alpha\rangle$ is a complicate Bethe eigenstate, in the thermodynamic limit all its local physics is encoded in an infinite set of scalar functions (i.e. the m -string root densities). These are much easier to work with, rather than the matrix $\hat{\rho}_{GGE}$, exponential of an extensive number of operators defined on an exponentially large Hilbert space. However, an a priori justification of its feasibility is definitely not trivial.

Thermalization, Microcanonical Description and Eigenstate Thermalization Hypothesis: In generic system, we expect thermalization to a GE which is only dependent on the energy of the initial state. This intuition is based on the *eigenstate thermalization hypothesis (ETH)*, proposed in [42, 128]: eigenstates with energy density in the same infinitesimal energy window are indistinguishable from the viewpoint of the expectation values of local operators. This implies that the late-time behaviour should agree with the predictions of the canonical Gibbs ensemble with inverse temperature β fixed by the energy. In this sense, the ETH states that thermalization is mainly a property of the Hamiltonian spectrum and eigenstates rather than the dynamics. Note that in this discussion nor the averaging over a microcanonical energy shell nor over time is involved. For a wide pedagogical review about the ETH, its consequences, and generalizations the reader is referred to [40].

³Once again, "generalized" is referred to the comparison with nonintegrable systems, where we expect thermalization to a Gibbs ensemble. Since there are not many differences in the discussion, for brevity here we view the non-generalized formulation as a particular case.

GGE, Generalized Microcanonical Description and Generalized ETH: In integrable systems, due to the presence of an extensive number of additional local conservation laws, the original ETH proves to fail: eigenstates with arbitrarily near or degenerate energies can display different values for the same local observable.

Nevertheless, one may wonder if a sufficiently "complete" set of local conserved charges exists such that all the simultaneous eigenstates of them and the Hamiltonian corresponding to the same set of eigenvalues are, as above, locally equivalent.

This hypothesis is called *generalized ETH*. It was first proposed for noninteracting systems in [28] and then adapted to interacting integrable models by Caux and Essler in [31].

According to the generalized ETH, in interacting integrable models the representative eigenstate is fixed by the expectation values of all the (quasi)local charges on the initial state $|\psi(0)\rangle$. Shifting to the thermodynamic limit, in §2 we showed that an increasing number of eigenstates corresponds to the same set of root densities. The latter encode all the information about the expectation values of local observables.

In particular, since different root densities lead to different expectation values for at least one observable, *if there is a one-to-one correspondence between the $\{\hat{H}_n\}$ of the initial state and the root densities, then $\{\hat{H}_n\}$ is certainly a "complete set" [93].*

3.4 CLASSIFICATION OF LOCAL OPERATORS

In the next chapter we are going to discuss the problem of finding a complete set of charges for the GGE in the XXZ model. As preliminary background, we now need to give a more formal notion of (quasi)local conserved charge [72]. Indeed, for the XXZ model, strictly extensive local charges turn out to be insufficient and a macroscopic family of hidden local conservation laws, referred to as the (extensive) quasi-local charges is uncovered.

For the sake of simplicity, we restrict a priori to infinite one-dimensional lattice systems.

$$\begin{aligned}\mathcal{H} & \quad \text{total Hilbert space of the lattice, with sites } \{j\} = \mathbb{Z} \\ \mathcal{H}_{[i,j]} & \quad \text{Hilbert space for sites from } i \text{ to } j\end{aligned}$$

We denote by $\mathfrak{U}_{[i,j]}$ the C^* -algebra of operators $\hat{a}_{[i,j]} : \mathcal{H}_{[i,j]} \rightarrow \mathcal{H}_{[i,j]}$, where we recall the product is represented by the ordinary composition of operators, while the involution by the Hermitian conjugation [23].

An arbitrary operator $\hat{a} : \mathcal{H} \rightarrow \mathcal{H}$ is called *local*, if a finite interval $[i, j] \subset \mathbb{Z}$ exists s.t.

$$\hat{a} = \hat{a}_{[i,j]} \otimes \hat{\mathbb{I}}_{\mathbb{Z} \setminus [i,j]} \quad \text{where} \quad \hat{a}_{[i,j]} \in \mathfrak{U}_{[i,j]}$$

This decomposition is clearly not unique. The smallest interval $[i, j]$ for which it is possible is referred to as the *support* of the local operator $\hat{a}$, while $|i - j| + 1$ is its *order of locality*.

For example, $\hat{h}_{j,j+1}$ is a local operator of order 2 with support $[j, j+1]$.

The corresponding quasilocal C^* -algebra $\mathfrak{U}$ of operators $\mathcal{H} \rightarrow \mathcal{H}$ is defined as the limit of the sequence

$$\{\mathfrak{U}_{[-n,n]} : n \in \mathbb{N}^{>0}\}$$

closed in the operator norm topology [23]. Every local operator is an element of $\mathfrak{U}$, but the infiniteness of this sequence opens the door to the existence of a new family of operators with infinite support while preserving a weaker notion of locality.

In order to define a suitable inner product, it is convenient to pass by the tracial state $\omega_0(\hat{a})$ of a local operator $\hat{a}$ with support $[i, j]$:

$$\omega_0(\hat{a}) := \frac{\text{Tr}_{[i,j]}[\hat{a}_{[i,j]}]}{\text{Tr}_{[i,j]}[\hat{\mathbb{I}}_{[i,j]}]}$$

This is extended over the entire $\mathfrak{U}$ by continuity.

$$\forall \hat{\alpha}, \hat{\beta} \in \mathfrak{U} \quad \langle \hat{\alpha}, \hat{\beta} \rangle := \omega_0(\hat{\alpha}^\dagger \hat{\beta}) - \omega_0(\hat{\alpha}^\dagger) \omega_0(\hat{\beta}) \quad \text{is an inner product}$$

Note that, if $\hat{a}$ and $\hat{b}$ have disjoint support, the trace of their product factorizes as the product of their traces, thus $\omega_0(\hat{a}\hat{b}) = \omega_0(\hat{a})\omega_0(\hat{b})$. In particular, if they are traceless, they are orthogonal.

The above inner product automatically induces the norm⁴ $\|\hat{\alpha}\| := \sqrt{\langle \hat{\alpha}, \hat{\alpha} \rangle}$.

The same holds for all the subalgebras $\mathfrak{U}_{[i,j]}$.

We denote by $\{\hat{\sigma}_j^x : x \in \{0, \dots, d^2 - 1\}\}$ an $\|\cdot\|$ -orthonormal basis for $\mathfrak{U}_{[j,j]}$, where d is the dimension of the local Hilbert space. For example, if $d = 2$ (e.g. for a lattice of $\frac{1}{2}$ -spins) we can choose $\hat{\sigma}_j^x = \hat{\mathbb{I}}_j$ and $\{\hat{\sigma}_j^1, \hat{\sigma}_j^2, \hat{\sigma}_j^3\}$ as the Pauli matrices.

This local basis automatically induces an orthonormal basis for every $\mathfrak{U}_{[i,j]}$, i.e.

$$\begin{aligned} \{\hat{\sigma}_{[i,j]}^{\vec{x}} : \vec{x} \in \{0, \dots, d^2 - 1\}^{|i-j|+1}\} \quad \text{where} \quad \hat{\sigma}_{[i,j]}^{\vec{x}} &:= \hat{\sigma}_i^{x_0} \otimes \hat{\sigma}_{i+1}^{x_1} \otimes \dots \otimes \hat{\sigma}_j^{x_{j-i}} \\ \langle \hat{\sigma}_{[i,j]}^{\vec{x}}, \hat{\sigma}_{[i,j]}^{\vec{x}'} \rangle &= \delta_{\vec{x}\vec{x}'} \end{aligned}$$

Once introduced a lattice shift automorphism of local operators $\hat{\mathcal{S}}_\Delta$ s.t.

$$\begin{aligned} \text{if } \hat{a} \text{ is a local operator} \\ \text{with support } [i, j], \text{ then:} \end{aligned} \quad \hat{\mathcal{S}}_\Delta[\hat{a}_{[i,j]} \otimes I_{\mathbb{Z} \setminus [i,j]}] := \hat{a}_{[i+\Delta, j+\Delta]} \otimes I_{\mathbb{Z} \setminus [i+\Delta, j+\Delta]}$$

(we do not specify further its formal definition since the meaning is intuitive) we can formally define extensive local operators in the traditional strong sense, i.e. as translationally invariant sums over local densities.

$$\begin{aligned} \text{If } \exists \hat{a}^{(n)} \in \mathfrak{U} \text{ local operator} \\ \text{of order } n \text{ s.t.} \end{aligned} \quad \hat{A}^{(n)} := \sum_{\Delta \in \mathbb{Z}} \hat{\mathcal{S}}_\Delta[\hat{a}^{(n)}] \quad \text{then } \hat{A}^{(n)} \text{ is a local extensive operator of order } n$$

For example, $\hat{H}_{XXZ}$ is an extensive local operator of order 2 with density $J\hat{h}_{j,j+1}$.

We stress that $\hat{A}^{(n)}$ is not an element of $\mathfrak{U}$ (its operator norm would be infinite). However, it can be viewed as representing a sequence of operators $\{\hat{A}^{(L,n)} \in \mathfrak{U}\}$ acting on finite lattices of increasing length⁵, namely

$$\hat{A}^{(L,n)} := \sum_{\Delta=-\lfloor L/2 \rfloor}^{\lfloor L/2 \rfloor - 1} \hat{\mathcal{S}}_\Delta[\hat{a}^{(n)}] \quad (3.7)$$

This sequence has the following properties:

1. Volume-scaling extensivity:

$$0 < \lim_{L \rightarrow +\infty} \frac{1}{L} \langle \hat{A}^{(L,n)}, \hat{A}^{(L,n)} \rangle < +\infty \quad (3.8)$$

For example, using $\hat{a}^{(1)} = \hat{S}_j^z$ one obtains $\langle \hat{A}^{(L,n)}, \hat{A}^{(L,n)} \rangle = L/4$.

2. Finite overlap with at least one local operator $\hat{b}$, e.g. $\hat{b} = \hat{a}^{(n)}$:

$$\lim_{L \rightarrow +\infty} \langle \hat{b}, \hat{A}^{(L,n)} \rangle \neq 0$$

⁴As remarked in [72], formally, $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ become respectively a proper inner product and norm only after the identity $\hat{\mathbb{I}}_Z$ has been removed from $\mathfrak{U}$. Otherwise, any operator proportional to identity has zero length. Moreover, note that these are supplementary structures with respect to the original C^* -topology.

⁵Note that this sequence does not converge in norm to $\hat{A}^{(n)}$ as $L \rightarrow +\infty$, thus it is not contradictory with the statement $\hat{A}^{(n)} \notin \mathfrak{U}$ despite $\mathfrak{U}$ is closed in the operator norm topology.

Finally, abandoning any reference to the sum over a shifted density, we define an *extensive quasilocal (or pseudolocal) operator* $\hat{\chi}$ as a sequence $\{\hat{\chi}^{(L)}\}$ satisfying these two properties.

In particular, any $\{\hat{\chi}^{(L)}\}$ given as in (3.7) by the sum over a shifted quasilocal density $\hat{\alpha} \in \mathfrak{U}$ s.t.

$$\hat{\alpha} = \sum_{r=0}^{+\infty} \hat{\alpha}_{[0,r]} \quad \text{where } \exists C, \xi > 0 \text{ s.t. } \|\hat{\alpha}_{[0,r]}\| < C e^{-r/\xi} \quad (3.9)$$

and the terms $\hat{\alpha}_{[0,r]}$ are mutually orthogonal, is an extensive quasilocal operator.

However, the above abstract definition is more general.

As final remark, in this section we made accurate distinction between (quasi)local operators and extensive (quasi)local operators. Nevertheless, as in this chapter, we will often omit "extensive" if it is clear from the context.

3.5 A COMPLETE SET OF CHARGES FOR THE XXZ MODEL

Quench-Action Approach as Test for the Completeness of a GGE: As we discussed at the end of §3.3, the GGE conserved charges must form a "complete set" in the sense that their initial state expectation values together with entropy maximization must uniquely fix root densities leading to a correct description of the stationary state.

However, to establish when a set of charges is truly complete is highly nontrivial.

Indeed, while maximization of entropy under these constraints generally leads to a unique solution, if the set of charges is not complete, the entropy maximization problem is underconstrained, yielding an incorrect LSS.

The reliability of a set of charges can only be checked by comparing predictions with experiments, numerical simulations, or other predictions which are known to be exact. These alternatives are possible only for a few special models and initial states.

Instead, our set of charges is clearly complete if expectation values displays a one-to-one correspondence with root densities, without any need of entropy minimization. Let us shelve for the moment this second very pleasant scenario.

We recall that mode occupations $\{\hat{n}_k\}$ prove to be a suitable complete set for a wide range of noninteracting Hamiltonians and initial states [126]. These can be linearly combined yielding a complete set of extensive strictly-local charges for the GGE.

However, now we turn to genuinely interacting integrable models, where strings solutions are present. In particular, here we focus on the XXZ model. In this framework, the main benchmark for the validity of the GGE is the *quench-action approach*, introduced in [31]. We do not explain the method here, since we do not employ it: for a pedagogical review the reader is referred to [30].

For our purposes, the important fact is that it enables a characterization of the stationary diagonal ensemble within the TBA formalism. The key idea is the introduction of an effective action whose saddle point corresponds to the LSS root densities. The quench action is defined in terms of both the Yang-Yang entropy and the overlaps between the initial state and the Bethe eigenstates. Given its extensivity, the saddle point solution is expected to be exact in the thermodynamic limit. Thus, predictions do not require the prior identification of a complete set of conserved charges.

Non-Completeness of local Charges in the XXZ Model: In §6 and §7 we will show how for the XXZ model, exploiting the relation between $\hat{H}_{XXZ}$ and the transfer matrix of the 6-vertex model, one is able to explicitly construct an extensive number of extensive local conserved charges.

Thus, one might expect these to form a suitable complete set for the GGE.

In [24, 93, 106, 149] the quench-action approach was used to exactly predict the equilibrium expectation values of some local observables in the XXZ model, after the quench from different initial states. These results were compared with the corresponding local-GGE predictions.

Surprisingly, local charges turned out to be insufficient for the characterization of the LSS. This generally holds in genuinely interacting models with string solutions of the Bethe equations.

This means that, in such integrable models, additional "effectively local" conserved charges exist which are linearly independent of the strictly local ones, but play a crucial role in constraining the relaxation dynamics. In [71], Ilievski, Medenjak and Prosen proposed a method to construct an infinite set of *quasilocal* conserved charges for the XXZ model. Then, in [73], a one-to-one correspondence was proved between their values on the initial state and the LSS root densities, called *string-charge duality*.

We will discuss this method in §7.

3.6 SYMMETRY RESTORATION

The spin $2n$ –rainbow state is invariant under both the $U(1)$ symmetry $\hat{U}(\theta) = \exp[(i/2)\theta \hat{M}^z]$ (it has null total magnetization) and global spin flip. Instead, unlike $\hat{H}_{XXZ}$, it is only invariant under $2n$ –site shifts. One may wonder whether the corresponding stationary state $\hat{\rho}_s$ is translationally invariant or not.

This question deals with the more general issue of *symmetry restoration*, i.e. whether a Hamiltonian symmetry $\hat{U}$ broken by the initial state is recovered by the LSS or not [51, 56].

$$[\hat{H}, \hat{U}] = 0 \quad \wedge \quad \hat{U} |\psi(0)\rangle \neq |\psi(0)\rangle \quad \xrightarrow{?} \quad [\hat{\rho}_s, \hat{U}] = 0$$

If we assume the stationary state to be described by a GGE with local charges $\{\hat{H}_n\}$ in involution, then any $\hat{U}$ symmetry is restored if and only if all conserved charges are invariant under the corresponding transformation, i.e. $[\hat{H}_n, \hat{U}] = 0 \quad \forall n$

For example, $\hat{M}^z$ has to be spanned by the GGE charges: since it is not invariant under global spin flip, we expect this symmetry not to be restored by the stationary state. This is consistent with the fact that $|\downarrow\downarrow\downarrow \dots\rangle$ and $|\uparrow\uparrow\uparrow\rangle$ evolve trivially (they are eigenstates). However, the $2n$ –rainbow state is invariant under global spin flip and its LSS inherits this symmetry.

Instead, all the quasilocal charges of the XXZ model are invariant under translation and $U(1)$, hence these symmetries are restored. In particular, *the $2n$ –rainbow state have to relax to a translationally invariant local stationary state*.

3.7 THERMALIZATION HYPOTHESIS FOR THE $2n$ –RAINBOW STATE

At the end of §7 we will present a conjecture proposed in [99], i.e. that "long random states" in the XXX model thermalize to infinite temperature as their unit cell becomes infinite-sized. Given the high structural similarity with these states, we propose to extend this conjecture to our family of initial states in the XXZ model.

More explicitly, as we mentioned in §1, the main question we address in this work is whether the $2n$ –rainbow state thermalizes to infinite temperature in the XXZ model as $n \rightarrow +\infty$.⁶ In next chapters we will try to answer this question with both analytical and numerical approaches.

Why Would it be Remarkable? According to the theoretical framework discussed in the present chapter, we expect our initial states to relax to a GGE expressed in terms of the quasilocal charges introduced in [71]. Instead, the statement that it *thermalizes* as $n \rightarrow +\infty$ is much stronger and highly nontrivial.

⁶The $2n$ –rainbow state satisfies macroscopic homogeneity in the $n \rightarrow +\infty$ limit as well. Indeed, we always require $L/n \rightarrow +\infty$ (infinite number of unit cells in the lattice). Thus our question is well defined.

Moreover, the $2n$ –rainbow state is pure and displays an elaborated, ordered, and regular Bell-pair entanglement structure. By contrast, the infinite temperature thermal state is maximally mixed and its density matrix is strikingly simple and featureless $\hat{\rho} \propto \hat{\mathbb{I}}$.

Note that highly entangled initial states such as the crosscap state (1.6) the state are locally equivalent to the infinite-temperature thermal state *from the beginning*. Due to the extensive entanglement range, every macroscopic subregion $\mathbb{A}$ of the initial state is already highly entangled with the environment $\bar{\mathbb{A}}$, yielding a massive loss of information when tracing out the complementary degrees of freedom.

Instead, in the $2n$ –rainbow state, every macroscopic subsystem $\mathbb{A}$ is assumed to contain an almost infinite number of unit cells, thus it is only slightly entangled with the environment. This is even more evident with the original states of the conjecture, where unit cells are random spin sequences with no entanglement.

Thus, we expect the high loss of information in our framework not to be due to an initial extensive entanglement range.

Rather, as in presence of unit cells which are random chains of spins, it is expected to come from the dynamic action of the hopping terms in the model, which finally lead to the restoring of translational invariance. Despite the extensive number of quasilocal conserved charges constrains this evolution, *the random nature of the unit cell, as $n \rightarrow +\infty$, guarantees that the initial expectation values of the conserved charges by definition agree, up to non-extensive corrections, with the infinite-temperature counterparts.*

As we will discuss more formally in §7, the unit cells of the $2n$ –rainbow state are very near to be a linear superposition of random chains of spins and we may expect them to share the above property.

Chapter 4

GENERALIZED HYDRODYNAMICS

As we will discuss later, apart from directly testing whether the stationary state root densities of the $2n$ –rainbow state converge to infinite-temperature thermal profiles upon increasing n , we will also study what kind of out-of-equilibrium dynamics develops by suddenly joining at $t = 0$ the pure $2n$ –rainbow state with the mixed infinite-temperature thermal state.

In particular, we are interested in the transport of conserved charges across the system and how their density profiles change over time.

In this chapter we introduce a mathematical framework yielding analytical predictions for this dynamic in a suitable scaling limit.

4.1 SETTING AND GENERAL INTUITION

Quench Protocols: The most elementary out-of-equilibrium protocol in which relaxation and thermalization of a quantum system (defined in §3) can be tested, both analytically, numerically and experimentally, is the *quantum quench* [105]: an initial state is suddenly left to evolve unitarily under a Hamiltonian $\hat{H}$ of which it is assumed not to be an eigenstate, nor the superposition of a finite number of eigenstates. In *homogeneous quenches*, both the initial state and the Hamiltonian are translationally invariant, at least at a macroscopic level.

By contrast, the simplest *inhomogeneous quench* employs an initial state built by joining two internally-homogeneous different halves. This is called *bipartitioning protocol*. Another common possibility, relevant for instance in trapped cold-atom experiments, is to consider an inhomogeneous background, typically a confining potential, leading to *trap quenches*.

We mainly consider homogeneous quenches and bipartitioning protocols here. Only in §11.2 we deal with a bipartite Hamiltonian $\hat{H} = \hat{H}_L + \hat{H}_R$.

We focus on analytical computations and numerical procedures. However, in the last two decades it has become possible to investigate quantum quenches experimentally in cold-atom systems. For this topic the reader is e.g. referred to [36, 67, 77, 78, 86, 137].

When Translational Invariance is Broken: As discussed in the previous chapter, once a suitable complete set of charges is provided, in (macroscopically) translationally invariant settings the corresponding GGE (or a representative eigenstate) determines the stationary values of local observables even in the presence of non-trivial interactions.

However, heuristically, when macroscopic translational invariance is broken, the very assumption that sufficiently late times t exist such that the expectation value of a given local observable $\hat{O}(x)$ reaches stationarity over the whole lattice is no longer acceptable.

Indeed, the expectation values of local observables in lattice models with short-range interactions are known to display a causal structure interpreted with the existence of a maximal velocity v_{max} for the information spreading. This causal structure emerges although the Schrödinger

equation, unlike relativistic theories, has no built-in speed limit. This behaviour is experimentally observed. Moreover, in 1972 Lieb and Robinson [90] formally proved the existence of the velocity v_{max} , only depending on the Hamiltonian and not on the initial state, such that out of the corresponding lightcone the upper bound for the norm of the commutator of two local observables decays exponentially. The theorem was originally stated for translationally invariant Hamiltonians with short-range interactions, but nowadays several generalizations are known.

This causal structure implies that a macroscopic inhomogeneity in either the initial state or the Hamiltonian induces a nontrivial spacetime dependence in the expectation values of local observables. For example, at every fixed t , the expectation values of local observables sufficiently far from the inhomogeneity have to relax in the same way as in the corresponding homogeneous quench. However, the region in which the relaxation dynamics is affected by the inhomogeneity is expected to expand with time.

In integrable systems this expansion is observed to be linear (leading to the so-called *light-cone dynamics*) and this phenomenon was rapidly connected to the existence of stable quasi-particle excitations [22]. The v_{max} of the Lieb-Robinson theorem is naturally interpreted as an upper-bound for the group velocities of these excitations in integrable systems satisfying its assumptions. However, remarkably, the maximal velocity underlying the observed lightcone effects turned out to depend on the specific initial state.

Apart from the purely theoretical interest, the development of suitable predictive methods for inhomogeneous quenches was an important challenge, since in many interesting settings translational invariance is violated. For example, this is necessary to observe a non-trivial transport of conserved charges, or in every cold-atoms experimental realization involving a trapping potential. Thus, *generalized hydrodynamics* (GHD) was introduced in [15, 29] to fill this gap.

It enables the formulation of a weaker concept of relaxation, still due to dephasing, in inhomogeneous settings, leading to *local quasi-stationary states* (LQSS). This yields analytical predictions of the spacetime profiles of local observables and currents in a suitable scaling limit. Moreover, LQSSs can once again be characterised by a GGE or, more conveniently, a representative eigenstate (3.6), which can be described within the TBA formalism. However, unlike true local stationary states, they are spacetime-dependent.

For a recent pedagogical review about GHD, the reader is referred to [3].

In this work, besides the homogeneous quench from the $2n$ -rainbow state in the XXZ model, we also analytically and numerically study the *bipartitioning protocol* between this state and the infinite-temperature state, or between $2n$ -rainbow states with different values of n .

Semiclassical Ballistic Carriers: At the core of GHD lies the following semiclassical argument, sharing the same intuition as the quasiparticle picture discussed later in §5.

Consider for simplicity a homogeneous quench: heuristically, from the perspective of the post-quench Hamiltonian and the stationary state, the initial system lies in a highly excited state. Semiclassically, one can imagine it as a gas of excitations, which then start to spread across the system. If the post-quench model is integrable (or a CFT, but we will not touch upon that here), these excitations are stable and can be assumed to undergo a ballistic motion.

As a result, the post-quench dynamics of expectation values of local observables is effectively described in terms of the ballistic spreading of these quasiparticles through the system.

In Bethe-ansatz solvable models, it is natural to make correspondence between these ballistic carriers and the quasiparticles (or bound states of m quasiparticles in models with string solutions). In noninteracting systems, spreading velocities are easily computed from the dispersion relation of quasiparticles. By contrast, in interacting systems quasiparticles are thought of as the low-lying particle/hole excitations around the stationary state. Their group velocities become state- and spacetime-dependent, as we will discuss in more detail in the following sections.

We just mention that the main difference in the case of non-integrable dynamics is that the quasiparticles representing excitations are no longer stable and display a finite lifetime.

4.2 NONINTERACTING SYSTEMS

Local Conserved Charges and Currents: As discussed in §3, in integrable systems the Hamiltonian can be rewritten (up to a constant) as sum over occupation number operators of free particles. A complete set of conserved local charges $\hat{H}^{(i)}$ exists, which can be written the same way by introducing the charge $\hat{h}^{(i)}(k)$ associated to a single particle of momentum k .

$$\hat{H} := \hat{H}^{(0)} = \sum_k \epsilon(k) \hat{n}_k \quad \hat{H}^{(i)} = \sum_k h^{(i)}(k) \hat{n}_k \quad [\hat{H}^{(i)}, \hat{H}^{(j)}] = 0 \quad (4.1)$$

Their explicit expression is displayed in (3.5).

Moreover, these charges are extensive in number and linearly independent.

Since the charges $\hat{H}^{(i)}$ are extensive local operators, they can be rewritten as sums of local densities $\hat{h}_x^{(i)}$. Thus the commutator $[\hat{H}, \hat{h}_x^{(i)}]$ is local as well.

$$\rightarrow \quad \exists \hat{j}_x^{(i)} \quad \text{local operator s.t.} \quad [\hat{H}, \hat{h}_x^{(i)}] = i(\hat{j}_x^{(i)} - \hat{j}_{x-1}^{(i)}) \quad \text{and} \quad \langle \Omega | \hat{j}_x^{(i)} | \Omega \rangle = 0 \quad \forall x \quad (4.2)$$

This is the definition of the *current (density) operator* (the prefactor i makes it Hermitian) and its existence can be physically interpreted as the fact that the corresponding charge obeys a continuity equation. Indeed, by adopting the Heisenberg picture:

$$\hat{h}_x^{(i)}(t) := e^{i\hat{H}t} \hat{h}_x^{(i)} e^{-i\hat{H}t} \quad \rightarrow \quad \partial_t \hat{h}_x^{(i)} = i[\hat{H}, \hat{h}_x^{(i)}] = -(\hat{j}_x^{(i)}(t) - \hat{j}_{x-1}^{(i)}(t)) \quad (4.3)$$

This highlights a fundamental difference between local and quasilocal integrals of motion: *a local conservation law can be recast in the form of a continuity equation involving a local current. This in general fails with quasilocal conservation laws, which have a more elusive physical meaning, despite displaying an equally crucial effect on the integrable dynamics.*

Completeness of these local charges translates functionally in the fact that, in the continuous approximation, $\{h^i(k)\}$ is a complete (but not orthogonal) set of functions in $L^2[0, 2\pi]$, i.e.

$$\int_0^{2\pi} h^{(i)}(k) f(k) dk = 0 \quad \leftrightarrow \quad f(k) = 0 \quad \forall k \quad (4.4)$$

Usually, in the noninteracting case it is possible to find exact analytic expressions for conserved charges and currents (such as (3.5)), while this generally fails in genuinely interacting models.

From another viewpoint, there is a one-to-one correspondence between the root density $\rho(\lambda)$ of the local stationary state and the expectation values of all charge densities on the initial state in the thermodynamic limit [3].

Homogeneous Quench: Consider the situation in which the initial state $|\psi_0\rangle$ of the quench is translationally invariant. As we discussed in §3, after the quench the system as a whole undergoes an eternal unitary evolution, without any loss of information about the initial state. Nevertheless, if we restrict to local observables in an infinite system, we expect to observe relaxation to a local stationary state $\hat{\rho}_s$ as defined in (3.3).

Moreover, exploiting translational invariance, we can always write

$$\lim_{t \rightarrow +\infty} \lim_{th} \langle \psi_0 | \hat{O}_x(t) | \psi_0 \rangle = \lim_{t \rightarrow +\infty} \lim_{th} \langle \psi_0 | \hat{O}_0(t) | \psi_0 \rangle \quad (4.5)$$

where the existence of this late-time limit for every possible local observable $\hat{O}_x$ is the very definition of relaxation, equivalent to (3.2).

The defining property of the LSS (3.3) is explicitly recast in terms of local observables as

$$[\hat{H}, \hat{\rho}_s] = 0 \quad \text{and} \quad \lim_{t \rightarrow +\infty} \lim_{th} \langle \psi_0 | \hat{O}_x(t) | \psi_0 \rangle = \lim_{th} \text{Tr} [\hat{O}_0 \hat{\rho}_s] \quad \forall \hat{O}_x \text{ local observable} \quad (4.6)$$

The most important advantage of assuming local relaxation, is that it becomes possible to find a representation of the stationary state without explicitly solving the dynamics.

This can be done in the following steps:

{A1} Assume that (4.6) holds for some (still unknown) stationary state $\hat{\rho}_s$.

{A2} Fix $\hat{\rho}_s$ by requiring conservation of the expectation value of all charge densities. Indeed, since $|\psi_0\rangle$ is translational invariant, taking the expectation value of (4.3) yields

$$\partial_t \langle \psi_0 | \hat{h}_x^{(i)} | \psi_0 \rangle = -\langle \psi_0 | \hat{j}_x^{(i)}(t) | \psi_0 \rangle + \langle \psi_0 | \hat{j}_{x-1}^{(i)}(t) | \psi_0 \rangle = 0 \quad \forall i$$

Therefore, *the local densities of the conserved charges are conserved, as well.*

Choosing $\hat{O}_x = \hat{h}_x^{(i)}$ and $x = 0$ in (4.6) and using time invariance to equal the initial expectation value to the asymptotic value, the above result implies that

$$\lim_{th} \text{Tr} [\hat{h}_0^{(i)} \hat{\rho}_s] = \lim_{t \rightarrow +\infty} \lim_{th} \langle \psi_0 | \hat{h}_0^{(i)}(t) | \psi_0 \rangle = \lim_{th} \langle \psi_0 | \hat{h}_0^{(i)}(0) | \psi_0 \rangle \quad \forall i$$

As discussed in §3, we expect to be able to characterize the LSS microcanonically, i.e. that an eigenstate $|\psi_s\rangle$ exists s.t. $\hat{\rho}_s = |\psi_s\rangle \langle \psi_s|$ satisfies (4.6). Let $\rho_s(k) = \langle \hat{n}_k \rangle_s / 2\pi$ be the root density in momentum space of the chosen eigenstate $|\psi_s\rangle$. Given $h^{(i)}(k)$ as defined in (4.1), the previous equation can be rewritten as

$$\lim_{th} \langle \psi_0 | \hat{h}_0^{(i)}(0) | \psi_0 \rangle = \int_0^{2\pi} h^{(i)}(k) \rho_s(k) dk \quad \forall i$$

The final step is to find the scalar function $\rho_s(k)$ which satisfies this set of integral equations. Note that usually for noninteracting systems an exact analytic expressions of every $h^{(i)}(k)$ and $\lim_{th} \langle \psi_0 | \hat{h}_0^{(i)}(0) | \psi_0 \rangle$ is available. $\rho_s(k)$ is fully equivalent to the knowledge of $\hat{\rho}_s$.

Bipartitioning Protocol: Consider now a quench in which the initial state is not translationally invariant, being obtained by joining two translationally invariant states, i.e. $|\psi_0\rangle = |\psi_L\rangle \otimes |\psi_R\rangle$.

The post-quench Hamiltonian is still invariant under translations but, crucially, the initial state is not. This means that the expectation values $\lim_{th} \langle \psi(t) | \hat{O}_x | \psi(t) \rangle$ at every time t have a nontrivial spatial profile. Thus, the first step (4.5) of the previous strategy to find their large-time limit fails and a notion of stationary state as in (4.6) becomes anachronistic.

Let us first consider {A1}. We aim to find a new formulation of (4.6), dealing with a weaker notion of relaxation. The key idea is that, if the dynamics is ruled by the spreading of ballistic carriers with velocities simply fixed by the dispersion relation of the model, i.e. $v(k) = \partial \epsilon(k) / \partial k$, then we should expect the value of an observable $\hat{O}_x(t)$ to depend only on whether carriers of momentum k coming from the junction have reached position x or not, separately for every k .

In particular, *assuming the effective semiclassical description to hold, the expectation value of every $\hat{O}_x(t)$ must depend only on the spacetime combination $\zeta = x/t$. Thus, we expect a weaker notion of stationarity, still due to dephasing, to emerge separately along each ζ -ray (i.e. along each spacetime line of constant ζ).*

Coming back to (4.6), now it is natural to think about relaxation separately for every value of ζ and together with the prescription to reach $t \rightarrow +\infty$ moving along the corresponding ζ -ray:

$$\lim_{sc} \lim_{th} \langle \psi_0 | \hat{O}_x(t) | \psi_0 \rangle = \lim_{th} \text{Tr} [\hat{O}_0 \hat{\rho}_s(\zeta)] \quad \text{LQSS} \quad (4.7)$$

where $\lim_{sc} : x, t \rightarrow +\infty$ with $\zeta = x/t$ fixed Scaling limit

The stationary state $\hat{\rho}_s(\zeta)$ now explicitly depends on ζ . At fixed t it can be different in every position $x = \zeta t$, and it evolves with time at fixed x . This is the LQSS mentioned in the first section.

The most evident consequence of this assumption is that, for models in which a maximum quasiparticle velocity v_{max} exists (e.g. for systems fulfilling the hypothesis of the Lieb-Robinson bound [90]), along ζ -rays with $|\zeta| > v_{max}$ the LQSS must coincide with the LSS of the homogeneous quench from $|\psi_{L/R}\rangle$ if $\zeta < 0$ or $\zeta > 0$, respectively.

The modification of {A2} is more direct. Now, in taking the expectation value of (4.3), it is no longer possible to exploit translational invariance to write $= 0$ on the right side and the equation keeps its general form.

$$\partial_t \langle \psi_0 | \hat{h}_x^{(i)} | \psi_0 \rangle + \langle \psi_0 | \hat{j}_x^{(i)}(t) | \psi_0 \rangle - \langle \psi_0 | \hat{j}_{x-1}^{(i)}(t) | \psi_0 \rangle = 0 \quad \forall i \quad (4.8)$$

In particular, *local densities are no longer conserved*. Putting everything together, we arrive at the following prescription to predict the late-time local physics of the system along ζ -rays:

{B1} Assume that, in the scaling limit $\lim_{sc}$, every local observable at spacetime positions along a ζ -ray can be asymptotically computed from $\hat{\rho}_s(\zeta)$, i.e. that (4.7) holds for every local observable $\hat{O}_x$.

{B2} Fix $\hat{\rho}_s(\zeta)$ (or the corresponding representative eigenstate) by requiring (4.8).

To show that this strategy is able to determine the stationary state, we proceed as in the homogeneous case. We consider the scaling limit in (4.8) and use (4.7) with $\hat{O} = \hat{h}^{(i)}$ and $\hat{O} = \hat{j}^{(i)}$:

$$\begin{aligned} \lim_{th} \partial_t \text{Tr} \left[\hat{h}_0^{(i)} \hat{\rho}_s(\zeta) \right] + \lim_{sc} \lim_{th} \partial_x \langle \psi_0 | \hat{j}_x^{(i)}(t) | \psi_0 \rangle &= 0 \\ \lim_{th} \left(-\zeta \partial_\zeta \text{Tr} \left[\hat{h}_0^{(i)} \hat{\rho}_s(\zeta) \right] + \partial_\zeta \text{Tr} \left[\hat{j}_0^{(i)} \hat{\rho}_s(\zeta) \right] \right) &= 0 \end{aligned}$$

Note that, by evaluating (4.8) in $x = 0$, we have not lost any information. Indeed, at every fixed ζ , the local stationary state $\hat{\rho}_s(\zeta)$ satisfies translational invariance.

Now we shift as above to the microcanonical description. We denote by $\langle \cdot \rangle_\rho$ the thermodynamic limit of these two expectation values in an eigenstate with root density $\rho(k)$. Let $\rho_\zeta(k)$ be the root density of our ζ -dependent representative eigenstate.

$$-\zeta \partial_\zeta \langle \hat{h}_0^{(i)} \rangle_{\rho_\zeta} + \partial_\zeta \langle \hat{j}_0^{(i)} \rangle_{\rho_\zeta} = 0 \quad (4.9)$$

$$\begin{aligned} \langle \hat{h}_0^{(i)} \rangle_\rho &= \int_0^{2\pi} dk h^{(i)}(k) \rho(k) \\ \langle \hat{j}_0^{(i)} \rangle_\rho &= \int_0^{2\pi} dk h^{(i)}(k) v(k) \rho(k) \quad \text{where} \quad v(k) = \frac{\partial \epsilon(k)}{\partial k} \quad \text{group velocity} \end{aligned} \quad (4.10)$$

The kinetic interpretation of this formula is transparent and $\langle \hat{j}_{n,0} \rangle_\rho$ is the local current density of the charge $\hat{H}^{(i)}$ transported by our ballistic carriers in spacetime positions along the ζ -ray. Expliciting this definition in the continuity equation yields

$$\int_0^{2\pi} h_0^{(i)}(k) [-\zeta \partial_\zeta \rho_\zeta(k) + v(k) \partial_\zeta \rho_\zeta(k)] dk = 0$$

Using the completeness (4.4) of the functions $\{h^{(i)}(k)\}$ we find

$$[v(k) - \zeta] \partial_\zeta \rho_\zeta(k) = 0 \quad (4.11)$$

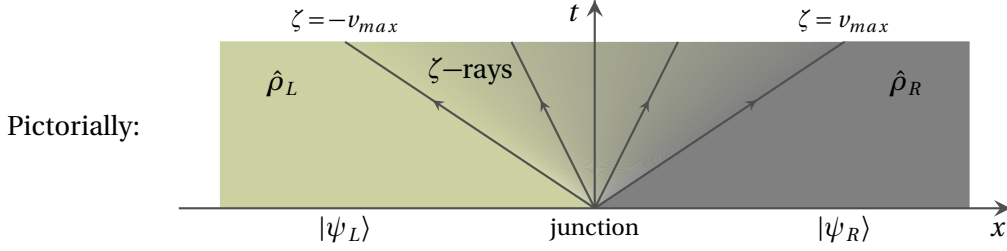
This is a first order differential equation and the solution of interest is selected by requiring as *boundary conditions* that observables along ζ -rays with $\zeta \rightarrow \pm\infty$ relax to the LSS of the homogeneous quench from $|\psi_{L/R}\rangle$, respectively, which we denote by $\hat{\rho}_{L/R}$.

$$\begin{aligned} \lim_{\zeta \rightarrow -\infty} \lim_{sc} \lim_{th} \langle \psi_0 | \hat{O}_x(t) | \psi_0 \rangle &= \lim_{th} \text{Tr}[\hat{O}_0 \hat{\rho}_L] \\ \lim_{\zeta \rightarrow +\infty} \lim_{sc} \lim_{th} \langle \psi_0 | \hat{O}_x(t) | \psi_0 \rangle &= \lim_{th} \text{Tr}[\hat{O}_0 \hat{\rho}_R] \end{aligned} \quad (4.12)$$

Given $\hat{\rho}_{L/R}(k)$ the root density of $\hat{\rho}_{L/R}$, the explicit solution is given by:

$$\rho_\zeta(k) = \rho_L(k) \Theta(v(k) - \zeta) + \rho_R(k) \Theta(\zeta - v(k)) \quad (4.13)$$

where $\Theta(x)$ is the Heaviside step function.



Note that, despite the Heaviside function is discontinuous, charge densities and currents profiles are in general continuous in ζ .

4.3 BIPARTITIONING PROTOCOLS IN INTERACTING SYSTEMS

We aim now to translate the above formulation in Yang-Baxter integrable models, in which, as discussed in the previous chapters:

- Owing to the Bethe ansatz, spectral properties can still be described in terms of stable quasiparticles corresponding to m -string solutions of the Bethe equations, with $m \in \mathbb{N}^{>0}$. These are interpreted as bound states of m quasiparticles. A Bethe eigenstate in the thermodynamic limit is fully characterized by its root densities as functions of the central rapidities of m -strings. However, we emphasize that $\rho_m^{tot}(\lambda)$ is state-dependent as well. Consequently, the knowledge of one of these densities is not sufficient by itself, and the BGT equations must be solved to reconstruct the remaining quantities.
- m -strings with (real) central rapidity λ can likewise be interpreted as stable carriers, but, due to interactions, with ζ -dependent "dressed" velocities $v_{m,\zeta}(\lambda)$. Indeed, as stable excitations around the (quasi) local stationary state, their velocities explicitly depend on the corresponding root densities, which characterize the local quasiparticle population and therefore determine the dressing. In interacting homogeneous quenches, velocities depend on the initial state, but they are still translationally invariant. By contrast, in bipartitioning protocols, they depend on the LQSS, thus they can be different for varying ζ .

From the semiclassical viewpoint, in bipartitioning protocols, owing to the nontrivial space-time dependence of root densities, carriers can follow curved trajectories. Generally, only quasiparticles spreading from the junction ($x = 0$) move along straight lines, since they propagate along a fixed ζ -ray, where the LQSS is constant. Instead, quasiparticles generated in the bulk can experience velocity variations while passing through different ζ -rays.

This implies that an observable at position x along a ζ -ray first receives contributions from λ -rapidity m -strings at time $x/v_{m,\zeta}(\lambda)$, since the corresponding carriers spreading from the junction move with constant velocity $v_{m,\zeta}(\lambda)$. Remarkably, this fact enables the connection with the noninteracting formulation, based on ζ -rays and lightcones.

- In opposition with noninteracting systems, as mentioned in §3, initial values of strictly local charges $\hat{H}^{(i)}$ are not a complete set for the GGE. Instead, this is obtained considering quasilocal charges $\{\hat{X}_s(\mu)\}$ with $s \in \frac{1}{2}\mathbb{Z}^+$, $\lambda \in \mathbb{R}$.

As above, if $\chi_{s,m}(\mu, \lambda)$ is the contribution to $\hat{X}_s(\mu)$ given by a single string of length m and rapidity λ , then completeness from a functional viewpoint means

$$\sum_{m=1}^{+\infty} \int_{\lambda_{min}}^{\lambda_{max}} \chi_{s,m}(\mu, \lambda) f_n(\lambda) d\lambda = 0 \quad \forall s, \mu \quad \leftrightarrow \quad f_m(\lambda) = 0 \quad \forall m, \lambda$$

where for the XXZ model in the thermodynamic limit $\lambda_{max/min} = \pm\pi/2$.

The expectation value of quasilocal charge densities and currents is the immediate generalization of (4.10) to interacting systems using (2.27)

$$\begin{aligned} \frac{\langle \hat{X}_s(\mu) \rangle_\rho}{L} &= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \chi_{s,m}(\mu, \lambda) \rho_m(\lambda) d\lambda \\ \langle \hat{j}_s(\mu) \rangle_\rho &:= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \chi_{s,m}(\lambda) v_m(\lambda) \rho_m(\lambda) d\lambda \end{aligned} \quad (4.14)$$

where now this is the very definition of the current, since there is no local density obeying (4.2) for quasilocal charges. Indeed, we are computing the total average semiclassical current density along the translationally invariant state $\hat{\rho}_s(\zeta)$ at fixed ζ . This is then interpreted as current density along the ζ -ray in our inhomogeneous system. In this sense $\langle \hat{j}_s(\mu) \rangle_\rho$ is an abuse of notation. Moreover, here we assume the charges $\hat{X}_s(\mu)$ to be extensive according to the definition of quasilocality in §3.4. In §7 we arbitrarily rescale all them by $1/L$ in order to have non-divergent expectation values. Thus, with the latter convention the factor $1/L$ is included in $\hat{X}_s(\mu)$, yet.

The main difficulty in the interacting framework is that now, as mentioned, the velocities to be used when evaluating (4.14) on the LQSS $\hat{\rho}_s(\zeta)$ are the "dressed" velocities $v_{m,\zeta}(\lambda)$. These depend on the LQSS root densities, which are still unknown, and vice versa.

The equation for dressed velocities corresponding to a given $\hat{\rho}_s(\zeta)$ was already derived in [22], before the introduction of GHD. The key idea is to start from the BT-numbers $\{I^{m,\alpha}\}_p$ (particles) and $\{I^{m,\alpha}\}_h$ (holes) of the representative eigenstate, satisfying (2.22), and to define particle-hole excitations around it. Finally, their value is given by solving the following integral consistence equation [1, 22], where $\epsilon_n(\lambda)$ is the energy of a bound state of rapidity λ explicitly written in (2.30):

$$v_{n,\zeta}(\lambda) \rho_{n,\zeta}^{tot}(\lambda) = \frac{\partial_\lambda \epsilon_n(\lambda)}{2\pi} - \sum_{m=1}^{+\infty} [a_{nm} * v_{m,\zeta}(\lambda) \rho_{m,\zeta}(\lambda)]_\lambda \quad \text{Dressed velocities} \quad (4.15)$$

$$\partial_\lambda \epsilon_n(\lambda) = \frac{J \sinh(\eta)}{2i} \left(\frac{1}{\sinh^2[(n\eta/2) - i\lambda]} - \frac{1}{\sinh^2[(n\eta/2) + i\lambda]} \right)$$

Let us temporarily shelve the issue of the presence of $\rho_{m,\zeta}(\lambda)$ in (4.15) and assume that dressed velocities are known. We have expressed the values of charges and currents through the ζ -LQSS root densities. Using the above discussed semiclassical picture, even if now $\hat{j}_s(\mu)$ is no longer a proper local current operator, we expect (4.9) to still hold, i.e.

$$-\zeta \partial_\zeta \frac{\langle \hat{X}_s(\mu) \rangle_{\rho_\zeta}}{L} + \partial_\zeta \langle \hat{j}_s(\mu) \rangle_{\rho_\zeta} = 0$$

Analogously, exploiting the functional completeness of $\{\chi_{s,m}(\lambda)\}$ yields

$$-\zeta \partial_\zeta \rho_{m,\zeta}(\lambda) + \partial_\zeta (v_{m,\zeta}(\lambda) \rho_{m,\zeta}(\lambda)) = 0$$

where the boundary conditions (4.12) are required.

Together with the BGT equations (2.29), this can be proven to be equivalent to [1]

$$[v_{m,\zeta}(\lambda) - \zeta] \partial_\zeta \theta_{m,\zeta}(\lambda) = 0 \quad \text{where} \quad \theta_{m,\zeta}(\lambda) = \frac{\rho_{m,\zeta}(\lambda)}{\rho_{m,\zeta}^{tot}(\lambda)}$$

This is the fundamental equation of the GHD approach to bipartitioning protocols and has the same shape as (4.11). Assuming that $\forall \lambda, m$ the equation $\zeta = v_{m,\zeta}(\lambda)$ has a unique solution (no exceptions are known), then as for noninteracting systems:

$$\theta_{m,\zeta}(\lambda) = \theta_{m,\zeta}^L(\lambda) \cdot \Theta(v_{m,\zeta}(\lambda) - \zeta) + \theta_{m,\zeta}^R(\lambda) \cdot \Theta(\zeta - v_{m,\zeta}(\lambda)) =$$

$$= \Theta(v_{m,\zeta}(\lambda) - \zeta) \cdot (\theta_{m,\zeta}^L(\lambda) - \theta_{m,\zeta}^R(\lambda)) + \theta_{m,\zeta}^R(\lambda) \quad (4.16)$$

However, we stress that this is still an implicit solution. The dressed velocities $\{v_{m,\zeta}(\lambda)\}$ are unknown functions depending on the ζ -LQSS root densities. Practically, *since (4.15), (4.16) and the BGT equations (2.29) are nontrivially coupled, root densities and dressed velocities have to be solved simultaneously, namely iteratively together by consistence.* Another necessary approximation comes from the maximal string-length truncation to m_{max} .

This necessarily leads to an approximate numerical solution, despite (4.15) and (4.16) from the theoretical viewpoint aim to predict the dynamics exactly in the hydrodynamic limit. These predictions are expected to be valid along ζ -rays as $\zeta \rightarrow +\infty$.

4.4 QUASI-DECOUPLING OF DRESSED VELOCITIES EQUATIONS

Remarkably, the equation for dressed velocities (4.15) has the form (2.34) as (2.29) and (2.33). Thus, it can be analogously quasi-decoupled.

$$\begin{aligned} \xi_n(\lambda) &= \frac{\partial_\lambda \epsilon_n(\lambda)}{2\pi} & f_n(\lambda) &= v_{n,\zeta}(\lambda) \rho_{n,\zeta}^{tot}(\lambda) \\ (f - g)_n(\lambda) &= v_{n,\zeta}(\lambda) (\rho_{n,\zeta}^{tot}(\lambda) - \rho_{n,\zeta}^h(\lambda)) = v_{n,\zeta}(\lambda) \rho_{n,\zeta}^h(\lambda) \end{aligned}$$

Plugging these into (2.38) yields, for every fixed λ ,

$$v_{n,\zeta} \rho_{n,\zeta}^{tot} = \frac{\partial_\lambda \epsilon_n}{2\pi} - s * \left(\frac{\partial_\lambda \epsilon_{n-1}}{2\pi} + \frac{\partial_\lambda \epsilon_{n+1}}{2\pi} \right) + s * [v_{n-1,\zeta} \rho_{n-1,\zeta}^h + v_{n+1,\zeta} \rho_{n+1,\zeta}^h]$$

Again, we can cancel the ϵ -term using (2.30) and (2.35). However, unlike the quasi-decoupling of the TBA equations, here we deal with $\partial_\lambda a_n$. Therefore, in order to apply (2.35) we have to integrate by parts the convolution, which enables the factorization of ∂_λ out of the entire ϵ -term:

$$\begin{aligned} \left[s * \frac{\partial_\lambda \epsilon_n}{2\pi} \right]_\lambda &= -\frac{J \sinh(\eta)}{2} \int_{-\pi/2}^{\pi/2} s(\lambda - \mu) \partial_\mu a_n(\mu) d\mu = \\ &= -\frac{J \sinh(\eta)}{2} [s(\lambda - \mu) a_n(\mu)]_{\mu=-\pi/2}^{\mu=\pi/2} + \frac{J \sinh(\eta)}{2} \int_{-\pi/2}^{\pi/2} \partial_\mu s(\lambda - \mu) a_n(\mu) d\mu \end{aligned}$$

With our convention (2.10), the phase shifts $\theta_n(\lambda)$ are antisymmetric, hence $a_n(\lambda) \propto \partial_\lambda \theta_n(\lambda)$ is symmetric. Thus the first term vanishes.

Moreover, by using $\partial_\mu s(\lambda - \mu) = -\partial_\lambda s(\lambda - \mu)$, we can factorize the derivative out of the integral:

$$\begin{aligned} \left[s * \frac{\partial_\lambda \epsilon_n}{2\pi} \right]_\lambda &= -\frac{J \sinh(\eta)}{2} \partial_\lambda [s * a_n]_\lambda \\ \rightarrow v_{n,\zeta} \rho_{n,\zeta}^{tot} &= -\frac{J \sinh(\eta)}{2} \partial_\lambda \{a_n - s * [a_{n-1} + a_{n+1}]\} + s * [v_{n-1,\zeta} \rho_{n-1,\zeta}^h + v_{n+1,\zeta} \rho_{n+1,\zeta}^h] \end{aligned}$$

Now we can use (2.35) for both $n = 1$ and $n > 1$, leading to

$$\begin{cases} v_{1,\zeta} \rho_{1,\zeta}^{tot} = -\frac{1}{2} J \sinh(\eta) \partial_\lambda s + s * [v_{2,\zeta} \rho_{2,\zeta}^h] \\ v_{n,\zeta} \rho_{n,\zeta}^{tot} = s * [v_{n-1,\zeta} \rho_{n-1,\zeta}^h + v_{n+1,\zeta} \rho_{n+1,\zeta}^h] \quad \forall n > 1 \end{cases} \quad (4.17)$$

As for the BGT equations, in the numerical implementation of (4.17) we have to decide which truncation scheme is more suitable (standard or reflected).

At least for $\Delta > 1$, for sufficiently long strings dressed velocities are generally known to display a weak dependence on λ . However, this constant value can be nonzero and depend on ζ . For

$\zeta \rightarrow \pm\infty$, it vanishes, in agreement with the observation that there is no preferential direction in a homogeneous quench (i.e. in the LSSs which are the boundary conditions of (4.16)).

This behaviour is explicitly shown in Fig.4.1.

In this work we consider the bipartitioning protocol between a $2n$ –rainbow state and the infinite-temperature thermal state.

For the infinite-temperature thermal state, where $\hat{\rho}_m^{tot} \approx \hat{\rho}_m^h$ decays as a power-law, we expect the reflected truncation to induce a smaller distortion in the large- m behaviour. Indeed, the ratio of two subsequent $v_m \hat{\rho}_m^h$ or $v_m \hat{\rho}_m^{tot}$ approaches one as $n \rightarrow +\infty$.

For the $2n$ –rainbow state, we can roughly distinguish two different regimes. The first in which, as for the majority of states, root densities decay exponentially with m . This would suggest to use the standard truncation. However, if we assume to use a sufficiently high truncation ceiling m_{max} which provides meaningful GHD predictions for the thermal side of the system, then we expect the effect of the particular truncation scheme to be negligible on the rainbow side, since root densities are exponentially suppressed.

In the second scenario, the LSS of the $2n$ –rainbow state approaches the infinite-temperature state as n increases. In this case, the same considerations discussed above apply.

Thus, we always perform a reflected truncation in the computation of dressed velocities.

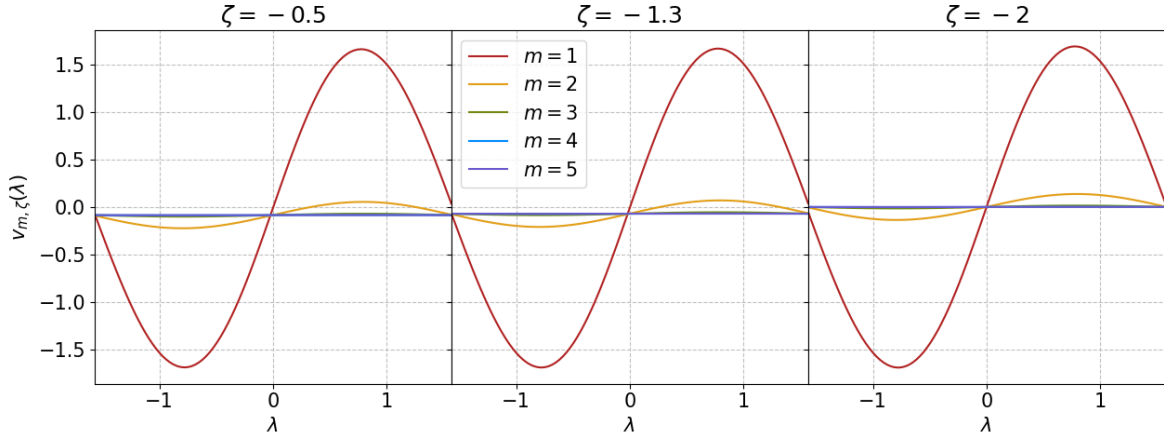


FIGURE 4.1: Dressed velocities for the bipartitioning protocol between the 4–rainbow state and the infinite-temperature thermal state in the XXZ model with $\Delta = 10$. The profiles are displayed for string lengths ranging from 1 to 5 and $\zeta \in \{-0.5, -1.3, -2\}$. For $\zeta \in \{-0.5, -1.3\}$, long strings exhibit an approximately constant, small but non-zero velocity.

4.5 SUMMARY OF THE PROCEDURE TO NUMERICALLY FIND THE LQSS

In this work, we are going to study the bipartitioning protocol between the quantum $2n$ –rainbow state (left) and the classical infinite-temperature thermal state (right).

The implemented numerical procedure can be summarized as:

1. Determine $\eta_m^{L/R}(\lambda)$ for the LSS corresponding to the homogeneous quench from the left/right initial state. For the infinite-temperature thermal state the η –ratios are analytically known (2.44). For the $2n$ –rainbow state we can exploit the mentioned string-charge duality (7.44), which returns the LSS root densities once the expectation values of quasilocal charges on the initial state are known. This fixes the boundary conditions (4.12).
2. $\forall \zeta$ solve (4.16). Numerically, this can be done by a simple iterative scheme:

- Start with an ansatz for the dressed velocities, e.g. $v_{m,\zeta}(\lambda) = \frac{1}{2}\partial_\lambda \epsilon_n(\lambda)$. This comes from the first term in (4.15) assuming $\rho_n^{tot} \approx 1/\pi$.
 - Plug the ansatz into (4.16) finding the corresponding $\theta_{m,\zeta}(\lambda)$.
 - Use the BGT equations (2.42) to compute all the root densities and use them to obtain a new ansatz for dressed velocities from (4.17).
 - Restart from point 2. finding from time to time new estimates of root densities and dressed velocities, till convergence up to the chosen tolerance.
3. Using the best estimate of root densities and dressed velocities, compute the desired charge and current profiles such as in (4.14).

4.6 OUR QUESTION REGARDING BALLISTIC TRANSPORT AND DENSITY PROFILES

Let us temporarily assume that the $2n$ -rainbow states exactly realize the infinite-temperature thermal root densities after a homogeneous quench as $n \rightarrow +\infty$.

Then, according to generalized hydrodynamics, when considering the bipartitioning protocol between this pure state and the infinite-temperature thermal state, in the scaling limit conserved charges display *exactly flat density profiles and no ballistic transport*.

Here, we wonder about what happens in bipartitioning protocols at finite n . In particular:

- Clearly, the boundary values of conserved charges, following from the boundary conditions of the GHD equations, are expected to converge towards one another upon increasing n . Indeed, as the rainbow root densities approach their infinite-temperature counterparts, the respective expectation values of corresponding conserved charges must converge. This leads to *a progressively shrinking vertical window in which a nontrivial profile of conserved-charge densities can develop*.
- Apart from the above expected shrinking, our main question is whether the increasing similarity between the $2n$ -rainbow state and the infinite-temperature thermal state upon increasing n causes ballistic transport to be suppressed sufficiently rapidly that density profiles already approach a *step function* while the relevant vertical window remains finite.

We emphasize that *this "avoided-mixing" would be interesting given the very different nature of the two initial halves of the bipartitioning protocol*, namely the pure $2n$ -rainbow state, with its highly structured microscopic entanglement pattern, and the mixed, completely uncorrelated and strikingly simple infinite-temperature thermal state.

Interestingly, it will turn out that, while in the *noninteracting XX model* ($\Delta = 0$) the energy density profile undergoes a smooth transition upon varying ζ between the two boundary values, in the *genuinely interacting XXZ antiferromagnet* ($\Delta > 1$) a sharp vertical jump at $\zeta \approx 0$ develops.

Chapter 5

EXACT RESULTS IN THE XX MODEL

In the previous chapters we started assembling a mathematical framework which will allow us to study the stationary state of a quench and the out-of-equilibrium dynamics after a bipartitioning protocol, in order to test our thermalization hypothesis for the $2n$ –rainbow state in the XXZ model. However, our final results will generally not be exact, and we will extrapolate them from numerical fits.

Thus, in this chapter, as a preliminary step, we consider a special limit of the XXZ model, where almost all is analytically accessible: the noninteracting XX model, corresponding to $\Delta = 0$.

Moreover, as an additional analysis, beyond the thermalization process, here we also study the entanglement dynamics in various settings concerning the $2n$ –rainbow state. We will do the same in the last chapter for the generic XXZ model.

5.1 XX MODEL AND CONVENTIONS

In order to use the same index convention as in (1.8), when considering a chain of $L = 2nN$ sites, we label them from $-\lceil n - (1/2) \rceil$ to $2n(N - 1) - (1/2)$. Thus we introduce the notation:

$$\sum_{j \in \Lambda} := \sum_{j=j_{\min}}^{j_{\min}+(L-1)} \quad \text{where} \quad j_{\min} = -\left(n - \frac{1}{2}\right)$$

In §1 we mentioned that the XX model can be viewed as particular case of the XXZ model with null anisotropy parameter $\Delta = 0$. Historically, this model was investigated in detail by Lieb, Schulz and Mattis in [87] and by Katsura in [76].

The spin Hamiltonian (1.1) becomes

$$\hat{H}_{XX} = J \sum_{j=1}^L (\hat{S}_j^x \hat{S}_{j+1}^x + \hat{S}_j^y \hat{S}_{j+1}^y) = \frac{J}{2} \sum_{j=1}^L (\hat{S}_j^+ \hat{S}_{j+1}^- + \hat{S}_j^- \hat{S}_{j+1}^+) \quad \text{PBC:} \quad \hat{S}_{L+1}^\alpha = \hat{S}_1^\alpha$$

while its spinless-fermion counterpart (1.5) is

$$\hat{H}_{XX} \leftrightarrow \frac{J}{2} \sum_{j=1}^L (\hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_{j+1}^\dagger \hat{c}_j) \quad \left\{ \begin{array}{ll} \hat{c}_{L+1} = \hat{c}_1 & \text{PBC if } N_{tot} \text{ odd} \\ \hat{c}_{L+1} = -\hat{c}_1 & \text{ABC if } N_{tot} \text{ even} \end{array} \right.$$

The JW transform (1.4) maps $\hat{H}_{XX}$ onto a model of *freely-hopping spinless fermions* with particle-number-dependent boundary conditions. In this work we are only interested in the time evolution of the $2n$ –rainbow states, which are *half filling states*, i.e. $N_{tot} = L/2 = nN$. However, we will take $N \rightarrow +\infty$ (thermodynamic limit and macroscopic homogeneity), thus we expect our final results not to depend crucially on the parity of nN and the boundary conditions.

Since we are going to perform several Fourier transforms, we find convenient to define:

$$\hat{d}_k := \frac{1}{\sqrt{L}} \sum_{j \in \Lambda} e^{-ijk} \hat{c}_j \quad \hat{c}_j := \frac{1}{\sqrt{L}} \sum_{k \in BZ_0^L} e^{ijk} \hat{d}_k$$

$$\sum_{k \in BZ_0^L} := \sum_{k \in k_0, \dots, k_{L-1}} \quad \text{where} \quad k_m = k_0 + \frac{2\pi}{L} m \quad k_0 = \begin{cases} 0 & \text{PBC, i.e. } N_{tot} = nN \text{ odd} \\ \pi/L & \text{ABC, i.e. } N_{tot} = nN \text{ even} \end{cases}$$

Note that the discretization of momenta depends on boundary conditions.

However, if $L \rightarrow +\infty$ then $k_0 \rightarrow 0$ and the set $\{k_m\}$ becomes dense inside $[0, 2\pi]$, independently of PBC/ABC. This is consistent with our above statement: all our results for the thermodynamic limit will not depend on boundary conditions.

Plugging the above definition into the Hamiltonian and using that

$$\frac{1}{L} \sum_{j \in \Lambda} e^{-ijk} = \delta_{k,0} \quad \text{if } k \in BZ_0^L \quad (5.1)$$

then it is automatically diagonalized:

$$\hat{H}_{XX} = \sum_{k \in BZ_0^L} \hat{n}_k \epsilon_k \quad \text{where} \quad \hat{n}_k = \hat{d}_k^\dagger \hat{d}_k \quad \text{and} \quad \epsilon_k = J \cos(k) \quad (5.2)$$

As discussed in §3, despite the $\{\hat{n}_k\}$ are not quasilocal operators, in the XX model they represent a suitable complete set of charges for the GGE. Indeed, up to a $\frac{1}{2\pi}$ rescaling, the set $\{\langle \hat{n}_k \rangle\}$ coincides with the root densities $\{\rho(k)\}$ of the representative eigenstate.

Eigenstates: We can view $\hat{d}_k^\dagger$ as the *creator for quasiparticles with quasimomentum k* .

In the above calculation we considered quasimomenta inside the first Brillouin zone BZ_0^L , but all the discussion still holds by choosing any other possible Brillouin zone, defined by

$$BZ^L(m_0) := \left\{ k_m := \frac{2\pi m}{L} \mid m \in \{m_0, \dots, m_0 + L - 1\} \right\}$$

with m_0 integer/semi-integer if L even/odd.

Then, the eigenstate with quasimomenta $\{m_1 < \dots < m_N\}$ is simply given by

$$|m_1, \dots, m_N\rangle := \left(\prod_{j=1}^N \hat{d}_{k_{m_j}}^\dagger \right) |\Omega\rangle$$

From a Bethe-ansatz perspective, $S_2 = -1$ and the Bethe wavefunction simply coincides with a Slater determinant, consistently with the above result.

Given the fermionic nature of $\hat{d}$, all m_j must be different and a permutation of them can only change the global sign. For simplicity, we assumed ordered sets $\{m_1 < \dots < m_N\}$. This way we build $\binom{L}{N}$ orthogonal eigenstates with N quasiparticles. For N ranging from 0 to L we obtain a complete orthogonal basis of eigenstates for our 2^L -dimensional Hilbert space.

5.2 CORRELATION MATRIX, FORM FACTOR AND DYNAMICS

Correlation Matrix: Let us define the *spatial correlation matrix* for the $2n$ -rainbow state as

$$C_{ij}^{(n)} = \langle R_n | \hat{c}_i^\dagger \hat{c}_j | R_n \rangle$$

Since $2n$ -rainbow states are Gaussian, as proven in §1, the knowledge of this matrix enables via Wick's theorem the computation of any other m -point correlation function. Following the notation of [120], we denote by $\sigma(i)$ the only other site entangled with the site i . Given

$$\Delta_i \quad \text{s.t.} \quad |\sigma(i) - i| = 2\Delta_i - 1$$

the initial correlation matrix of the $2n$ -rainbow state is straightforwardly

$$C_{ij}^{(n)} = \frac{1}{2} (\delta_{ji} + (-1)^{\Delta_i+1} \delta_{j\sigma(i)}) \quad (5.3)$$

Form Factor: The above correlation matrix is particularly simple. However, in order to study its time evolution, it is convenient to recast the description in momentum space.

$$F_{kk'}^{(n)} := \langle R_n | \hat{a}_k^\dagger \hat{a}_{k'} | R_n \rangle = \frac{1}{L} \sum_{j,w \in \Lambda} e^{ikj} e^{-ik'w} C_{jw}^{(n)} \quad \text{Form factor} \quad (5.4)$$

The advantage in using this object for the dynamics will be clear in the next paragraph.

Moreover, it will allow us to compute the Lagrange multipliers of the GGE associated to the $\hat{n}_k$ operators, since $\langle \hat{n}_k \rangle = F_{kk}$.

We now begin by plugging (5.3) into (5.4) and using (5.1), which leads to

$$F_{kk'}^{(n)} = \frac{\delta_{kk'}}{2} + S_{kk'}^{(n)} \quad \text{where} \quad S_{kk'}^{(n)} = \frac{1}{2L} \sum_{j \in \Lambda} e^{ikj} e^{-ik'\sigma(j)} (-1)^{\Delta_j+1}$$

In order to compute this sum, it is convenient to recast it in terms of the index $\ell \in \{0, \dots, N-1\}$, which labels the rainbows along the chain, and $\Delta \in \{1, \dots, n\}$, which labels the arcs of a single $2n$ -rainbow from the innermost to the outermost. Note that the notation $(j, \sigma(j))$ for a Bell pair admits both $j > \sigma(j)$ and $j < \sigma(j)$. Thus the above $\sum_j$ is double-counting each pair. Instead, the label (ℓ, Δ) uniquely identifies a rainbow arc. Hence, for every arc we have to separately sum over the two possibilities $(j, \sigma(j)) = (2n\ell \mp [\Delta - 1/2], 2n\ell \pm [\Delta - 1/2])$.

Nicely, $\sum_{\ell, \Delta}$ factorizes as two independent sums

$$\begin{aligned} S_{kk'}^{(n)} &= \frac{1}{2L} \sum_{\ell=0}^{N-1} \sum_{\Delta=1}^n (-1)^{\Delta+1} e^{i2n\ell(k-k')} \left[e^{-i[\Delta-\frac{1}{2}](k+k')} + e^{i[\Delta-\frac{1}{2}](k+k')} \right] = \\ &= -\frac{1}{L} \left[\sum_{\ell=0}^{N-1} e^{i2n\ell(k-k')} \right] \left\{ \sum_{\Delta=1}^n (-1)^\Delta \cos \left[\left(\Delta - \frac{1}{2} \right) (k+k') \right] \right\} \end{aligned}$$

Using that $\sum_{\Delta} \cos[\alpha(\Delta)] = \text{Re}[\sum_{\Delta} e^{i\alpha(\Delta)}]$ and that $L = 2nN$ finally yields

$$\begin{aligned} S_{kk'}^{(n)} &= \left[\frac{1}{N} \sum_{\ell=0}^{N-1} e^{i\ell 2n(k-k')} \right] \cdot \text{Re} \left[-\frac{1}{2n} \sum_{\Delta=1}^n (-1)^\Delta e^{i[\Delta-\frac{1}{2}](k+k')} \right] = \\ &= \left[\frac{1}{N} \sum_{\ell=0}^{N-1} e^{i\ell 2n(k-k')} \right] \cdot \text{Re} \left[\frac{e^{i\frac{1}{2}(k+k')}}{2n} \sum_{\Delta=0}^{n-1} \left(-e^{i(k+k')} \right)^\Delta \right] \end{aligned} \quad (5.5)$$

(notice the global -1 shift in the sum over Δ in the second line)

- *First Sum:* In the *proper rainbow state* (i.e. $N = 1$ and $n \rightarrow +\infty$) there is no sum over ℓ and the first [...] simply makes 1. Instead, for the $2n$ -rainbow state we would like to use (5.1). To this end, we need to observe that, even if k and k' are individually L positive values

$k_m = k_0 + (2\pi/L)m$ with $m \in \{0, \dots, L-1\}$ in the first Brillouin zone BZ_0^L of an L lattice with ABC/PBC conditions, now

$$k - k' \in \left\{ \frac{2\pi}{L} m, m \in \{-(L-1), \dots, 0, \dots, L-1\} \right\} \subset (B_{-1}^L \cup B_0^L)$$

$$2n(k - k') \in \left\{ \frac{2\pi}{N} m, m \in \{-(L-1), \dots, 0, \dots, L-1\} \right\} \subset \bigcup_{\alpha=-2n}^{2n-1} B_\alpha^N$$

Thus we have to extend (5.1) over these $2n$ Brillouin zones of an N -site lattice:

$$\frac{1}{N} \sum_{\ell=0}^{N-1} e^{i\ell 2n(k-k')} = \sum_{\alpha=0}^{2n-1} \delta_{2n|k-k'|, 2\pi\alpha}$$

- *Second Sum:* For finite n , this is simply the partial sum of a geometric succession with ratio $r = -e^{i(k+k')}$, which satisfies $(1-r) \sum_{\Delta=0}^{n-1} r^\Delta = 1 - r^n$. Given $q = \frac{k+k'}{2}$, if $r = -e^{i2q} \neq 1$,

$$\begin{aligned} \frac{e^{iq}}{2n} \sum_{\Delta=0}^{n-1} (-e^{i2q})^\Delta &= \frac{e^{iq}}{2n} \left[\frac{1 - e^{i(2q+\pi)n}}{1 - e^{i2q}} \right] = \\ &= \frac{e^{i(q+\frac{\pi}{2})n}}{2n} \left[\frac{e^{-i(q+\frac{\pi}{2})n} - e^{i(q+\frac{\pi}{2})n}}{e^{-iq} + e^{iq}} \right] = \frac{e^{i[(q+\frac{\pi}{2})n-\frac{\pi}{2}]}}{2n} \left[\frac{\sin[(q+\frac{\pi}{2})n]}{\cos(q)} \right] \\ \text{Re} \left[\frac{e^{iq}}{2n} \sum_{\Delta=0}^{n-1} (-e^{i2q})^\Delta \right] &= \frac{1}{2n} \frac{\sin^2[(q+\frac{\pi}{2})n]}{\cos(q)} \end{aligned} \quad (5.6)$$

In the limit $-e^{i2q} \rightarrow 1$, i.e. $q = \pi m - \frac{\pi}{2} + \epsilon$ with $n\epsilon \ll 1$, this ratio converges linearly to zero:

$$\frac{1}{2n} \frac{\sin^2[(q+\frac{\pi}{2})n]}{\cos(q)} = \frac{(-1)^m \sin^2(n\epsilon)}{2n \sin(\epsilon)} = (-1)^m \frac{\epsilon}{2} + o(\epsilon^2)$$

For $r = 1$ we cannot divide $(1-r) \sum_{\Delta=0}^{n-1} r^\Delta = 1 - r^n$ by $1-r$ and our derivation of (5.6) fails. However, $-e^{i2q} = 1$ means $e^{iq} = \pm i$, i.e. the object inside $\text{Re}[\dots]$ is purely imaginary and the final result is exactly zero.

Therefore, (5.6) can be extended by continuity over $-e^{i2q} = 1$, where it vanishes.

However, since we aim to study the case $n \rightarrow +\infty$, we need to verify if this term can be regarded as $O(1/2n)$ or not. This can be false only when $\cos(q) \rightarrow 0$, i.e. $q = \pi m - \frac{\pi}{2} + \epsilon$.

Nevertheless, this is the situation we analysed above. For $|\epsilon| \ll 1/n$ the function grows linearly as $\approx (-1)^m \epsilon/2$. Then it has to vanish again when $\epsilon = \pm\pi/n$ due to a zero of the numerator. For $1/n \leq \epsilon \leq \pi/n$ we estimate the maximal magnitude at fixed ϵ as $\leq \frac{1}{2n\epsilon}$, corresponding to $\sin^2(n\epsilon) = 1$. This means that we expect our function to develop two opposite peaks in $\epsilon \in [-\pi/n, 0]$ and $\epsilon \in [0, \pi/n]$, with amplitude ≤ 0.5 .

Summarizing, while $n \rightarrow +\infty$, $\text{Re}[\dots]$ can have peaks of finite amplitude ≈ 0.5 , but their horizontal width vanishes as $1/n$. Thus,

In $F_{kk'}$ the $\text{Re}[\dots]$ factor, as function of $q = \frac{k-k'}{2}$, converges both pointwise and distributionally to zero as $2n \rightarrow +\infty$

Our analysis is consistent with the numerical plot in Fig.5.1.

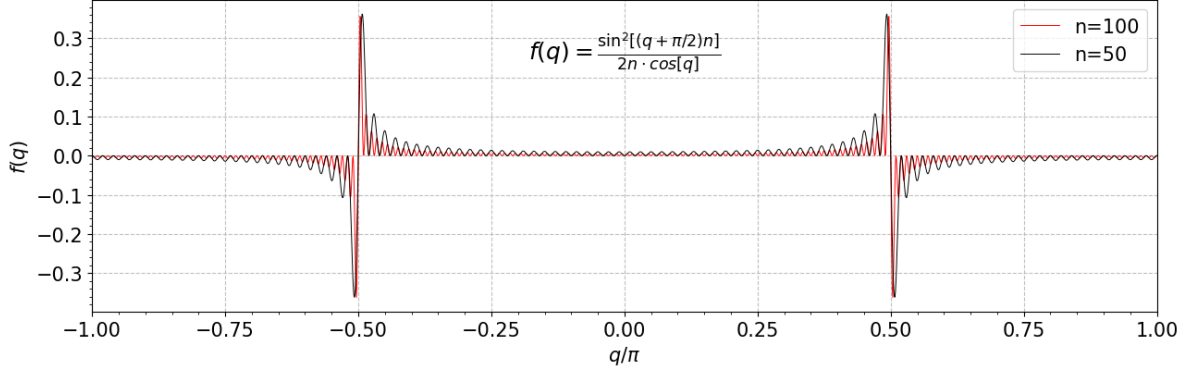


FIGURE 5.1: The function in (5.6) converges pointwise and distributionally to zero as $2n \rightarrow +\infty$. It develops pairs of opposite peaks near $q = \pi m + \pi/2$ with amplitude ≤ 0.5 but width $\propto 1/n$, since the numerator vanishes every $\Delta q = \pi/n$.

Putting everything together, the *form factor of the (true) rainbow state supported on $2n$ sites* is

$$F_{kk'}^{(n)} = \frac{\delta_{kk'}}{2} + \frac{1}{2n} \left\{ \frac{\sin^2 \left[\left(\frac{k+k'+\pi}{2} \right) n \right]}{\cos \left[\frac{k+k'}{2} \right]} \right\} = \frac{\delta_{kk'}}{2} + O\left(\frac{1}{2n}\right)$$

while the *form factor of the $2n$ -rainbow state* is

$$F_{kk'}^{(n)} = \frac{\delta_{kk'}}{2} + \frac{1}{2n} \left\{ \frac{\sin^2 \left[\left(\frac{k+k'+\pi}{2} \right) n \right]}{\cos \left[\frac{k+k'}{2} \right]} \right\} \left(\sum_{\alpha=0}^{2n-1} \delta_{|k-k'|, \alpha\pi/n} \right) = \frac{\delta_{kk'}}{2} + O\left(\frac{1}{2n}\right) \quad (5.7)$$

Note that this formula seems to be independent of the number N of $2n$ -rainbows in the lattice, however, this information is implicitly encoded in the available values of k .

Probably, the most important fact about this result is that, as $n \rightarrow +\infty$ the coefficient of each $\delta_{|k-k'|, \alpha\pi/n}$ vanishes as $1/2n$, i.e. *modes with $k \neq k'$ become uncorrelated*.

This is true for the proper rainbow state as well.

Free-Fermion Dynamics: Shifting to Heisenberg representation, owing to the fact that $\hat{d}_k^\dagger$ is the construction operator for a quasiparticle of energy ϵ_k , the time-evolved form factor is easily obtained.

$$\begin{aligned} F_{kk'}^{(n)}(t) &= \langle R_n | \hat{d}_k^\dagger(t) \hat{d}_{k'}(t) | R_n \rangle = e^{i(\epsilon_k - \epsilon_{k'})t} F_{kk'}^{(n)} = \\ &= \frac{\delta_{kk'}}{2} \left\{ 1 + \frac{1}{2n} \left[\frac{\sin^2 \left[\left(k + \frac{\pi}{2} \right) n \right]}{\cos[k]} \right] \right\} + \left(\sum_{\alpha=1}^{2n-1} \delta_{|k-k'|, \alpha\pi/n} \right) \left\{ \frac{e^{i(\epsilon_k - \epsilon_{k'})t}}{2n} \frac{\sin^2 \left[\left(\frac{k+k'+\pi}{2} \right) n \right]}{\cos \left[\frac{k+k'}{2} \right]} \right\} \end{aligned}$$

Here the time-independent diagonal term expresses $\{\hat{n}_k\}$ conservation during time evolution.

If now we are interested in the *time-evolution of spatial correlations*, since the initial state is Gaussian and the Hamiltonian is quadratic/noninteracting, every p -point correlation function can be obtained from the 2-point function $C_{jw}(t)$, i.e. the evolved correlation matrix. The latter can be straightforwardly expressed in terms of the above form factor as

$$\begin{aligned} C_{jw}^{(n)}(t) &= \langle R_n | \hat{c}_j^\dagger(t) \hat{c}_w(t) | R_n \rangle = \frac{1}{L} \sum_{k, k' \in BZ_0^L} e^{-i(kj - k'w)} e^{i(\epsilon_k - \epsilon_{k'})t} F_{kk'}^{(n)} = \\ &= \frac{\delta_{jw}}{2} + \frac{1}{L} \sum_{k, k' \in BZ_0^L} e^{-i(kj - k'w)} e^{i(\epsilon_k - \epsilon_{k'})t} \frac{1}{2n} \left\{ \frac{\sin^2 \left[\left(\frac{k+k'+\pi}{2} \right) n \right]}{\cos \left[\frac{k+k'}{2} \right]} \right\} \left(\sum_{\alpha=0}^{2n-1} \delta_{|k-k'|, \alpha\pi/n} \right) \end{aligned} \quad (5.8)$$

5.3 PROOF OF THERMALIZATION OF THE $2n$ –RAINBOW STATE IN THE XX MODEL

We recall that the main question we address in the present work is whether the $2n$ –rainbow state thermalizes to infinite temperature in the XXZ model as $n \rightarrow +\infty$.

In this section we provide an exact proof in the XX model, where the thermalization statement can be recast as

$$\exists \beta \quad \text{s.t.} \quad \hat{\rho}_{GGE} = e^{-\beta \hat{H}_{XXZ}} = e^{-\sum_k \beta \epsilon_k \hat{n}_k}$$

Note that, while till here we were able to transfer our results to the proper rainbow state, up to little modifications, now this is no longer possible. As discussed in §3 and §4, in order for the LSS to exist and to be described by a GGE, the initial state must be macroscopically homogeneous. While the $2n$ –rainbow state satisfies this property as $L/N \rightarrow +\infty$, the rainbow state does not.

As discussed in §3, we can use the set $\{\hat{n}_k\}$ to completely fix the stationary state through the GGE Lagrange multipliers $\{\beta_k\}$:

$$\hat{\rho}_s = \frac{1}{Z_{GGE}} \exp \left(- \sum_{k \in BZ_0^L} \hat{n}_k \beta_k \right)$$

It is convenient to compute traces employing the eigenbasis $|\{n_k\}\rangle$, which immediately yields

$$Z_{GGE} = \text{Tr} \left[e^{-\sum_k \beta_k \hat{n}_k} \right] = \sum_{\{\vec{n}_k\}} \prod_k e^{-\beta_k n_k} = \prod_k (1 + e^{-\beta_k})$$

$$\text{Tr} \left[\hat{n}_k e^{-\sum_q \beta_q \hat{n}_q} \right] = e^{-\beta_k} \prod_{k \neq q} (1 + e^{-\beta_q})$$

$$\rightarrow \quad \langle \hat{n}_k \rangle = \frac{e^{-\beta_k}}{(1 + e^{-\beta_k})} = \frac{1}{e^{\beta_k} + 1}$$

As expected, if $\beta_k = \beta \epsilon_k$ this is the Fermi-Dirac distribution. By plugging $\langle \hat{n}_k \rangle = F_{kk}^{(n)}$ in the last equation and using the exact result in (5.7), one completely fixes the GGE lagrange multipliers:

$$\langle \hat{n}_k \rangle = F_{kk}^{(n)} = \frac{1}{2} + \frac{1}{2n} \left(\frac{\sin^2[(k + \pi/2)n]}{\cos[k]} \right) = \frac{\delta_{kk'}}{2} + O\left(\frac{1}{2n}\right) \quad (5.9)$$

$$\beta_k = \log \left(\frac{1}{F_{kk}^{(n)}} - 1 \right) = \log \left(\frac{1 - \frac{1}{n} \left(\frac{\sin^2[(k + \pi/2)n]}{\cos[k]} \right)}{1 + \frac{1}{n} \left(\frac{\sin^2[(k + \pi/2)n]}{\cos[k]} \right)} \right)$$

This result is exact $\forall n \in \mathbb{N}$. However, in general there is clearly no value of β s.t. $\beta_k = \beta \epsilon_k \quad \forall k$. Instead, as $n \rightarrow +\infty$ one can expand:

$$\beta_k = \log \left[1 - \frac{2}{n} \left(\frac{\sin^2[(k + \frac{\pi}{2})n]}{\cos[k]} \right) + o\left(\frac{1}{n}\right) \right] \rightarrow 0 \quad \forall k$$

i.e. as $n \rightarrow +\infty$ the stationary state is thermal with $\beta = 0$

The aim of next chapters will be to numerically prove that this still holds in the interacting gapped phase $\Delta > 1$ of the XXZ model.

Consistency Test - Energy Density: We proved that the energy density (1.10) of the $2n$ -rainbow state approaches zero as $1/n$. In agreement with the above result, it is straightforward to show that the infinite-temperature state has null energy.

Indeed, $\hat{H}$ has null trace given that the same holds for every $\hat{S}_j$.

$$E_{\beta=0} = \text{Tr}[\hat{H}\hat{\rho}_{\beta=0}] = \frac{J}{2L} \sum_{j=1}^L \left(\text{Tr}[S_j^x] \cdot \text{Tr}[S_{j+1}^x] + \text{Tr}[S_j^y] \cdot \text{Tr}[S_{j+1}^y] + \Delta \text{Tr}[S_j^z] \cdot \text{Tr}[S_{j+1}^z] \right) = 0 \quad (5.10)$$

Note that, while the energy density decays to zero as $1/n$, the total energy diverges in the thermodynamic limit as $L/n \propto N$.

5.4 ENERGY DENSITY PROFILE AND BALLISTIC TRANSPORT

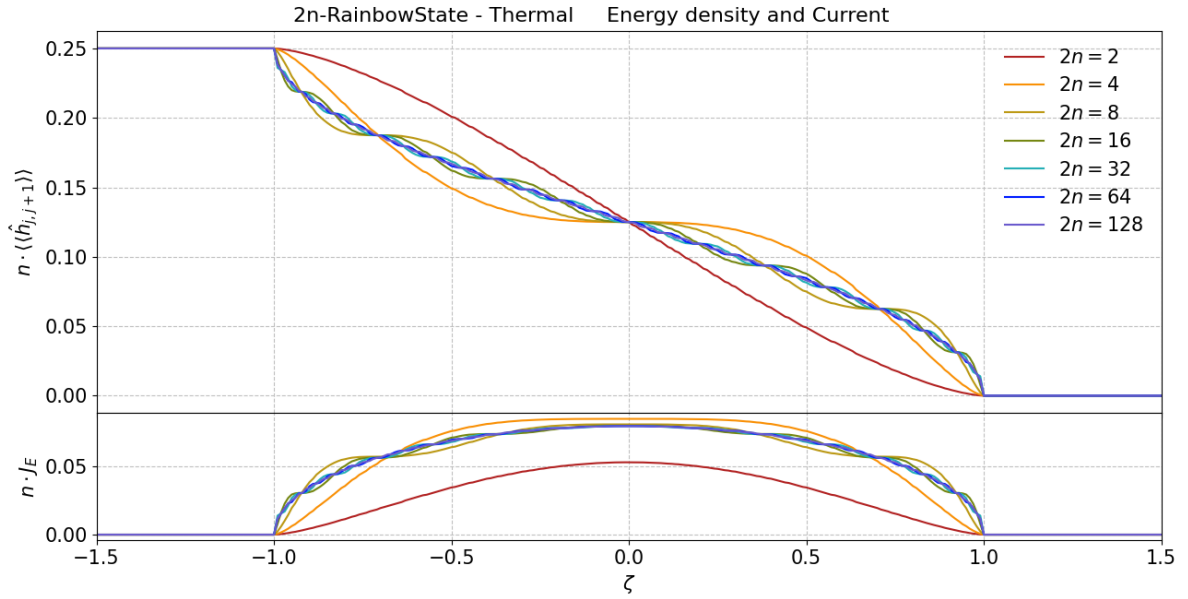


FIGURE 5.2: Bipartitioning protocol from the $2n$ -rainbow state (upon increasing $2n$ from 2 to 128) and the infinite-temperature thermal state. The upper subplot shows the GHD prediction for the local energy density $\langle\langle\hat{h}\rangle\rangle(\zeta)$ in the scaling limit. The lower subplot shows the corresponding GHD current. Both the energy and current profiles are multiplied by n , to observe a profile collapse as $n \rightarrow +\infty$.

In free models, GHD yields an extremely simple prescription to obtain analytic predictions for density and current profiles of local conserved charges.

Indeed, by putting together (4.10), (4.13), (5.9), $\rho_\zeta(k) = \langle n_k \rangle_\zeta / 2\pi$, and the dispersion relation (5.2), where we assume $J = 1$, one obtains

$$\begin{aligned} \langle n_k \rangle^L &= \frac{1}{2} + \frac{1}{2n} \left(\frac{\sin^2[(k + \pi/2)n]}{\cos[k]} \right) & \langle n_k \rangle^R &= \frac{1}{2} \\ \langle n_k \rangle_\zeta &= \langle n_k \rangle^L \Theta(-\sin(k) - \zeta) + \langle n_k \rangle^R \Theta(\zeta + \sin(k)) & \epsilon_k &= \cos(k) \\ \langle\langle\hat{h}\rangle\rangle(\zeta) &= \int_0^{2\pi} \frac{dk}{2\pi} \epsilon_k \langle n_k \rangle_\zeta & J_E(\zeta) &= - \int_0^{2\pi} \frac{dk}{2\pi} \sin(k) \epsilon_k \langle n_k \rangle_\zeta \end{aligned}$$

The notation $\langle\langle\hat{h}\rangle\rangle(\zeta)$ for the local energy density will be more formally defined in §10.

In Fig.5.2 we display the obtained curves where we multiplied both the energy density and the corresponding current by n . The reason for this rescaling is that, as discussed at the end of the previous chapter, we already know that the interesting energy window shrinks as $\propto 1/n$ and we expect a corresponding suppression in currents. Instead, we are interested in the profiles beyond this shrinking. Here the convergence to an n -independent smooth transition between the two boundary values is clearly visible.

5.5 TAXONOMICAL QUASIPARTICLE REFORMULATION

In (5.7) we assume a single quasiparticle species with $k \in (0, 2\pi]$. As $L \rightarrow +\infty$ the form factor becomes an infinite-dimensional matrix. However, due to the presence of a $2n$ -site unit cell s.t. $|R_n\rangle = |\psi\rangle^{L/2n}$, only particles with $|k - k'| = \alpha\pi/n$ and $\alpha \in \{0, \dots, 2n-1\}$ can be entangled.

It follows that only an infinitesimal subset of the $F_{kk'}$ entries is nonzero.

To exploit this property, it is rather convenient to introduce $2n$ quasiparticle species [9, 10, 18, 25] (labelled by capital letters $A \in \{0, \dots, 2n-1\}$), each with a reduced Brillouin zone of width π/n .

$$k \in BZ_n := BZ_0^L \cap \left(0, \frac{\pi}{n}\right] \quad k_A := k + \frac{\pi}{n}A$$

This requires a different dispersion relation and a creation operator for each species.

$$\hat{\eta}_{A,k} := \hat{d}_{k_A} \quad \epsilon_A(k) := \epsilon[k_A] = J \cos\left(k + \frac{\pi}{n}A\right)$$

A useful rewriting when studying time evolution in this framework is:

$$\epsilon_A(k) - \epsilon_B(k) = -2J \sin\left[\frac{\pi}{n}\left(\frac{A-B}{2}\right)\right] \sin\left[k + \frac{\pi}{n}\left(\frac{A+B}{2}\right)\right] \quad (5.11)$$

In the thermodynamic limit, momenta densely populate the Brillouin zone. The Hamiltonian can be expressed using (2.27), where the integration interval is $k \in [0, 2\pi]$ instead of $\lambda \in [-\pi/2, \pi/2]$ and the corresponding root density is $\rho(k) = \langle \hat{n}_k \rangle / 2\pi$.

$$\hat{H}_{XX} = L \int_0^{2\pi} \frac{dk}{2\pi} \epsilon_A(k) \hat{d}_k^\dagger \hat{d}_k = L \int_{B_n} \frac{dk}{2\pi} \sum_{A=0}^{2n-1} \epsilon_A(k) \hat{\eta}_{A,k}^\dagger \hat{\eta}_{A,k}$$

This taxonomical picture allows us to rewrite the form factor $F_{kk'}^{(n)}$ as a $2n \times 2n$ matrix indexed by species $F_{AB}^{(n)}(k)$, since now only quasiparticles with the same momentum can be entangled.

$$F_{AB}^{(n)}(k) := \langle R_n | \hat{\eta}_{A,k}^\dagger \hat{\eta}_{B,k} | R_n \rangle = \frac{\delta_{AB}}{2} + \frac{1}{2n} \left[\frac{\sin^2\left[\left(\frac{k_A + k_B + \pi}{2}\right)n\right]}{\cos\left[\frac{k_A + k_B}{2}\right]} \right] = \frac{\delta_{AB}}{2} + S_{\left(\frac{A+B}{2}\right)}^{(n)}(k)$$

$$\text{where} \quad S_X^{(n)}(k) := \frac{1}{2n} \left[\frac{\sin^2\left[n\left(k + \frac{\pi}{2}\right) + \pi X\right]}{\cos\left[k + \frac{\pi}{n}X\right]} \right] = O\left(\frac{1}{2n}\right) \quad (5.12)$$

Consistently with our results in terms of $F_{kk'}^{(n)}$, as $n \rightarrow +\infty$ the number of species grows linearly and becomes infinite, but correlations between different species vanish.

In the thermodynamic limit, the spatial correlation matrix (5.8) translates as:

$$\begin{aligned} C_{jw}^{(n)}(t) &= \frac{1}{L} \sum_{A,B} \sum_{k \in BZ_n} e^{-ik(j-w)} e^{-i(Aj-Bw)\pi/n} e^{i[\epsilon_A(k) - \epsilon_B(k)]t} F_{AB}^{(n)}(k) \approx \\ &\approx \frac{\delta_{jw}}{2} + \sum_{A,B=0}^{2n-1} \frac{e^{-i(Aj-Bw)\pi/n}}{2n} \int_{BZ_n} \frac{dk}{2\pi} e^{-ik(j-w)} e^{i[\epsilon_A(k) - \epsilon_B(k)]t} \left[\frac{\sin^2\left[n\left(k + \frac{\pi}{2}\right) + \pi\left(\frac{A+B}{2}\right)\right]}{\cos\left[k + \frac{\pi}{n}\left(\frac{A+B}{2}\right)\right]} \right] \end{aligned}$$

where $\approx$ simply deals with the replacement of $\sum_k$ by the integral.

Then, using the beat rewriting (5.11):

$$R := \frac{A-B}{2} \quad X := \frac{A+B}{2}$$

$$C_{jw}^{(n)}(t) = \frac{\delta_{jw}}{2} + \sum_{R=-(2n-1)}^{2n-1} e^{-i\frac{\pi}{n}R(j-w)} \int_{BZ_n} \frac{dk}{2\pi} \sum_{X=|R|}^{2n-1-|R|} e^{-2iJ \sin[\frac{\pi}{n}R] \sin[k+\frac{\pi}{n}X]t} e^{-i[k+\frac{\pi}{n}X](j-w)} S_X^{(n)}(k)$$

5.6 ENTANGLEMENT EVOLUTION: QUASIPARTICLE PICTURE

Entanglement and Thermodynamic Entropy: Entanglement and thermodynamic entropy are pivotal concepts in quantum statistical mechanics. Recently, the study of quantum quenches revealed that they are intricately intertwined.

Let us come back to the notations of §3, where a system was prepared in a pure quantum state $|\psi\rangle$, with density matrix $\hat{\rho} = |\psi\rangle\langle\psi|$, and then time-evolved with a Hamiltonian $\hat{H}$ of which it was assumed not to be an eigenstate. We considered its bipartition $\mathbb{A} \mid \bar{\mathbb{A}}$, whose reduced density matrices are $\hat{\rho}_{\mathbb{A}} = \text{Tr}_{\bar{\mathbb{A}}}[\hat{\rho}]$ and $\hat{\rho}_{\bar{\mathbb{A}}} = \text{Tr}_{\mathbb{A}}[\hat{\rho}]$.

Thus, the corresponding *entanglement entropy* is simply the von Neumann entropy of $\hat{\rho}_{\mathbb{A}}$ (or equivalently $\hat{\rho}_{\bar{\mathbb{A}}}$).

$$S_{\mathbb{A}} := -\text{Tr}[\hat{\rho}_{\mathbb{A}} \log \hat{\rho}_{\mathbb{A}}] \quad [S_{\mathbb{A}} = S_{\bar{\mathbb{A}}}] \quad (5.13)$$

This definition and why this object can be thought of as a measure of entanglement will be discussed later in §9. Here we just anticipate that for an unentangled product state $S_{\mathbb{A}} = 0$, while it reaches its maximum for a maximally entangled state.

A crucial observation is that, *while the von Neumann entropy of the whole system remains zero, the entanglement entropy of the bipartition in general undergoes a nontrivial evolution $S_{\mathbb{A}}(t)$. In particular, it typically grows linearly with time until it saturates to a constant value [26].*

From (3.3) it follows that this asymptotic value must coincide with the von Neumann entropy of the reduced density matrix of the stationary state $\hat{\rho}_s$.

$$S_{\mathbb{A}}^{\infty} := \lim_{t \rightarrow +\infty} S_{\mathbb{A}}(t) = -\text{Tr}[\rho_{s,\mathbb{A}} \log \rho_{s,\mathbb{A}}] := S_{s,\mathbb{A}}$$

Remarkably, in general $\hat{\rho}_s$ is not a pure state, thus $S_{s,\mathbb{A}}$ cannot be interpreted as an entanglement entropy. We will analyse its nature in more detail in §9.

In §3 we introduced the issue of the relaxation to a (generalized) Gibbs ensemble in isolated quantum systems prepared in pure states.

Given the above relation between $S_{\mathbb{A}}^{\infty}$ and $S_{s,\mathbb{A}}$, *it has been proposed to view the thermodynamic entropy density of a statistical ensemble (GE or GGE) as the asymptotically accumulated entanglement entropy density of a large subsystem $\mathbb{A}$ in an infinite system prepared in a pure state [4, 43, 77].* In to order make this reasoning more transparent, we now explicitly list relations between the various notions of entropy involved in our framework:

- $S_s^{(th)} = -\text{Tr}[\hat{\rho}_s \log \hat{\rho}_s]$: It is the von Neumann entropy of $\hat{\rho}_s$, which here is assumed to be $\hat{\rho}_{GE}$ or $\hat{\rho}_{GGE}$. If we think of $\hat{\rho}_s$ as the mixed state of an isolated system, $S_s^{(th)}$ originates from a classical superposition of microstates, i.e. it is its *thermodynamic entropy*. We added the label $^{(th)}$ to highlight this fact.

Note that in the thermodynamic limit, where the partition function

$$Z = \text{Tr}[e^{-\sum_n \beta_n \hat{H}_n}] = \int [D\{\rho_n\}] e^{S_{\text{SY}}[\{\rho_n\}]} e^{-\sum_n \beta_n \langle \hat{H}_n \rangle[\{\rho_n\}]} = \int [D\{\rho_n\}] e^{-\tilde{F}[\{\rho_n\}]}$$

$$\tilde{F}[\{\rho_n\}] := \left[\sum_n \beta_n \langle \hat{H}_n \rangle [\{\rho_n\}] \right] - S_{YY}[\{\rho_n\}]$$

can be computed by saddle-point approximation, the Von Neumann entropy of $\hat{\rho}_{GGE}$ coincides with the Yang-Yang entropy of the set $\{\bar{\rho}_n\}$ minimizing the free energy $\tilde{F}_{\{\beta_n\}}$.

$$-\text{Tr}[\hat{\rho}_{GGE} \log \hat{\rho}_{GGE}] = \log Z + \sum_n \beta_n \langle \hat{H}_n \rangle = -\tilde{F}[\{\bar{\rho}_n\}] + \sum_n \beta_n \langle \hat{H}_n \rangle [\{\bar{\rho}_n\}] = S_{YY}[\{\bar{\rho}_n\}]$$

- $S_{s,\mathbb{A}} = -\text{Tr}_{\mathbb{A}}[\hat{\rho}_{s,\mathbb{A}} \log \hat{\rho}_{s,\mathbb{A}}]$: Naively, it quantifies the degree of mixedness of $\hat{\rho}_{s,\mathbb{A}}$. If the local stationary state $\hat{\rho}_s$ is thermal (GGE or GE, as discussed above), its Von Neumann entropy $S_s^{(th)}$ has to be extensive. We should therefore expect $S_{s,\mathbb{A}}$ to be mainly thermal and extensive as well, thus

$$\frac{S_s^{(th)}}{L} = \frac{S_{s,\mathbb{A}}}{\ell}$$

- Since $\hat{\rho}(t) = e^{-i\hat{H}t} |\psi\rangle \langle \psi| e^{i\hat{H}t}$ is pure, its von Neumann entropy is trivially $S(t) = 0 \ \forall t$.
- $S_{\mathbb{A}}^{\infty} = -\text{Tr}_{\mathbb{A}}[\hat{\rho}_{\mathbb{A}} \log \hat{\rho}_{\mathbb{A}}]$: It is the *entanglement entropy* of the bipartition $\mathbb{A} | \bar{\mathbb{A}}$, which is purely due to quantum correlations between the two subsystems, since $\hat{\rho}(t)$ is a pure state. However, plugging $S_{\mathbb{A}}^{\infty} = S_{s,\mathbb{A}}$ in the above assumption, we obtain that thermodynamic and entanglement entropy densities are equal

$$\frac{S_s^{(th)}}{L} = \frac{S_{\mathbb{A}}^{\infty}}{\ell} \quad (5.14)$$

which is the mathematical reformulation of the above statement. To date, this equivalence has been analytically proven only for noninteracting models [39, 55, 83].

Summarizing, although the whole system is preserved in a pure state $\hat{\rho}(t) = |\psi(t)\rangle \langle \psi(t)|$ by the unitary time evolution, local observables are asymptotically described by a statistical ensemble $\hat{\rho}_{GGE}$ with nonzero thermodynamic entropy density $S_s^{(th)}/L$. The latter coincides with the entanglement entropy density accumulated during the dynamics.

Even if only one of $\left[\begin{array}{l} \text{the pure and time-dependent } \hat{\rho}(t) \\ \text{the thermal and stationary } \hat{\rho}_s \end{array} \right]$ is the "true" state of the whole system, locally it is not possible to make any distinction between them. In this sense, understanding the entanglement evolution unveils how thermodynamics emerges in isolated systems, since the boundary between the notions of entanglement entropy and thermodynamic entropy becomes evanescent from a local perspective.

Proper Quasiparticle Picture: As mentioned above, the typical observed behaviour of entanglement entropy after homogeneous quenches is a linear growth followed by saturation to a constant value $\propto \ell$, if $\mathbb{A}$ is a segment of length $[\mathbb{A}] = \ell$.

The quasiparticle picture was originally proposed (within a CFT framework) in [26] by Calabrese and Cardy as a semiclassical effective description to predict entanglement dynamics. More precisely, by definition it describes the scaling part of entanglement spreading, as formally defined in [10]. The corresponding procedure is sketched in Fig.5.3. For more recent reviews about the quasiparticle picture and its generalizations the reader is e.g. referred to [3, 5].

To date, *the quasiparticle picture is a physical conjecture, without any first-principles proof.*

Indeed, its validity has been analytically proven for noninteracting models [55] and for rational CFTs (conformal field theories) in [6].

Even if without a validity proof, in Yang-Baxter integrable models the quasiparticle picture has been successfully used to describe the out-of-equilibrium dynamics after inhomogeneous

quenches. Moreover, the same idea of ballistic carriers is at the foundation of the generalized hydrodynamics discussed in §4, which enables the study of transport in integrable models.

Despite its many generalizations, for the sake of clarity we start by describing its original formulation, dealing with homogeneous noninteracting systems. Indeed, here the behaviour of entanglement entropy has the simplest heuristic physical interpretation:

- As depicted in §4, the fundamental idea is that from the viewpoint of the post-quench Hamiltonian and the stationary state, the quench at $t = 0$ produces an extensive number of quasiparticle excitations. In free models, these are stable quasiparticles which propagate semiclassically (with both defined position and momentum) with constant velocity given by the dispersion relation

$$v(k) = \frac{d\epsilon(k)}{dk} \quad \text{Group velocity}$$

As a result, the post-quench dynamics of entanglement is effectively described in terms of the ballistic spreading of these quasiparticles through the system.

- Only particles generated in a same position can be entangled. In particular, the correlations between two particles with k and k' are fixed by the form factor $F_{kk'}$ of the initial state.

This assumption is highly based on the fact that "positions" are associated to *mesoscopic cells* rather than sites (*coarse grained description*). The size Δ of such cells is by definition much larger than the characteristic length of inhomogeneities a . A system containing many mesoscopic cells appears homogeneous. In order for the subsystem $\mathbb{A}$ to appear homogeneous as well, it must be $a \ll \Delta \ll \ell \ll L \rightarrow +\infty$. If we assume the correlation range to be of the same order as a , distinct cells can be approximated as non-entangled at $t = 0$.

Quasiparticle-picture results are expected to be valid for $\ell, t \rightarrow +\infty$ s.t. $t/\ell \approx \text{const}$, which defines the corresponding *scaling limit* leading to data collapse for $S_\ell(t)/\ell$.

- Only pairs of entangled quasiparticles are produced (not generic multiplets).

Given these assumptions, *the entanglement $S_{\mathbb{A}}(t)$ of the bipartition can be measured by counting the number of quasiparticle pairs that, emitted at the same point in space, are shared between $\mathbb{A}$ and $\bar{\mathbb{A}}$ at time t , where only the corresponding "weight" $f(k, k')$ is left unknown.*

$$\begin{aligned} S_{\mathbb{A}}(t) &\approx \int_{\mathbb{A}} dx' \int_{\bar{\mathbb{A}}} dx'' \int_{\mathbb{R}} dx \int \frac{dk}{2\pi} \frac{dk'}{2\pi} f(k, k') \delta[x' - (x + v(k)t)] \delta[x'' - (x + v(k')t)] = \\ &= \int_{\mathbb{A}} dx' \int_{\bar{\mathbb{A}}} dx'' \int \frac{dk}{2\pi} \frac{dk'}{2\pi} f(k, k') \delta[(x'' - x') + (v(k) - v(k'))t] \end{aligned}$$

We further assume that only particles with opposite velocity are entangled.

$$\begin{aligned} \int \frac{dk'}{2\pi} f(k, k') \delta[(x'' - x') + (v(k) - v(k'))t] &:= f(k) \delta[(x'' - x') + 2v(k)t] \\ \rightarrow S_{\mathbb{A}}(t) &\approx \int \frac{dk}{2\pi} [\Omega(\mathbb{A}, v(k), t)] \cdot f(k) \end{aligned} \quad (5.15)$$

where $[\Omega]$ is the area of $\Omega(\mathbb{A}, v, t) := \{x' \in \mathbb{A} \text{ s.t. } [x' - 2vt] \in \bar{\mathbb{A}}\}$

Thus, if $\mathbb{A}$ is simply a single segment of length ℓ , for every k s.t. $2|v(k)|t < \ell$ the subregion of $\mathbb{A}$ such that its $2v(k)t$ -shift falls out of $\mathbb{A}$ has clearly area $2|v(k)|t$, while if $2|v(k)|t \geq \ell$ then $\mathbb{A}$ and the entire $\mathbb{A} - 2v(k)t$ are disjoint intervals.

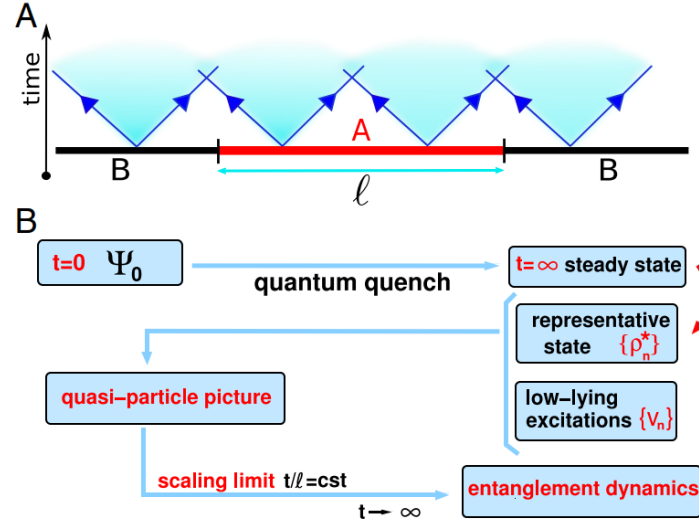


FIGURE 5.3: Reproduced from [4]. (A) - Space-time picture illustrating how the entanglement between an interval $[A] = \ell$ and the rest of the system grows linearly and then saturates, due to oppositely moving coherent quasiparticles. Solid blue lines denote quasiparticles with maximum velocity emitted at $t = 0$. Shaded cones show the halo of slower quasiparticles. (B) - Main steps to calculate the entanglement dynamics using Bethe ansatz and the quasiparticle picture: Bethe-ansatz techniques enable the characterization of the stationary state and the corresponding dressed velocities of low-lying excitations; then the quasiparticle picture employs this result to determine the scaling part of entanglement spreading, without explicitly solving the dynamics.

$$S_\ell(t) = \left[2t \int_{\{k: |v(k)| < \frac{\ell}{2t}\}} \frac{dk}{2\pi} |v(k)| \cdot f(k) \right] + \left[\ell \int_{\{k: |v(k)| > \frac{\ell}{2t}\}} \frac{dk}{2\pi} f(k) \right] \quad (5.16)$$

Note that here the linear growth and saturation is realized exactly by the contribution of each momentum k , separately. However, if a maximum quasiparticle velocity v_{max} exists (e.g. for systems fulfilling the hypothesis of the Lieb-Robinson bound [90]), this behaviour is approximately inherited by $S_\ell(t)$. Indeed, as long as $t < \ell/2v_{max}$ the second term is null while the first integral is constant. When $t \gg \ell/2v_{max}$ the first term does not contribute and the second asymptotically saturates to a value $\propto \ell$.

Moreover, by studying S_ℓ/ℓ vs. $\tau = 2t/\ell$, the following universal behaviour emerges:

$$\frac{S_\ell(\tau)}{\ell} = \left[\tau \int_{\{k: |v(k)| < \frac{1}{\tau}\}} \frac{dk}{2\pi} |v(k)| \cdot f(k) \right] + \int_{\{k: |v(k)| > \frac{1}{\tau}\}} \frac{dk}{2\pi} f(k) \quad \tau = \frac{2t}{\ell} \quad (5.17)$$

This was essentially the idea originally proposed in [26], which correctly predicts a light-cone spreading of quantum-correlations (entanglement).

Apart from extensions of this picture to more complicated settings (e.g. interacting systems or inhomogeneous quenches), the main remaining challenge to analytically predict entanglement dynamics was the evaluation of $f(k)$.

Generalization to Integrable Interacting Models and Evaluation of $f(k)$: In Bethe-ansatz interacting integrable models, quasiparticles simply labelled by their real quasimomentum are replaced by bound states of m -particles with real rapidities, labelled by the pair (m, λ) .

The natural generalization of the above proposal is obtained by considering separately entangled pairs of m -quasiparticle bound states, each with its own group velocity $v_m(\lambda)$ as in GHD.

As we discussed in §4, it is natural to identify entangled carriers with particle/hole low-lying excitations around the stationary state. We recall that, unlike noninteracting systems, in presence of interactions their $\{v_m(\lambda)\}$ are *dressed velocities* depending on the stationary state itself. These are found by solving (4.15). However, we recall that here the quench is homogeneous and there is no dependence on ζ .

As a result (5.15) and (5.16) are recast as

$$S_{\mathbb{A}}(t) \approx \sum_{m=1}^{+\infty} \int d\lambda [\Omega(\mathbb{A}, v_m(\lambda), t)] \cdot f_m(\lambda) \quad (5.18)$$

$$S_{\mathbb{A}}(t) = \sum_{m=1}^{+\infty} \left\{ \left[2t \int_{\{\lambda: |v_m(\lambda)| < \frac{\ell}{2t}\}} |v_m(\lambda)| \cdot f_m(\lambda) d\lambda \right] + \left[\ell \int_{\{\lambda: |v_m(\lambda)| > \frac{\ell}{2t}\}} f_m(\lambda) d\lambda \right] \right\}$$

where the disappearance of $1/2\pi$ is arbitrary and its convenience will be transparent soon.

In [4] it was proposed to evaluate $f_m(\lambda)$ from (5.14), where in the TBA formalism $S_s^{(th)}$ is the Yang-Yang entropy of the local stationary state, which can be expressed in terms of (2.32).

Therefore, by considering the late-time limit of the above expression, one expects

$$\sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \frac{f_m(\lambda)}{\Delta} d\lambda = \frac{S_{\ell}^{\infty}}{\ell} = \frac{S_s^{(th)}}{L} = \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} s_m(\lambda) d\lambda \quad \rightarrow \quad f_m(\lambda) = s_m(\lambda)$$

It is convenient to rewrite explicitly this result in terms of an integral over momenta in noninteracting models. We will use it several times. Here (2.32) becomes:

$$\begin{aligned} \rho_1(\lambda) = \frac{\langle n_{k(\lambda)} \rangle}{2\pi} &\rightarrow s_1(\lambda(k)) = -\frac{1}{2\pi} [\langle n_k \rangle \log(\langle n_k \rangle) + (1 - \langle n_k \rangle) \log(1 - \langle n_k \rangle)] := \frac{1}{2\pi} s(k) \\ &\rightarrow \frac{S_s^{(th)}}{L} = \int s_1(\lambda) d\lambda = \int \frac{dk}{2\pi} s(k) \\ &\rightarrow f(k) = s(k) = -[\langle n_k \rangle \log(\langle n_k \rangle) + (1 - \langle n_k \rangle) \log(1 - \langle n_k \rangle)] \end{aligned} \quad (5.19)$$

As we mentioned at the beginning, the validity of this conclusion has been proven in free models [55], where, by exactly solving the dynamics, one can show that (5.18) with $f_m = s_m$ becomes exact in the scaling limit $\ell, t \rightarrow +\infty$ with $\ell/t = \text{const}$. Moreover, it has been numerically tested in the XXZ model [4, 5] and nested spin chains [94]. A remarkable analytical proof has been obtained for some solvable quenches in the "Rule 54" quantum cellular automaton [80–82].

The striking versatility of this method has enabled its application to inhomogeneous quenches as well [1, 2, 16]. Other extremely recent generalizations [33, 34] deal with highly entangled states, where quantum correlations exist between sites much sites much farther apart than the characteristic length of inhomogeneities. The main difference is that here entangled particles originate in macroscopically distant sites of the lattice.

5.7 GENERALIZED QUASIPARTICLE PICTURE

Spreading Multiplets: As discussed, the original formulation of the quasiparticle picture assumes excitations to be produced in pairs of opposite velocities.

However, the heuristic idea behind the method can be easily generalized to less constrained scenarios, for example with multiplets of m symmetrically-spreading quasiparticles. The first formal steps in this direction have been pursued in [9, 18]. Later reviews are [10, 16, 25].

In our taxonomical rewriting of the form factor, we introduced $2n$ quasiparticle species, corresponding to n pairs of species with opposite velocities.

$$v_A(k) := \frac{d\epsilon_A(k)}{dk} = -J \sin(k_A) = -J \sin\left(k + A \frac{\pi}{n}\right) = v(k_A) \quad (5.20)$$

Only quasiparticles with same k can be entangled.

Thus, we can imagine that *at the quench instant, in every cell of the system, a multiplet of $2n$ entangled quasiparticles with same momentum k (one of each species) is generated*. This happens for every possible value of k separately.

As before, this description is expected to be valid in the homogeneous hydrodynamic regime, where "positions" are associated to mesoscopic cells containing many $2n$ -site unit cells.

$$2n \ll \Delta \ll \ell \ll L \rightarrow +\infty \quad t \rightarrow +\infty \quad \text{s.t.} \quad \frac{t}{\ell} \approx \text{const.}$$

Particles of every $2n$ -uplet start propagating symmetrically with $\{v_A(k) \forall A\}$. Indeed, with our convention for A -labels, in (5.20) particles with $A \in \{0, \dots, n-1\}$ propagate leftward while particles with $A \in \{n, \dots, 2n-1\}$ rightward, with the relation

$$|A - B| = n \rightarrow v_A(k) = -v_B(k)$$

Instead, the relation between the labels $A \in \{0, \dots, n-1\}$ and the magnitude of the corresponding velocities is not stable for varying k and it is possible for two species of left-moving particles to display crossing velocities as functions of k , since $\sin(k_A)$ is not monotonic in $[0, \pi]$.

Generalization of (5.15) to Entangled Multiplets: A pivotal object in (5.15) is the *dynamic factor* $[\Omega(\mathbb{A}, v(k), t)]$, which is the area of the set $\Omega(\mathbb{A}, v(k), t)$ of positions x such that a pair of particles generated in x and spreading with $\pm v(k)$ has exactly one particle in $\mathbb{A}$ and one in $\bar{\mathbb{A}}$ at time t .

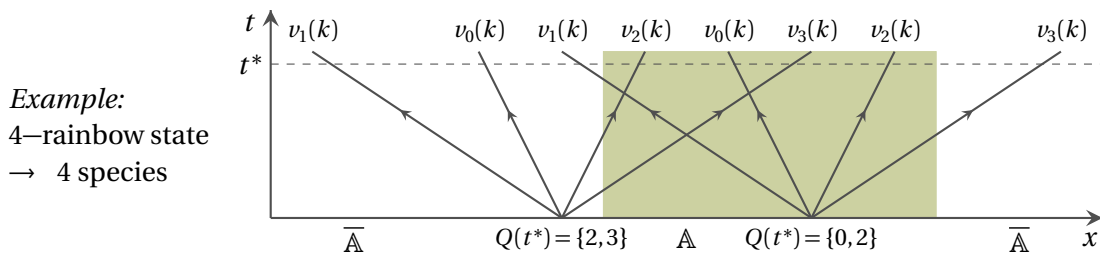
Now entanglement receives contributions from shared multiplets in which at least one particle is not in the same subsystem as the others. The number of possible nontrivial partitions Q of $2n$ particles between two subsystems are $2^{2n} - 2$.

Many simplifications come from the assumption that $\mathbb{A}$ is a segment of length ℓ or the region $(-\infty, 0]$. Indeed, this allows one to exclude partitions Q which are dynamically impossible. However, for the moment we do not make assumptions about $\mathbb{A}$.

Let $\Omega(\mathbb{A}, Q, k, t)$ be the set of positions x such that at time t a k -momentum multiplet generated in x finds itself in the configuration Q . We expect

$$S_{\mathbb{A}}(t) = \int_{BZ_n} \frac{dk}{2\pi} \sum_Q [\Omega(\mathbb{A}, Q, k, t)] \cdot s(k, Q) \quad (5.21)$$

where the only unknown is again the entanglement content $s(k, Q)$ of such a shared multiplet.



Entanglement Contribution of a Shared Multiplet: Here we just summarize the proposals introduced in [18] and [9]. The reader is referred to these articles for further details.

First, in general, together with the form factor (5.12) we would have to introduce

$$G_{AB}^{(n)} := \langle R_n | \hat{\eta}_{A,k} \hat{\eta}_{B,k} | R_n \rangle = 0$$

which is zero for the $2n$ -rainbow state. Indeed, it is a linear superposition of Fock-basis eigenstates with the same number $L/2$ of particles. Then, if we define vectors of creation and annihilation operators of the only particles which can be entangled (i.e. with same momentum),

$$\hat{\Gamma}(k) := \begin{bmatrix} \hat{\eta}_{0,k} \\ \dots \\ \hat{\eta}_{2n-1,k} \\ \hat{\eta}_{0,k}^\dagger \\ \dots \\ \hat{\eta}_{2n-1,k}^\dagger \end{bmatrix} \quad \hat{\Gamma}^\dagger(k) := \begin{bmatrix} \hat{\eta}_{0,k}^\dagger & \dots & \hat{\eta}_{2n-1,k}^\dagger & \hat{\eta}_{0,k} & \dots & \hat{\eta}_{2n-1,k} \end{bmatrix}$$

the corresponding correlation matrix is the following object:

$$\begin{aligned} \mathcal{C}_{AB}^{(n)}(k) &:= \langle R_n | \hat{\Gamma}_A^\dagger(k) \hat{\Gamma}_B(k) | R_n \rangle = \langle R_n | \begin{pmatrix} \hat{\eta}_{A,k}^\dagger \hat{\eta}_{B,k} & \hat{\eta}_{A,k} \hat{\eta}_{B,k} \\ \hat{\eta}_{A,k}^\dagger \hat{\eta}_{B,k}^\dagger & \hat{\eta}_{A,k} \hat{\eta}_{B,k}^\dagger \end{pmatrix} | R_n \rangle = \\ &= \begin{pmatrix} F_{AB}^{(n)}(k) & G_{AB}^{(n)}(k) \\ G_{BA}^{(n)}(k)^* & \delta_{AB} - F_{BA}^{(n)}(k) \end{pmatrix} \end{aligned}$$

We used the anticommutation relation $\{\hat{\eta}_{A,k}, \hat{\eta}_{B,k}^\dagger\} = \delta_{A,B}$ for the last rewriting. Notice the inversion of species indices in the second line of the matrix.

For every fixed k this can be viewed as a 2×2 matrix $\mathcal{C}^{(n)}(k)$

$$\mathcal{C}^{(n)}(k) := \langle R_n | \hat{\Gamma}^\dagger(k) \hat{\Gamma}(k) | R_n \rangle = \begin{pmatrix} F^{(n)}(k) & G^{(n)}(k) \\ G^{(n)}(k)^\dagger & \mathbb{I}_{2n} - F^{(n)}(k)^\tau \end{pmatrix}$$

whose elements are $2n \times 2n$ matrices. We denoted by † the Hermitian conjugation, while by $^\tau$ the simple transposition.

The main idea for the evaluation of $s(k, Q)$ is the introduction of an ancillary Hilbert space $\mathcal{H}_{x,k}$ in every position x and for every momentum k , which has the role of "Hilbert space for the k -multiplet generated at position x ". This picture is somewhat reminiscent of the *vierbein* construction of fermionic Hamiltonians in curved spacetime [20].

The corresponding states are built by applying $\{\hat{\eta}_{A,k}^\dagger\}$ with fixed k to the vacuum $|\Omega_{x,k}\rangle$. Since these operators are fermionic and only distinguished by the quantum number A , $\dim \mathcal{H}_{x,k} = 2^{2n}$. If we assume the state of our multiplet to be Gaussian and we require its two point correlations to be that encoded in $\mathcal{C}^{(n)}$, then its density matrix can be fully expressed in terms of $\mathcal{C}^{(n)}$ [100].

Let us consider the situation in which at time t only a subset $Q_\mathbb{A} = \{A_1, \dots, A_m\}$ of the $2n$ species of this multiplet is in the subregion $\mathbb{A}$.

The entanglement entropy of this subsystem can be proven to be [9, 37, 100, 101]

$$\begin{aligned} s(k, Q) &= -\text{Tr} \left[\mathcal{C}_Q^{(n)} \cdot \log \mathcal{C}_Q^{(n)} \right] = \\ &= -\text{Tr} \left\{ F_Q^{(n)}(k) \cdot \log \left[F_Q^{(n)}(k) \right] + \left[\mathbb{I} - F_Q^{(n)}(k)^\tau \right] \cdot \log \left[\mathbb{I} - F_Q^{(n)}(k)^\tau \right] \right\} \end{aligned} \quad (5.22)$$

where we denoted by $\mathcal{C}_Q^{(n)}$ the matrix $\mathcal{C}^{(n)}$ restricted to its indices falling within Q . In particular, if a subset Q of the species is not entangled with the others of the multiplet (or if all the particles are in the same subsystem), then $s(k, Q) = 0$.

Two immediate tests for the consistency of this proposal are:

- When $n = 1$, the result (5.19) and the original quasiparticle picture are recovered;
- From the known properties of the von Neumann entropy of the reduced density matrix, if we define as $\bar{Q} := \{0, \dots, 2n-1\} \setminus Q$ the complementary bipartition of Q , then it can be proven that $s(k, Q) = s(k, \bar{Q})$. If we plug (5.22) into (5.21), then correctly $S_\mathbb{A}(t) = S_\mathbb{A}(t)$.

We stress that (5.22) gives exactly the entanglement entropy of the subsystem Q of the multiplet in the setting we depicted above. The true assumption deals with the fact that the entanglement entropy of our original system is obtained by summing such contributions.

This way the generalization of (5.18) to a noninteracting multispecies scenario is complete.

Nondiagonal Contributions to Entanglement Entropy at Finite Times: A remarkable property of (5.19) is that $s(k)$ only depends on the diagonal elements of $F_{AB}^{(n)}(k)$, i.e. that of $F_{kk'}$, which are the mode occupation numbers $\langle \hat{n}_k \rangle$. This implies that the thermodynamic/Yang-Yang entropy of the stationary state only depends on the diagonal elements of the form factor.

By contrast, in a multiplet scenario only the entanglement contributions of those partitions in which Q or $\bar{Q}$ contain a single particle satisfy this property: in general, $s(k, Q)$ receives contributions from nondiagonal correlations (i.e. between different species) [18, 25].

This paradox is solved by recalling that, as just said, nondiagonal contributions correspond to partitions Q in which more than one particle of the multiplet is in the subregion $\mathbb{A}$.

If Δv_{min} is the minimum difference between two same-momentum velocities, then the distance between two particles of the multiplet at time t is $\Delta x \geq t \Delta v_{min}$. This means that, for $t \geq \ell / \Delta v_{min}$, only configurations Q in which at most a single quasiparticle is in $\mathbb{A}$ can have nonzero dynamic factor $[\Omega(\mathbb{A}, Q, k, t)]$.

Setting these observations in our framework, we summarize that:

- While as $n \rightarrow +\infty$ only diagonal elements of $F_{kk'}^{(n)}$ are nonzero, the $2n$ -rainbow state for every finite n displays nontrivial correlations $F_{AB}^{(n)}(k)$ between the $2n$ different species of quasiparticles with momentum k . The number of entangled species grows $\propto n$.
- *The complex off-diagonal correlation pattern in $F_{kk'}^{(n)}$ causes the entanglement entropy (5.21) to receive contributions which are absent in the thermodynamic entropy of the stationary GGE.* This could seem in conflict with the fact that a consistent generalization of the quasiparticle picture requires the entanglement entropy density to coincide with the thermodynamic one as $t \rightarrow +\infty$.
- However, this is exactly the case. Indeed, *correlations between different species (i.e. off-diagonal correlations in $F_{kk'}^{(n)}$) only contribute to the $s(k, Q)$ of partitions Q in which more than one particle of the multiplet is in the subregion $\mathbb{A}$. As we discussed above, these partitions can correspond to a nonzero dynamic factor $[\Omega(\mathbb{A}, Q, k, t)]$ only at finite times.*

Recovering of the Proper Quasiparticle Picture for the 2-Rainbow State: In the last part of this chapter we are going to apply the generalized quasiparticle picture to some particular frameworks involving $2n$ -rainbow states. However, in general, here correlations between species (5.12) are sufficiently complex to prevent any particular further simplification of (5.21).

First, we show that for $n = 1$ the ordinary quasiparticle picture is recovered and write the corresponding explicit result for the triplet-dimer initial state.

For $n = 1$ there are only two species, corresponding to

$$v_0(k) = -J \sin(k) < 0 \quad v_1(k) = -v_0(k) > 0 \quad k \in BZ_0^L \cap (0, \pi]$$

The only nontrivial contributions to $s(k, Q_{\mathbb{A}})$ (i.e. exactly one particle in each subsystems) come from $Q_{\mathbb{A}} = \{0\}$ and $Q_{\mathbb{A}} = \{1\}$. In both cases $F^{(1)}(k)_{\mathbb{A}}$ has one single element.

$$F_{AA}^{(1)}(k) = \frac{1 + (-1)^A \cos(k)}{2} \quad \rightarrow \quad \langle \hat{n}_k \rangle = F_{00}^{(1)}(k) = 1 - F_{11}^{(1)}(k) = 1 - \langle \hat{n}_{k+\pi} \rangle$$

$$s(k, \{0\}) = s(k, \{1\}) = -[\langle n_k \rangle \log(\langle n_k \rangle) + (1 - \langle n_k \rangle) \log(1 - \langle n_k \rangle)]$$

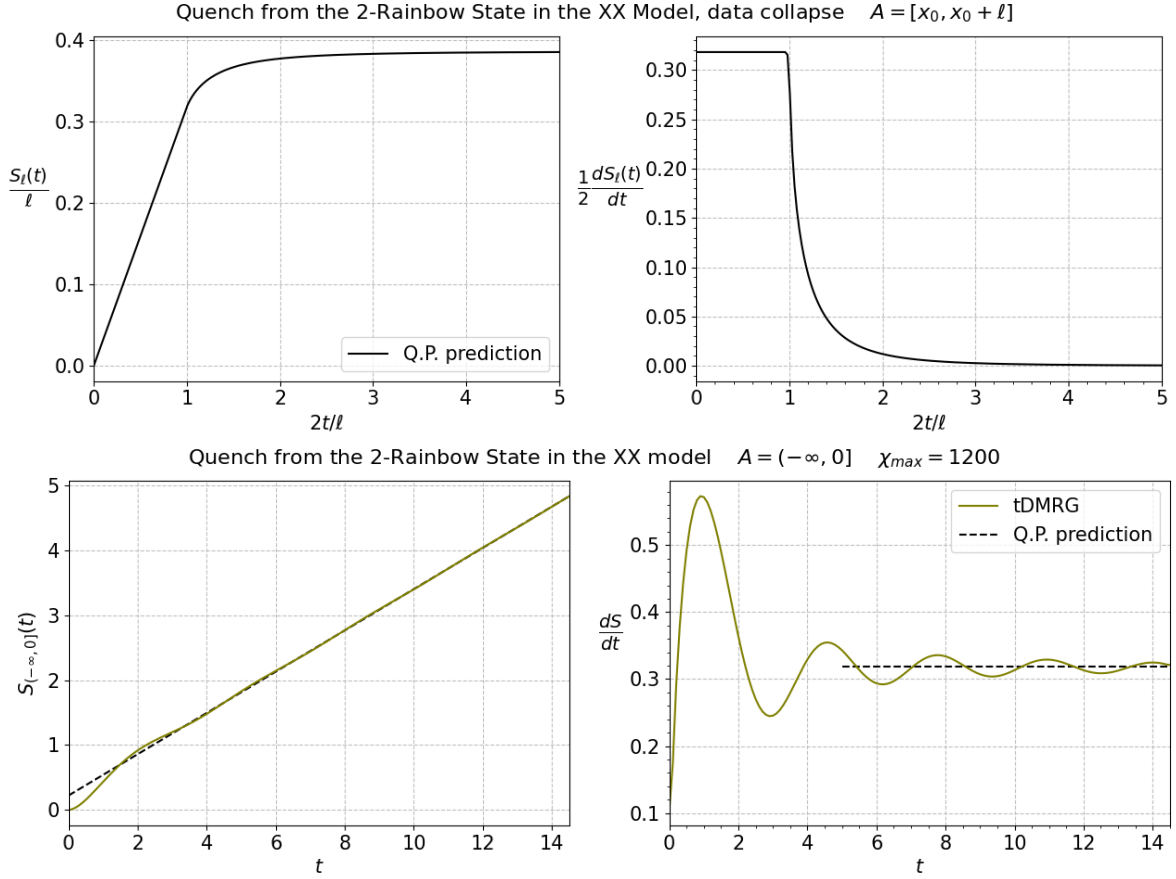


FIGURE 5.4: ABOVE: Quasiparticle-picture prediction for the scaling behaviour of S_ℓ/ℓ vs. $\tau = 2t/\ell$ for the homogeneous quench from the 2–rainbow state in the XX model, computed from (5.17) using that $f(k) = s(k)$. The left panel shows one half of the derivative of the entropy, corresponding to $dS/d\tau$. The entropy grows linearly when $\tau < 1$, i.e. $t < \ell/2$, while asymptotically saturates to a value $\propto \ell$. BELOW: comparison between (5.23) (black dashed line, up to a suitable time translation in the left panel) and the corresponding direct tDMRG simulation of the homogeneous quench (green solid line). Note that the quasiparticle prediction for the slope is one half of that in the above panel for $\tau < 1$, as expected. Throughout, $J = 1$.

If we consider a region $\mathbb{A}$ which is a segment of length ℓ , the derivation of the kinematic factor (which is the same for $Q_{\mathbb{A}} = \{0\}$ and $Q_{\mathbb{A}} = \{1\}$) is that of the proper quasiparticle picture in (5.16):

$$[\Omega(\mathbb{A}, \{0\}, k, t)] = [\Omega(\mathbb{A}, \{1\}, k, t)] = 2t |v_0(k)| \cdot \Theta \left[\frac{\ell}{2t} - |v_0(k)| \right] + \ell \cdot \Theta \left[|v_0(k)| - \frac{\ell}{2t} \right]$$

Note that, even if the sum over both $Q_{\mathbb{A}} = \{0\}$ and $Q_{\mathbb{A}} = \{1\}$ seems to lead to a factor 2 in front of this contribution in $S_{\ell}(t)$, for every fixed Q the momentum integral is only over $BZ_1 = [0, \pi]$ instead of $[0, 2\pi]$. Thus the quasiparticle-picture result is fully recovered.

An alternative common and simpler choice is $\mathbb{A} = (-\infty, 0]$. Here only right-moving quasiparticles of $\mathbb{A}$ can reach the left half $\bar{\mathbb{A}}$ and vice versa, hence

$$\begin{aligned} \frac{S_{(-\infty, 0]}(t)}{|J|t} &= \frac{1}{2|J|t} \lim_{\ell \rightarrow +\infty} S_{\ell}(t) = \\ &= -2 \int_0^{\pi} \frac{dk}{2\pi} \sin(k) \left[\left(\frac{1 + \cos(k)}{2} \right) \log \left(\frac{1 + \cos(k)}{2} \right) + \left(\frac{1 - \cos(k)}{2} \right) \log \left(\frac{1 - \cos(k)}{2} \right) \right] \approx 0.318 \end{aligned} \quad (5.23)$$

Fig.5.4 shows both the scaling behaviour noticed in (5.17), computed for the homogeneous quench from the 2-rainbow state in the XX model, and the agreement between (5.23) and the corresponding direct tDMRG simulation.

We will discuss in detail our tDMRG procedures in §9.

However, when considering the situation $n > 1$, the dynamic factor for the length- ℓ segment becomes soon highly complicated.

5.8 ENTANGLEMENT EVOLUTION IN THE BIPARTITIONING PROTOCOL BETWEEN $2n$ -RAINBOW STATES WITH DIFFERENT n

As discussed in [3], when considering an inhomogeneous setting such as a bipartitioning protocol, we can still imagine entangled pairs/multiplets to be created in the two halves of the system as in the homogeneous case. The only difference is that now the number of species and their correlations depend on the half where a specific multiplet is generated.

Therefore, we have to split (5.21) in two separate contributions:

$$\begin{aligned} S_{(-\infty, 0]}(t) &= S_L(t) + S_R(t) \quad \text{where} \\ S_{L/R}(t) &= \int_{BZ_{n_{L/R}}} \frac{dk}{2\pi} \sum_{Q_{L/R}} [\Omega_{L/R}((-\infty, 0], Q_{L/R}, k, t)] s_{L/R}(k, Q_{L/R}) \end{aligned}$$

If the number of species is different in the two halves, then the admitted partitions $Q_{L/R}$ are different as well. Unlike the previous homogeneous case, $\Omega_{L/R}$ only includes positions of multiplets in the left/right half of the system, respectively.

Fortunately, in free models these contributions can be computed separately in a purely geometric way, since quasiparticles propagate with constant velocities. As example, we now consider the bipartitioning protocol between a $2n_L$ - and $2n_R$ -rainbow state.

The main subtlety is that, while quasiparticles $\{0, \dots, n_L - 1\}$ are always left-moving and the others are right-moving with opposite velocities, inside these two subsets the functions $\{v_A(k)\}$ can cross each others upon varying $k \in BZ_{n_{L/R}}$.

To sidestep this problem we define two maps $\sigma_{L/R}(A_{L/R}, k)$ s.t.

$$v_{\sigma(1,k)}(k) \leq \dots \leq v_{\sigma(n-1,k)}(k) \leq 0 \leq v_{\sigma(n,k)}(k) \leq \dots \leq v_{\sigma(2n-1,k)}(k)$$

This way the only dynamically admitted partitions are that for which a threshold $q_{L/R}$ exists s.t.

$$Q_L(q_L) := \{\sigma_L(q_L, k), \dots, \sigma_L(2n_L - 1, k)\} \quad Q_R(q_R) := \{\sigma_R(0, k), \dots, \sigma_R(q_R, k)\}$$

and after some geometric considerations one can write

$$\begin{aligned} [\Omega_L((-\infty, 0], Q_L(q_L), k, t)] &= t \left[\nu_{\sigma_L(q_L, k)}^{(L)}(k) - \theta[q_L - n] \nu_{\sigma_L(q_L - 1, k)}^{(L)}(k) \right] \\ [\Omega_R((-\infty, 0], Q_R(q_R), k, t)] &= t \left[\theta[(n - 1) - q_R] \nu_{\sigma_L(q_R, k)}^{(R)}(k) - \nu_{\sigma_L(0, k)}^{(R)}(k) \right] \end{aligned}$$

In particular, the entanglement entropy grows linearly and does not saturate, due to the infinite size of $\mathbb{A} = (-\infty, 0]$.

Example - Bipartitioning Protocol Between the $2n = 2$ and $2n = 4$ Rainbow States: For the sake of clarity we assume $J = 1$. If we set the left part of the system in the 2–rainbow state, then the dynamic factor $[\Omega_L]$ is one half of that of (5.23).

$$\frac{S_L(t)}{t} = - \int_0^\pi \frac{dk}{2\pi} \sin(k) \left[\left(\frac{1 + \cos(k)}{2} \right) \log \left(\frac{1 + \cos(k)}{2} \right) + \left(\frac{1 - \cos(k)}{2} \right) \log \left(\frac{1 - \cos(k)}{2} \right) \right] \approx 0.159$$

The right side with $n = 2$ requires additional care. The only left-moving quasiparticles, which are able to reach $\mathbb{A} = (-\infty, 0]$, correspond to species 0 and 1.

$$BZ_2 = \left(0, \frac{\pi}{2} \right] \quad \nu_0(k) = -\sin(k) \quad \nu_1(k) = -\cos(k) \quad |\nu_1(k)| > |\nu_0(k)| \leftrightarrow k < \frac{\pi}{4}$$

The dynamically possible partitions are

$$Q = \{0\} \leftrightarrow k > \frac{\pi}{4} \quad Q = \{1\} \leftrightarrow k < \frac{\pi}{4} \quad Q = \{0, 1\}$$

$$[\Omega_R((-\infty, 0], \{0\}, k, t)] = t \cdot \theta \left[k - \frac{\pi}{4} \right] [\sin(k) - \cos(k)]$$

$$[\Omega_R((-\infty, 0], \{1\}, k, t)] = t \cdot \theta \left[\frac{\pi}{4} - k \right] [\cos(k) - \sin(k)]$$

$$[\Omega_R((-\infty, 0], \{0, 1\}, k, t)] = t \cdot \min[\sin(k), \cos(k)]$$

The form factor restricted to the first two species counts the following elements:

$$F_{\{0,1\}}^{(2)}(k) = \begin{bmatrix} F_{00}^{(2)}(k) & F_{01}^{(2)}(k) \\ F_{01}^{(2)}(k) & F_{11}^{(2)}(k) \end{bmatrix}$$

$$F_{00}^{(2)}(k) = \frac{1}{2} + \sin^2(k) \cos(k) \quad F_{11}^{(2)}(k) = \frac{1}{2} - \sin(k) \cos^2(k) \quad F_{01}^{(2)}(k) = \frac{1}{4} \frac{\cos^2(2k)}{\cos[k + \frac{\pi}{4}]}$$

The eigenvalues of $F_{\{0,1\}}^{(2)}(k)$ are given by

$$\lambda_1 + \lambda_2 = F_{00}^{(2)}(k) + F_{11}^{(2)}(k) \quad \lambda_1 \lambda_2 = F_{00}^{(2)}(k) F_{11}^{(2)}(k) - [F_{01}^{(2)}(k)]^2$$

$$\rightarrow \lambda_{1,2}(k) = \frac{1}{2} \left[F_{00}^{(2)}(k) + F_{11}^{(2)}(k) \pm \sqrt{(F_{00}^{(2)}(k) + F_{11}^{(2)}(k))^2 + 4([F_{01}^{(2)}(k)]^2 - F_{00}^{(2)}(k) F_{11}^{(2)}(k))} \right]$$

The entropy content of the involved multiplet configurations are

$$s(k, \{0\}) = -[F_{00}^{(2)}(k) \log F_{00}^{(2)}(k) + (1 - F_{00}^{(2)}(k)) \log(1 - F_{00}^{(2)}(k))]$$

$$s(k, \{1\}) = -[F_{11}^{(2)}(k) \log F_{11}^{(2)}(k) + (1 - F_{11}^{(2)}(k)) \log(1 - F_{11}^{(2)}(k))]$$

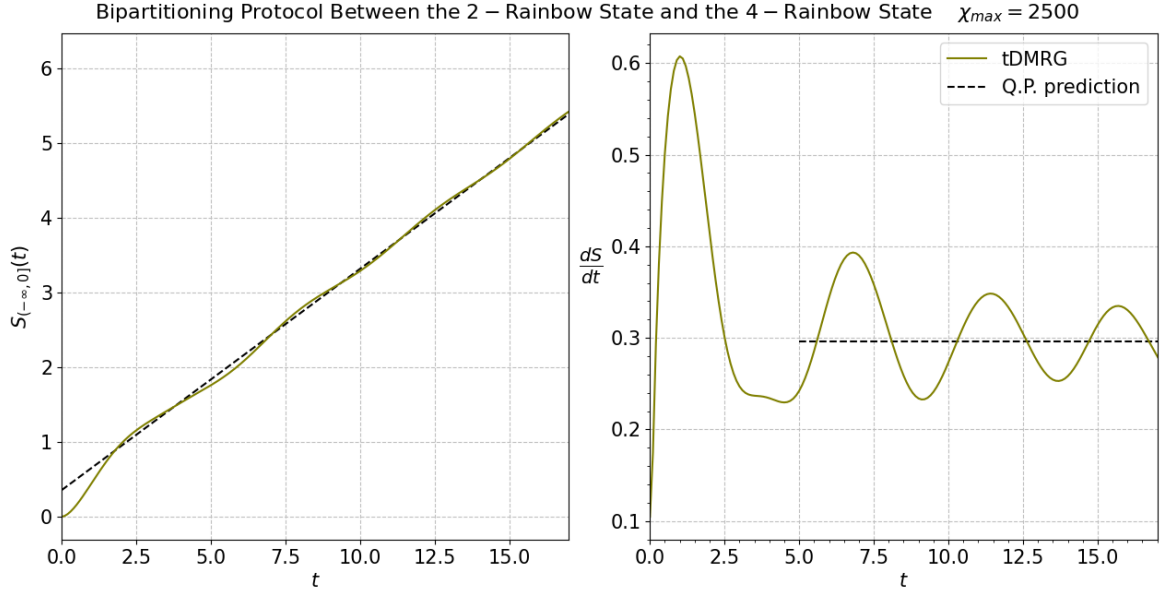


FIGURE 5.5: Comparison between $S_{(-\infty,0]}(t) \approx t \cdot 0.297$ (black dashed line), up to a suitable time translation in the left panel, and the corresponding direct tDMRG simulation.

$$s(k, \{0, 1\}) = - \left[\sum_{i \in \{1, 2\}} \lambda_i(k) \log \lambda_i(k) + (1 - \lambda_i(k)) \log(1 - \lambda_i(k)) \right]$$

Finally, the right contribution to entanglement entropy is

$$\begin{aligned} \frac{S_R(t)}{t} = & - \left\{ \int_0^{\pi/2} \frac{dk}{2\pi} \min[\sin(k), \cos(k)] s(k, \{0, 1\}) \right\} - \\ & - \left\{ \int_0^{\pi/4} \frac{dk}{2\pi} [\cos(k) - \sin(k)] s(k, \{1\}) \right\} - \left\{ \int_{\pi/4}^{\pi/2} \frac{dk}{2\pi} [\sin(k) - \cos(k)] s(k, \{0\}) \right\} \approx 0.137 \end{aligned}$$

The contribution of multiplet partitions in which one or two particles are in $\mathbb{A}$ are almost equal, namely 0.0738 and 0.0635, respectively.

$$\rightarrow \frac{S_{(-\infty,0]}(t)}{t} = \frac{S_L(t) + S_R(t)}{t} \approx 0.297$$

This result is consistent with the corresponding tDMRG simulation in Fig. 5.5.

Chapter 6

2D CLASSICAL 6-VERTEX MODEL AND 1D QUANTUM XXZ MODEL

To clarify some of the mathematical objects and procedures that will be introduced in §7, let us consider temporarily the 2D classical 6-vertex model [11, 62]. Indeed, objects such as transfer or monodromy matrices and intertwiner operators do not have an immediate interpretation within a 1D quantum system, while the idea behind their definition comes from the transfer matrix formalism introduced in the framework of 2D classical models.

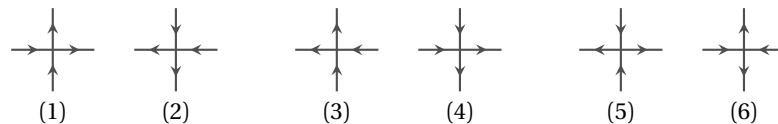
In particular, the integrability of the 2D classical *6-vertex model* is proven through the construction of an infinite set of conserved charges, generated by *the transfer matrix: this will be the same operator as in the algebraic solution of the XXZ chain and one of the conserved charges is the Hamiltonian of the quantum model*. Within a quantum-to-classical mapping, the solution of the classical model automatically gives a solution for a corresponding 1D quantum system. For a detailed description of the 6-vertex model and its solution the reader is referred to [11, 62]: here we just briefly sketch the shift from the 2D classical theory to the 1D quantum theory, in order to make next chapter mathematical objects more intuitive.

6.1 THE 6-VERTEX MODEL

The 6-vertex model lives on a rectangular lattice with M horizontal rows and L vertical columns. Its degrees of freedom (dofs) are vertical and horizontal bonds connecting sites, each taking two possible values as classical $1/2$ -spins.

Pictorially, we represent the two states of each bond by the two possible verses of an arrow on its line. We require:

- PBC in both directions (*toroidal boundary conditions*).
- *Ice rule*: arrows can meet at each site forming only the following six possible configurations (the six possible "vertices"):



The ice rule is equivalent to impose "arrow-flow conservation in any vertex". Moreover, remarkably, for every vertex its arrow-flipped counterpart is included, as well.

Physically, this rule was proposed by Slater in 1941 to describe hydrogen bonding in ice crystals, where the oxygen atoms form a lattice of coordination number four, and between each adjacent pair of atoms is an hydrogen ion, which can be nearer to one or the other

atom. The above arrows can be viewed as representing this dislocation and the available vertices are motivated by electric neutrality.

In particular, by horizontal PBC, in a given row type 5 and 6 vertices must appear in equal number, since they represent point in which the horizontal arrow flux changes direction.

Consider two adjacent rows of vertical bonds, connected by a row of vertices. Type $\{1, 2, 3, 4\}$ vertices does not change the state of a vertical bond, while $\{5, 6\}$ invert it.

However, since $n_5 = n_6$, these inversions preserve the total number $n_\uparrow$ of $\uparrow$, which must be the same in the two rows of vertical bonds.

$$n_\uparrow \in \{0, 1, \dots, L\} \quad n_\uparrow = n'_\uparrow \quad n_\downarrow = n'_\downarrow = L - n_\uparrow \quad (6.1)$$

The model has some *free continuous parameters*, i.e. the energy ϵ_i individually associated to each of the 6 vertices. By requiring *bond-flip symmetry* ("zero-field condition"), it must be

$$\epsilon_1 = \epsilon_2 := \epsilon_a \quad \epsilon_5 = \epsilon_6 := \epsilon_c$$

so that the free parameters are just 3.

If we mean these energies as yet reduced by the energy scale given by the temperature, then the Boltzmann weight associated with the presence of a vertex of type i is $e^{-\epsilon_i}$. Then, the Boltzmann weight W_c of a single configuration c of the whole lattice is given by the product of the weights of all its vertices and the final partition function is

$$Z = \sum_c W_c$$

6.2 TRANSFER MATRIX AND LAX MATRIX

Transfer Matrix: Let us label by $r^{(i)}$ the full horizontal configuration of L vertical bonds immediately under the i -th row. A key role is played by the *transfer matrix* T , defined s.t., given a lattice configuration with $(r^{(j+1)}, r^{(j)})$, $T_{r^{(j)}}^{r^{(j+1)}} \equiv \langle r^{(j+1)} | \hat{T} | r^{(j)} \rangle$ returns the weight contribution to the state in the partition function Z of these two adjacent vertical-bond rows, i.e.

$$T_r^{r'} \equiv \langle r' | \hat{T} | r \rangle := \left(\begin{array}{ccccccc} & \uparrow & & \uparrow & & \uparrow & \\ r'_L & \bullet & & \bullet & & \bullet & \\ & \downarrow & & \downarrow & & \downarrow & \\ & r_L & & r_3 & & r_2 & r_1 \end{array} \right) = \sum_a^* e^{-\beta \sum_{i=1}^6 n_i \epsilon_i} \quad \text{Transfer Matrix}$$

$$Z = \left(\begin{array}{c} \left(\begin{array}{cccccc} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \end{array} \right) \left. \vphantom{\begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{array}} \right\} M \\ \left. \vphantom{\begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{array}} \right\} L \end{array} \right) = \quad (6.2)$$

$$= \sum_{\{r^{(i)}\}} \langle r^{(M)} | \hat{T} | r^{(M)} \rangle \langle r^{(M)} | \dots | r^{(3)} \rangle \langle r^{(3)} | \hat{T} | r^{(2)} \rangle \langle r^{(2)} | \hat{T} | r^{(1)} \rangle = \text{Tr} [\hat{T}^M] \quad \text{where:}$$

- the trace comes from the constraint of vertical PBC;
- α identifies a configuration of horizontal bonds in the row between r and r' ;

- $\star$ means that the sum is done over the only α s.t. (r', α, r) is a vertex-row configuration which is acceptable according to ice rule and horizontal PBC;
- $\{n_i\}$ is the number of vertices of each kind in the configuration (r', α, r) ;
- $\langle r' | \hat{T} | r \rangle$ is null if r and r' are non-compatible configurations, so that in Z the $\sum_{r^{(i)}} |r^{(i)}\rangle \langle r^{(i)}|$ are sums over complete sets, equal to $\hat{\mathbb{I}}$ for completeness.

A first example of the convenience of this rewriting is that, in the $M \rightarrow +\infty$ limit, given the highest eigenvalue λ_1 of the transfer matrix, $Z \approx \lambda_1^M$, while λ_2 notoriously encodes the correlation length of the system. Two properties of the T -operator/matrix are:

- Given (6.1), the transfer matrix has a block diagonal structure with blocks corresponding to configurations with the same $n_\uparrow/2 \in \{0, \dots, L\}$.

From an 1D Bethe-ansatz perspective, this is equivalent to the symmetry corresponding to quasiparticles/magnetization conservation in scattering, as discussed in the next chapter.

- It is a real symmetric matrix of Boltzmann weights $\langle r' | \hat{T} | r \rangle = \langle r | \hat{T} | r' \rangle$.

Since any vertical-bond row of the system can assume the same set of states $\{r\}$, with an improper quantum reformulation in which the T -matrix is interpreted as an operator $\hat{T}$ expressed in a basis " $\{|r\rangle\}$ ", one can define

$\mathcal{H}$ "Hilbert" space of a single vertical-bond row
 $\mathcal{H}_j$ local "Hilbert" space of the j -th vertical bond in the row
 ν "Hilbert" space a single horizontal bond

$$\hat{T} : \mathcal{H} \rightarrow \mathcal{H}$$

Lax Matrix: We stress that $(r^{(i+1)}, r^{(i)})$ does not fully specify the state of the intermediate row of horizontal bonds $\alpha^{(i)} = (\alpha_1^{(i)}, \dots, \alpha_L^{(i)})$. In order for (6.2) to hold, $\hat{T}$ must implicitly contain the sum and trace over these "ancillary" horizontal dofs. To explicit this sum, one can introduce, for every row, an horizontal-propagation transfer matrix, the so-called *Lax matrix* $\mathcal{L}^{(j)}$, s.t.

$$\mathcal{L}^{(j)}_{r'_j \beta} \equiv \langle r'_j \beta | \mathcal{L}^{(j)} | r_j \alpha \rangle := \left[\begin{array}{c} r'_j \in \mathcal{H}_j \\ \beta \in \nu \leftarrow \bullet \rightarrow \alpha \in \nu \\ r_j \in \mathcal{H}_j \end{array} \right] \quad \mathcal{L}^{(j)} : \mathcal{H}_j \otimes \nu \rightarrow \mathcal{H}_j \otimes \nu \quad \text{Lax matrix}$$

$$T_{r'}^{r'} \equiv \langle r' | \hat{T} | r \rangle := \left(\begin{array}{c} r'_L \\ \alpha_1 \uparrow \bullet \alpha_L \uparrow \bullet \uparrow \uparrow \uparrow \bullet \alpha_4 \uparrow \bullet \alpha_3 \uparrow \bullet \alpha_2 \uparrow \bullet \alpha_1 \end{array} \right)_{r_L} = \mathcal{L}^{(L)}_{r'_L \alpha_1} \left(\prod_{j=L-1}^1 \mathcal{L}^{(j)}_{r'_j \alpha_{j+1}} \right)_{r_L \alpha_L}$$

As mentioned, the ice rule can be viewed as the fact that the total outgoing arrow-flux from each vertex is zero. Moreover, we recall the arrow-flip symmetry of the model. Thus, if we denote by $+$ an ingoing arrow and $-$ an outgoing arrow, then in the basis $|r, \alpha\rangle \in \{|++\rangle, |+-\rangle, |-+\rangle, |--\rangle\}$ the Lax matrix has the following structure:

$$\mathcal{L}^{(j)} = \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & c & 0 \\ 0 & c & b & 0 \\ 0 & 0 & 0 & a \end{pmatrix} \quad (6.3)$$

Tracing over $\nu \otimes \nu'$ (and exploiting trace cyclicity) to make transfer matrices appear:

$$\text{i.e. } [\hat{T}, \hat{T}'] = 0$$

Spectral Parameter of the Family (and Anisotropy Parameter): At this point $\mathcal{L}$ and $\mathcal{L}'$, according to (6.3), are defined by their (a, b, c) and (a', b', c') unknown parameters, while R is a general 4×4 unknown matrix. In order to find a solution R for (6.4) we make the ansatz that the R -matrix has the same structure (6.3) as an $\mathcal{L}$ -matrix, with (a'', b'', c'') .

Of these 9 parameters, only 6 are relevant, since YBE equations are homogeneous, i.e. the normalization of involved objects is arbitrary. Note that $\hat{R}$ and $\mathcal{L}^{(j)}$ act on different spaces, but this ansatz is equivalent to assume that the ice rules applies to the tilted R -vertex as well.

As described in more detail in [62], to find solutions of (6.4) it is convenient to reparametrize $(a, b, c) \rightarrow (\rho, \Delta, \lambda)$, where ρ encodes the normalization and is out of interest.

$$\begin{aligned} \Delta &:= \cosh(\eta) & a &= \rho \sinh(\lambda + \eta) & b &= \rho \sinh(\lambda) & c &= \rho \sinh(\eta) \\ \rightarrow \mathcal{L}^{(j)}(\lambda) &= \rho \begin{pmatrix} \sinh(\lambda + \eta) & 0 & 0 & 0 \\ 0 & \sinh(\lambda) & \sinh(\eta) & 0 \\ 0 & \sinh(\eta) & \sinh(\lambda) & 0 \\ 0 & 0 & 0 & \sinh(\lambda + \eta) \end{pmatrix} \end{aligned} \quad (6.6)$$

Note that, if $\lambda = 0$, then $\mathcal{L}^{(j)}(0) = \rho \sinh(\eta) \hat{\pi}_{j\nu}$ where $\hat{\pi}_{j\nu}$ is the *permutation operator*, which exchanges the state of the physical and ancillary dofs.

Finally one finds that, in order for a solution R of (6.4) to exist:

$$\Delta = \Delta' = \Delta'' \quad \text{and} \quad \lambda'' = \lambda' - \lambda$$

For every fixed "anisotropy parameter" Δ , assumed equal and implicit in each of the following objects, one automatically obtains a family of R operators and Lax matrices satisfying

$$\mathcal{L}_\nu^{(j)}(\lambda) \mathcal{L}_{\nu'}^{(j)}(\lambda') \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) = \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) \mathcal{L}_{\nu'}^{(j)}(\lambda') \mathcal{L}_\nu^{(j)}(\lambda) \quad \text{RLL relation} \quad (6.7)$$

In order to come back to transfer matrices, we first take the product of all the L columns of the system, obtaining the above defined monodromy matrices.

$$\begin{aligned} \hat{M}_\nu(\lambda) &:= \mathcal{L}_\nu(\lambda)^{\otimes L} := \mathcal{L}_\nu^{(L)}(\lambda) \otimes \dots \otimes \mathcal{L}_\nu^{(1)}(\lambda) : \mathcal{H} \otimes \nu \rightarrow \mathcal{H} \otimes \nu & \text{Monodromy matrix} \\ \hat{M}_\nu(\lambda) \hat{M}_{\nu'}(\lambda') \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) &= \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) \hat{M}_{\nu'}(\lambda') \hat{M}_\nu(\lambda) \end{aligned} \quad (6.8)$$

Apart from the explicit spectral parameters, this is exactly the relation proven in (6.5).

We stress that (as clear from the pictorial representation of monodromy matrices) the tensor product $\mathcal{L}^{(\nu)}(\lambda)^{\otimes L}$ only refers to horizontal bonds $\mathcal{H}_i$, while every $\mathcal{L}^{(i)}$ acts on the same ancillary space ν . Vertical-bond dofs undergo ordinary matrix multiplication.

Tracing over the ancillary dof ν , i.e. requiring horizontal PBC:

$$\hat{T}(\lambda) := \text{Tr}_\nu[\hat{M}_\nu(\lambda)] : \mathcal{H} \rightarrow \mathcal{H} \quad \text{Transfer matrix}$$

As pictorially proven above, multiplying (6.8) on the left by R^{-1} and tracing over the two ancillary spaces, transfer matrices of the defined family commute with each other:

$$[\hat{T}(\lambda), \hat{T}(\mu)] = 0$$

Yang-Baxter Equation (YBE): In the next chapter, we will develop a fully abstract method to simultaneously define 1D quantum models and their solution, that is the *quantum inverse scattering method*, or *algebraic Bethe ansatz*. The starting point is an R -matrix which, by requiring the RLL relation, enables the construction of an infinite family of commuting transfer matrices.

To this end, it is convenient to recast (6.8) as a condition on R -matrices only. The method begins from the identification of an R -matrix which satisfy this necessary requirement. To obtain such a condition, we consider the product of three monodromy matrices and require its associativity. For the sake of readability here we omit the spectral parameter. However, the correspondence between ancillary spaces and spectral parameters is $\nu \leftrightarrow \lambda$, $\nu' \leftrightarrow \lambda'$ and $\nu'' \leftrightarrow \lambda''$.

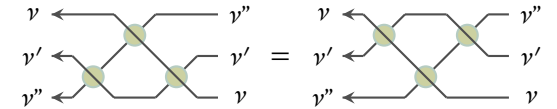
$$\text{We can rewrite (6.8) as } \hat{M}_\nu \hat{M}_{\nu'} = \hat{R}^{(\nu' \otimes \nu)} \hat{M}_{\nu'} \hat{M}_\nu [\hat{R}^{(\nu' \otimes \nu)}]^{-1}$$

There are two ways to plug this relation into the product of three monodromy matrices, yielding to two different expressions:

$$\begin{aligned} [A] \bullet \quad & \hat{M}_\nu (\hat{M}_{\nu'} \hat{M}_{\nu''}) = \hat{R}^{(\nu'' \otimes \nu')} (\hat{M}_\nu \hat{M}_{\nu''}) \hat{M}_{\nu'} [\hat{R}^{(\nu'' \otimes \nu')}]^{-1} = \\ & = \hat{R}^{(\nu' \otimes \nu')} \hat{R}^{(\nu'' \otimes \nu)} \hat{M}_{\nu''} (\hat{M}_\nu \hat{M}_{\nu'}) [\hat{R}^{(\nu'' \otimes \nu)}]^{-1} [\hat{R}^{(\nu' \otimes \nu')}]^{-1} = \\ & = \hat{R}^{(\nu' \otimes \nu')} \hat{R}^{(\nu'' \otimes \nu)} \hat{R}^{(\nu' \otimes \nu)} (\hat{M}_{\nu''} \hat{M}_{\nu'} \hat{M}_\nu) [\hat{R}^{(\nu' \otimes \nu)}]^{-1} [\hat{R}^{(\nu'' \otimes \nu)}]^{-1} [\hat{R}^{(\nu' \otimes \nu')}]^{-1} \\ [B] \bullet \quad & (\hat{M}_\nu \hat{M}_{\nu'}) \hat{M}_{\nu''} = \hat{R}^{(\nu' \otimes \nu)} \hat{M}_{\nu'} (\hat{M}_\nu \hat{M}_{\nu''}) [\hat{R}^{(\nu' \otimes \nu)}]^{-1} = \\ & = \hat{R}^{(\nu' \otimes \nu)} \hat{R}^{(\nu'' \otimes \nu)} (\hat{M}_{\nu''} \hat{M}_{\nu'}) \hat{M}_\nu [\hat{R}^{(\nu'' \otimes \nu)}]^{-1} [\hat{R}^{(\nu' \otimes \nu)}]^{-1} = \\ & = \hat{R}^{(\nu' \otimes \nu)} \hat{R}^{(\nu'' \otimes \nu)} \hat{R}^{(\nu'' \otimes \nu')} (\hat{M}_{\nu''} \hat{M}_{\nu'} \hat{M}_\nu) [\hat{R}^{(\nu'' \otimes \nu')}]^{-1} [\hat{R}^{(\nu'' \otimes \nu)}]^{-1} [\hat{R}^{(\nu' \otimes \nu)}]^{-1} \end{aligned}$$

Since by associativity $[A] = [B]$, the R -matrix must satisfy

$$\hat{R}^{(\nu'' \otimes \nu')}(\lambda'' - \lambda') \hat{R}^{(\nu'' \otimes \nu)}(\lambda'' - \lambda) \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) = \hat{R}^{(\nu' \otimes \nu)}(\lambda' - \lambda) \hat{R}^{(\nu'' \otimes \nu)}(\lambda'' - \lambda) \hat{R}^{(\nu'' \otimes \nu')}(\lambda'' - \lambda')$$

Pictorially:  YBE (6.9)

where the pictorial representation is remarkably the same as (2.12).

This last observation is consistent with the fact that the permutation operator $\hat{\pi}_{\nu\nu'}$, which exchanges two ancillary dofs, has to be a solution of the Yang-Baxter equation, as follows from (6.6) for $\lambda = 0$ (we assumed $\hat{\mathcal{L}}$ and $\hat{R}$ to have the same shape).

6.4 CONSERVED CHARGES AND XXZ HAMILTONIAN

We now relax the hypothesis for T -elements to be real Boltzmann weights, extending λ to the entire $\mathbb{C}$. We redefine in (6.10):

$$\begin{aligned} \lambda &\rightarrow i \left(\lambda - i \frac{\eta}{2} \right) \quad \eta \rightarrow -\eta \quad \rho = \frac{-i}{\sinh(\eta)} \\ \hat{\mathcal{L}}^{(j)}(\lambda) &= \frac{1}{\sinh(\eta)} \begin{pmatrix} \sin^{[+]}(\lambda) & 0 & 0 & 0 \\ 0 & \sin^{[-]}(\lambda) & i \sinh(\eta) & 0 \\ 0 & i \sinh(\eta) & \sin^{[-]}(\lambda) & 0 \\ 0 & 0 & 0 & \sin^{[+]}(\lambda) \end{pmatrix} = \\ &= \frac{1}{\sinh(\eta)} \begin{pmatrix} \sin(\lambda + i\eta \hat{s}^z) & i \sinh(\eta) \hat{s}^- \\ i \sinh(\eta) \hat{s}^+ & \sin(\lambda - i\eta \hat{s}^z) \end{pmatrix} \end{aligned} \quad (6.10)$$

where $\{\hat{s}^z, \hat{s}^\pm\}$ are the spin generators in the ancillary space ν of spin $s = 1/2$.

We call *shift point* the value of the spectral parameter λ where the Lax matrix becomes proportional to the exchange operator $\hat{\pi}$, which replace the state of the physical spin with that of the ancillary one and vice versa.

$$\mathcal{L}^{(j)}(i\eta/2) = i\hat{\pi}_{j\nu} \rightarrow \lambda = \frac{i\eta}{2} \text{ shift point}$$

In particular, here the Lax matrix is unitary.

At the shift point the transfer matrix has a particularly simple form as well. Indeed, it is proportional the one-step translation operator.

$$\hat{T}(i\eta/2) = i^L \text{Tr}_\nu[\hat{\pi}_{L\nu} \dots \hat{\pi}_{1\nu}] = i^L \hat{\pi}_{L(L-1)} \dots \hat{\pi}_{21} \hat{\pi}_{1L} = i^L e^{i\hat{K}}$$

where $\hat{K}$ is the lattice momentum operator already discussed in §1 and §2.

The main result of the above construction is the definition of a family $\{\hat{T}(\lambda)\}$ of commuting transfer matrices $[\hat{T}(\lambda), \hat{T}(\mu)] = 0$, parametrized by λ .

If we expand these transfer matrices (or their logarithm, as is discussed below) in powers of $(\lambda - \lambda_0)$ around an arbitrarily chosen point λ_0 , since transfer matrices with different spectral parameters commute, all the operators in this expansion must commute with each other. It follows that (the logarithms of) $\hat{T}(\mu)$ can be interpreted as the generator of an infinite set of commuting integrals of motion. With regard to the choice of expanding the logarithm of the transfer matrix, the reason is that in this way, if λ_0 is the shift point, then these conserved operators turn out to be Hermitian extensive local charges as defined in §3.4.

We thus define the expansion of the transfer matrix logarithm around the shift point:

$$\begin{aligned} \log \hat{T}_{\frac{1}{2}}\left(\lambda + \frac{i\eta}{2}\right) &= 2\pi L i \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \hat{H}^{(k)} \\ \hat{H}^{(k)} &:= \frac{1}{L} \frac{1}{2\pi i} \frac{\partial^{k-1}}{\partial \lambda^{k-1}} \log \left[\hat{T}_{\frac{1}{2}}^{[+]}(\lambda) \right] \Big|_{\lambda=0} \quad k \in \mathbb{N}^{\geq 2} \\ \log \hat{T}\left(\lambda + \frac{i\eta}{2}\right) &= i \sum_{k=0}^{\infty} \lambda^k \hat{H}^{(k)} \rightarrow \hat{H}^{(k)} = -i \frac{\partial^{k-1}}{\partial \lambda^{k-1}} \log \hat{T}\left(\lambda + \frac{i\eta}{2}\right) \Big|_{\lambda=0} \quad k \in \mathbb{N}^{>0} \end{aligned}$$

Note that we factorized L out of $\hat{H}^{(k)}$, to guarantee finite eigenvalues in the thermodynamic limit. Thus, they are extensive local operators up to this rescaling.

As mentioned, plugging this into $[\hat{T}(\lambda), \hat{T}(\lambda')] = 0$ yields

$$[\hat{H}^{(k)}, \hat{H}^{(m)}] = 0 \quad \forall k, m \in \mathbb{N}^{\geq 2}$$

Connection with the XXZ Model: It can be proven [62] that, with the mapping $\Delta = \cosh(\eta) > 1$, $\hat{H}^{(2)}$ is proportional to the Hamiltonian of the XXZ model plus a constant. This implies that all $\hat{H}^{(k)}$, which commute with $\hat{H}^{(2)}$ are local conserved charges in involution of the XXZ model. More in general, the 6-vertex model transfer matrix and the XXZ chain share the same eigenvectors and the solution of one model translates into the other.

For the isotropic $\Delta = 0$ and critical $|\Delta| < 1$ phases, the mapping is different but still holds, as will be mentioned in the next chapter. In particular, to obtain the XXX model one has first to rescale $\lambda \rightarrow \eta\lambda$ and only after to take the limit $\eta \rightarrow 0$ [72]. This way the obtained Lax matrix is

$$\mathcal{L}^{(j)}(\lambda) = \lambda \cdot \hat{\mathbb{I}} + i\hat{\pi}_{j\nu} \quad (6.11)$$

Chapter 7

QUANTUM INVERSE SCATTERING METHOD

To briefly recall some historical background, as summarized by Sklyanin in [121], the *quantum inverse scattering method (QISM)* [53, 62, 72, 121, 123], or *algebraic Bethe ansatz (ABA)*, was developed in 1978-79 by the Leningrad school, in particular by Faddeev, Sklyanin, and Takhtadzhyan.

In their seminal work [122], they proposed a "quantum-mechanical variant" of the classical inverse scattering method (introduced in [63]) and applied it to obtain an exact solution of the sine-Gordon model.

Closely related ideas were developed almost independently during the same years at Fermilab (by Thacker, Creamer, and Wilkinson) and in Germany by Honerkamp and collaborators.

The key breakthrough of these studies was the uncovering of a connection between Bethe's formulation and the classical inverse scattering method: two traditions in the theory of integrability which, up to that date, were developing quite independently.

In this work we are not going to discuss the classical inverse scattering method, which is not required to understand the QISM.

7.1 THE IDEA OF AN INVERSE APPROACH

In the previous chapter we used an infinite family of mutually commuting transfer matrices $\{\hat{T}(\lambda)\}$ to generate an infinite set of local conserved charges $\{\hat{H}^{(n)}\}$ in involution.

Interpreting one of these charges (or an arbitrary linear superposition of them) as the Hamiltonian of a 1D quantum system, then the others represent an infinite set of local integrals of motion. It follows that the corresponding model is integrable.

In particular, we showed that transfer matrices of the 2D classical 6-vertex model (for different free parameters) lead to conserved charges including the XXZ model Hamiltonian and this allowed us to find the local conserved charges of the latter. At this stage, if local charges were a complete set, we would have been immediately able to build the GGE of the LSS.

From a more general and abstract perspective, we showed that *the family of transfer matrices can be constructed from an R -matrix acting on two ancillary spaces v and v' and which satisfies the Yang-Baxter equation (6.9). With respect to §6, this viewpoint does not require auxiliary spaces to have the same dimension as the physical local Hilbert space. This additional freedom is the key for the generalization of the QISM enabling the computation of quasilocal charges in involution as well.* In order to generalize the previous setting, we define the notations

- $\mathcal{H}$ total Hilbert space of the physical lattice
- $\mathcal{H}_i$ local Hilbert space of the i -th site of the lattice, with $\dim(\mathcal{H}_i) = d$
- v_s Hilbert space of the ancillary dof of "spin s ", i.e. $\dim v = 2s + 1$

where now the ancillary dof has no direct physical interpretation in the physical quantum system of interest. The spin s of the auxiliary space $\mathcal{V}_s$ is arbitrarily fixed before looking for the corresponding solution of the YBE (under some ansatz or symmetry requirement).

The connection with physical dofs emerges only after, by fixing the dimension of their local Hilbert space $\dim \mathcal{H}_j = 2$ and then looking for a family of Lax matrices satisfying an intertwining relation mediated by R on a fixed physical site j (the RLL relation (6.7)).

Before starting with the depiction of the method, we find instructive to point out the two faces of its "inverse" nature, being simultaneously its strength and weakness:

- As mentioned, once an R -matrix solution of the YBE is found, this immediately induces a broad class of integrable models together with their (quasi)local integrals of motions.
- While the derivation $\hat{R} \rightarrow \hat{H}$ follows a standard procedure defined by the method, there is no prescription about how to point to the R -matrix leading to the solution of a specific quantum Hamiltonian. For the XXZ model, this problem is solved owing to the previously discussed connection with the 6-vertex transfer matrix.

In our work, we employ the QISM for the computation of the XXZ model quasilocal charges. Thus, in this chapter we do not discuss deeply technical and formal aspects of its general proof, preferring rather a more intuitive overview, focussed on the derivation and evaluation of the quasilocal charges.

7.2 FROM THE ABSTRACT YBE TO THE FAMILY OF ARBITRARY-SPIN TRANSFER MATRICES

The Algebraic Structure Underlying Integrability: The abstract construction we are going to sketch was introduced by Drinfeld [46], but we will mainly follow the overviews in [53, 108]. We will mention some concepts of Drinfeld's *Hopf algebras* perspective later.

One of the pivotal insights of this formulation of the QISM is that the infinite number of (quasi)local conserved charges in an integrable model can be traced back to an underlying symmetry, which can be encoded within an appropriate algebraic structure $\mathcal{A}$. In this chapter we consider the universal enveloping algebra $U(\mathfrak{sl}(2))$ and the corresponding quantum group $U_q(\mathfrak{sl}(2))$, but the method is fully general.

Then, *both the local physical space $\mathcal{H}_j$ and the auxiliary space $\mathcal{V}_s$ (of arbitrary spin s) are assumed to be representations (usually irreducible) of $\mathcal{A}$. This way the method enables the construction of a fundamental monodromy matrix whose off-diagonal entries are operators acting on $\mathcal{H}$ connecting the common eigenstates of the local conserved charges $\{\hat{H}^{(n)}\}$, including the Hamiltonian, similarly to creation and annihilation operators in the second quantization formalism. It follows that Bethe eigenstates of the corresponding integrable models are algebraically generated by repeated applications of these operators to a vacuum state, as we will mention in §7.9.*

Abstract Yang-Baxter Equation: Given that, to find a generalization of the consistency requirement (6.9), we start from the algebra $\mathcal{A}$ encoding the symmetry of the integrable physical models we are going to generate (and simultaneously solve).

Then, we define the abstract $\mathcal{R}$ -matrix $\mathcal{R} \in \mathcal{A} \otimes \mathcal{A}$ as a solution of

$$\mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12} = \mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} \quad \text{Abstract YBE} \quad (7.1)$$

where this relation holds in $\mathcal{A} \otimes \mathcal{A} \otimes \mathcal{A}$ and $\{1, 2, 3\}$ are labels for the three (still not defined) spaces where $\mathcal{A}$ is represented.

The chosen algebra $\mathcal{A}$ must have a family of representations $r(a, \lambda)$, parametrized by a discrete label a and a continuous parameter λ , called *spectral parameter*.

As we will discuss in §7.3, for the XXX and XXZ model, $\mathcal{A}$ is $U(\mathfrak{sl}(2))$ and its quantum deformation $U_q(\mathfrak{sl}(2))$, respectively. Thus, for simplicity, in this work we assume both physical and ancillary spaces to be unitary semi-integer spin s representations of these specific algebras. However, the procedure can also be generalized to non-unitary complex spin representations [72] of $U_q(\mathfrak{sl}(2))$ or other algebras.

Once $\mathcal{A}$ is chosen, the concrete R -matrices are defined by fixing the dimensions¹ of the ancillary spaces on which they act and considering the relative representations of $\mathcal{R}$.

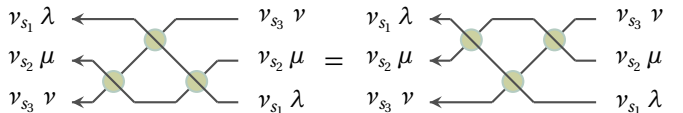
$$\hat{R}_{ss'}(\lambda - \mu) := [r(s, \lambda) \otimes r(s', \mu)] \mathcal{R} : \nu_s \otimes \nu_{s'} \rightarrow \nu_s \otimes \nu_{s'}$$

$$\hat{R}_{ss'}(\lambda - \mu) = \begin{array}{ccc} & \nearrow \nu_s \lambda & \\ & \bullet & \\ & \nwarrow \nu_{s'} \mu & \\ \nu_{s'} \mu & & \nu_s \lambda \end{array} \quad \text{R-Matrix}$$

The assumed dependence of $R_{ab}^{(s)}$ on the difference $\lambda - \mu$ rather than (λ, μ) separately reflects some homogeneity required to the family of representations $r(s, \lambda)$ and their parametrization.

This is the very definition of the *intertwiner operator* (or *R-matrix*), analytically depending on the *spectral parameter* $\lambda \in \mathbb{C}$: this operator is the main protagonist in Yang-Baxter integrability.

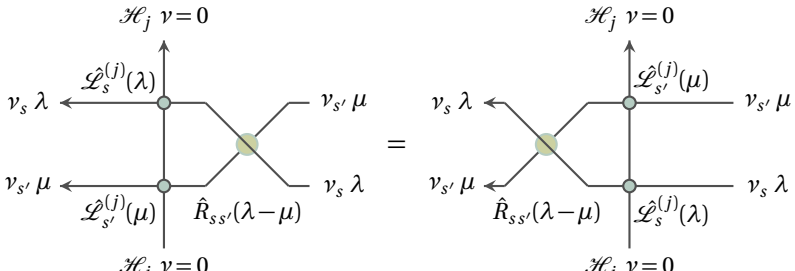
By construction, $\hat{R}$ satisfies

$$\hat{R}_{s_2 s_3}(\mu - \nu) \hat{R}_{s_1 s_3}(\lambda - \nu) \hat{R}_{s_1 s_2}(\lambda - \mu) = \hat{R}_{s_1 s_2}(\lambda - \mu) \hat{R}_{s_1 s_3}(\lambda - \nu) \hat{R}_{s_2 s_3}(\mu - \nu)$$

YBE (7.2)

Once $\hat{R}_{ss'}$ is fixed, the s -spin Lax matrices, responsible of the bridging with the physical space $\mathcal{H}$, are similarly obtained by specifying the dimension of the local Hilbert space $\dim \mathcal{H}_j$.

$$\hat{\mathcal{L}}_s^{(j)}(\lambda - \mu) \in r(s, \lambda) \cdot \mathcal{A} \otimes r(j, \mu) \cdot \mathcal{A} \quad \hat{\mathcal{L}}_s^{(j)}(\lambda - \mu) : \nu_s \otimes \mathcal{H}_j \rightarrow \nu_s \otimes \mathcal{H}_j$$

$$\hat{\mathcal{L}}_{s'}^{(j)}(\mu) \hat{\mathcal{L}}_s^{(j)}(\lambda) \hat{R}_{ss'}(\lambda - \mu) = \hat{R}_{ss'}(\lambda - \mu) \hat{\mathcal{L}}_s^{(j)}(\lambda) \hat{\mathcal{L}}_{s'}^{(j)}(\mu) \quad (7.3)$$


RLL relation

For the sake of clarity we specify that in $r(j, \mu)$ the label j simply denotes the local Hilbert space $\mathcal{H}_j$, which has fixed dimension d .

Note that $\hat{\mathcal{L}}_s^{(j)}(\lambda - \mu) := [r(s, \lambda) \otimes r(j, \mu)] \mathcal{R}$ is automatically a good solution. Indeed, by representing the abstract YBE (7.1) on $r(s, \lambda) \otimes r(s', \mu) \otimes r(\frac{1}{2}(d-1), \nu=0)^{(j)}$ one obtains the RLL relation. However, the above definition leaves a bit more freedom in choosing Lax matrices, as we will explicitly show in §7.3 within the XXX model.

¹More precisely, in the original ABA literature the R -matrix is defined on two ancillary spaces of the same dimension. Here, since we aim to prove that transfer matrices obtained from ancillary spaces with $s \neq s'$ commute as well, we consider a "generalized" R -matrix. The extreme convention in this direction, that is to view Lax matrices as R -matrices acting between a physical and an ancillary space, is for example adopted in [84, 151].

Formally, the RLL relation expresses the commutation of two Lax matrices acting on the same physical site (but corresponding to potentially different spectral parameters or ancillary spaces) up to the action of a suitable intertwiner $\hat{R}$. Thus, the RLL relation is the fundamental property of Lax matrices which enables their use in the construction of an infinite family of commuting transfer matrices and, from these, an infinite set of quasilocal charges in involution. For this reason, the mathematical structure arising from the Yang-Baxter and RLL equations is sometimes regarded as the most general and abstract definition of integrability, but the topic is still discussed [32].

A Glimpse into Hopf Algebras: The above discussion can be recast in a more formal and elegant form after the introduction of Hopf algebras [44, 46, 47, 79].

Essentially, a Hopf algebra is an associative algebra $\mathcal{A}$ with a homomorphism

$$\Delta : \mathcal{A} \rightarrow \mathcal{A} \otimes \mathcal{A} \quad \text{Comultiplication}$$

with some properties which we do not mention here. For example, the most important is

$$(\mathbb{I} \otimes \Delta) \circ \Delta = (\Delta \otimes \mathbb{I}) \circ \Delta$$

Consider two representations (r_1, r_2) of $\mathcal{A}$ on two vector spaces (v_1, v_2) , respectively. Then, remarkably, the comultiplication Δ allows us to define a representation r_3 of $\mathcal{A}$ on $v_1 \otimes v_2$.

In this abstract picture, the existence of an abstract intertwiner operator $\mathcal{R} \in \mathcal{A} \otimes \mathcal{A}$ satisfying (7.1) defines the so-called *quasitriangular Hopf algebras*. In this case, the intertwining relation means by definition that, $\forall (v_1, v_2)$, the representations $v_1 \otimes v_2$ and $v_2 \otimes v_1$ are isomorphic.

Later we will employ some practical applications of these definitions.

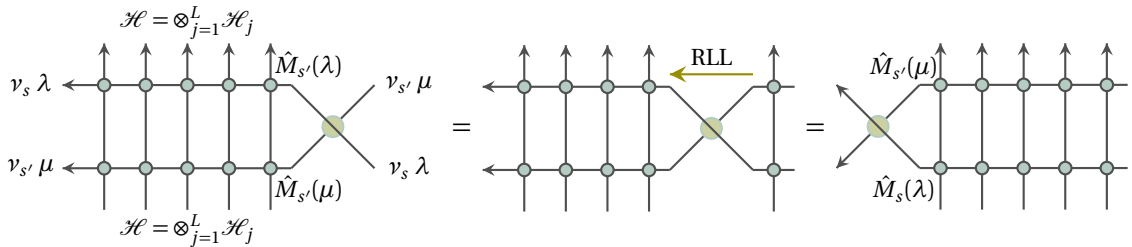
Monodromy Matrices and Commuting Transfer Matrices: Once we have Lax matrices and R -matrices satisfying the RLL relation and the YBE, we can go over the steps of §6.3 again to build the monodromy matrices and prove the commutation of the corresponding transfer matrices.

$$\hat{M}_s(\lambda) := \hat{\mathcal{L}}_s(\lambda)^{\otimes L} := \hat{\mathcal{L}}_s^{(L)}(\lambda) \otimes \dots \otimes \hat{\mathcal{L}}_s^{(1)}(\lambda) : v_s \otimes \mathcal{H} \rightarrow v_s \otimes \mathcal{H} \quad \text{Monodromy matrices} \quad (7.4)$$

We recall that these tensor products deal with physical spaces $\mathcal{H}_j$, while every $\hat{\mathcal{L}}_s^{(j)}$ acts on the same ancillary space v_s . Auxiliary dofs undergo ordinary matrix multiplication.

As in §6.3, starting from $M_s(\mu) M_{s'}(\lambda) R_{ss'}(\lambda - \mu)$ and then leftshifting the intertwiner exploiting the RLL relation for every lattice site, yields an intertwining relation for monodromy matrices.

Pictorially:



$$\hat{M}_s(\mu) \hat{M}_{s'}(\lambda) \hat{R}_{ss'}(\lambda - \mu) = \hat{R}_{ss'}(\lambda - \mu) \hat{M}_{s'}(\lambda) \hat{M}_s(\mu) \quad (7.5)$$

Now, if we define transfer matrices $\hat{T}_s(\lambda)$ as the trace of $\hat{M}_s(\lambda)$ over the ancillary space

$$\hat{T}_s(\lambda) := \text{Tr}_{v_s} \hat{M}_s(\lambda) : \mathcal{H} \rightarrow \mathcal{H} \quad \text{Transfer matrices} \quad (7.6)$$

then by applying $\hat{R}_{ss'}^{-1}(\lambda - \mu)$ at the right of (7.5) and tracing over v_s and $v_{s'}$, one is able to prove that they commute with each other:

i.e. $[\hat{T}_s(\lambda), \hat{T}_{s'}(\mu)] = 0 \quad \forall s, s' \in \frac{1}{2}\mathbb{Z}^+ \quad \forall \lambda, \mu \in \mathbb{C}$

It follows that the spectral parameter and the ancillary semi-integer spin s parametrize a family of commuting transfer matrices $\{\hat{T}_s(\lambda)\}$.

In §7.8 we will discuss how to use this result to build a suitable complete set of quasilocal conserved charges in involution for the GGE.

7.3 THE $U_q(\mathfrak{sl}(2))$ QUANTUM GROUP AND THE XXZ LAX OPERATOR

The Universal Enveloping Algebra $U(\mathfrak{sl}(2))$: $\mathfrak{sl}(2)$ is the *special linear Lie algebra*.

Its defining (fundamental) representation $r_{s=\frac{1}{2}}$ is obtained by considering the vector space over the field $\mathbb{C}$ spanned by

$$r_{s=\frac{1}{2}}[\hat{J}^z] = \frac{\hat{\sigma}^z}{2} \quad r_{s=\frac{1}{2}}[\hat{J}^\pm] = \hat{\sigma}^\pm = \frac{\hat{\sigma}^x \pm i\hat{\sigma}^y}{2} \quad (7.7)$$

and where the abstract Lie brackets $[\cdot, \cdot]$ are represented by the commutator. It follows that, from the abstract viewpoint, it is defined by the following relations:

$$[\hat{J}^+, \hat{J}^-] = 2\hat{J}^z \quad [\hat{J}^z, \hat{J}^\pm] = \pm\hat{J}^\pm \quad (7.8)$$

For the sake of clarity, henceforth we will denote by $\hat{\sigma}^\alpha$ the Pauli matrices, by $\hat{J}^\alpha$ the abstract generators and by $\hat{s}^\alpha$ their representations on a $(2s+1)$ -dimensional space, i.e. $r_s[\hat{J}^\alpha] = \hat{s}^\alpha$.

Consider for the moment a more general Lie algebra $\mathcal{L}$ over a field $\mathbb{F}$, with Lie brackets denoted by $[\cdot, \cdot]$. Formally, a *universal enveloping algebra* of $\mathcal{L}$ [69] is a pair $(U(\mathcal{L}), \psi)$, where $U(\mathcal{L})$ is a unital associative algebra over $\mathbb{F}$ with product $\cdot$ and $\psi : \mathcal{L} \rightarrow U(\mathcal{L})$ is a linear map satisfying

$$\psi([a, b]) = \psi(a) \cdot \psi(b) - \psi(b) \cdot \psi(a) \quad \forall a, b \in \mathcal{L} \quad (7.9)$$

Moreover, the following universality property is required: for every unital associative algebra $\mathcal{U}$ and linear map $\psi' : \mathcal{L} \rightarrow \mathcal{U}$ satisfying (7.9), a unique homomorphism $\phi : U(\mathcal{L}) \rightarrow \mathcal{U}$ exists such that $\phi \circ \psi = \psi'$.

Note that the Lie brackets of $\mathcal{L}$ define commutation relations between the generators of $U(\mathcal{L})$. Moreover, the Poincaré-Birkhoff-Witt theorem [69] implies that if $\{\hat{J}_1, \dots, \hat{J}_D\}$ are the generators of $\mathcal{L}$, then a basis of $U(\mathcal{L})$ is given by all the ordered monomials

$$\{\hat{J}_{n(1)} \cdot \dots \cdot \hat{J}_{n(m)} \text{ s.t. } [1 \leq n(1) \leq \dots \leq n(m) \leq D] \wedge m \in \mathbb{N}\}$$

where the monomial with zero elements is meant to be the identity.

In the case of our interest $\mathbb{F} = \mathbb{C}$ and $\mathcal{L} = \mathfrak{sl}(2)$, with generators $\{\hat{J}^z, \hat{J}^\pm\}$.

It is possible to view $U(\mathfrak{sl}(2))$ as a Hopf algebra by defining the following comultiplication [44] on the generators of $\mathfrak{sl}(2)$:

$$\Delta[\hat{J}^\alpha] := \hat{J}^\alpha \otimes \hat{\mathbb{1}} + \hat{\mathbb{1}} \otimes \hat{J}^\alpha \quad \forall \alpha \in \{z, \pm\}$$

Then the generators $\{\Delta[\hat{J}^\alpha]\}$ acting on $v_1 \otimes v_2$ satisfy (7.8).

$U(\mathfrak{sl}(2))$ and the XXX Model: Let us start by looking for a solution of the Yang-Baxter equation using $\mathcal{A} = U(\mathfrak{sl}(2))$. Apart from identity, the exchange operator $\hat{\pi}_{ij}$ is a solution of the YBE represented on $s = \frac{1}{2}$, as made pictorially transparent in (2.12) and (7.2). It can be expressed as

$$\hat{\pi}_{ij} = \frac{1}{2}(\hat{\mathbb{I}} + \hat{\sigma}_i \otimes \hat{\sigma}_j)$$

Thus, we should expect to be able to find a family of solutions for the $s = \frac{1}{2}$ YBE with the ansatz $\hat{R}_{\frac{1}{2}\frac{1}{2}}(\lambda) = f(\lambda) \cdot \hat{\mathbb{I}} + i\hat{\pi}_{12}$, where 1 and 2 are generic labels for the two spaces where these operators act [123]. The i is chosen in order to have unitary matrices: this is arbitrary, but a different notation leads to the opposite change in the formula for conserved charges. Plugging this ansatz inside (7.2), the only constraint turns out to be

$$f(\lambda - \mu) + f(\mu - \nu) - f(\lambda - \nu) = 0 \quad \text{which is only satisfied by } f(\lambda) = \alpha\lambda$$

$$\hat{R}_{\frac{1}{2}\frac{1}{2}}(\lambda) = \alpha\lambda \hat{\mathbb{I}} + i\hat{\pi}_{12} = \left(\alpha\lambda + \frac{i}{2}\right) \hat{\mathbb{I}} + \frac{i}{2} \hat{\sigma}_1 \otimes \hat{\sigma}_2$$

This solution, $\forall \alpha$, also satisfies the RLL relation (7.3), thus represents a good Lax matrix. However, looking for more general Lax matrices, but still of the form $\mathcal{L}_{\frac{1}{2}}^{(j)}(\lambda) = g(\lambda) \cdot \hat{\mathbb{I}} + i\hat{\pi}_{js}$, requiring (7.3) we obtain

$$g(\lambda) - \alpha\lambda = g(\mu) - \alpha\mu \quad \text{which is only satisfied by } g(\lambda) = \alpha(\lambda - \lambda_0)$$

$\lambda = \lambda_0$ is the *shift point* in which the fundamental ($s = 1/2$) Lax matrix becomes proportional to $\hat{\pi}$. If we arbitrarily choose $\alpha = 1$ and $\lambda_0 = i/2$:

$$\mathcal{L}_{\frac{1}{2}}^{(j)}(\lambda) = \left(\lambda - \frac{i}{2}\right) \hat{\mathbb{I}} + i\hat{\pi}_{12} = \lambda \hat{\mathbb{I}} + \frac{i}{2} \hat{\sigma}_1 \otimes \hat{\sigma}_2 \quad (7.10)$$

Remarkably, this coincides with the Lax matrix (6.11) which is known to return the XXX model Hamiltonian spanned by its local charges. Thus, till here the explicit connection with $U(\mathfrak{sl}(2))$ was not necessary and we would have been able to compute local charges using the Lax matrix obtained from the 6-vertex model, as sketched in §6.4.

However, as mentioned, to compute quasilocal charges we need higher-spin Lax matrices.

With the new more abstract approach these are immediately obtained by replacing $\{\frac{1}{2}\hat{\sigma}^z, \hat{\sigma}^\pm\}$ with their counterparts $\{\hat{s}^z, \hat{s}^\pm\}$ in higher-spin representations r_s of $\mathcal{A} = U(\mathfrak{sl}(2))$, i.e.

$$\mathcal{L}_s^{(j)}(\lambda) = \lambda \hat{\mathbb{I}} + i\hat{\sigma} \otimes \hat{s} = \begin{pmatrix} \lambda + i\hat{s}^z & i\hat{s}^- \\ i\hat{s}^+ & \lambda - i\hat{s}^z \end{pmatrix} \quad (7.11)$$

Practical Outline of the Quantum Group $U_q(\mathfrak{sl}(2))$: Here we are not going to formally introduce Hopf algebras and quantum groups (or equivalently deformed algebras).

A pedagogical introduction to the topic can be found in [44].

We just mention that the quantum group $U_q(\mathfrak{sl}(2))$ is obtained by "deforming" the universal enveloping algebra $U(\mathfrak{sl}(2))$ in terms of an additional continuous parameter q .

Explicitly, the q -deformed commutation relations are

$$[\hat{f}^z, \hat{f}^\pm] = \pm \hat{f}^\pm \quad [\hat{f}^+, \hat{f}^-] = [2\hat{f}^z]_q \quad \text{where } [x]_q := \frac{q^x - q^{-x}}{q - q^{-1}} = \frac{\sinh(x \log q)}{\sinh(\log q)}$$

$[x]_q$ is called the q -deformation of the object x and is periodic of period $2\pi/\log(q)$ in the imaginary part of x . Note that, owing to $(\hat{\sigma}^z)^2 = \hat{\mathbb{I}}$, when considering the fundamental representation $r_{\frac{1}{2}}$ these become effectively equivalent to (7.8) and $\{\hat{f}^z, \hat{f}^\pm\}$ are represented by (7.7).

A family of semi-integer representations [72, 108] parametrized by $s \in \frac{1}{2}\mathbb{Z}^{>0}$, which are unitary and finite-dimensional with $\dim \nu_s = 2s + 1$, is obtained by defining as $|k\rangle$, with $k \in \{0, \dots, 2s\}$, respectively the $2s + 1$ eigenstates of $\hat{s}^z$ with eigenvalues $s^z \in \{s, \dots, -s\}$:

$$\begin{aligned}\hat{s}^z &:= r_s[\hat{f}^z] = \sum_{k=0}^{2s} (s-k) |k\rangle \langle k| \\ \hat{s}^+ &:= r_s[\hat{f}^+] = \sum_{k=0}^{2s-1} \sqrt{[2s-k]_q [k+1]_q} |k\rangle \langle k+1| \\ \hat{s}^- &:= r_s[\hat{f}^-] = \sum_{k=0}^{2s-1} \sqrt{[2s-k]_q [k+1]_q} |k+1\rangle \langle k|\end{aligned}\tag{7.12}$$

These concrete representations are parity symmetric, in the sense that

$$\hat{P}_s \hat{s}^\pm \hat{P}_s^{-1} = \hat{s}^\mp \quad \hat{P}_s \hat{s}^z \hat{P}_s^{-1} = -\hat{s}^z$$

where $\hat{P}_s : \nu_s \rightarrow \nu_s$ is the spin-flip operator

$$\hat{P}_s := \sum_{k=0}^{2s} |k\rangle \langle 2s-k|$$

Before proceeding, we just mention that $U_q(\mathfrak{sl}(2))$ has a family of infinite-dimensional "complex spin" $s \in \mathbb{C}$ representations as well. These lead to solutions of the YBE (7.2) of physical interest in the study of non-equilibrium states with currents [72, 108]. However, here we will not use them and we will always think about the above discussed $s \in \frac{1}{2}\mathbb{Z}^{>0}$ representations.

Finally, as for $U(\mathfrak{sl}(2))$ it is possible to define the following comultiplication [44, 47] for $U_q(\mathfrak{sl}(2))$:

$$\begin{aligned}\Delta[\hat{s}^z] &:= \hat{s}^z \otimes \hat{1} + \hat{1} \otimes \hat{s}^z \quad \text{or equivalently} \quad \Delta[q^{\pm \hat{s}^z}] := q^{\pm \hat{s}^z} \otimes q^{\pm \hat{s}^z} \\ \Delta[\hat{s}^\pm] &:= \hat{s}^\pm \otimes q^{\hat{s}^z} + q^{-\hat{s}^z} \otimes \hat{s}^\pm\end{aligned}$$

Again, these operators acting on $\nu_1 \otimes \nu_2$ satisfy the deformed commutation relations of $U_q(\mathfrak{sl}(2))$.

$U_q(\mathfrak{sl}(2))$ and the Lax Matrices of the XXZ Model: In §6 we found that the XXZ Hamiltonian is spanned by the local conserved charges generated by the 6-vertex model Lax matrix. Moreover, by replacing $\lambda \rightarrow \eta\lambda$ and taking the limit $\eta \rightarrow 0$, the above Lax matrix for the XXX model is recovered. In this section we showed that, by plugging into the latter the generators $\{\hat{s}^z, \hat{s}^\pm\}$ of higher-spin representations of $U(\mathfrak{sl}(2))$, a family of higher-spin R - and Lax matrices satisfying the YBE and the RLL relation was found.

Similarly, it can be proven [44, 53] that by replacing the fundamental representation spin operators in the general 6-vertex model Lax matrix (6.10) with their higher-spin counterparts $\hat{s}^\alpha = r_s[\hat{f}^\alpha]$ with deformation parameter² given by

$$\begin{cases} q := e^\eta & \Delta := \cosh(\eta) & \eta > 0 & \text{if } |\Delta| \geq 1 & \text{(gapped AFM or FM regime)} \\ q := e^{i\eta} & \Delta := \cos(\eta) & \eta \in (0, \pi) & \text{if } |\Delta| < 1 & \text{(Luttinger liquid)} \end{cases}$$

then Lax matrices corresponding to YBE solutions for $\mathcal{A} = U_q(\mathfrak{sl}(2))$ are obtained.

$$\mathcal{L}_s^{(j)}(\lambda) = \frac{1}{\sinh(\eta)} \begin{pmatrix} \sin(\lambda + i\eta \hat{s}^z) & i \sinh(\eta) \cdot \hat{s}^- \\ i \sinh(\eta) \cdot \hat{s}^+ & \sin(\lambda - i\eta \hat{s}^z) \end{pmatrix} \quad \hat{s}^\alpha = r_s[\hat{f}^\alpha] \quad \mathcal{A} = U_q(\mathfrak{sl}(2)) \tag{7.13}$$

²In the critical regime $|\Delta| < 1$ the replacement $\eta \rightarrow -i\eta$ and $\lambda \rightarrow -i\lambda$ is needed in the Lax matrix to restore the most common notation [72]. The case $\Delta = 1$, namely $\eta \rightarrow 0$, has already been discussed at the beginning of the section.

where $\sin(\lambda \pm i\eta \hat{s}^z) = \sin(\lambda) \cosh(\eta \hat{s}^z) \pm i \cos(\lambda) \sinh(\eta \hat{s}^z)$

This writing highlights that $\mathcal{L}_s(\lambda)$ is a 2×2 matrix with respect to its physical indices, but each of its four elements is a $(2s+1) \times (2s+1)$ matrix on the auxiliary space $\mathcal{V}_s$.

Moreover, note that in the $\{|k\rangle\}$ basis $\hat{s}^z$ is diagonal, e.g. $\sin(\lambda \pm i\eta \hat{s}^z)$ is simply

$$\sin(\lambda \pm i\eta \hat{s}^z) = \sum_{k=0}^{2s} \sin[\lambda \pm i\eta(s-k)] |k\rangle \langle k|$$

Another way to write (7.13) is

$$\mathcal{L}_s^{(j)}(\lambda) = \frac{\hat{\mathbb{I}} + \hat{\sigma}^z}{2} \otimes \sin(\lambda + i\eta \hat{s}^z) + \frac{\hat{\mathbb{I}} - \hat{\sigma}^z}{2} \otimes \sin(\lambda - i\eta \hat{s}^z) + i \sinh(\eta) \cdot (\hat{\sigma}^- \otimes \hat{s}^+ + \hat{\sigma}^+ \otimes \hat{s}^-)$$

which makes transparent that $\mathcal{L}_s^{(j)}(\lambda)$ *preserves the total (physical+ancillary) magnetization* not only for $s = 1/2$, as already observed in §6, but for higher-spin auxiliary spaces as well.

$$\left[\mathcal{L}_s^{(j)}(\lambda), \hat{\mathbb{I}}_s \otimes \frac{\hat{\sigma}_j^z}{2} + \hat{s}^z \otimes \hat{\mathbb{I}}_j \right] = 0 \quad \forall \lambda, j, s \quad (7.14)$$

Moreover, if we define as $\hat{P}_j$ the spin-flip operator on the physical single-site space $\mathcal{H}_j$, then from (7.13) it is clear that the Lax matrix is invariant under *global spin-flip*.

$$\hat{P}_j \hat{P}_s \mathcal{L}_s^{(j)}(\lambda) \hat{P}_s^{-1} \hat{P}_j^{-1} = \mathcal{L}_s^{(j)}(\lambda) \quad (7.15)$$

Before proceeding, as $\hat{s}^+ = \hat{s}^- = \hat{s}^z = 0$ for $s = 0$ (trivial representation), $\mathcal{L}_0^{(j)} \propto \hat{\mathbb{I}}_j \otimes \hat{\mathbb{I}}_{\mathcal{V}_0}$. Thus, it is convenient to directly define it as a scalar

$$\mathcal{L}_0(\lambda) := \frac{\sin(\lambda)}{\sinh(\eta)} \quad (7.16)$$

This allows one to write $\hat{M}_0(\lambda) = [\mathcal{L}_0(\lambda)]^L \hat{\mathbb{I}}_{\mathcal{H}} \otimes \hat{\mathbb{I}}_{\mathcal{V}_0}$, where $\hat{T}_0(\lambda) \equiv \hat{M}_0(\lambda)$ (the trace over a single element is trivial). Thus we introduce the analogous scalar:

$$T_0(\lambda) := [\mathcal{L}_0(\lambda)]^L = \left(\frac{\sin(\lambda)}{\sinh(\eta)} \right)^L \quad (7.17)$$

7.4 HIROTA EQUATION AND Y-SYSTEM

It can be proven [68, 84, 85, 151] that the family of commuting transfer matrices $\{\hat{T}_s(\lambda)\}$ built from Lax and R -matrices which are solutions of the YBE and the RLL relation satisfies

$$\hat{T}_s^{[+]} \hat{T}_s^{[-]} = \hat{\mathbb{I}} T_0^{[2s+1]} T_0^{[-(2s+1)]} + \hat{T}_{s-\frac{1}{2}} \hat{T}_{s+\frac{1}{2}} \quad \left[\hat{T}_{-1/2} := 0 \right] \quad (7.18)$$

For every fixed spectral parameter λ this is a discrete system of bilinear relations called *Hirota equation, or T-system*. However, it displays some gauge symmetry and it is convenient to recast it in terms of Y -operators given by gauge-invariant combinations of the transfer matrices.

The mapping is nonlinear:

$$\hat{Y}_s := \frac{\hat{T}_{s-\frac{1}{2}} \hat{T}_{s+\frac{1}{2}}}{T_0^{[2s+1]} T_0^{[-(2s+1)]}} = \frac{\hat{T}_s^{[+]} \hat{T}_s^{[-]}}{T_0^{[2s+1]} T_0^{[-(2s+1)]}} - \hat{\mathbb{I}} \quad (7.19)$$

where the two definitions are equivalent only once the Hirota equation is required.

This way, the T -system can be rewritten as

$$\hat{Y}_s^{[+]} \hat{Y}_s^{[-]} = \left(\hat{\mathbb{I}} + \hat{Y}_{s-\frac{1}{2}} \right) \left(\hat{\mathbb{I}} + \hat{Y}_{s+\frac{1}{2}} \right) \quad \text{Y-system} \quad \left[\hat{Y}_0 := 0 \right]$$

7.5 Q-OPERATOR: EMERGENCE OF THE BETHE EQUATIONS SPECTRUM OF THE TRANSFER MATRICES, AND PROOF OF THE INVERSION RELATION

The XXZ fundamental transfer matrix³ $\hat{T}_{\frac{1}{2}}$ satisfies

$$\hat{T}_{\frac{1}{2}} \hat{Q} = T_0^{[+]} \hat{Q}^{[-2]} + T_0^{[-]} \hat{Q}^{[+2]} \quad \text{TQ-equation} \quad (7.20)$$

where $\hat{Q}$ is the so-called *Baxter's Q-operator* [13, 14, 72], since was first introduced by Baxter in his seminal paper about the 8-vertex model [12], and has the additional property that

$$[\hat{T}_s(\lambda), \hat{Q}(\mu)] = 0 \quad \forall \lambda, \mu \in \mathbb{C} \wedge \forall s \in \frac{1}{2}\mathbb{Z}^{\geq 0}$$

By virtue of commutativity we can work at the spectral level, i.e. by regarding (7.20) as a set of equations in which $\hat{Q}$ and $\hat{T}$ operators are replaced by the corresponding eigenvalues for every common eigenvector.

Q-Operator and Bethe Equations: A first remarkable fact which makes (7.20) very important, is that the Bethe equations, whose solutions correspond to the possible sets of rapidities of Bethe eigenstates, can be derived from it as the condition for the cancellation of superficial poles in the eigenvalues of $\hat{T}_{\frac{1}{2}}$ (see e.g. [13]). Indeed, if $\mathcal{A} = U(\mathfrak{sl}(2))$, eigenvalues of the Q -operator on Bethe states are N -degree polynomials whose roots are the rapidities $\{\lambda_j\}$ of the state. In the general case $\mathcal{A} = U_q(\mathfrak{sl}(2))$, they are given by the following product [72, 73]:

$$Q_{\vec{\lambda}}(\mu) := \langle \vec{\lambda} | \hat{Q}(\mu) | \vec{\lambda} \rangle \propto \prod_{j=1}^N \sin(\mu - \lambda_j) \quad (7.21)$$

Spectrum of Higher-Spin Transfer Matrices and Inversion Relation: Moreover, plugged into the Hirota equation (7.18), the TQ -equation (7.20) enables one to express the eigenvalues of $\hat{T}_s$ explicitly in terms of that of Q -operators (7.21):

$$\frac{\langle \vec{\lambda} | \hat{T}_s^{[+]} | \vec{\lambda} \rangle}{T_0^{[2s+1]}} = Q_{\vec{\lambda}}^{[2(s+1)]} Q_{\vec{\lambda}}^{[-2s]} \sum_{s'=0}^s \frac{\zeta_{s,s'}^L}{Q_{\vec{\lambda}}^{[2(2s'-s)]} Q_{\vec{\lambda}}^{[2(2s'-s+1)]}} \quad (7.22)$$

$$\text{where } \zeta_{s,s'}(\mu) = \frac{\sin^{[4s'-2s+1]}(\mu)}{\sin^{[2s+1]}(\mu)} \rightarrow \zeta_{s,s'}^L = \frac{T_0^{[4s'-2s+1]}}{T_0^{[2s+1]}}$$

$$\text{Note that } \left| \frac{\sin(\mu + i\alpha)}{\sin(\mu + i\beta)} \right|^2 = \frac{\cosh(2\alpha) - \sin(2\mu)}{\cosh(2\beta) - \sin(2\mu)} \leq 1 \quad \forall \alpha \leq \beta$$

and it is 1 only when $\alpha = \beta$.

This implies that the largest-magnitude $\zeta_{s,s'}$ in the above sum corresponds to $s' = s$.

(7.22) turns out to play a pivotal role in the proof of quasilocality of the charges obtained from higher-spin transfer matrices and in the derivation of the string-charge duality.

Indeed, in these applications we are interested in the thermodynamic limit $L \rightarrow +\infty$. Assuming that the large- L behaviour of (7.22) can be determined by considering the only $\zeta_{s,s'}$ of largest magnitude, corresponding to $s' = s$ as just proved,

$$\lim_{L \rightarrow +\infty} \frac{\langle \vec{\lambda} | \hat{T}_s^{[+]}(\mu) | \vec{\lambda} \rangle}{T_0^{[2s+1]}(\mu)} = \lim_{L \rightarrow +\infty} \frac{Q_{\vec{\lambda}}^{[-2s]}(\mu)}{Q_{\vec{\lambda}}^{[2s]}(\mu)} \quad (7.23)$$

³This is more in general true for every integrable model with $U_q(\mathfrak{sl}(2))$ symmetry.

Plugging this result into (7.19) at the spectral level, it follows that [72, 73]:

$$\begin{aligned} \lim_{L \rightarrow +\infty} 1 + \hat{Y}_s(\lambda) &= \lim_{L \rightarrow +\infty} \frac{\hat{T}_s^{[+]}(\lambda)}{T_0^{[2s+1]}(\lambda)} \frac{\hat{T}_s^{[-]}(\lambda)}{T_0^{[-(2s+1)]}(\lambda)} = \hat{\mathbb{I}}_{\mathcal{H}} \quad \forall s \in \frac{1}{2}\mathbb{Z}^{\geq 0} \quad \forall \lambda \in \mathcal{P}_\eta \\ &\rightarrow \lim_{L \rightarrow +\infty} \left(\frac{\hat{T}_s^{[+]}(\lambda)}{T_0^{[2s+1]}(\lambda)} \right)^{-1} = \lim_{L \rightarrow +\infty} \frac{\hat{T}_s^{[-]}(\lambda)}{T_0^{[-(2s+1)]}(\lambda)} \quad \text{Inversion relation} \end{aligned} \quad (7.24)$$

7.6 LOCAL AND QUASILOCAL CHARGES FROM THE ALGEBRAIC BETHE ANSATZ

Local Charges from the Fundamental Transfer Matrix: Let us restrict for the moment to the fundamental transfer matrices only. As we discussed in §6.4, at the shift point $\lambda = i\eta/2$ the fundamental Lax matrix becomes proportional to the operator $\hat{\pi}$ exchanging the state of the physical and ancillary dof, i.e. $\hat{\mathcal{L}}_{\frac{1}{2}}^{(j)}(i\eta/2) = i\hat{\pi}$. This implies that the corresponding transfer matrix is proportional the one-step translation operator. If we define the following expansion of the logarithm of the fundamental transfer matrix around the shift point

$$\begin{aligned} \log \hat{T}_{\frac{1}{2}} \left(\lambda + \frac{i\eta}{2} \right) &= 2\pi L i \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \hat{H}^{(k)} \\ \hat{H}^{(k)} &:= \frac{1}{L} \frac{1}{2\pi i} \frac{\partial^{k-1}}{\partial \lambda^{k-1}} \log \left[\hat{T}_{\frac{1}{2}}^{[+]}(\lambda) \right] \Big|_{\lambda=0} \quad k \in \mathbb{N}^{\geq 2} \quad \text{Local charges} \end{aligned} \quad (7.25)$$

it can be proven that *the charges $\{L\hat{H}^{(k)}\}$ are strictly local extensive operators of order k .*

In particular, as mentioned in §6.4, $\hat{H}^{(2)}$ turns out to be proportional to the XXZ Hamiltonian plus a constant. Moreover from $[\hat{T}_{\frac{1}{2}}(\lambda), \hat{T}_{\frac{1}{2}}(\mu)] = 0$ it follows that

$$[\hat{H}^{(k)}, \hat{H}^{(m)}] = 0 \quad \forall k, m \in \mathbb{N}^{\geq 2}$$

This means that these are all local integrals of motion of the XXZ model.

However, as yet discussed in §3, local charges turn out to be insufficient to completely characterize the LSS in generic interacting integrable systems. This suggests the existence of additional "effectively local" conserved charges, linearly independent of the strictly local one but with crucial effects on the integrable dynamics underlying relaxation.

QuasilocaI Charges from Higher-Spin Transfer Matrices: For the XXZ model, this challenge was solved in 2015 by the Ljubljana statistical physics group, since

- in [71] an infinite set of quasilocaI conserved charges in involution was built from higher-spin transfer matrices $\{\hat{T}_s(\lambda)\}$, as we will discuss in this section;
- in [73] it was analytically proven that the expectation values of these charges on the initial state are in one to one correspondence with the LSS root densities (this result is known as string-charge duality and will be the topic of §7.8).

We recall that in this work we restrict to the gapped regime $|\Delta| \geq 1$ and unitary semi-integer spin representations of $U_q(\mathfrak{sl}(2))$. For variations corresponding to the critical regime $|\Delta| < 1$ or non-unitary complex spin representations of $U_q(\mathfrak{sl}(2))$ the reader is referred to [72].

QuasilocaI charges from higher-spin transfer matrices are given by [71, 73]

$$\hat{X}_s(\lambda) := \frac{1}{L} \frac{1}{2\pi i} \frac{\partial}{\partial \lambda} \log \left(\frac{\hat{T}_s^{[+]}(\lambda)}{T_0^{[2s+1]}(\lambda)} \right) \quad \lambda \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad \text{QuasilocaI charges} \quad (7.26)$$

where $T_0^{[2s+1]}(\lambda)$ is the scalar defined in (7.17).

This expression can be analytically continued preserving quasilocality as defined in §3.4 to the entire physical strip $\mathcal{P}_\eta$ of (2.41). However, charges are Hermitian only for $\mu \in \mathbb{R}$ [72].

The proof of quasilocality can be found in [71, 72]. The main idea is to check (3.8) starting from the following Hilbert-Schmidt kernel K and demonstrating that it is finite and nonzero.

$$K_{s,s'}(\lambda, \mu) := \lim_{L \rightarrow +\infty} \frac{1}{L} \langle L \hat{X}_s(\lambda), L \hat{X}_{s'}(\mu) \rangle$$

Note that, in both this expression and the previous, while i guarantees charges to be Hermitian for $\lambda \in \mathbb{R}$ and $1/L$ that their eigenvalues remain finite in the thermodynamic limit⁴ the prefactor $1/2\pi$ is arbitrary, and we use it following the notations of [73]. Indeed, this choice is convenient in view of string-charge duality formulas, as it will be soon clear.

The local charges of (7.25) have not to be added to this set for the GGE, as the family $\{\hat{X}_{\frac{1}{2}}(\lambda)\}$, upon varying λ , is fully equivalent. Indeed, expanding the charges $\{\hat{X}_{\frac{1}{2}}(\lambda)\}$ about the origin:

$$\hat{X}_{1/2}(\lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \hat{H}^{(k+2)} \quad \rightarrow \quad \hat{H}^{(k+2)} = \partial_\lambda^k \hat{X}_{1/2}(\lambda) \Big|_{\lambda=0}$$

In particular, it follows that the XXZ Hamiltonian is given by $X_{1/2}(0)$, up to an overall rescaling and shift of the spectrum, since it coincides with $\hat{H}^{(2)}$.

Nevertheless, we recall that, *in principle, any charge taken from $\{\hat{X}_s(\lambda)\}$ can be regarded as a Hamiltonian with the same S -matrix*. Clearly, the fundamental family $\{\hat{X}_{\frac{1}{2}}(\lambda)\}$ returns Hamiltonians expressed as sum of local densities, while higher-spin charges imply long-range interactions, even if with rapidly decreasing intensity, say exponentially, by definition of quasilocality.

Alternative Rewritings of (7.26): In view of the following sections, we now present two different ways to recast (7.26) in the thermodynamic limit.

The first is the *more convenient in the proof of quasilocality and for the explicit evaluation of charges on a state* (e.g. the initial state of a quench protocol). It exploits the inversion relation (7.24), which allows us to remove the logarithm:

$$\hat{X}_s(\lambda) := \frac{1}{L} \frac{1}{2\pi i} \left[\frac{\partial}{\partial \mu} \left(\frac{\hat{T}_s^{[-]}(\lambda)}{T_0^{[-(2s+1)]}(\lambda)} \frac{\hat{T}_s^{[+]}(\mu)}{T_0^{[2s+1]}(\mu)} \right) \right]_{\lambda=\mu} \quad \lambda \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad [L \rightarrow +\infty] \quad (7.27)$$

The intermediate splitting $\lambda \rightarrow (\lambda, \mu)$ is necessary to bring the first fraction back into the derivative (since this way it does not depend on λ).

The second is obtained by plugging (7.23) into (7.26). It is *convenient for the derivation of the string-charge duality, given its direct connection with the values of charges on Bethe states*.

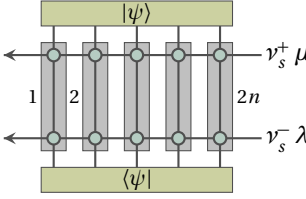
$$\langle \vec{\lambda} | \hat{X}_s(\mu) | \vec{\lambda} \rangle = \frac{1}{L} \frac{1}{2\pi i} \partial_\mu \log \frac{Q_{\vec{\lambda}}^{[-2s]}(\mu)}{Q_{\vec{\lambda}}^{[+2s]}(\mu)} \quad \mu \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad [L \rightarrow +\infty]$$

Parity Symmetry of Local and Quasiloal Charges: Arbitrary-spin monodromy matrices inherit parity symmetry from the corresponding Lax matrices (7.15). The trace operation preserves this property (restricted to the physical space only). Thus all transfer matrices are invariant under global spin flip. This means that all QISM local and quasiloal charges are parity symmetric.

This highlights a fundamental point: the total magnetization operator $\hat{S}^z$ is not spanned by these charges. In order to have a true complete set for the GGE, we have to add it to the quasiloal charges $\{\hat{X}_s(\lambda)\}$. Therefore, when in other parts of this work we say that quasiloal charges represent a complete set, it is always meant "once $\hat{S}^z$ has been included".

⁴As clear from the definition of the Hilbert-Schmidt kernel, formally speaking the true charges satisfying quasilocality are the $\{L \hat{X}_s(\lambda)\}$, i.e. require the multiplication by L to cancel the factor $1/L$ in (7.26). The reason of this mismatch is that in the following sections to work with finite-eigenvalued "densities" will turn out more convenient.

We highlight that $L/2n$ is not the tensor product of $L/2n$ operators acting on different segments of the lattice, but simply the successive application on $\nu_s^- \otimes \nu_s^+$ of the following operator, *computable from a single physical unit cell* $|\psi\rangle$:



$$\hat{T}_s(\lambda, \mu) := \langle \psi | \hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes 2n} | \psi \rangle = \quad : \nu_s^- \otimes \nu_s^+ \rightarrow \nu_s^- \otimes \nu_s^+ \quad (7.31)$$

$$\rightarrow \langle \Psi | \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes L}] | \Psi \rangle = \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{T}_s(\lambda, \mu)^{L/2n}]$$

Before proceeding, it is convenient to observe that, since Lax matrices preserve total magnetization in ancillary and physical spaces, the same must hold for the $\hat{\mathbb{L}}$ operator built out of them. Therefore:

$$\hat{s}_{aux}^z := \hat{s}_{\nu_s^-}^z \otimes \hat{\mathbb{I}}_{\nu_s^+} + \hat{\mathbb{I}}_{\nu_s^-} \otimes \hat{s}_{\nu_s^+}^z \rightarrow [\hat{T}_s(\lambda, \mu), \hat{s}_{aux}^z] = 0 \quad (7.32)$$

Plugging (7.31) into (7.30), then coming back to (7.27), yields a simpler expression for the computation of mean values of quasilocal charges, reduced to a single $2n$ —site unit cell:

$$\chi_s(\lambda) := \frac{1}{2\pi i} \left[\frac{\partial}{\partial \mu} \left\{ \frac{1}{L} \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{T}_s(\lambda, \mu)^{L/2n}] \right\} \right]_{\lambda=\mu} \quad \lambda \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad L \rightarrow +\infty \quad (7.33)$$

It follows that the object in $\{\}$ can be formally identified as generating function for the quasilocal charges mean values on the initial state.

Indeed, it is sometimes called *boundary partition function* [73].

Recursive Computation of $\hat{T}_s^{(n)}(\lambda, \mu)$ for the $2n$ —Rainbow State: If $|\Psi\rangle$ is the $2n$ —rainbow state, hence $|\psi\rangle$ is a single $2n$ —rainbow, we can build $\hat{T}_s^{(n)}(\lambda, \mu)$ recursively, by adding every time two new entangled sites at the boundary (i.e. a new external rainbow arc):

$$\begin{aligned} \hat{T}_s^{(n)} &:= \langle R_n | \hat{\mathbb{L}}_s^{\otimes 2n} | R_n \rangle = \\ &= \langle 1, 0 |_{(-n+1/2)}^{(n-1/2)} \otimes \langle R_{n-1} | \hat{\mathbb{L}}_s^{(-n+1/2)} \otimes \hat{\mathbb{L}}_s^{\otimes 2(n-1)} \otimes \hat{\mathbb{L}}_s^{(n-1/2)} | R_{n-1} \rangle \otimes | 1, 0 |_{(-n+1/2)}^{(n-1/2)} = \\ &= \langle 1, 0 |_{(-n+1/2)}^{(n-1/2)} \hat{\mathbb{L}}_s^{(-n+1/2)} \otimes \hat{T}_s^{(n-1)} \otimes \hat{\mathbb{L}}_s^{(n-1/2)} | 1, 0 |_{(-n+1/2)}^{(n-1/2)} \end{aligned}$$

where we denoted as $|1, 0\rangle_i^j$ the triplet Bell-pair corresponding to sites i and j .

By introducing the notation $\hat{\mathbb{L}}_{s,ij}$ with indices (i, j) referred to the four elements of $\hat{\mathbb{L}}$ as matrix on the single-site physical space $\mathcal{H}_k$ (each of which is an operator acting on the two ancillary spaces), finally:

$$\hat{T}_s^{(1)} = \frac{1}{2} \left[\hat{\mathbb{L}}_{s,11}^{(-\frac{1}{2})} \hat{\mathbb{L}}_{s,22}^{(\frac{1}{2})} + \hat{\mathbb{L}}_{s,22}^{(-\frac{1}{2})} \hat{\mathbb{L}}_{s,11}^{(\frac{1}{2})} + \hat{\mathbb{L}}_{s,12}^{(-\frac{1}{2})} \hat{\mathbb{L}}_{s,21}^{(\frac{1}{2})} + \hat{\mathbb{L}}_{s,21}^{(-\frac{1}{2})} \hat{\mathbb{L}}_{s,12}^{(\frac{1}{2})} \right] \quad (7.34)$$

$$\hat{T}_s^{(n)} = \frac{1}{2} \left[\hat{\mathbb{L}}_{s,11}^{(-n+\frac{1}{2})} \hat{T}_s^{(n-1)} \hat{\mathbb{L}}_{s,22}^{(n-\frac{1}{2})} + \hat{\mathbb{L}}_{s,22}^{(-n+\frac{1}{2})} \hat{T}_s^{(n-1)} \hat{\mathbb{L}}_{s,11}^{(n-\frac{1}{2})} + \hat{\mathbb{L}}_{s,12}^{(-n+\frac{1}{2})} \hat{T}_s^{(n-1)} \hat{\mathbb{L}}_{s,21}^{(n-\frac{1}{2})} + \hat{\mathbb{L}}_{s,21}^{(-n+\frac{1}{2})} \hat{T}_s^{(n-1)} \hat{\mathbb{L}}_{s,12}^{(n-\frac{1}{2})} \right]$$

Computationally-Friendly Form of (7.33): In order to avoid the power computation $\hat{T}_s^{L/2n}$, it is useful to note that, given $t_s^{(\alpha)}(\lambda, \mu)$ with $\alpha \in \{1, \dots, 2(2s+1)\}$ the eigenvalues of $\hat{T}_s(\lambda, \mu)$,

$$\text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{T}_s(\lambda, \mu)^{L/2n}] = \sum_{\alpha=1}^{2(2s+1)} [t_s^{(\alpha)}(\lambda, \mu)]^{L/2n} \quad (7.35)$$

$$\frac{1}{L} \partial_\mu \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{T}}_s(\lambda, \mu)^{L/2n}] = \frac{1}{2n} \sum_{a=1}^{2(2s+1)} [t_s^{(a)}(\lambda, \mu)]^{(L/2n)-1} \partial_\mu t_s^{(a)}(\lambda, \mu)$$

Since we assume that $L \gg 2n$,

$$\lim_{L/2n \rightarrow +\infty} \frac{1}{L} \partial_\mu \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{T}}_s(\lambda, \mu)^{L/2n}] = \lim_{L/2n \rightarrow +\infty} \frac{1}{2n} [t_s^{(max)}(\lambda, \mu)]^{(L/2n)-1} \partial_\mu t_s^{(max)}(\lambda, \mu) \quad (7.36)$$

where $t_s^{(max)}(\lambda, \mu)$ is assumed to have nonzero derivative.

Only now one can equal $\mu = \lambda$ in $[t_s^{(max)}(\lambda, \mu)]^{(L/2n)-1}$: note that $t_s^{(max)}(\lambda, \lambda) = 1$ is the only possibility to have well defined and nontrivial quasilocal charges.

However, this is necessarily true, since (7.24) implies

$$\lim_{L \rightarrow +\infty} \langle \Psi | \frac{\hat{T}_s^{[+]}(\lambda)}{T_0^{[2s+1]}(\lambda)} \frac{\hat{T}_s^{[-]}(\lambda)}{T_0^{[-(2s+1)]}(\lambda)} | \Psi \rangle = 1$$

If we connect this property to the definition of $\hat{\mathbb{T}}$ using (7.30) and (7.31), then

$$\begin{aligned} 1 &= \lim_{L \rightarrow +\infty} \langle \Psi | \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{L}}_s(\lambda, \lambda)^{\otimes L}] | \Psi \rangle = \\ &= \lim_{L \rightarrow +\infty} \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\langle \psi | \hat{\mathbb{L}}_s(\lambda, \lambda)^{\otimes 2n} | \psi \rangle^{L/2n}] = \lim_{L \rightarrow +\infty} \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{T}}_s(\lambda, \lambda)^{L/2n}] \end{aligned}$$

Here we recognise (7.35) and, since to guarantee macroscopic homogeneity of the initial state $L \rightarrow +\infty$ is always meant as $L/2n \rightarrow +\infty$, this limit is well defined if and only if

$$\begin{aligned} \hat{\mathbb{T}}_s(\lambda, \lambda) &\text{ has one and only one eigenvalue equal to 1} \\ &\text{ and all other eigenvalues have } |t_s^{(\alpha)}| < 1. \\ \text{Therefore } t_s^{(max)} &= 1. \end{aligned}$$

Another numerically important property is that, since $|\psi\rangle$ is a product state of many triplet Bell-pairs, it is invariant under spin-flip on the $2n$ -site unit cell, implemented by $\hat{P}_{ph} = \otimes_k \hat{P}_k$.

Then, recalling (7.15), we conclude that $\hat{\mathbb{L}}_s^{(k)}(\mu, \lambda)$ is invariant under $\hat{P}_k \hat{P}_s$ as well. Therefore, if we define $\hat{P}_{aux} = \hat{P}_{\nu_s^-} \otimes \hat{P}_{\nu_s^+}$, from the very definition of $\hat{\mathbb{T}}_s(\lambda, \mu)$:

$$\begin{aligned} \hat{\mathbb{T}}_s(\lambda, \mu) &= \langle \psi | \hat{P}_{ph} \hat{P}_{aux} \hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes 2n} \hat{P}_{aux}^{-1} \hat{P}_{ph}^{-1} | \psi \rangle = \\ &= \hat{P}_{aux} \langle \psi | \hat{P}_{ph} \hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes 2n} \hat{P}_{ph} | \psi \rangle \hat{P}_{aux}^{-1} = \\ &= \hat{P}_{aux} \langle \psi | \hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes 2n} | \psi \rangle \hat{P}_{aux}^{-1} = \hat{P}_{aux} \hat{\mathbb{T}}_s(\lambda, \mu) \hat{P}_{aux}^{-1} \end{aligned}$$

This means that $\hat{\mathbb{T}}_s(\lambda, \mu)$ is invariant under global ancillary spin-flip. Moreover, in (7.32) we proved that it preserves the total ancillary magnetization.

Let $|a\rangle$ be an eigenstate of $\hat{\mathbb{T}}_s(\lambda, \mu)$ with eigenvalue t_s . If $\langle \hat{s}_{aux}^z \rangle \neq 0$, then $\hat{P}_{aux} |a\rangle$ is a different state with magnetization $-\langle \hat{s}_{aux}^z \rangle$ but same eigenvalue t_s . Consider now $\mu = \lambda$. Since the eigenspace $t_s^{(max)} = 1$ is nondegenerate, it must be a subspace with zero magnetization.

$$\hat{\mathbb{T}}_s(\lambda, \lambda) |\psi\rangle = |\psi\rangle \rightarrow \hat{s}_{aux}^z |\psi\rangle = 0$$

Therefore, the eigenvalue $t_s^{(max)} = 1$ falls within the zero-magnetization block of $\hat{\mathbb{T}}_s(\lambda, \lambda)$.

Coming back to (7.36) and plugging into it the last results, we obtain:

$$\chi_s(\lambda) = \frac{1}{2n} \frac{1}{2\pi i} [\partial_\mu t_s^{(max)}(\lambda, \mu)]_{\lambda=\mu} \quad \lambda \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad \left[\frac{L}{2n} \rightarrow +\infty \right] \quad (7.37)$$

Let us denote by $\hat{\mathbb{T}}_s^{(0)}$ the operator $\hat{\mathbb{T}}_s$ restricted to the space $\mathcal{V}_0$ defined as the subspace of $\mathcal{V}_s^- \otimes \mathcal{V}_s^+$ with zero total magnetization of the two spins. Indeed, given (7.32), $\hat{\mathbb{T}}_s$ is simultaneously diagonalizable with $\hat{s}_{aux}^z$ and in particular it is block diagonal in the computational basis.

We assume the existence of a sufficiently small neighbourhood of λ s.t., for μ within it (namely " $\mu \approx \lambda$ "), the eigenvector of $\hat{\mathbb{T}}_s(\mu, \nu)$ with maximal eigenvalue has zero magnetization as well and t_s is a continuous function of μ in this interval.

This allows us to compute $\partial_\mu t_s^{(max)}(\lambda, \mu)_{\lambda=\mu}$ using the simpler $\hat{\mathbb{T}}_s^{(0)}$ instead of the entire $\hat{\mathbb{T}}_s$.

To avoid the diagonalization and explicit evaluation of $\partial_\mu t_s^{(max)}(\lambda, \mu)$, we can use that [56]

$$\partial_\mu \det \hat{A}(\mu) = \text{Tr}[\text{adj}(\hat{A}(\mu)) \partial_\mu \hat{A}(\mu)] \quad \text{Jacobi's Formula}$$

where $\text{adj}(\hat{A})$ denotes the *adjugate*, i.e. the transpose of the matrix of $\hat{A}$ cofactors.

Indeed, given that $t_s^{(max)}$ for $\lambda \approx \mu$ is an eigenvalue of $\hat{\mathbb{T}}_s^{(0)}$, it follows that $0 = \det(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)})$.

$$\begin{aligned} 0 &= \partial_\mu \det(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)}) = \text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)}) \partial_\mu (\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)})] = \\ &= \text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)}) \partial_\mu \hat{\mathbb{T}}_s^{(0)}] - \text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)})] \cdot \partial_\mu t_s^{(max)} \end{aligned} \quad (7.38)$$

By expressing $\partial_\mu t_s^{(max)}$ in (7.37) through this result, the obtained final numerically-convenient formula for quasilocal charges is

$$\chi_s(\lambda) = \frac{1}{2n} \frac{1}{2\pi i} \left[\frac{\text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)}) \partial_\mu \hat{\mathbb{T}}_s^{(0)}]}{\text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)} - \hat{\mathbb{T}}_s^{(max)})]} \right]_{\lambda=\mu}$$

where the only point where the splitting is necessary remains the derivative. Therefore [73]

$$\chi_s(\lambda) = \frac{1}{2n} \frac{1}{2\pi i} \frac{\text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)}(\lambda, \lambda) - \hat{\mathbb{T}}) (\partial_\mu \hat{\mathbb{T}}_s^{(0)}(\lambda, \mu))_{\lambda=\mu}]}{\text{Tr}_{\mathcal{V}_0}[\text{adj}(\hat{\mathbb{T}}_s^{(0)}(\lambda, \lambda) - \hat{\mathbb{T}})]} \quad \lambda \in \mathbb{R} \quad s \in \frac{1}{2}\mathbb{Z}^+ \quad (7.39)$$

We recall that this optimized formula is valid for every spin-flip invariant periodic initial state. Moreover, for the $2n$ -rainbow state, $\hat{\mathbb{T}}_s^{(0)}$ can be built by recursion from (7.34).

7.8 QUANTUM QUENCHES: STRING-CHARGE DUALITY

Retrieving two main results discussed up to now:

- The ABA method allows one to build an infinite set of quasilocal conserved charges in involution through (7.27);
- The representative eigenstate, which enables the expression of expectation values on the LSS according to the GETH, is defined by its m -string root densities in the TBA framework.

Our aim is now to show how that in the XXZ model there exist a one-to-one mapping between the quasilocal charges expectation values on the initial state $\{\chi_s\}$ and the LSS root densities. This relation is called *string-charge duality* and was introduced in [73].

This connection makes so that the ABA with non-fundamental unitary spin representations turns out to be a powerful method for the study of relaxation.

Action of the Discrete d'Alembert Operator on $a_n(\lambda)$ inside $\mathcal{P}_\eta$: Before introducing the string-charge duality, we need some preliminary definitions and results.

First, we define a *discrete d'Alembert operator* $\square$

$$\square_s f_s := f_s^{[+]} + f_s^{[-]} - f_{s-\frac{1}{2}} - f_{s+\frac{1}{2}} \quad \text{Discrete d'Alembert operator} \quad (7.40)$$

whose action is defined on any set of objects $f_s \equiv f_s(\lambda)$ (with $s \in \frac{1}{2}\mathbb{Z}_+$) which are analytic inside the physical strip $\mathcal{P}_\eta$ defined in (2.41). Apart from the $\epsilon \rightarrow 0$ limit, the above definition of $\square$ allows one to rewrite the action of $s^{-1}*$ expressed (2.40) as

$$[s^{-1} * f_s]_\lambda = \square_s f_s + f_{s-\frac{1}{2}} + f_{s+\frac{1}{2}}$$

If we apply $s^{-1}*$ to (2.35) while interpreting the string length as $n := 2s - \alpha$, with $\alpha \in \mathbb{N}^{<2s}$:

$$\begin{aligned} \begin{cases} a_2 = -\delta(\lambda) + [s^{-1} * a_{2s-\alpha}]_{2s=\alpha+1} = -\delta(\lambda) + \square_s a_{2s-\alpha}|_{2s=\alpha+1} + a_2 \\ a_{2s-\alpha-1} + a_{2s-\alpha+1} = s^{-1} * a_{2s-\alpha} = \square_s a_{2s-\alpha} + a_{2s-\alpha-1} + a_{2s-\alpha+1} \quad \forall s > 1/2 \end{cases} \\ \rightarrow \begin{cases} \square_s a_{2s-\alpha}|_{2s=\alpha+1} = \delta(\lambda) \\ \square_s a_{2s-\alpha} = 0 \quad \forall s > 1/2 \end{cases} \end{aligned} \quad (7.41)$$

String-Charge Duality: Owing to the Hirota equation, which holds in any $U_q(\mathfrak{sl}(2))$ integrable model, we are able to express the spectrum of higher-spin transfer matrices in the thermodynamic limit through (7.23).

By plugging this result into (7.26), we obtain the spectrum of quasilocal charges:

$$\langle \vec{\lambda} | \hat{X}_s(\mu) | \vec{\lambda} \rangle = \frac{1}{L} \frac{1}{2\pi i} \frac{\partial}{\partial \mu} \log \frac{Q_{\vec{\lambda}}^{[-2s]}(\mu)}{Q_{\vec{\lambda}}^{[+2s]}(\mu)}, \quad \mu \in \mathcal{P}_\eta$$

These are the values of quasilocal charges over the Bethe eigenstates, where we recall that the explicit form of $Q_{\vec{\lambda}}$ is given by (7.21). Then, the expectation value of $\hat{X}_s(\lambda)$ on the initial state of a quench leading to the LSS root densities $\{\rho_n\}$ can be proven [73] to be

$$\chi_s(\lambda) = \sum_{n=1}^{+\infty} (G_{2s,n} * \rho_n)(\lambda) \quad \text{String-Charge Duality "}\rightarrow\text{"} \quad (7.42)$$

where the kernels $G_{2s,n}$ can be expressed in terms of the $a_n(\lambda)$ defined in (2.26) as

$$G_{2s,n}(\lambda) = \sum_{k=1}^{\min(2s,n)} a_{|2s-n|-1+2k}(\lambda) \quad (7.43)$$

Note that $G_{2s,n}$ is symmetric under $2s \leftrightarrow n$, but in the above equation the two indices have very different nature: $2s$ is twice the auxiliary spin, while n is the length of a string.

Moreover, using (7.41), it is immediate to observe that, by applying the discrete d'Alembertian of (7.40) $\square_s$ to $G_{2s,n}$, the only nontrivial contributions come from the terms $a_{|2s-n|-1+2k}$ in which $|2s-n|-1+2k = 1$, i.e. $k = 1 - |s - n/2|$. Inside the sum $k \geq 1$, the only solution is given by $s = n/2$.

$$\rightarrow \square_s G_{2s,n}(\lambda) = \delta_{2s,n} \delta(\lambda)$$

Heuristically, this result allows us to interpret, in (7.42), the root densities as input, returning as output the charges eigenvalues associated to the stationary state, by means of a discrete Green function $G_{2s,k}(\lambda)$ corresponding to a system undergoing the "wave operator" $\square_s$.

Indeed, we can immediately invert (7.42) by applying $\square_s$ and using the above equation:

$$\rho_{2s}(\lambda) = \square_s \chi_s(\lambda) \quad \text{String-Charge Duality "}\leftarrow\text{"} \quad (7.44)$$

Furthermore, using BGT equations (2.42) and the relation $\rho_n^{tot} = \rho_n + \rho_n^h$, one arrives to

$$\begin{aligned} \rho_{2s}^h &= a_{2s} - \chi_s^{[+]} - \chi_s^{[-]} \\ \rho_{2s}^{tot} &= \rho_{2s} + \rho_{2s}^h = a_{2s} - \chi_{s+\frac{1}{2}}^{[+]} - \chi_{s-\frac{1}{2}}^{[-]} \end{aligned}$$

Final Remarks: Before proceeding, we emphasize that:

- Even when η_n ratios are exactly known, such as for the infinite-temperature state, in order to find root densities we have to use the BGT equations. These, despite their local form (2.42), where only three string lengths at a time are involved, necessarily require the discussed maximal string-length truncation approximation to be solved. It follows that for every finite maximal string length, all root densities are approximated.

By contrast, *when using (7.44), every set $\{\rho_n, \rho_n^h, \rho_n^{tot}\}$ for every $n \in \mathbb{N}^{>0}$ only requires the evaluation of the first $n+1$ charges $\{\chi_s\}$, in order to be computed exactly.*

- As the transfer matrices, the charges $\hat{X}_s$ are defined up to some arbitrariness arising from the gauge symmetry of the Hirota equations. However, string densities must be independent of this⁵. Indeed, we just mention that, in the large- L limit, the string densities can be expressed in an explicitly gauge-invariant form in terms of the spectrum of the Y -operators defined in (7.19) as [73]:

$$\rho_{2s}(\mu) = \frac{1}{L} \frac{1}{4\pi i} \frac{\partial}{\partial \mu} \log \left(\frac{1 + 1/Y_{s,\vec{\lambda}}^{[+]}(\mu)}{1 + 1/Y_{s,\vec{\lambda}}^{[-]}(\mu)} \right) \quad \rho_{2s}^h(\mu) = \frac{1}{L} \frac{1}{4\pi i} \frac{\partial}{\partial \mu} \log \left(\frac{1 + Y_{s,\vec{\lambda}}^{[-]}(\mu)}{1 + Y_{s,\vec{\lambda}}^{[+]}(\mu)} \right)$$

where $\vec{\lambda}$ is the vector of rapidities of a representative eigenstate.

- As we discussed at the end of §7.6, the quasilocal charges $\{\hat{X}_s(\lambda)\}$ are all parity symmetric. However, eigenstates connected by a global spin-flip have the same root densities once the vacuum state has been transformed as well. Indeed, we recall that, if $|\vec{\lambda}\rangle$ is a Bethe eigenstate, then $\hat{P}|\vec{\lambda}\rangle$ must be an eigenstate as well, with degenerate energy.

Explicitly, assume that the root densities of an eigenstate with $\langle \hat{S}^z \rangle \geq 0$ are known. The spin-flipped state is another eigenstate with $\langle \hat{S}^z \rangle \geq 0$.

If we modify (2.1) by replacing $\hat{S}^+ \leftrightarrow \hat{S}^-$ and $|\downarrow\downarrow\downarrow\dots\rangle \leftrightarrow |\uparrow\uparrow\uparrow\dots\rangle$, keeping the Bethe wavefunction unchanged, then this new state has the same root densities as the original one.

This explains why $\hat{S}^z$ is not necessary to define root densities according to string-charge duality. To fully characterize the LSS microcanonically it is sufficient to supplement the set $\{\rho_n(\lambda)\}$ with a binary variable $f = \pm$, which bears information about the sign of the magnetization, i.e. $\{\rho_n^f(\lambda)\}$ [102].

In our case, the $2n$ -rainbow state has $\langle \hat{S}^z \rangle = 0$, so this correction has no effect.

An important consequence for zero-magnetization states is that the $f = \pm$ descriptions must return equal results. This is trivially satisfied for all QISM quasilocal charges, which are parity-symmetric. By contrast, the GHD magnetization current would change sign. This implies that *bipartitioning protocols between states with null magnetization cannot display ballistic spin transport*.

7.9 OUTLINE OF EIGENSTATE CONSTRUCTION IN THE QISM

In this work we are not going to use the QISM to write Bethe eigenstate and compute correlation functions. Nevertheless, this represents an important achievement of the method, and we find it worthwhile to give a brief overview of this application [62, 123].

Indeed, it can be proven that, within this construction, while the two diagonal elements of the fundamental monodromy matrix give rise to transfer matrices through $\{\hat{T}_{\frac{1}{2}}(\lambda) = \text{Tr}_V[\hat{M}_{\frac{1}{2}\frac{1}{2}}(\lambda)]\}$, its

⁵This implies that the one-to-one correspondence of string-charge duality is meant modulo this freedom, which must be absorbed by the kernel of $\square_s$. Analogously, as in usual Green function problems, (7.44) returns a particular solution for the charges, lacking the gauge arbitrariness.

off-diagonal entries act on $\mathcal{H}$ connecting common eigenstates of the local conserved charges $\{\hat{H}^{(n)}\}$, including $\hat{H}$. It follows that Bethe eigenstates of the corresponding integrable models are algebraically generated by repeated applications of these operators to a reference state, called *pseudovacuum*. Thus, *heuristically, the ABA can be regarded as a second-quantization-inspired reformulation of the standard coordinate Bethe ansatz* [62]. This is useful since one of the main limitations to the practical use of the latter is that the eigenstates with N quasiparticles are expressed as a superposition of $N!$ terms, which is an exponentially large number as $N \rightarrow +\infty$.

This is the same problem encountered in the first quantization formalism when dealing with many-body systems of fermions or bosons.

The starting point is to rewrite the fundamental monodromy matrix as a 2×2 matrix whose entries are operators acting on $\mathcal{H}$.

$$\hat{M}_{1/2}(\lambda) = \mathcal{L}_{\frac{1}{2}}(\lambda)^{\otimes L} = \leftarrow \begin{array}{c} \uparrow \uparrow \uparrow \uparrow \uparrow \\ \bullet \bullet \bullet \bullet \bullet \\ \downarrow \downarrow \downarrow \downarrow \downarrow \end{array} \xrightarrow{\nu_{\frac{1}{2}} \lambda} := \begin{pmatrix} \hat{A}(\lambda) & \hat{B}(\lambda) \\ \hat{C}(\lambda) & \hat{D}(\lambda) \end{pmatrix} \quad (7.45)$$

$\mathcal{H}$

$$\text{where} \quad \hat{A}(\lambda), \hat{B}(\lambda), \hat{C}(\lambda), \hat{D}(\lambda) : \mathcal{H} \rightarrow \mathcal{H}$$

Since Lax matrices preserve the total (physical+ancillary) magnetization (7.14), this property is inherited by arbitrary-spin monodromy matrices as well.

$$\left[\hat{M}_s, \hat{s}^z \otimes \hat{\mathbb{I}}_{\mathcal{H}} + \sum_{j=1}^L \hat{\mathbb{I}}_s \otimes \frac{\hat{\sigma}^z}{2} \right] = 0$$

Therefore, when acting on $|\psi\rangle \in \mathcal{H}$, the operators $\hat{A}$ and $\hat{D}$ preserve the number of Bethe quasiparticles, $\hat{B}$ adds an excitation, while $\hat{C}$ has the opposite effect. In particular it can be proven that, if one identifies a state $|\Omega\rangle$ which is an eigenstate of all $\{\hat{A}(\lambda)\}$ and $\{\hat{D}(\lambda)\}$ and falls within the kernel of all the "annihilators" $\{\hat{C}(\lambda)\}$ (here it coincides with $|\downarrow\downarrow\downarrow\dots\rangle$ in the spin picture)

$$\hat{C}(\lambda)|\Omega\rangle = 0 \quad \forall \lambda \quad \text{Pseudo-vacuum}$$

then the Bethe state corresponding to the set of (all different) rapidities $\{\lambda_1, \dots, \lambda_N\}$ is given by

$$|\vec{\lambda}\rangle = \prod_{j=1}^N \hat{B}(\lambda_j)|\Omega\rangle \quad (7.46)$$

In this framework, the Bethe equations are obtained by requiring that these states are common eigenvectors of all the fundamental transfer matrices:

$$\exists \Lambda(\mu, \vec{\lambda}) \quad \text{s.t.} \quad \hat{T}_{\frac{1}{2}}(\mu)|\vec{\lambda}\rangle = \Lambda(\mu, \vec{\lambda})|\vec{\lambda}\rangle \quad \forall \mu$$

In practice, as in the second quantization method, these operators are never written explicitly, while every state and operator of interest is expressed in terms of them and correlation functions are computed exploiting generalized commutation rules (see next paragraph).

Unlike the coordinate Bethe ansatz, here states are expressed in a very simple way, while the recasting of operators in terms of $\{\hat{A}(\lambda), \hat{B}(\lambda), \hat{C}(\lambda), \hat{D}(\lambda)\}$ is in general cumbersome, apart from special cases. For example, typically one is interested in local observables, while the operators appearing in the monodromy matrix are nonlocal.

Commutation Relations and Coincident Rapidities: Unlike the second quantization creation and annihilation operators, the monodromy matrix entries satisfy the following more complicated commutation relations, obtained by explicitly writing the intertwining relation (7.5):

$$[\hat{A}(\lambda), \hat{A}(\mu)] = [\hat{B}(\lambda), \hat{B}(\mu)] = [\hat{C}(\lambda), \hat{C}(\mu)] = [\hat{D}(\lambda), \hat{D}(\mu)] = 0$$

$$[\hat{A}(\lambda), \hat{D}(\mu)] = g(\lambda - \mu) \{ \hat{C}(\lambda) \hat{B}(\mu) - \hat{C}(\mu) \hat{B}(\lambda) \}$$

$$[\hat{D}(\lambda), \hat{A}(\mu)] = g(\lambda - \mu) \{ \hat{B}(\lambda) \hat{C}(\mu) - \hat{B}(\mu) \hat{C}(\lambda) \}$$

$$[\hat{B}(\lambda), \hat{C}(\mu)] = g(\lambda - \mu) \{ \hat{D}(\lambda) \hat{A}(\mu) - \hat{D}(\mu) \hat{A}(\lambda) \}$$

$$[\hat{C}(\lambda), \hat{B}(\mu)] = g(\lambda - \mu) \{ \hat{A}(\lambda) \hat{D}(\mu) - \hat{A}(\mu) \hat{D}(\lambda) \}$$

$$\hat{A}(\lambda) \hat{B}(\mu) = f(\mu - \lambda) \hat{B}(\mu) \hat{A}(\lambda) + g(\lambda - \mu) \hat{B}(\lambda) \hat{A}(\mu)$$

$$\hat{B}(\lambda) \hat{A}(\mu) = f(\mu - \lambda) \hat{A}(\mu) \hat{B}(\lambda) + g(\lambda - \mu) \hat{A}(\lambda) \hat{B}(\mu)$$

$$\hat{A}(\lambda) \hat{C}(\mu) = f(\lambda - \mu) \hat{C}(\mu) \hat{A}(\lambda) + g(\mu - \lambda) \hat{C}(\lambda) \hat{A}(\mu)$$

$$\hat{C}(\lambda) \hat{A}(\mu) = f(\lambda - \mu) \hat{A}(\mu) \hat{C}(\lambda) + g(\mu - \lambda) \hat{A}(\lambda) \hat{C}(\mu)$$

$$\hat{D}(\lambda) \hat{B}(\mu) = f(\lambda - \mu) \hat{B}(\mu) \hat{D}(\lambda) + g(\mu - \lambda) \hat{B}(\lambda) \hat{D}(\mu)$$

$$\hat{B}(\lambda) \hat{D}(\mu) = f(\lambda - \mu) \hat{D}(\mu) \hat{B}(\lambda) + g(\mu - \lambda) \hat{D}(\lambda) \hat{B}(\mu)$$

$$\hat{D}(\lambda) \hat{C}(\mu) = f(\mu - \lambda) \hat{C}(\mu) \hat{D}(\lambda) + g(\lambda - \mu) \hat{C}(\lambda) \hat{D}(\mu)$$

$$\hat{C}(\lambda) \hat{D}(\mu) = f(\mu - \lambda) \hat{D}(\mu) \hat{C}(\lambda) + g(\lambda - \mu) \hat{D}(\lambda) \hat{C}(\mu)$$

$$\text{where} \quad f(\lambda) := \frac{\sin(\lambda/2 + i\eta)}{\sin(\lambda/2)} \quad g(\lambda) := \frac{\sin(i\eta)}{\sin(\lambda/2)}$$

Note that, from the first relation, creation operators commute, thus the definition (7.46) is independent of the order of rapidities. This may seem in contrast with the antisymmetry of the Bethe wavefunction under exchange of quasimomenta. However, as we discussed in §2, this antisymmetry is unphysical and Bethe eigenstate can be defined as symmetric as well.

A more problematic apparent paradox comes from the fact that *the Pauli-like property of Bethe eigentates, forbidding coincident rapidities, cannot arise from these commutation relations.*

It neither comes from the Bethe equations themselves, as already pointed out in §2. In that chapter we solved this issue recalling that the Pauli-like property was employed in the derivation of the Bethe equations starting from Bethe's ansatz for the wavefunction. Thus, when one looks for solutions, the distinctness of rapidities is an additional external constraint.

Here the explicit wavefunction was not involved, thus the Pauli-like property must have an alternative origin. This problem is discussed in detail in [123] for the XXX model.

Analogously, the key observation is that, if the existence of two equal rapidities is assumed a priori of the derivation of the Bethe equations, one ends up with the same number of independent equations, while the number of variables has been reduced by one, leading to an overdetermined system, which is in general unsolvable.

Chapter 8

FIRST TEST FOR THE LONG RANDOM STATE CONJECTURE IN THE XXZ MODEL

We begin this chapter by expressing our "long random states conjecture" for the XXZ model more formally, justifying the statement within the algebraic Bethe ansatz framework.

In particular, within our setting, we expect the $2n$ -rainbow state quasilocal charges to converge to their expectation values on the infinite-temperature thermal state as $n \rightarrow +\infty$.

We also explicitly write an exact and numerically-convenient expression for the infinite-temperature expectation values, analogous to (7.39).

Then, as a first test for our thermalization hypothesis (apart from the exact preliminary results in the XX model), in this chapter we directly compare the stationary state root densities after a homogeneous quench from the $2n$ -rainbow state upon increasing n with the infinite-temperature thermal root densities. In particular, we show that the series converges to a profile which is compatible with the thermal root densities.

8.1 ORIGINAL CONJECTURE IN THE XXX MODEL

Consider a state $|\Psi_n\rangle$ periodically built from unit cells $|\psi\rangle$ of length n which are linear superpositions of random states $|r_k\rangle$.

$$|\Psi_n\rangle = |\psi\rangle^{\otimes L/n} \quad |\psi\rangle = \sum_{k=1}^{k_{max}} C_k |r_k\rangle \quad \vec{C} \in \mathbb{C}^{k_{max}} \quad (8.1)$$

Each $|r_k\rangle$ is a random product state for n spins (e.g. $|\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow\uparrow\rangle$ if $n = 10$), while the coefficients C_k are not necessarily random and their number can be any $k_{max} \in \{1, \dots, 2^n\}$.

In a relatively recent paper by Pawełczyk [99], it is conjectured that, *in the XXX model, all such "long random states" $|\Psi_n\rangle$ exhibit local stationary states approaching the infinite-temperature thermal state as $n \rightarrow +\infty$.*

Since for the infinite-temperature thermal state $\hat{\rho} = \hat{\mathbb{I}}/2^L$, the corresponding expectation values of quasilocal charges are

$$\chi_s^{th}(\lambda) = \text{Tr}_{\mathcal{H}} [\hat{X}_s(\lambda)] = \frac{1}{2^L} \frac{1}{2\pi i} \frac{\partial}{\partial \mu} \text{Tr}_{\mathcal{H} \otimes \mathcal{V}_s^- \otimes \mathcal{V}_s^+} [\hat{\mathbb{L}}_s(\lambda, \mu)^{\otimes L}] \Big|_{\lambda=\mu}$$

where we plugged (7.30) into (7.27).

Then, we can apply $\text{Tr}_{\mathcal{H}}$ before $\text{Tr}_{\nu_s^- \otimes \nu_s^+}$ and exploit that the trace of a tensor product of matrices is the product of their traces.

$$\chi_s^{th}(\lambda) = \frac{1}{L} \frac{1}{2\pi i} \frac{\partial}{\partial \mu} \text{Tr}_{\nu_s^- \otimes \nu_s^+} \left\{ \left[\frac{1}{2} \text{Tr}_{\mathcal{H}_j} \hat{\mathbb{L}}_s^{(j)}(\lambda, \mu) \right]^L \right\} \Big|_{\lambda=\mu} \quad (8.2)$$

Using the explicit XXX Lax matrix (7.11), then one finds

$$\frac{1}{2} \text{Tr}_{\mathcal{H}_j} \hat{\mathbb{L}}_s^{(j)}(\lambda, \mu) = \frac{\hat{\mathbb{L}} \lambda^{[+]} \mu^{[-]} - \hat{s}^z \otimes \hat{s}^z - (1/2)(\hat{s}^- \otimes \hat{s}^+ + \hat{s}^+ \otimes \hat{s}^-)}{\lambda^{[-(2s+1)]} \mu^{[2s+1]}} \quad (8.3)$$

where $\lambda^{[k]} = \lambda + (ik/2)$. At this point, as in our proof of (7.39), one has to find the highest-magnitude eigenvalue of the object inside $[\]$ in (8.2). For this purpose, the Hopf algebra perspective for $U(\mathfrak{sl}(2))$ turns out particularly useful.

Indeed, the Casimir operator on $U(\mathfrak{sl}(2))$, up to constants or a global normalization, is

$$\hat{C} = (\hat{s}^z)^2 + \frac{1}{2}(\hat{s}^+ \hat{s}^- + \hat{s}^- \hat{s}^+) + \frac{\hat{\mathbb{I}}}{4} = \left(\hat{s}^z + \frac{\hat{\mathbb{I}}}{2} \right)^2 + \hat{s}^+ \hat{s}^-$$

When represented on an s -spin irreducible representation, $\hat{C}$ is proportional to identity (Schur's lemma) and its eigenvalue is easily found by evaluation on the highest-weight state $s^z = s$:

$$\hat{C} = C_s \hat{\mathbb{I}} \quad C_s = \left(s + \frac{1}{2} \right)^2$$

The comultiplication $\Delta[\hat{C}]$ yields a representation of the Casimir operator on $\nu_s^- \otimes \nu_s^+$. It has a particularly nice form which allows one to express (8.3) in terms of Casimir operators, immediately yielding its eigenvalues.

$$\begin{aligned} \Delta[\hat{C}] &= \hat{C} \otimes \hat{\mathbb{I}} + \hat{\mathbb{I}} \otimes \hat{C} - \frac{\hat{\mathbb{I}}}{4} + 2\hat{s}^z \otimes \hat{s}^z + \hat{s}^- \otimes \hat{s}^+ + \hat{s}^+ \otimes \hat{s}^- \\ \rightarrow \quad \frac{1}{2} \text{Tr}_{\mathcal{H}_j} \hat{\mathbb{L}}_s^{(j)}(\lambda, \mu) &= \frac{\left[\lambda^{[+]} \mu^{[-]} + \left(s + \frac{1}{2} \right)^2 - \frac{1}{8} \right] \hat{\mathbb{I}} - \frac{1}{2} \cdot \Delta[\hat{C}]}{\lambda^{[-(2s+1)]} \mu^{[2s+1]}} \end{aligned}$$

The smallest eigenvalue of $\Delta[\hat{C}]$ when composing two equal s -spins is $C_0 = 1/4$. Thus, the maximum eigenvalue of (8.3) is

$$\Lambda_s^{(max)}(\lambda, \mu) = \frac{\lambda^{[+]} \mu^{[-]} + \left(s + \frac{1}{2} \right)^2 - \frac{1}{4}}{\lambda^{[-(2s+1)]} \mu^{[2s+1]}} = \frac{4\lambda\mu + 2i(\mu - \lambda) + (2s+1)^2}{4\lambda\mu + 2i(\mu - \lambda)(2s+1) + (2s+1)^2}$$

It is consistently $= 1$ for $\lambda = \mu$, since otherwise the charges would diverge. Finally:

$$\chi_s^{th}(\lambda) = \frac{1}{2\pi i} \frac{\partial}{\partial \mu} \Lambda_s^{(max)}(\lambda, \mu) \Big|_{\lambda=\mu} = \frac{1}{\pi} \frac{2s}{[(2s+1)^2 + 4\lambda^2]}$$

Remarkably, by plugging this into (7.44) one obtains very simple expressions for the infinite-temperature root densities in the XXX model.

Simultaneously, some analytical and numerical results were provided corroborating the conjecture that states of the form (8.1) display expectation values of the quasilocal charges approaching the above expression as $n \rightarrow +\infty$.

8.2 EXTENSION TO THE XXZ MODEL AND THE $2n$ -RAINBOW STATE

The $2n$ -rainbow state shares a strong structural similarity with the states defined in (8.1). Indeed, if we denote by $|r_k\rangle$ the spin configuration obtained by the mapping $0 \leftrightarrow \downarrow$ and $1 \leftrightarrow \uparrow$ of the number k in basis 2 and by $|\overline{r_k}\rangle$ its "conjugate", i.e. the state where $\uparrow$ and $\downarrow$ are exchanged, then

$$|R_n\rangle = |\psi\rangle^{\otimes L/2n} \quad |\psi\rangle = \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |r_k\rangle \otimes |\overline{r_k}\rangle \quad (8.4)$$

Clearly, the states of this sum are not random and some of them surely display charges which are far from being close to infinite temperature values. These special states are called non-generic in [99], since the probability that a random state coincides with one of them is assumed to be negligible. However, the crucial fact is that, as $n \rightarrow +\infty$, only an infinitesimal fraction of the states included in the sum (8.4) are non-generic. The others are expected to behave as true random states. Therefore, *the above rewriting raises the question of whether the LSS of the $2n$ -rainbow state in the XXZ model as $n \rightarrow +\infty$ is thermal with infinite temperature*. We recall that we have already analytically proven that this is true in the XX model.

However, here the simple expressions of charges and root densities computed in the previous paragraph fail and we have to restart from (8.2).

$$\begin{aligned} \frac{1}{2} \text{Tr}_{\mathcal{H}_j} \hat{\mathbb{L}}_s^{(j)}(\lambda, \mu) &= \frac{\sin^{[-]}(\lambda + i\eta \hat{s}^z) \otimes \sin^{[+]}(\mu + i\eta \hat{s}^z) + \sin^{[-]}(\lambda - i\eta \hat{s}^z) \otimes \sin^{[+]}(\mu - i\eta \hat{s}^z)}{\mathcal{N}(\lambda, \mu)} - \\ &- \frac{(\hat{s}^- \otimes \hat{s}^+ + \hat{s}^+ \otimes \hat{s}^-) \sinh^2(\eta)}{\mathcal{N}(\lambda, \mu)} \quad \text{where} \quad \mathcal{N}(\lambda, \mu) = 2 \sin^{[-(2s+1)]}(\lambda) \sin^{[2s+1]}(\mu) \end{aligned} \quad (8.5)$$

Once the highest-magnitude eigenvalue $\Lambda_s^{(max)}(\lambda, \mu)$ of this object is found (at least for $\mu \approx \lambda$), since its value for $\lambda = \mu$ must be $= 1$ to avoid divergent charges, the infinite-temperature quasiloal charges must be again given by

$$\chi_s^{th}(\lambda) = \frac{1}{2\pi i} \left. \frac{\partial}{\partial \mu} \Lambda_s^{(max)}(\lambda, \mu) \right|_{\lambda=\mu}$$

Evaluation of the Infinite-Temperature Quasiloal Charges in the XXZ Model: First, one may wonder whether we can reduce (8.5) to Casimir operators, as in the XXX model. The $U_q(\mathfrak{sl}(2))$ Casimir operator is [44, 47, 79]

$$\hat{C}_q = \frac{\cosh[\eta(2\hat{s}^z + 1)]}{2 \sinh^2(\eta)} + \hat{s}^- \hat{s}^+$$

Its comultiplication, yielding a representation of $\hat{C}_q$ on $v_s^- \otimes v_s^+$ yields

$$\Delta[\hat{C}_q] = \hat{C}_q \otimes q^{2\hat{s}^z} + q^{-2\hat{s}^z} \otimes \hat{C}_q - \frac{\cosh(\eta)}{2 \sinh^2(\eta)} q^{-2\hat{s}^z} \otimes q^{2\hat{s}^z} + (\hat{s}^- q^{-\hat{s}^z} \otimes q^{\hat{s}^z} \hat{s}^+ + q^{-\hat{s}^z} \hat{s}^+ \otimes \hat{s}^- q^{\hat{s}^z})$$

which is not trivially recognised inside (8.5). However, if we define

$$\hat{\mathbb{T}}_s^{th}(\lambda, \mu) := \frac{1}{2} \text{Tr}_{\mathcal{H}_j} \hat{\mathbb{L}}_s^{(j)}(\lambda, \mu) = \langle \hat{\mathbb{L}}_s(\lambda, \mu) \rangle_{\beta=0}$$

this operator shares all the property of the previously introduced $\hat{\mathbb{T}}_s(\lambda, \mu)$. In particular, *if $\lambda = \mu$ then $\Lambda_s^{(max)}(\lambda, \lambda) = 1$ and this (nondegenerate) eigenvalue falls within the zero-magnetization*

Hilbert subspace $\mathcal{V}_0$. For example, if $s = 1/2$ the corresponding eigenstate is the Clebsch-Gordan singlet $(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$.

Thus, for the computation of conserved charges, it is convenient to employ

$$\hat{\mathbb{T}}_s^{th(0)}(\lambda, \mu) := \hat{\mathbb{T}}_s^{th}(\lambda, \mu)|_{\gamma_0}$$

This matrix is tridiagonal, symmetric, and invariant under global spin-flip.

We have not still been able to obtain a general analytic characterization of the highest-magnitude eigenvalue $\Lambda_s^{(max)}(\lambda, \mu)$ for arbitrary spin s , at least for $\lambda \approx \mu$. Otherwise, this would have allowed us to analytically obtain the infinite-temperature quasilocal charges for the XXZ model and then the corresponding root densities through string-charge duality, as for the XXX model.

However, at this point we can easily compute numerically the exact profile of root densities, avoiding the approximation due to the string-length truncation in the BGT equations, as previously discussed for periodic product states.

Indeed, observe that (8.2) is the same expression of (7.33):

$$\chi_s^{th}(\lambda) := \frac{1}{L} \frac{1}{2\pi i} \left[\frac{\partial}{\partial \mu} \left\{ \text{Tr}_{\nu_s^- \otimes \nu_s^+} [\hat{\mathbb{T}}_s^{th}(\lambda, \mu)^L] \right\} \right]_{\lambda=\mu}$$

This is due to the fact that, *despite the infinite temperature thermal state is nota a product state, it is translationally invariant and operators acting on different sites are completely uncorrelated.*

Since $\hat{\mathbb{T}}_s^{th}(\lambda, \mu)$ shares all the general properties of $\hat{\mathbb{T}}_s(\lambda, \mu)$, this is sufficient to replicate all the formalism leading to

$$\chi_s^{th}(\lambda) = \frac{1}{2\pi i} \frac{\text{Tr}_{\gamma_0} \left[\text{adj} \left(\hat{\mathbb{T}}_s^{th(0)}(\lambda, \lambda) - \hat{\mathbb{I}} \right) \left(\partial_\mu \hat{\mathbb{T}}_s^{th(0)}(\lambda, \mu) \right) \right]_{\lambda=\mu}}{\text{Tr}_{\gamma_0} \left[\text{adj} \left(\hat{\mathbb{T}}_s^{th(0)}(\lambda, \lambda) - \hat{\mathbb{I}} \right) \right]} \quad (8.6)$$

8.3 COMPARISON BETWEEN THE $2n$ —RAINBOW STATE AND THE INFINITE-TEMPERATURE ROOT DENSITIES

We recall that, by evaluating the first $\tilde{m} + 1$ quasilocal charges $\chi_s(\lambda)$ of an initial state, we can exactly compute the first $\tilde{m}$ stationary state m —string root densities by string-charge duality (7.44). For the $2n$ —rainbow state and the infinite-temperature thermal state this can be done employing (7.39) and (8.6), respectively. As a preliminary qualitative check, we sampled $\{\rho_m(\lambda), \theta_m(\lambda)\}$ for both states, increasing the rainbow size $2n$ up to 250, for both $\Delta = 3$ and $\Delta = 10$.

The obtained profiles are displayed in Fig.8.1. The figure qualitatively shows the convergence of the $2n$ —rainbow state LSS to the infinite-temperature root densities as $n \rightarrow +\infty$.

Fitting Functions for ρ_m^V and θ_m^V vs. the Rainbow Size $2n$: To quantitatively characterize the convergence of the $2n$ —rainbow state root densities to their thermal counterparts, we performed fits of the maximal values of root densities $\{\rho_m^V, \theta_m^V\}$ upon varying $2n$ up to 5000.

We found the following argument motivating a suitable fitting function for ρ_m^V .

In (1.10), we proved that $E_n/L \propto n^{-1}$. Moreover, in (2.30) we showed that $\epsilon_m \approx -J \sinh(\eta)$ independently of m if $\eta m \gg 1$. Therefore, if we define $\rho_m = \rho_m^\infty + \delta\rho_m$,

$$\frac{E_n}{L} = \left[\frac{J\Delta}{4} + \sum_m \int_{-\pi/2}^{\pi/2} \epsilon_m \rho_m^\infty d\lambda \right] + \sum_m \int_{-\pi/2}^{\pi/2} \epsilon_m \delta\rho_m^\infty d\lambda \approx -J \sinh(\eta) \sum_m \int_{-\pi/2}^{\pi/2} \delta\rho_m^\infty d\lambda$$

Since the energy density decays as $1/n$, we expect

$$\text{Fitting function for } \rho_m^V: \quad \rho_m^V(2n) = \rho_m^{V,\infty} + \frac{A\rho}{2n} \quad (8.7)$$

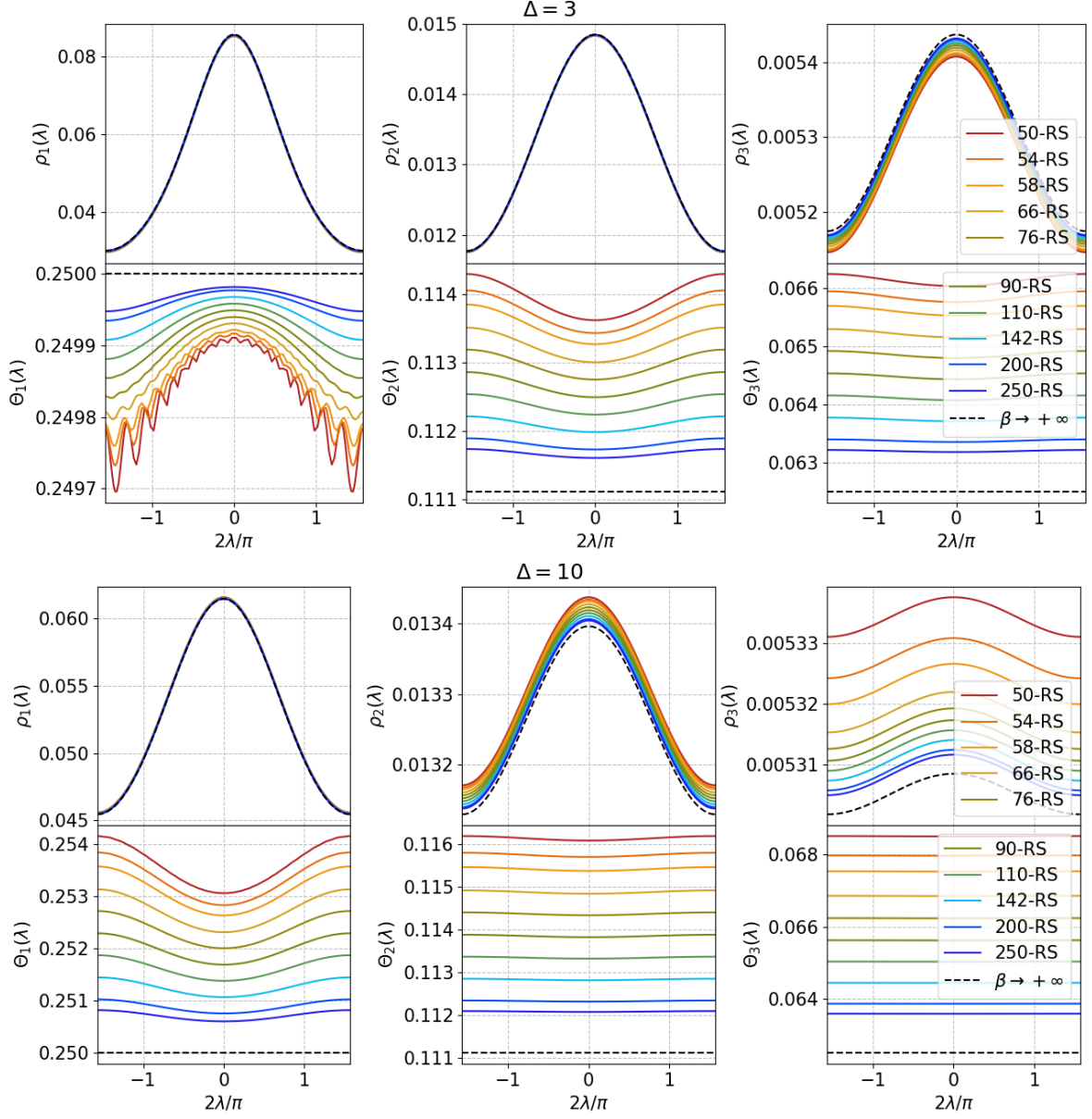


FIGURE 8.1: Coloured solid lines represent $\{\rho_m(\lambda), \theta_m(\lambda)\}$ for rainbow sizes $2n \in \{50, \dots, 250\}$, computed using string-charge duality (7.44) for both $\Delta = 3$ and $\Delta = 10$.

Decreasing colour wavelength means increasing rainbow size. Black dashed lines correspond to the infinite-temperature root densities.

With regard to θ_m^V , we can estimate

$$\theta_m = \frac{\rho_m}{\rho_m^{tot}} = \frac{\rho_m^\infty + \delta\rho_m}{\rho_m^{tot,\infty} + \delta\rho_m^{tot}} \approx \theta_m^\infty + \frac{1}{\rho_m^{tot,\infty}} [\delta\rho_m - \theta_m^\infty \delta\rho_m^{tot}]$$

Until $\delta\rho_m$ is dominant, we thus expect $\delta\theta_m$ to decay as $1/n$ with same sign as A_ρ . Then, the simplest hypothesis we can make is that $\theta_m^\infty \delta\rho_m^{tot}$ displays the same dependence on n as well. In this case, the presence of the second term may change the sign of the proportionality factor.

$$\text{Fitting function for } \theta_m^V: \quad \theta_m^V(2n) = \theta_m^{(\beta=0)} + \frac{A_\theta}{2n} \quad (8.8)$$

Now the asymptotic value is analytically known and there remains only one free parameter A_θ .

$$\text{Exact values:} \quad \theta_1^{(\beta=0)} = \frac{1}{4} = 0.25 \quad \theta_2^{(\beta=0)} = \frac{1}{9} = 0.\bar{1} \quad \theta_3^{(\beta=0)} = \frac{1}{16} = 6.25 \cdot 10^{-2}$$

Fit Results: For both ρ_m^V and θ_m^V , the data and the corresponding fit curves are shown in Fig.8.2: qualitatively, our data seem to be well described by the above fitting functions.

Depending on the specific plot, we had to remove some of the lower- n data to achieve a satisfactory fit. However, this procedure is reasonable because our ansatz for the fitting function in meant to be a leading-order approximation for high n .

The resulting best-fit parameters (where the uncertainty is estimated by performing a new fit after excluding the leftmost data point and considering the variation of parameters) are:

Δ	m	$\rho_m^{V,\infty}$	A_ρ	A_θ
3	1	$(8.57445 \pm 0.00001) \cdot 10^{-2}$	$(-2.0854 \pm 0.0016) \cdot 10^{-2}$	$(-4.578 \pm 0.007) \cdot 10^{-3}$
	2	$(1.484899 \pm 0.000001) \cdot 10^{-2}$	$(-7.93 \pm 0.02) \cdot 10^{-4}$	$(1.2503 \pm 0.0008) \cdot 10^{-1}$
	3	$(5.43748 \pm 0.00002) \cdot 10^{-3}$	$(-1.291 \pm 0.005) \cdot 10^{-3}$	$(1.745 \pm 0.002) \cdot 10^{-1}$
Δ	m	$\rho_m^{V,\infty}$	A_ρ	A_θ
10	1	$(6.144376 \pm 0.000004) \cdot 10^{-2}$	$(6.922 \pm 0.006) \cdot 10^{-3}$	$(1.5248 \pm 0.0003) \cdot 10^{-1}$
	2	$(1.33963751 \pm 0.00000013) \cdot 10^{-2}$	$(2.0539 \pm 0.0002) \cdot 10^{-3}$	$(2.453 \pm 0.003) \cdot 10^{-1}$
	3	$(5.308492 \pm 0.000007) \cdot 10^{-3}$	$(7.91 \pm 0.02) \cdot 10^{-4}$	$(2.806 \pm 0.009) \cdot 10^{-1}$

Conclusions: The extrapolated values of $\rho_m^{V,\infty}$ are consistent (within 1–2 times their uncertainty) the $\lambda = 0$ values of the infinite temperature thermal states, directly computed by (8.6) and string-charge duality, which are

Δ	m	$\rho_m^{V,th}$	Δ	m	$\rho_m^{V,th}$
3	1	$8.57444 \cdot 10^{-2}$	10	1	$6.144370 \cdot 10^{-2}$
	2	$1.484897 \cdot 10^{-2}$		2	$1.33963767 \cdot 10^{-2}$
	3	$5.43745 \cdot 10^{-3}$		3	$5.308501 \cdot 10^{-3}$

The numerical error associated to these values is negligible compared to the uncertainty on the above fit parameters. Thus we assume them to be exact and the truncation is performed to match the number of meaningful digits of the fit parameters.

Analogously, even if we did not leave $\theta_m^{V,\infty}$ as free parameter, Fig.8.2 qualitatively shows that the scaling of θ_m^V with the rainbow size (for sufficiently large $2n$) is consistent with the assumption that asymptotic values are given by (2.44).

Therefore, *these results are in agreement with our hypothesis that the LSS for the $2n$ –rainbow state approaches the infinite-temperature thermal state as $n \rightarrow +\infty$.*

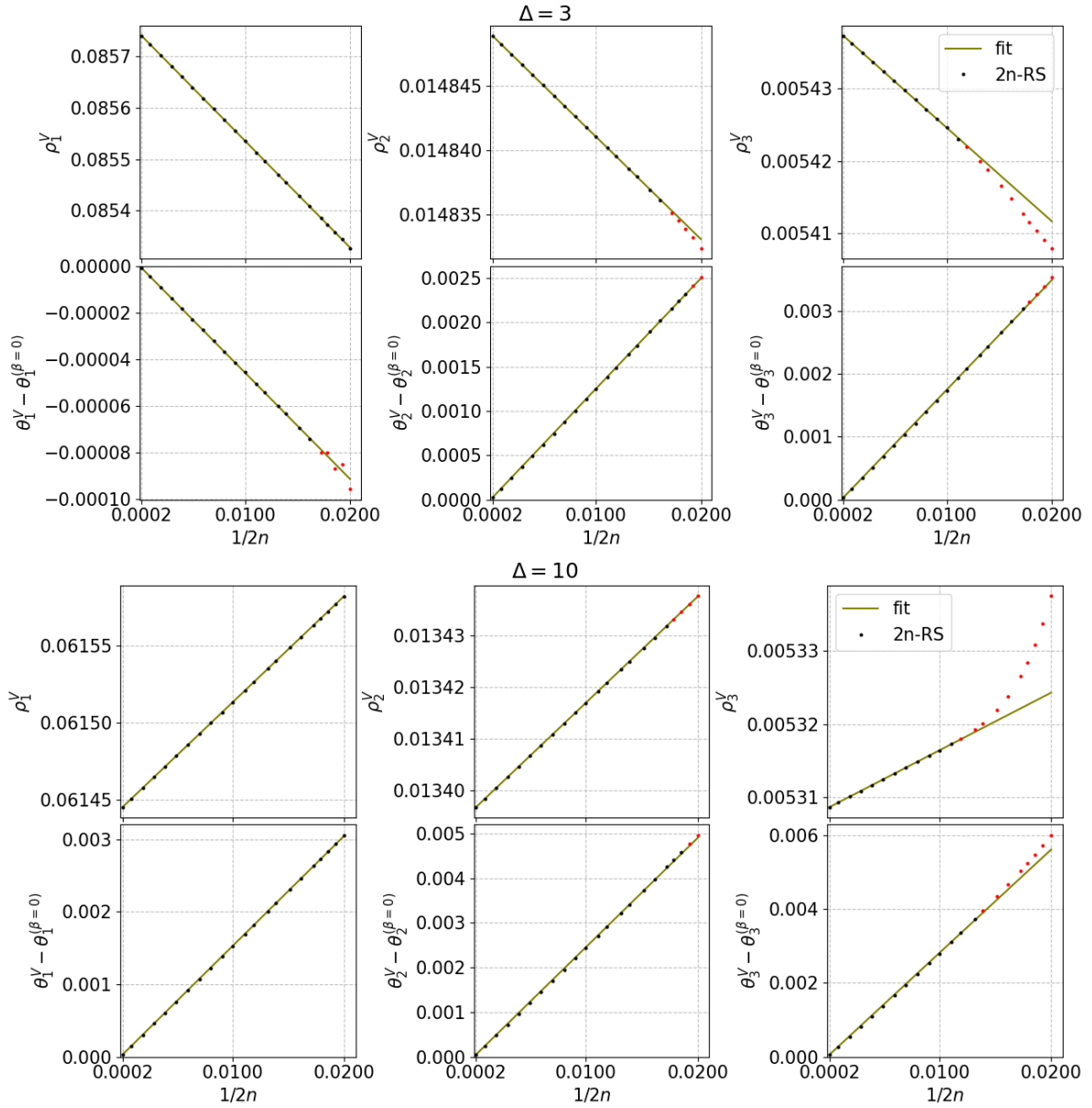


FIGURE 8.2: Vertex values (black dots) of $\rho_m(\lambda)$ and $\theta_m(\lambda)$ for $m \in \{1, 2, 3\}$ and rainbow sizes ranging from 50 to 5000. Green solid lines are the best fit curves corresponding to (8.7) and (8.8). Red dots indicate data excluded from the fit procedure.

8.4 MAJUMDAR-GHOSH STATE

As final test for the reliability of our numerical procedure, we computed the root densities for the Majumdar-Ghosh (MG) state, i.e. the ground state of the Majumdar-Ghosh Hamiltonian [91].

It has the same dimer structure as the 2-rainbow state, with the difference that its dimers correspond to singlet rather than triplet Bell pairs:

$$|MG\rangle := \left(\frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} \right)^{\otimes L/2}$$

The MG state is a convenient benchmark because its root densities can be analytically computed using the quench-action method [93, 106] or the Hirota difference equation [73]. They are proven to obey the following recursive relations [1, 5, 56, 93]:

$$\eta_n(\lambda) = \frac{\eta_{n-1}^{[-]}(\lambda) \eta_{n-1}^{[+]}(\lambda)}{1 + \eta_{n-2}(\lambda)} - 1$$

$$\rho_n^h(\lambda) = \hat{\rho}_n^{tot, [+]} + \rho_n^{tot}(\lambda) - \rho_{n-1}^h(\lambda)$$

with base steps given by

$$\eta_0(\lambda) := 0 \quad \eta_1(\lambda) = \frac{\cos(4\lambda) - \cosh(2\eta)}{\cos^2(\lambda)(\cos(2\lambda) - \cosh(2\eta))} - 1$$

$$\rho_0^h(\lambda) := 0 \quad \rho_1^h(\lambda) = a_1(\lambda) - \omega^{[+]}(\lambda) - \omega^{[-]}(\lambda) \quad (8.9)$$

where
$$\omega(\lambda) = \frac{\sinh \eta}{2\pi} \left[\frac{4 \cos(2\lambda) [\sinh^2 \eta - \cosh \eta] + \cosh \eta + 2 \cosh(2\eta) + 3 \cosh(3\eta) - 2}{4 [\cosh(2\eta) - \cos(2\lambda)]} \right]$$

Note that the expression of ρ_1^h has the same shape as (7.44) with the correspondence $\chi_{\frac{1}{2}} \leftrightarrow \omega$. The above result comes from a quench-action framework and precedes the introduction of the string-charge duality. However, we expect ω to coincide with $\chi_{\frac{1}{2}}$ up to corrections falling within the kernel of the operator $f^{[+]} + f^{[-]}$.

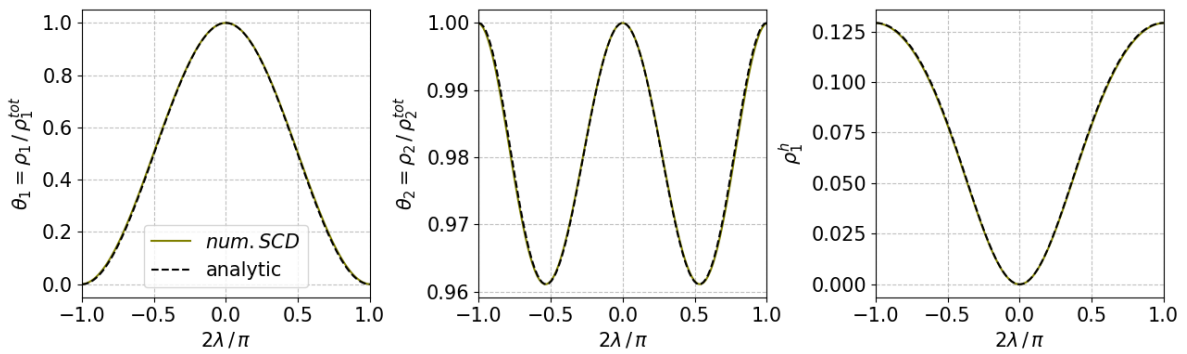


FIGURE 8.3: Comparison between the exact analytical expressions of $\theta_1(\lambda)$, $\theta_2(\lambda)$ and $\rho_1^h(\lambda)$ in (8.9) and (8.10) (black dashed lines) and the numerical evaluation of string-charge duality (green solid lines), for $\Delta = 10$.

The comparison between these exact analytical root densities and those obtained from string-charge duality is shown in Fig.8.3. In particular, we plotted with black dashed lines $\hat{\rho}_1^h(\lambda)$ and:

$$\theta_1(\lambda) = \frac{\rho_1(\lambda)}{\rho_1^{tot}(\lambda)} = \frac{1}{1 + \eta_1(\lambda)} = \frac{\cos^2(\lambda)(\cos(2\lambda) - \cosh(2\eta))}{\cos(4\lambda) - \cosh(2\eta)} \quad (8.10)$$

$$\theta_2(\lambda) = \frac{\rho_2(\lambda)}{\rho_2^{tot}(\lambda)} = \frac{1}{1 + \eta_2(\lambda)} = \left[1 + \frac{4 \sin^2(2\lambda) (2 \cos(2\lambda) + \cosh(\eta) + \cosh(3\eta))}{(\cos(2\lambda) + \cosh(\eta))^2 (\cos(2\lambda) - \cosh(3\eta))} \right]^{-1}$$

Their numerical counterparts are shown as green solid lines. In all panels the two curves seem to be in agreement.

Chapter 9

MATRIX PRODUCT STATES, PURIFICATION, ENTANGLEMENT AND TIME EVOLUTION

We already discussed our interest in the out-of-equilibrium dynamics after the bipartitioning protocol between the $2n$ –rainbow state and the infinite temperature thermal state.

GHD provides a powerful instrument for the characterization of the LQSS and ballistic transport in the scaling limit, at late times.

In this chapter we introduce a complementary tool, enabling a numerical direct simulation of the finite-time evolution and the study of both ballistic and non-ballistic phenomena. It is the time-dependent density matrix renormalization group algorithm (tDMRG), which has the additional beautiful feature to automatically return the entanglement dynamics.

We also define the "half-purification" applied to the right initially-thermal half of our lattice only. It allows us to represent our mixed initial state and its evolution in terms of an artificial pure state. We find interesting to note that the bipartite entanglement computed on the half-purified state coincides with the von Neumann entropy of the reduced density matrix of the initially-pure half of the system. This observation will turn out useful in §11, where we will focus on the entanglement dynamics in our setting.

As a corollary test, we check that in our half-purified setting there is no direct connection between the bipartite entanglement and the operator entanglement of the non-purified density matrix, which is another interesting quantity in the dynamics from mixed initial states.

9.1 PURIFICATION OF A MIXED STATE

Consider the set of all Hermitian operators on a d –dimensional Hilbert space $\mathcal{H}$ with non-negative eigenvalues and unit trace. Each such operator can be interpreted as a mixed state $\hat{\rho}$.

If $\{|\lambda_k\rangle\}$ are the eigenvectors of $\hat{\rho}$, then the latter can be spectrally decomposed as

$$\hat{\rho} = \sum_k \lambda_k |\lambda_k\rangle \langle \lambda_k| : \mathcal{H} \rightarrow \mathcal{H} \quad \lambda_k \geq 0 \quad \forall k \in \{1, \dots, d\} \quad \sum_k \lambda_k = 1$$

If we now introduce an ancillary copy $\tilde{\mathcal{H}}$ of the Hilbert space and define the pure state

$$|\rho\rangle\rangle := \sum_k \sqrt{\lambda_k} |\lambda_k\rangle \otimes |\lambda_k\rangle \in \mathcal{H} \otimes \tilde{\mathcal{H}} \quad \langle\langle \rho | \rho \rangle\rangle = \text{Tr}[\hat{\rho}] = \sum_k \lambda_k = 1$$

then $|\rho\rangle\rangle$ is equivalent to $\hat{\rho}$ for the evaluation of observables on $\mathcal{H}$. Indeed $\hat{\rho}$ is exactly the reduced density matrix of $|\rho\rangle\rangle$ with respect to the original space:

$$\text{Tr}_{\tilde{\mathcal{H}}} [|\rho\rangle\rangle \langle\langle\rho|] = \sum_{i,j,k} \sqrt{\lambda_i \lambda_j} \langle\lambda_k|\lambda_i\rangle |\lambda_i\rangle \langle\lambda_j| \langle\lambda_j|\lambda_k\rangle = \hat{\rho}$$

More generally, we can consider any pure state $|\psi\rangle\rangle \in \mathcal{H} \otimes \tilde{\mathcal{H}}$ such that $\text{Tr}_{\tilde{\mathcal{H}}} [|\psi\rangle\rangle \langle\langle\psi|] = \hat{\rho}$, since they are equivalent for the computation of observables supported on $\mathcal{H}$:

$$\begin{aligned} \langle\hat{O} \otimes \hat{\mathbb{I}}\rangle_{|\psi\rangle\rangle} &= \langle\langle\psi|(\hat{O} \otimes \hat{\mathbb{I}})|\psi\rangle\rangle = \sum_{i,j} \langle\langle\psi|(\hat{O} \otimes \hat{\mathbb{I}})|\lambda_i\rangle \otimes |\lambda_j\rangle \langle\lambda_i| \otimes \langle\lambda_j|\psi\rangle\rangle = \\ &= \sum_i \langle\lambda_i| \left[\sum_j \langle\lambda_j|\psi\rangle\rangle \langle\langle\psi|\lambda_j\rangle \right] \hat{O} |\lambda_i\rangle = \text{Tr} [\hat{\rho} \hat{O}] = \langle\hat{O}\rangle_{\hat{\rho}} \end{aligned} \quad (9.1)$$

In summary, the interpretation of a generic mixed state $\hat{\rho} : \mathcal{H} \rightarrow \mathcal{H}$ as the reduced density matrix of a pure state $|\psi\rangle\rangle$ in an enlarged Hilbert space $\mathcal{H} \otimes \tilde{\mathcal{H}}$ and the use of the latter for the computation of observables according to (9.1) is called *purification*.

Application to Bipartitioning Protocols with Mixed States: In the context of a bipartitioning protocol between a pure state $|\psi_L\rangle \in \mathcal{H}_L$ and a mixed state $\hat{\rho}_R : \mathcal{H}_R \rightarrow \mathcal{H}_R$, we can simply introduce an ancillary copy of $\mathcal{H}_R$, define the corresponding $|\psi_R\rangle\rangle \in \mathcal{H}_R \otimes \tilde{\mathcal{H}}_R$ and then consider as initial state $|\psi_L\rangle \otimes |\psi_R\rangle\rangle \in \mathcal{H}_L \otimes \mathcal{H}_R \otimes \tilde{\mathcal{H}}_R$.

$$\text{Tr}_{\tilde{\mathcal{H}}_R} [|\psi_R\rangle\rangle \langle\langle\psi_R|] = \hat{\rho}_R \quad \rightarrow \quad \text{Tr}_{\tilde{\mathcal{H}}_R} [|\psi_L\rangle \langle\psi_L| \otimes |\psi_R\rangle\rangle \langle\langle\psi_R|] = |\psi_L\rangle \langle\psi_L| \otimes \hat{\rho}_R$$

However, since our aim is to study the post-quench dynamics, we emphasize that the Hamiltonian is an operator supported only on the original physical space $\mathcal{H}_L \otimes \mathcal{H}_R$ and acts trivially on the ancillary copy $\tilde{\mathcal{H}}_R$. This will be discussed in more detail later.

9.2 PRELIMINARIES ON MATRIX PRODUCT STATES (MPS)

In this overview of MPSs we mainly follow the approach of [38] and [118], focused on the connection with the *density matrix renormalization group* (DMRG) algorithm and vice versa.

Let us consider a lattice system of L sites. Its Hilbert space is thus the tensor product of local spaces associated to sites: $\mathcal{H} = \otimes_{j=1}^L \mathcal{H}_j$. If $\dim \mathcal{H}_j = d$, then $\dim \mathcal{H} = d^L$.

To focus on the setting of our interest, we immediately restrict to 1D systems, where adjacent sites can always be assigned consecutive labels. Since we are considering short-range interactions, this automatically implies that interacting sites are labelled by nearby integers. This is one of the key requirements for the efficient application of DMRG techniques. Indeed, for example, a 2D lattice with nearest-neighbour interactions can be labelled using different paths of varying efficiency [117], but in general the average range of interactions in the resulting one-dimensional Hamiltonian diverges with the system size.

If we denote by σ_j the local degree of freedom described in $\mathcal{H}_j$, then $\{|\sigma_1, \dots, \sigma_L\rangle\}$ is an orthonormal basis of $\mathcal{H}$. Every pure state $|\psi\rangle \in \mathcal{H}$ can be uniquely decomposed as

$$|\psi\rangle = \sum_{\vec{\sigma}} C_{\sigma_1, \dots, \sigma_L} |\sigma_1, \dots, \sigma_L\rangle \quad (9.2)$$

We say that $|\psi\rangle$ admits a *matrix product state* representation if the tensor C is factorized as a product of L tensors $\{A^{[j]}\}$, each with only one site-state index σ_j :

$$\begin{array}{c} C \\ \hline \sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \dots \quad \sigma_L \end{array} = C_{\sigma_1, \dots, \sigma_L} = A_{a_1}^{[1]\sigma_1} \left(\prod_{j=2}^{L-1} A_{a_{j-1} a_j}^{[j]\sigma_j} \right) A_{a_{L-1}}^{[L]\sigma_L} = \begin{array}{c} A^{[1]} \quad A^{[2]} \quad A^{[3]} \quad A^{[4]} \quad \dots \quad A^{[L]} \\ \sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \dots \quad \sigma_L \end{array}$$

If the intermediate "artificial" index α_j runs over $\alpha_j \in \{1, \dots, D_j\}$, then D_j is said to be the *bond dimension* associated with the pair of sites j and $j+1$.

This factorization of a tensor C is always possible, but not unique (there still exists some gauge freedom). By contrast, the bond dimension of a pair of adjacent sites is gauge-independent.

Indeed, if we consider a pair $(A^{[j]}, A^{[j+1]})$, any invertible matrix $X \in \text{Mat}_{D_j \times D_j} \mathbb{C}$ can be used to define an equivalent but different pair $(A'^{[j]}, A'^{[j+1]}) := (A^{[j]}X, X^{-1}A^{[j+1]})$.

The bond dimension D_j is clearly left untouched by this transformation.

As we will discuss later, for a generic $|\psi\rangle$, D_j grows symmetrically as $d^{\min[j, L-j]}$ toward the centre of the MPS, so that its maximal value is $d^{\lfloor L/2 \rfloor}$. However, the term MPS is often more strictly referred to states with the computationally-friendly property that their maximal bond dimension stays below a threshold $D_{max} \ll d^L$ while $L \rightarrow +\infty$.

Singular Value Decomposition: Consider an arbitrary rectangular matrix $M \in \text{Mat}_{a \times b} \mathbb{C}$. It can always be decomposed as

$$M = \tilde{U} \tilde{S} \tilde{V}^\dagger \quad \begin{array}{ll} M \in \text{Mat}_{a \times b} \mathbb{C} & \tilde{U} \in \text{Mat}_{a \times m} \mathbb{C} \\ \tilde{S} = \text{diag}[s_1, \dots, s_m] & \tilde{V}^\dagger \in \text{Mat}_{m \times b} \mathbb{C} \end{array} \quad \begin{array}{l} m := \min\{a, b\} \\ s_i \geq 0 \quad \forall i \quad \text{Singular Values} \end{array}$$

- $\tilde{U}$ is a left-orthogonal matrix, i.e. its columns are m orthonormal vectors called *left singular vectors*, $(\tilde{U}^\dagger \tilde{U} = \mathbb{I})$;
- $\tilde{V}^\dagger$ is a right-orthogonal matrix i.e. its rows are m orthonormal vectors called *right singular vectors*, or equivalently $\tilde{V}$ is left-orthogonal $(\tilde{V}^\dagger \tilde{V} = \mathbb{I})$;
- $\tilde{S}$ is a diagonal matrix with non-negative entries $\{s_1, \dots, s_m\}$, called singular values, which we assume ordered s.t. $s_1 \geq \dots \geq s_m$.
- The number r of non-zero singular values is the *Schmidt rank* of M . A useful property is that, if we define $U \in \text{Mat}_{a \times r} \mathbb{C}$ and $V^\dagger \in \text{Mat}_{r \times b} \mathbb{C}$ by keeping only the first r right/left singular vectors, then

$$M = U S V^\dagger \quad \text{where} \quad S = \text{diag}[s_1, \dots, s_r] \quad (9.3)$$

The diagram shows the SVD decomposition of matrix M . Matrix M is represented as a green rectangle with dimensions a (height) and b (width). It is equal to the product of three matrices: U (a green rectangle with dimensions a and r), S (a grey square with dimensions r and r), and $V^\dagger$ (a yellow rectangle with dimensions r and b). The dimensions are labeled on the rectangles.

Henceforth, since numerical procedures aim to minimize computational costs, we will always refer to this second "compact" SVD [7].

Note that, in general, *the SVD of M is not unique*, since some freedom persists in choosing $V^\dagger$ and U . However, the fundamental fact is that the S matrix (i.e. singular values and their multiplicity) is one-to-one associated to M . Moreover, it is sometimes useful to observe that *singular values are the square roots of the nonzero eigenvalues of $M^\dagger M$, or equivalently $M M^\dagger$.*

MPS Decomposition of a Generic Pure State: Let $C_{\sigma_1, \dots, \sigma_L}$ be the tensor one-to-one associated to $|\psi\rangle$ according to (9.2). It can be viewed as a $d \times d^{L-1}$ matrix with σ_1 as input and $(\sigma_2, \dots, \sigma_{2n})$ as output. Performing the SVD of this matrix we obtain

$$C_{\sigma_1, \dots, \sigma_L} = U_{\sigma_1 \alpha_1} [S V^\dagger]_{\alpha_1 \sigma_2, \dots, \sigma_L} := A_{\sigma_1 \alpha_1}^{[1]} C'_{\alpha_1 \sigma_2, \dots, \sigma_L}$$

Analogously, one can proceed by applying a new SVD to C' on the left with respect to the indices bipartition $(\alpha_1 \sigma_2 \mid \sigma_3, \dots, \sigma_L)$ or on the right with respect to $(\alpha_1, \sigma_2, \dots, \sigma_{L-1} \mid \sigma_L)$.

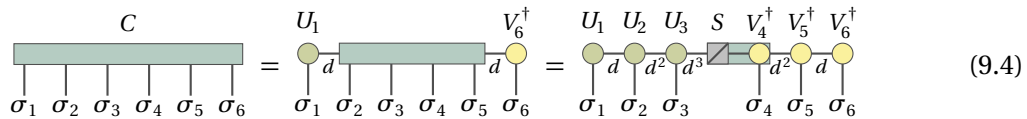
Finally, after $2n - 1$ SVDs, the $2n$ -site tensor C is completely factorized as a product of $2n$ single-site matrices $A^{[j]}$. SVDs performed on the left lead to left-orthogonal A -matrices, while SVDs performed on the right to right-orthogonal A -matrices. The last SVD is performed on a two-site C -tensor and the obtained $[SV^\dagger]$ is the only not-orthogonal (nor left nor right) A -matrix of the MPS. The corresponding site is called the *orthogonality centre*, since it separates the left-orthogonal matrices (on the left) from the right-orthogonal ones (on the right).

However, note that $[SV^\dagger]$'s rows, even if not orthonormal, are orthogonal, since

$$[SV^\dagger]VS = S^2 = \text{diag}[s_1^2, \dots, s_r^2]$$

In the following picture we illustrate as example the decomposition of a C -tensor on 6 sites obtained by applying SVDs alternatively on the left and on the right.

The orthogonality centre is at site 4.



As in all following pictures, we denote by green/yellow right/left-orthogonal tensors.

Clearly, we can use as not-orthogonal matrix US instead of $[SV^\dagger]$: these two choices are linked by a gauge transformation with $X = S$.

Moreover, our construction leads to an MPS which is left/right-orthogonal on the left/right of the orthogonality centre. This is a very convenient form in many situations and is always possible. Other gauge choices, for example, lead to many orthogonality centres or to A -matrices with no particular orthogonality property at all. We will discuss later how using the gauge freedom to shift the orthogonality centre along the MPS is a very useful and common procedure in the computation of strictly-local observables.

Example - MPS Decomposition of the Triplet/Singlet Dimer: The triplet/singlet ($\pm$) dimer has the following C tensor defined on two sites:

$$|\Psi^\pm\rangle = \sum_{\sigma_1\sigma_2} C_{\sigma_1\sigma_2} |\sigma_1\sigma_2\rangle \quad \text{where} \quad C = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix}$$

An instructive observation is that here the eigenvalues are given by $0 = \text{Tr}[C] = \lambda_1 + \lambda_2$ and $\mp 1 = \det[C] = \lambda_1\lambda_2$. They are purely imaginary for the singlet dimer and real but with opposite signs for the triplet dimer. Thus they do not coincide with singular values, which must be positive. We recall that singular values coincide with the square root of the nonzero eigenvalues of $C^\dagger C$ (which here is exactly equal to $CC^\dagger$):

$$C^\dagger C = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ \mp 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix} = \frac{\mathbb{I}}{2} \quad \rightarrow \quad s_1 = s_2 = \frac{1}{\sqrt{2}}$$

The decomposition is not unique, since, for any given solution (U', V') , every unitary matrix X is such that $(U, V) = (U'X, V'X)$ is a good decomposition as well. Moreover, (U', V') are not only left-orthogonal, but also unitary, since C is a square matrix and $C^\dagger C = CC^\dagger \propto \mathbb{I}$. This means that it is always possible to choose $X = (U')^{-1} = (U')^\dagger$, leading to $U = U'X = \mathbb{I}$.

Therefore, we can look for a solution $(\mathbb{I}, V)$ and, by the definition (9.3), it is simply $V^\dagger = \sqrt{2}C$.

Finally, the MPS decomposition (with the usual gauge freedom) is given by

$$\rightarrow \quad A_{\alpha_1}^{[1]\sigma_1} = U_{\sigma_1\alpha_1} = \delta_{\sigma_1\alpha_1} \quad A_{\alpha_1}^{[2]\sigma_2} = s_{\alpha_1} V_{\alpha_1\sigma_2}^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix} \quad (9.5)$$

Maximal Bond Dimension: We mentioned before that the maximal bond dimension D_j in an MPS starts from d at both ends of the chain and then grows exponentially moving towards the centre up to $d^{\lfloor L/2 \rfloor}$. In order to prove this statement, since the bond dimension does not depend on the gauge choice, we can imagine a symmetric SVD procedure as described above. The boundary maximal Schmidt rank is d . Then, the next left-SVD is performed on a matrix with input index $\alpha_1 \sigma_2$, which can take d^2 values, while the right index can take many more values. Therefore, the maximal Schmidt rank is d^2 . The same argument applies symmetrically from the right.

Therefore, this exponential growth sets an upper bound for the bond dimension along the MPS and a generic state $|\psi\rangle \in \mathcal{H}$ saturates it. Only an infinitesimal corner of the Hilbert space corresponds to sufficiently low entangled states in which the bond dimension keeps always under this threshold and in particular stays finite in the thermodynamic limit. For example, in a product state (e.g. Néel state) $D_j = 1 \ \forall j$.

Orthogonality Centre in the Computation of Local Observables: The left/right orthogonality of A -matrices leads to remarkable simplifications when computing an expectation value $\langle\psi|\hat{O}|\psi\rangle$.

The simplest case corresponds to $\hat{O} = \hat{\mathbb{I}}$, namely a normalization-check on $|\psi\rangle$. Owing to left/right-orthogonality on the left/right of the orthogonality centre, respectively:

$$\langle\psi|\psi\rangle = \text{Diagram} = \text{Tr}[S^2] = \sum_{j=1}^r s_j^2$$

Thus, if the state is normalized s.t. $\langle\psi|\psi\rangle = 1$, then the singular values must satisfy $\sum_j s_j^2 = 1$.

In a more general situation, $\hat{O}^{[j]}$ is a k -site operator, i.e. has support only on a finite segment of the lattice of length k , say $\{j, \dots, j+k-1\}$. Here it is convenient to use the gauge freedom to shift the orthogonality centre into this interval. This way the contractions between sites out of it become trivial. In the following picture we consider as example $k=2$:

$$\begin{aligned} \langle\psi|\hat{O}^{[j]}|\psi\rangle &= \text{Diagram} = \text{Diagram} \\ &= (V^{[j+1]}S)_{\sigma'_{j+1}}^{\alpha_{j+1}\alpha'_j} (U^{[j]\dagger})_{\sigma'_j}^{\alpha'_j\alpha_{j-1}} (O^{[j]})_{\sigma_j\sigma_{j+1}}^{\sigma'_j\sigma'_{j+1}} (U^{[j]})_{\alpha_{j-1}\alpha_j}^{\sigma_j} (SV^{[j+1]\dagger})_{\alpha_j\alpha_{j+1}}^{\sigma_{j+1}} \end{aligned}$$

In presence of extensive local operators expressed as $\hat{O} = \sum_j \hat{O}^{[j]}$, it is possible to choose the most suitable orthogonality centre for each term in $\langle\psi|\hat{O}|\psi\rangle = \sum_j \langle\psi|\hat{O}^{[j]}|\psi\rangle$.

9.3 OUR BIPARTITE INITIAL STATE AS AN MPS

Infinite-Temperature Thermal State as MPS: To implement the thermal-side purification, we need a state $|\psi\rangle\rangle$ satisfying

$$\text{Tr}_{\mathcal{H}} [|\psi\rangle\rangle\langle\langle\psi|] = \frac{\hat{\mathbb{I}}}{\dim \mathcal{H}} \quad \langle\langle\psi|\psi\rangle\rangle = 1$$

Clearly, the solution is not unique and, for a spin chain of L sites, we can exploit the fact that

$$|\Psi^\pm\rangle := \frac{|\uparrow\downarrow\rangle \pm |\downarrow\uparrow\rangle}{\sqrt{2}} \quad |\Phi^\pm\rangle := \frac{|\uparrow\uparrow\rangle \pm |\downarrow\downarrow\rangle}{\sqrt{2}} \quad (9.6)$$

$$\text{are s.t.} \quad \text{Tr}_{\tilde{s}} [|\Psi^\pm\rangle\langle\Psi^\pm|] = \text{Tr}_{\tilde{s}} [|\Phi^\pm\rangle\langle\Phi^\pm|] = \frac{\hat{\mathbb{I}}}{2}$$

Thus, $|\psi\rangle\rangle$ can be constructed defining L Bell pairs (of any kind) composed by a site of the original space $\mathcal{H}$ and one (arbitrarily chosen) of its ancillary copy.

More concretely, since we aim to represent the thermal state with the smallest possible initial bond dimensions (i.e. with entangled sites as near as possible) while keeping the Hamiltonian short-ranged,

- we index $\mathcal{H}_R \otimes \tilde{\mathcal{H}}_R$ with staggered physical and ancillary sites

$$s'_1 := s_1 \quad s'_2 := \tilde{s}_1 \quad \dots \quad s'_{2L-1} := s_L \quad s'_{2L} := \tilde{s}_L$$

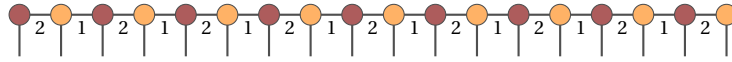
- and entangle each physical site with the ancillary site immediately to its right.

Moreover, since the spin 2–rainbow state is a chain of $|\Psi^+\rangle$ Bell pairs, we choose to define the infinite-temperature $|\psi\rangle\rangle$ using Bell pairs of this same kind. This way, the full initial state on $\mathcal{H}_L \otimes \mathcal{H}_R \otimes \tilde{\mathcal{H}}_R$ when $|\psi_L\rangle = |R^{(2)}\rangle$ is simply a chain of identical dimers.

In order to implement the so-defined $|\psi\rangle\rangle$ as MPS, we can exploit the MPS decomposition of the triplet dimer in (9.5). We define the two A –tensors associated to the j –th pair of physical–ancillary sites $(2j-1, 2j)$ as

$$\begin{aligned} \alpha_{2(j-1)} \begin{array}{c} \text{---} \text{---} \\ | \quad | \\ 1 \quad 2 \\ | \quad | \\ \sigma_{2j-1} \end{array} \alpha_{2j-1} &= [A^{[2j-1]}]_{\alpha_{2(j-1)}, \alpha_{2j-1}}^{\sigma_{2j-1}} = \delta_{1, \alpha_{2(j-1)}} U_{\sigma_{2j-1} \alpha_{2j-1}} \\ \alpha_{2j-1} \begin{array}{c} \text{---} \text{---} \\ | \quad | \\ 2 \quad 1 \\ | \quad | \\ \sigma_{2j} \end{array} \alpha_{2j} &= [A^{[2j]}]_{\alpha_{2j-1} \alpha_{2j}}^{\sigma_{2j}} = [S V^\dagger]_{\alpha_{2j-1} \sigma_{2j}} \delta_{1, \alpha_{2j}} \end{aligned}$$

where $\delta_{1, \alpha}$ is a superfluous notation for the trivial role of the indices $\alpha_{2(j-1)}$ and α_{2j} , which connect two adjacent non-entangled dimers and have unit bond dimension.



Two Ways to Build the $2n$ –Rainbow State as MPS in Numerical Practice: We have just discussed how to write the infinite-temperature state as MPS. It has the role of right-half initial state $|\psi_R\rangle\rangle$ in the bipartitioning protocol, while the left-half initial state $|\psi_L\rangle$ corresponds to a $2n$ –rainbow state. Thus the full purified initial state is $|\psi_L\rangle \otimes |\psi_R\rangle\rangle$.

As mentioned above, if $n = 1$ the 2–rainbow state is a chain of dimers, identical to our purified thermal state. Instead, the two states become different for $n \geq 2$.

We now propose two different methods to numerically implement the $2n$ –rainbow state.

In *the first*, we explicitly construct the tensor $C_{\sigma_1, \dots, \sigma_{2n}}$ such that the k –th unit cell of the $2n$ –rainbow state is given by

$$|R_n^{[2nk+1, 2n(k+1)]}\rangle = C_{\sigma_{2nk+1}, \sigma_{2nk+2}, \dots, \sigma_{2n(k+1)}} |\sigma_{2nk+1}, \sigma_{2nk+2}, \dots, \sigma_{2n(k+1)}\rangle$$

To this end, we observe that, even if it has 2^{2n} total elements, its nonzero entries are only 2^n and all have the value $2^{-n/2}$ leading to the correct normalization $C^\dagger C = 1$.

A single element is specified by the spin values of $2n$ indices: the main idea is that nonzero entries are characterized by having any possible spin combination in the first n indices, but the exactly specular configuration in the remaining n ones, as explicitly written in (8.4).

Therefore, we can generate the indices of all nonzero entries by considering numbers from 0 to $2^n - 1$, writing them down in basis 2 and then, if d_i is the i –th least meaningful digit, defining:

$$d_i = 0 \leftrightarrow \begin{cases} \sigma_i = \downarrow \\ \sigma_{2n-(i+1)} = \uparrow \end{cases} \quad d_i = 1 \leftrightarrow \begin{cases} \sigma_i = \uparrow \\ \sigma_{2n-(i+1)} = \downarrow \end{cases} \quad \forall i \in \{1, \dots, n\}$$

Once C is constructed, it can be factorized as MPS with orthogonality centre on the $(n+1)$ –th site by performing successive symmetric SVDs as described in §9.2.

Von Neumann Entropy of $\hat{\rho}_A$ as a Measure of Bipartite Entanglement: Before proceeding, it is important to understand in what sense $S_N(\hat{\rho}_A)$ can be regarded as a "measure of the bipartite entanglement". In the previous paragraph we studied the situation in which the whole system is prepared in a pure state. As mentioned, if and only if $r = 1$ then $|\psi\rangle$ is a product state and $\hat{\rho}_A$ is a pure state. The corresponding entanglement entropies are zero $S_A^{(\alpha)}(|\psi\rangle) = S_N(\hat{\rho}_A) = 0$.

The Schmidt rank enables a distinction between exact product states and states in which A and B are entangled. However, from the physical perspective this is not the most relevant classification. Indeed, it is possible to have maximal Schmidt rank $r = \min[\dim \mathcal{H}_A, \dim \mathcal{H}_B]$ while $|\psi\rangle$ is "almost" a product state, i.e. $s_1 \approx 1$ and $s_j \ll 1 \ \forall j > 1$.

Entanglement entropies correctly classify these states as being very close to product states. For example, if the only other nonzero singular value is $s_2 = \epsilon \ll 1$, while $s_1 = 1 + o(\epsilon)$:

$$S_A^{(\alpha)}(|\psi\rangle) := \frac{\epsilon^{2\alpha} + o(\epsilon^{2\alpha})}{1 - \alpha} \quad S_N(\hat{\rho}_A) = o(\epsilon)$$

By contrast, the maximal value of $S_N(\hat{\rho}_A)$ is reached (assuming $\langle\psi|\psi\rangle = \sum_j s_j^2 = 1$) when all singular values are equal, i.e. $r = r_{max} = \min[\dim \mathcal{H}_A, \dim \mathcal{H}_B]$ and $s_j^2 = 1/r_{max}$.

$$S_N(\hat{\rho}_A)_{max} = \log(r_{max})$$

This is consistent with the idea that in this case A and B are maximally entangled. Indeed, if A is the smaller subsystem, then $r_{max} = \dim \mathcal{H}_A$ and $\hat{\rho}_A \propto \hat{\mathbb{I}}_A$.

Thus, *if the whole system is in a pure state, then $S_N(\hat{\rho}_A)$ continuously increases from 0 to $\log(r_{max})$ as quantum correlations (entanglement) between A and B become stronger.*

We recall that the global pure-state-nature is preserved by the unitary time evolution.

However, in our work we choose as initial state a mixed state with $\hat{\rho} = |R_n\rangle\langle R_n| \otimes \hat{\rho}_{\beta=0}$, where the right side is thermal with infinite temperature. *The fact that $\hat{\rho}$ is no longer pure gives introduces several important differences with respect to the previous setting $\hat{\rho} = |\psi\rangle\langle\psi|$:*

- Although $S_N(\hat{\rho}_A)$ is still a measure of how much $\hat{\rho}_A$ is a pure state, *it no longer measures how close the entire system is to a product state.* Indeed, the state of the system is not pure.

More formally, $S_N(\hat{\rho}_A)$ *is not a measure of correlations between A and B .* It measures how much $\hat{\rho}_A$ is mixed. If the whole system is in a pure state, mixedness is only due to the presence of entanglement, while this fails when $\hat{\rho}$ is a mixed state. For example, if $\hat{\rho} = \hat{\rho}_A \otimes \hat{\rho}_B$, then the two subsystems are uncorrelated, but if $\hat{\rho}_A$ is a mixed state then $S_N(\hat{\rho}_A) \neq 0$.

- Physically, the very *correlations between A and B , if present, are no longer of purely quantum nature*, since they also contain classical contributions from the incoherent superposition in $\hat{\rho}$. $S_N(\hat{\rho}_A)$ is a "mixture" of entanglement entropy and thermodynamic entropy.
- While, if the whole system is in a pure state, then the von Neumann and Rényi entropies do not depend on which subsystem is used to compute them, now in general $S_N(\hat{\rho}_A) \neq S_N(\hat{\rho}_B)$. This is evident for states $\hat{\rho} = \hat{\rho}_A \otimes \hat{\rho}_B$ where $\hat{\rho}_A$ is pure while $\hat{\rho}_B$ mixed.
- Independently of the nature (pure or mixed) of $\hat{\rho}$, the von Neumann entropy satisfies

$$S_N(\hat{\rho}) \leq S_N(\hat{\rho}_A) + S_N(\hat{\rho}_B) \quad \text{Subadditivity}$$

with $=$ corresponding to the two subsystems being completely uncorrelated, i.e.

$$S_N(\hat{\rho}_A \otimes \hat{\rho}_B) = S_N(\hat{\rho}_A) + S_N(\hat{\rho}_B)$$

Note that for Rényi entropies subadditivity is generally violated except for $\alpha \rightarrow 1$, where the von Neumann entropy is recovered.

In presence of mixed initial states, it is sometimes more interesting to study the *operator entanglement*, which measures the distance of $\hat{\rho}$ from a product operator $\hat{\rho}_A \otimes \hat{\rho}_B$, where the two subsystems are free of both classical and quantum correlations (we will introduce it later).

However, in our particular case, we start with the subsystem A in a pure state and then perform a tDMRG evolution on the purified state of the whole system. Thus:

- $S_N(\hat{\rho}_A)$ measures how much $\hat{\rho}_A$ is taken far from its pure state initial condition;
- it is the entanglement entropy of the artificial purified state and its increase in general reflects that of computational resources requested to reach longer time intervals.

Overview About Some Relevant Norms: In the next paragraph we will define the operator entanglement. However, while operators such as density matrices are usually referred to as "normalized" when they have unit trace, in this context we need to introduce a different norm for operators. These concepts will turn out useful in §9.6 as well.

To see the problem from a wider perspective, highlighting some deep differences between state and operator entanglement, let us have a brief digression about norms.

A *quantum state* is by definition a vector in a d -dimensional Hilbert space, i.e. it can be represented by an element $\vec{v} \in \mathbb{C}^d$. We recall that in $\mathbb{C}^d$ it is possible to define

$$\|\vec{v}\|_p := \left(\sum_{j=1}^d |v_j|^p \right)^{1/p} \quad \ell_p\text{-norm} \quad [\|\vec{v}\|_\infty := \max\{|v_j|\}]$$

and the 2-norm corresponds to the usual *Euclidean norm*.

Remarkably, *in the definition of Rényi/von-Neumann entropies the 2-normalization of the state $|\psi\rangle$ of the system was assumed*. Otherwise, these entropies would not be zero for pure states. However, this is precisely the usual normalization required to quantum states.

Let us now make a comparison with operators.

The norms of our interest on $\text{Mat}_{m \times n} \mathbb{C}$ are called Schatten p -norms [7] and defined as the ℓ_p -norm of the vector $\vec{s}$ of the singular values of the considered matrix.

$$\|M\|_p := \|\vec{s}\|_p = \left(\sum_{j=1}^r s_j^p \right)^{1/p}$$

We recall that singular values are uniquely associated to a matrix, independently of the basis in which it is represented and the particular U and V matrices used in the SVD (gauge freedom).

The Schatten 2-norm corresponds to the

$$\|M\|_F := \sqrt{\sum_{i=1}^m \sum_{j=1}^n |M_{ij}|^2} = \sqrt{\text{Tr}[M^\dagger M]} = \sqrt{\sum_{j=1}^r s_j^2} = \|M\|_2 \quad \text{Frobenius norm}$$

since, we recall, the squares of singular values s_j^2 of M are the nonzero eigenvalues of $M^\dagger M$, which is Hermitian by construction, thus always diagonalizable.

The other situation of our interest is $p = 1$, i.e.

$$\|M\|_N := \text{Tr}[\sqrt{M^\dagger M}] = \sum_{j=1}^r s_j = \|M\|_1 \quad \text{Nuclear norm, or trace norm}$$

A remarkable difference with quantum states, is that density matrices $\hat{\rho}$ are required to have unit trace, i.e. are normalized with respect to the Schatten 1-norm, while, as discussed in the next paragraph, operator entanglement requires normalization of operators with respect to the

$$\hat{\rho}_A = \text{Tr}_{\mathcal{H}_B} [\hat{\rho}] = \text{Tr}_{\mathcal{H}_B \otimes \tilde{\mathcal{H}}} [|\psi\rangle\rangle \langle\langle\psi|]$$

we introduce here the following object, which is the reduced density matrix of $|\psi\rangle\rangle \langle\langle\psi|$ with respect to the "artificial" subsystem $\mathcal{H}_A \otimes \tilde{\mathcal{H}}_A$:

$$\hat{\eta}_A := \text{Tr}_{\mathcal{H}_B \otimes \tilde{\mathcal{H}}_B} [|\psi\rangle\rangle \langle\langle\psi|] \rightarrow \hat{\rho}_A = \text{Tr}_{\tilde{\mathcal{H}}_A} [\hat{\eta}_A]$$

We denote by $\vec{\tau}_{A/B}$ the quantum numbers of the ancillary sites. Then, performing an SVD on $|\psi\rangle\rangle$ and using (9.10):

$$\begin{aligned} \hat{\eta}_A &= \sum_{j=1}^r \sum_{\vec{\sigma}_A, \vec{\sigma}'_A} \sum_{\vec{\tau}_A, \vec{\tau}'_A} s_j^2 \left(C_{\vec{\sigma}'_A \vec{\tau}'_A}^{(A)j} \right)^* C_{\vec{\sigma}_A \vec{\tau}_A}^{(A)j} |\vec{\sigma}_A \vec{\tau}_A\rangle \langle \vec{\sigma}'_A \vec{\tau}'_A| \\ \hat{\rho}_A &= \sum_{j=1}^r \sum_{\vec{\sigma}_A, \vec{\sigma}'_A} \left(\sum_{\vec{\tau}_A} s_j^2 \left(C_{\vec{\sigma}'_A \vec{\tau}_A}^{(A)j} \right)^* C_{\vec{\sigma}_A \vec{\tau}_A}^{(A)j} \right) |\vec{\sigma}_A\rangle \langle \vec{\sigma}'_A| \end{aligned}$$

Therefore, in general an SVD on $|\psi\rangle\rangle$ only returns eigenvalues and eigenvectors of $\hat{\eta}_A \neq \hat{\rho}_A$ and

$$-\sum_{j=1}^r s_j^2 \log s_j^2 = S_N(\hat{\eta}_A) \neq S_N(\hat{\rho}_A)$$

However, in our particular setting, where $\tilde{\mathcal{H}}_A$ is a trivial Hilbert space (only the right thermal part of the initial state requires purification),

$$\hat{\rho} = |R_n\rangle \langle R_n| \otimes \hat{\rho}_B \rightarrow \hat{\rho}_A = \hat{\eta}_A \rightarrow -\sum_{j=1}^r s_j^2 \log s_j^2 = S_N(\hat{\rho}_A)$$

This means that, in general, if $\chi \in \{A, B\}$ does not contain ancillary sites, then the SVD of the purified global state returns eigenvalues and eigenvectors of $\hat{\rho}_\chi$, enabling for example an immediate computation of its von Neumann entropy.

However, it is important to retrieve that, as discussed in the previous paragraph:

$$S_N(\hat{\rho}_A) = S_N(\hat{\eta}_A) = S_N(\hat{\eta}_B) \neq S_N(\hat{\rho}_B)$$

Relation with the Operator Entanglement: Again, we start from the case in which the whole system is in a pure state $|\psi\rangle$, i.e. $\hat{\rho} = |\psi\rangle \langle\psi|$. The Schmidt decomposition of $|\psi\rangle$ automatically induces an operator Schmidt decomposition of $\hat{\rho}$.

$$|\psi\rangle \langle\psi| = \sum_{i,j=1}^r s_i s_j |\phi_i^A\rangle \langle\phi_j^A| \otimes |\phi_i^B\rangle \langle\phi_j^B| \quad (9.15)$$

Indeed, as required by (9.13), the operators $\{\hat{O}_{ij}^\chi = |\phi_i^\chi\rangle \langle\phi_j^\chi|\}$, with $\chi \in \{A, B\}$, are orthonormal and $\hat{\rho} = |\psi\rangle \langle\psi|$ is correctly normalized, as proven in (9.12).

This means that (9.15) is a true operator Schmidt decomposition of $\hat{\rho}$, with singular values $\tilde{s}_{ij} := s_i s_j$, hence the corresponding operator entanglement is by definition

$$S_{op}^{(\alpha)}(|\psi\rangle \langle\psi|) := \frac{1}{1-\alpha} \log \left[\sum_{i,j=1}^r \tilde{s}_{ij}^{2\alpha} \right] = \frac{1}{1-\alpha} \log \left[\left(\sum_{j=1}^r s_j^{2\alpha} \right)^2 \right] = \frac{2}{1-\alpha} \log \left[\sum_{j=1}^r s_j^{2\alpha} \right] = 2S_A^{(\alpha)}(|\psi\rangle)$$

Thus, the relation between operator entanglement and von Neumann entropy of the reduced density matrix is extremely simple:

$$S_{op}(|\psi\rangle \langle\psi|) = 2S_N(\hat{\rho}_A)$$

Let us now shift to the situation in which *the whole system is in a generic mixed state*. We wonder whether the singular values of the purified state $|\psi\rangle$ encode some information about the operator entanglement of $\hat{\rho}$.

We have just proven that the Schmidt decomposition of $|\psi\rangle$ naturally returns the operator Schmidt decomposition (9.15) of $|\psi\rangle\langle\psi|$. By tracing over ancillary sites, this leads to a decomposition of $\hat{\rho}$. We have to test if this satisfies the definition (9.13).

$$\hat{\rho} = \text{Tr}_{\mathcal{H}}[|\psi\rangle\langle\psi|] = \sum_{i,j=1}^r s_i s_j \text{Tr}_{\mathcal{H}_A}[|\phi_i^A\rangle\langle\phi_j^A|] \otimes \text{Tr}_{\mathcal{H}_B}[|\phi_i^B\rangle\langle\phi_j^B|]$$

This seems a harmless expression, analogous to (9.15). However there are two important differences, which lead the previous conclusion to fail.

First, the density matrix $\hat{\rho}$ is no longer a projector, i.e. $\hat{\rho}^2 \neq \hat{\rho}$, hence $\|\hat{\rho}\|_2 \neq \|\hat{\rho}\|_1 = 1$. Thus (9.13) should rather be compared with

$$\frac{\hat{\rho}}{\|\hat{\rho}\|_2} = \frac{\text{Tr}_{\mathcal{H}}[|\psi\rangle\langle\psi|]}{\|\text{Tr}_{\mathcal{H}}[|\psi\rangle\langle\psi|]\|_2} = \sum_{i,j=1}^r \frac{s_i s_j}{\|\hat{\rho}\|_2} \text{Tr}_{\mathcal{H}_A}[|\phi_i^A\rangle\langle\phi_j^A|] \otimes \text{Tr}_{\mathcal{H}_B}[|\phi_i^B\rangle\langle\phi_j^B|]$$

Second, the operators $\{\hat{O}_{ij}^\chi = \text{Tr}_{\mathcal{H}_\chi}[|\phi_i^\chi\rangle\langle\phi_j^\chi|]\}$, with $\chi \in \{A, B\}$, are in general not $\|\cdot\|_2$ -normalized (while they are $\|\cdot\|_1$ -normalized), nor orthogonal.

This unpleasant conclusion still holds in our case, where the subsystem A is free of ancillary sites. Indeed, while the set $\{\hat{O}_{ij}^A\}$ is correctly 2-orthonormal, this fails for $\{\hat{O}_{ij}^B\}$.

9.5 INFINITESIMAL EVOLUTOR IN A PURIFIED SYSTEM

As mentioned, since the ancillary copy $\mathcal{H}_R$ is purely artificial, the Hamiltonian can be promoted to an operator on the purified Hilbert space by requiring a trivial action on auxiliary sites, i.e.

$$\hat{H}' := \hat{H} \otimes \hat{\mathbb{I}} : \mathcal{H}_L \otimes \mathcal{H}_R \otimes \tilde{\mathcal{H}}_R \rightarrow \mathcal{H}_L \otimes \mathcal{H}_R \otimes \tilde{\mathcal{H}}_R \quad \text{where} \quad \hat{H} : \mathcal{H} \rightarrow \mathcal{H}$$

Note that any other choice would be equally admissible as long as the physical and ancillary subsystems remain decoupled, i.e. $\hat{H}' := \hat{H} \otimes \hat{\mathbb{I}} + \hat{\mathbb{I}} \otimes \hat{H}_{aux}$. Indeed, even if auxiliary sites undergo a nontrivial evolution, owing to the cyclicity of the trace this does not affect $\hat{\rho}(t)$. Thus, for the sake of simplicity, we choose $\hat{H}_{aux} = 0$ (null operator).

Concretely, if we define, using as natural units $J = 1$,

$$\begin{aligned} \hat{h}_{i,j} &= \frac{(\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+)}{2} + \Delta \hat{S}_i^z \hat{S}_j^z \\ \rightarrow \quad \hat{H}' &= \sum_{j=1}^L \hat{h}_{j,j+1} + \sum_{j=1}^{L-1} \hat{h}_{L+2j-1, L+2j+1} := \hat{H}'_{odd} + \hat{H}'_{even} \\ \hat{H}'_{odd} &= \sum_{k=1}^{L/2} \hat{h}_{2k-1, 2k} + \sum_{k=1}^{L/2} \hat{h}_{L+4k-3, L+4k-1} \quad \hat{H}'_{even} = \sum_{k=1}^{L/2} \hat{h}_{2k, 2k+1} + \sum_{k=1}^{L/2-1} \hat{h}_{L+4k-1, L+4k+1} \end{aligned}$$

In the above picture, blue circles denote the A -tensors of $|\psi_L\rangle$ as MPS, while red/orange circles the A -tensors associated to physical/ancillary sites of $|\psi_R\rangle$ as MPS.

Violet/green dashed arcs represent the interactions in $\hat{H}'_{odd}$ and $\hat{H}'_{even}$, respectively.

Note that orange sites are not involved in any interaction.

Clearly, the purity of the purified initial state $|\psi_L\rangle\otimes|\psi_R\rangle$ is preserved by time evolution. Moreover, since the extended Hamiltonian acts trivially on ancillary sites, we can always obtain the physical $\hat{H}$ -evolved mixed state

$$\hat{\rho}(t) := e^{-i\hat{H}t} (|\psi_L\rangle\langle\psi_L| \otimes \hat{\rho}_R) e^{i\hat{H}t}$$

from the artificial $\hat{H}'$ -evolved purified state by simply tracing over the ancillary space $\mathcal{H}_R$:

$$\begin{aligned} |\psi(t)\rangle &:= e^{-i\hat{H}'t} |\psi_L\rangle \otimes |\psi_R\rangle \\ \text{Tr}_{\mathcal{H}_R} [|\psi(t)\rangle\langle\psi(t)|] &= \text{Tr}_{\mathcal{H}_R} [e^{-i\hat{H}'t} |\psi_L\rangle\langle\psi_L| \otimes |\psi_R\rangle\langle\psi_R| e^{i\hat{H}'t}] = \\ &= e^{-i\hat{H}t} (|\psi_L\rangle\langle\psi_L| \otimes \text{Tr}_{\mathcal{H}_R} [|\psi_R\rangle\langle\psi_R|]) e^{i\hat{H}t} = \hat{\rho}(t) \end{aligned} \quad (9.16)$$

Approximated One-Step Time Evolution: The $\hat{h}_{ij}$ terms of the Hamiltonian do not commute with each other. This implies that we cannot separately evolve each dimer by $e^{-i\hat{h}_{ij}t}$. However, the exponentiation of the Hamiltonian $e^{i\hat{H}'t}$, which requires the exact diagonalization of an exponentially large matrix or an infinite series expansion in powers of $\hat{H}'t$, is not numerically feasible.

Therefore, it is convenient to act as in path-integral techniques and to recast the time evolution as $\mathcal{N} \gg 1$ successive applications of the one-step evolver $e^{i\hat{H}'\tau}$, with $\tau = t/\mathcal{N}$. Note that no approximation is involved.

If τ is sufficiently small, we can perform an n -th order Trotter decomposition [124, 131, 132] of $e^{i(\hat{H}'_{\text{odd}} + \hat{H}'_{\text{even}})\tau}$. For example, a second-order Trotter decomposition is given by

$$\begin{aligned} e^{-i\hat{H}'\tau} &= \hat{U}(\tau) + o(\tau^2) \\ \text{where} \quad \hat{U}(\tau) &:= e^{-i\hat{H}'_{\text{odd}}\tau/2} e^{-i\hat{H}'_{\text{even}}\tau} e^{-i\hat{H}'_{\text{odd}}\tau/2} \end{aligned}$$

In our particular numerical procedures we employ the following fourth-order Trotter decomposition, due to Suzuki [130]:

$$e^{-i\hat{H}'\tau} = \hat{U}(\tau_1)\hat{U}(\tau_1)\hat{U}(\tau_2)\hat{U}(\tau_1)\hat{U}(\tau_1) + o(\tau^4) \quad (9.17)$$

where $\hat{U}(t)$ is the above-defined operator and

$$\tau_1 = \frac{\tau}{4-4^{1/3}} \quad \tau_2 = \tau - 4\tau_1 = \tau \left(1 - \frac{1}{1-4^{-2/3}}\right)$$

Remarkably, the above-defined $\hat{H}'_{\text{odd}}$ and $\hat{H}'_{\text{even}}$ are such that, separately, their terms commute with each other, hence the even/odd evolvers can be factorized as the product of single-dimer evolvers, without any additional approximation.

Swap Gates for the Purified Side: While on the left side of the system we deal with nearest-neighbour interactions, on the right side interactions occur between staggered sites.

We recall that the numerical indices associated to the sites when defining the MPS matrices and the Hamiltonian are purely arbitrary: it is in general convenient to choose them in order to minimize the range of both interactions and initial-state-entanglement. We decided to alternate physical and ancillary sites in the thermal side in order not to have long-range entanglement in the initial state, but this led to range-2 interactions.

However, instead of applying $e^{-i\hat{h}_{i,i+2}t}$ as three-sites operator, it is still possible to exchange sites $j+1 \leftrightarrow j+2$ temporarily, in order to have a nearest-neighbour interaction, then apply the evolver, which now acts on two adjacent sites only, and finally swap them back.

The operators performing this exchange are commonly called *swap gates* [129].

Note that this procedure for operators acting on two non-adjacent sites is by default implemented in the "apply" function of ITensor library [60] in Julia programming language, which we used in our implementation of the post-quench dynamics.

9.6 SVD COMPRESSION OF AN MPS

Before proceeding with the direct implementation of the dynamics on our initial state, we introduce the pivotal concept of *approximability of a generic state $|\psi\rangle$ by an MPS with all bond dimensions under a fixed threshold χ_{max}* . We refer the reader e.g. to [119] for a more formal and detailed discussion. This is a very fundamental property when dealing with numerical simulations, since, while every state can be recast as an MPS, its bond dimension generally blows up exponentially with the system size, but exponentially big matrices are numerically intractable.

Let us first define the *truncation of the bond dimension of an MPS in a single fixed bipartition*. Consider the bipartition (9.8), where the SVD of the tensor C with respect to the bipartitioning point is performed. The most natural idea in order to reduce the corresponding bond dimension r to a threshold χ_{max} is to keep only the first χ_{max} singular values, i.e. to truncate the Schmidt decomposition (9.9) of $|\psi\rangle$ to the first χ_{max} states:

$$|\psi\rangle \approx \sum_{j=1}^{\min\{r, \chi_{max}\}} s_j |\phi_j^A\rangle \otimes |\phi_j^B\rangle \quad (9.18)$$

Clearly, the quality of every approximation depends on the metric used to define the distance between the original and the new object. Here it is transparent that *this approximation is optimal with respect to the Frobenius norm* in the space of C -matrices, but this remarkably coincides with the Euclidean norm on the corresponding $|\psi\rangle$ -vectors, which, as discussed, is the most meaningful norm when dealing with quantum states.

After that, if normalization is requested, the retained singular values can be suitably rescaled.

Since *our aim is to truncate all the bond dimensions to χ_{max}* , we can simply repeat this procedure $L - 1$ times, successively shifting the orthogonality centre in all bonds of the lattice: this is called *SVD compression*. If we define, for the truncation of the k -th bond-link with singular values $\{s_1^{(k)}, \dots, s_{r_k}^{(k)}\}$,

$$\epsilon_k(\chi_{max}) := \sum_{j=\chi_{max}+1}^{r_k} s_j^{(k)}$$

then it can be proven that, after the levelling of the entire lattice [138]

$$\| |\psi\rangle - |\psi_{\chi_{max}}\rangle \|_2^2 \leq 2 \sum_{k=1}^{L-1} \epsilon_k(\chi_{max})$$

This relation shows that, heuristically, if with respect to every possible bipartition singular values decay sufficiently rapidly (e.g. exponentially) before χ_{max} , truncations are little invasive. By contrast, if there exist bipartitions where the two subsystems are highly entangled, then the smaller singular values are about the same order as the higher ones and we are probably neglecting some crucial physical information.

One now may wonder whether this simple global compression is optimal with respect to $\|\cdot\|_2$ such as that performed on single bond-links.

In general the answer is not. Indeed, by performing truncations by sweeping from one edge of the system to the other (for the sake of clarity we assume from left to right), an oriented interdependence of truncations occurs: first we truncate the bond dimension between sites $j - 1$ and j , then multiply the S^{trunc} diagonal matrix with the site matrix $A^{[j]}$ obtaining $A'^{[j]}$; after this, in order to right-shift the orthogonality centre, we perform an SVD on $A'^{[j]}A^{[j+1]}$. This new SVD and the subsequent truncation involve the object $A'^{[j]}$ which was already affected by the previous cut. Thus, truncations performed further to the right depend on those to the left but not vice versa. Hence, in general the final state is not the same returned by a right-to-left compression.

While the single bond dimension truncation is optimal with respect to $\|\cdot\|_2$, when one looks at the state obtained after a complete levelling, the compression can be far from optimal.

However, this effect is expected to be higher for stronger truncations, e.g. for increasing time in a tDMRG procedure, while negligible at sufficiently short times. Thus, once pointed out this problem, for our purposes this compression method will suffice.

For the more sophisticated *variational compression* the reader is referred to [118].

Approximability and Von Neumann Entropy: From above, we should expect the truncation to work well while $S_N(\hat{\rho}_A) \ll \log(\chi_{max})$, whereas caution may be required when $S_N(\hat{\rho}_A) \approx \log(\chi_{max})$. Nevertheless, from a mathematically rigorous perspective, bipartite entanglement is not directly connected to approximability by an MPS.

Indeed, in [119] the authors prove that an at most logarithmic scaling of Rényi entropies $S_A^{(\alpha)}$ with $\alpha < 1$, implies approximability by MPS, while a faster than logarithmic increase when $\alpha > 1$, as well as a linear growth of the von Neumann entropy, rules it out. For all other cases, the scaling of entanglement entropies with system size does not enable any conclusions about approximability. In particular, the authors bring concrete examples for the case of constant von Neumann entropy, proving that area law is not a sufficient condition for approximability.

Summarizing, in typical states of interest, far from such pathological cases, area law in general corresponds to efficient MPS approximability and in our numerical study of the early-time dynamics we will have a look on von Neumann entropy. However, this will not be our only indicator of the reliability of the results and we will perform some other complementary tests.

9.7 TIME-DEPENDENT DMRG

The *density-matrix renormalization group* (DMRG) was originally introduced by Steve White in two subsequent articles [147, 148] dated 1992-93. The algorithm was initially proposed for the study of ground states in one-dimensional quantum lattices.

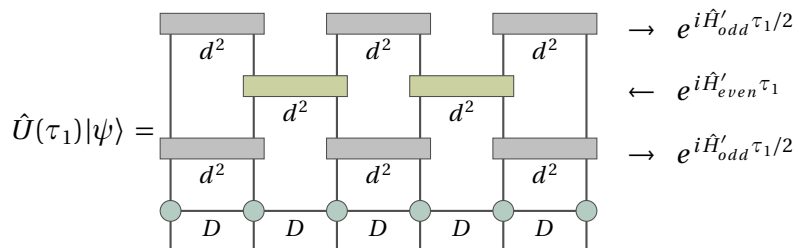
At that time, the study of MPSs had already started developing separately, mostly in an analytical framework. Some years after White's articles a connection was noticed between MPS theory and the DMRG mechanism [49, 97]. A true systematic reformulation of DMRG in terms of MPS was only developed in 2003-2004, enabling deeper insights and leading to extensions beyond exclusive investigation of ground states, e.g. the post-quench real-time evolution [140, 141].

The algorithm used in this work, the *time-dependent DMRG* (tDMRG), was introduced in [41, 139]. As we are going to discuss in more detail, the accessible time range within a tDMRG framework is limited by entanglement growth.

General Sketch of the Algorithm: The key idea is the decomposition of the global time evolution ruled by $e^{-i\hat{H}'t}$ into one-step evolutors $e^{i\hat{h}_{ij}\tau}$ applied to single dimers of the lattice and corresponding to the infinitesimal interval τ , as formally discussed in §9.5. Simultaneously, as in the proper *finite-system DMRG*, the orthogonality centre performs many successive sweeps of the lattice, with each position being optimal for a different evolver.

Consider the rightmost $\hat{U}$ operator in (9.17). First we have to apply $e^{-i\hat{H}'_{odd}\tau_1/2}$, which can be decomposed as a product of odd-dimer evolutors. These can be applied in a single sweep of the lattice, each one when the orthogonality centre reaches its sites.

Then the same is performed during the back-sweep for $e^{-i\hat{H}'_{even}\tau_1}$ and so on.



Note that, to reduce the number of required sweeps, it is also possible to compose the two $e^{-i\hat{H}'_{odd}\tau_1/2}$ of adjacent $\hat{U}$ -operators.

After applying each dimer evolver $e^{-i\hat{h}_{ij}\tau}$, a new SVD is performed to recover the MPS structure and the bond dimension is truncated to the requested χ_{max} . This way the MPS stays compressed as described in §9.6. Indeed, a two-site operator can be represented as an MPO with maximal bond dimension d^2 and its application to a dimer in general implies an increase of its maximal bond dimension from D to $d^2 D$. Apart from this upper ceiling, this argument provides a heuristic explanation for why the bond dimension and the entanglement entropy typically grows exponentially with time.

Finally, a new SVD, which does not lead to any additional approximation, is simply used to shift the orthogonality centre.

Once the entire one-step evolver (9.17) has been applied, since truncations change the normalization of $|\psi\rangle$ (even if only slightly provided the approximation is not too invasive), it is good practice to renormalize the state.

Before the next application, one or more sweeps can be devoted to the evaluation of some relevant observables (e.g. local magnetization and energy density).

Entanglement Growth and Limitations: Every state can be decomposed as an MPS through successive SVDs. However, in the thermodynamic limit, only an infinitesimal corner of the Hilbert space corresponds to "strict MPSs", i.e. states which can be written as (or efficiently approximated by) MPS with finite bond dimension. Our initial state, obtained by joining the $2n$ -rainbow state and the purified infinite-temperature thermal state, falls within this corner, since its maximal bond dimension is 2^n independently of L .

Nevertheless, as discussed in the previous paragraph, in general time evolution induces an exponential growth of the bond dimension and the entanglement entropy. Every area-law state gradually becomes a generic volume-law state, with bond dimension exponentially increasing towards the centre. This means that *only the short-time dynamics can be exactly studied with finite computational resources*.

Truncations become more and more invasive as time progresses, making the error associated with the Trotter decomposition negligible in most cases.

However, since we expect observables to maintain some physical consistence a few time after the beginning of the bond dimension saturation to χ_{max} , it is convenient to test their reliability by checking conservation laws or comparisons with estimates or analytical results when available.

Chapter 10

NUMERICAL RESULTS FOR THE BIPARTITIONING PROTOCOL

In §8 we showed that the $2n$ –rainbow state realizes the infinite-temperature thermal root densities after a homogeneous quench as $n \rightarrow +\infty$. Thus, as discussed at the end of §4, we expect *exactly flat density profiles and no ballistic transport* in the bipartitioning protocol between this asymptotic pure state and the infinite-temperature thermal state.

However, in this chapter we investigate density profiles and ballistic transport in the scaling limit at finite n . We already expect that, due to the convergence of the left and right GHD boundary conditions upon increasing n , the vertical window in which a nontrivial profile of conserved-charge densities can develop shrinks as $\propto 1/n$.

Our main question is whether the increasing similarity between the $2n$ –rainbow state and the infinite-temperature thermal state causes ballistic transport to be suppressed sufficiently rapidly that density profiles approach a *step function*, besides the $1/n$ rescaling.

In §5 we displayed that this does not happen in the noninteracting XX model ($\Delta = 0$), where the energy density profile undergoes a smooth transition upon varying ζ between the two boundary values. Instead, in this chapter we show that in the *genuinely interacting XXZ antiferromagnet* ($\Delta > 1$) a sharp vertical jump at $\zeta \approx 0$ develops.

Throughout this work, for both the BGT and dressed velocities equations, we always performed convolutions using the *Gaussian quadrature method* with 100 points in $[-\pi/2, \pi/2]$. Only for the upper-left panel of Fig.10.1, corresponding to the GHD prediction for $\Delta = 3$ and $2n = 2$, the number of quadrature points was increased to 250 in order to suppress numerical artefacts.

10.1 ENERGY DENSITY EVOLUTION

Energy Density Profile from tDMRG, to Compare with the GHD Prediction: In our tDMRG evolution (with $\tau = 0.1$) of the initial purified state, after each step we devoted a sweep to the computation of local energy densities $\langle \hat{h}_{x,x+1} \rangle$. Hereafter, we denote positions by x instead of j to underline that we only refer to physical sites, neglecting the presence of the ancillary subsystem.

In this section, we aim to compare the short-time evolution of energy density obtained from tDMRG with the corresponding GHD prediction in the hydrodynamic scaling limit along the ζ –rays. To bridge these two descriptions, we need to perform a data collapse of our tDMRG results in terms of the ray variable ζ .

The GHD energy density profile is obtained by plugging into (2.31) the solution for the LQSS root densities and dressed velocities at every fixed ζ . The ballistic transport of energy is described

by the corresponding GHD current (4.14).

$$\frac{E(\zeta)}{L} := \frac{J\Delta}{4} + \sum_{n=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \epsilon_n(\lambda) \rho_{n,\zeta}(\lambda) d\lambda$$

$$J_E(\zeta) := \sum_{n=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \epsilon_n(\lambda) v_{n,\zeta}(\lambda) \rho_{n,\zeta}(\lambda) d\lambda$$

However, we cannot directly compare $\langle \hat{h}_{\zeta t, \zeta t+1} \rangle(t)$ with $E(\zeta)/L$. Indeed GHD deals with two homogeneous subsystems. By contrast, while the $2n$ -rainbow state is defined to be macroscopically homogeneous, it is clearly inhomogeneous at microscopic scales. This means that, in the rainbow side of our initial state, the binding energy oscillates strongly in space at fixed t .

The pivotal idea to solve this problem is that, if we neglect the first part of time evolution, then for sufficiently large t a small variation $\delta\zeta$ covers some subsequent oscillations of the microscopic energy profile.

Therefore, we define in the following way a more reasonable counterpart of $E(\zeta)/L$:

- Select a suitable *time interval* $t \in [t_{min}, t_{max}]$, where t_{max} is sufficiently small for our tDMRG data to be considered reliable, while both t_{min} and $t_{max} - t_{min}$ are large enough to enable an efficient average over microscopic inhomogeneities.
- Choose the *amplitude of a "bin"* $\delta\zeta$, big enough to grant an efficient averaging, but small enough to leave a good resolution of the local energy density profile.
- In our case it is convenient to define a ζ -*dependent amplitude*, e.g. to define two distinct regions $\delta\zeta_{L/R}$ divided by a *critical value* ζ_c . Indeed, the rainbow (left) part of our initial state is highly inhomogeneous. To observe the boundary convergence to (1.10) we need a larger $\delta\zeta_L$. By contrast, near the junction the GHD profile shows steep jumps and angled points: in order to perform a meaningful comparison, we cannot have a $\delta\zeta_R$ larger than the characteristic width of these variations. Finally, the thermal rightmost part is highly homogeneous, thus a variation of $\delta\zeta$ has no effect on the resulting profile and we can, for the sake of simplicity, keep the value $\delta\zeta_R$ used in the vicinity of the junction.
- For every value ζ of interest, average over all the tDMRG data with

$$(x, t) \text{ s.t. } t \in [t_{min}, t_{max}] \text{ and } \frac{x}{t} \in [\zeta - \delta\zeta, \zeta + \delta\zeta]$$

To distinguish this new averaged observable from the ordinary $\langle \hat{h}_{\zeta t, \zeta t+1} \rangle(t)$, we denote it by double angular parenthesis, i.e. $\langle\langle \hat{h} \rangle\rangle(\zeta)$.

Collapse Plots and Numerical Results: Collapse plots (and their comparison with GHD predictions) for the 2, 4, 6, 8 and 10-rainbow state are displayed in Fig.10.1 and 10.2, for $\Delta \in \{3, 10\}$.

Far enough from the junction (in terms of ζ), energy transport is expected to vanish due to the presence of a maximal velocity ζ_{max} for quasiparticle motion (which can be potentially different for left/right-moving carriers).

Out of the nontrivial interval nearby the junction, the density profile (after averaging over microscopic inhomogeneities) becomes constant, due to the boundary conditions which lead to (4.16). Remarkably, we know the exact analytical values of boundary energy density, which we indicate by a light-blue line in Fig.10.1 and Fig.10.2:

- on the right it must be $\lim_{\zeta \rightarrow +\infty} E/L = 0$, since the infinite-temperature state has null energy, as we proved in (5.10);

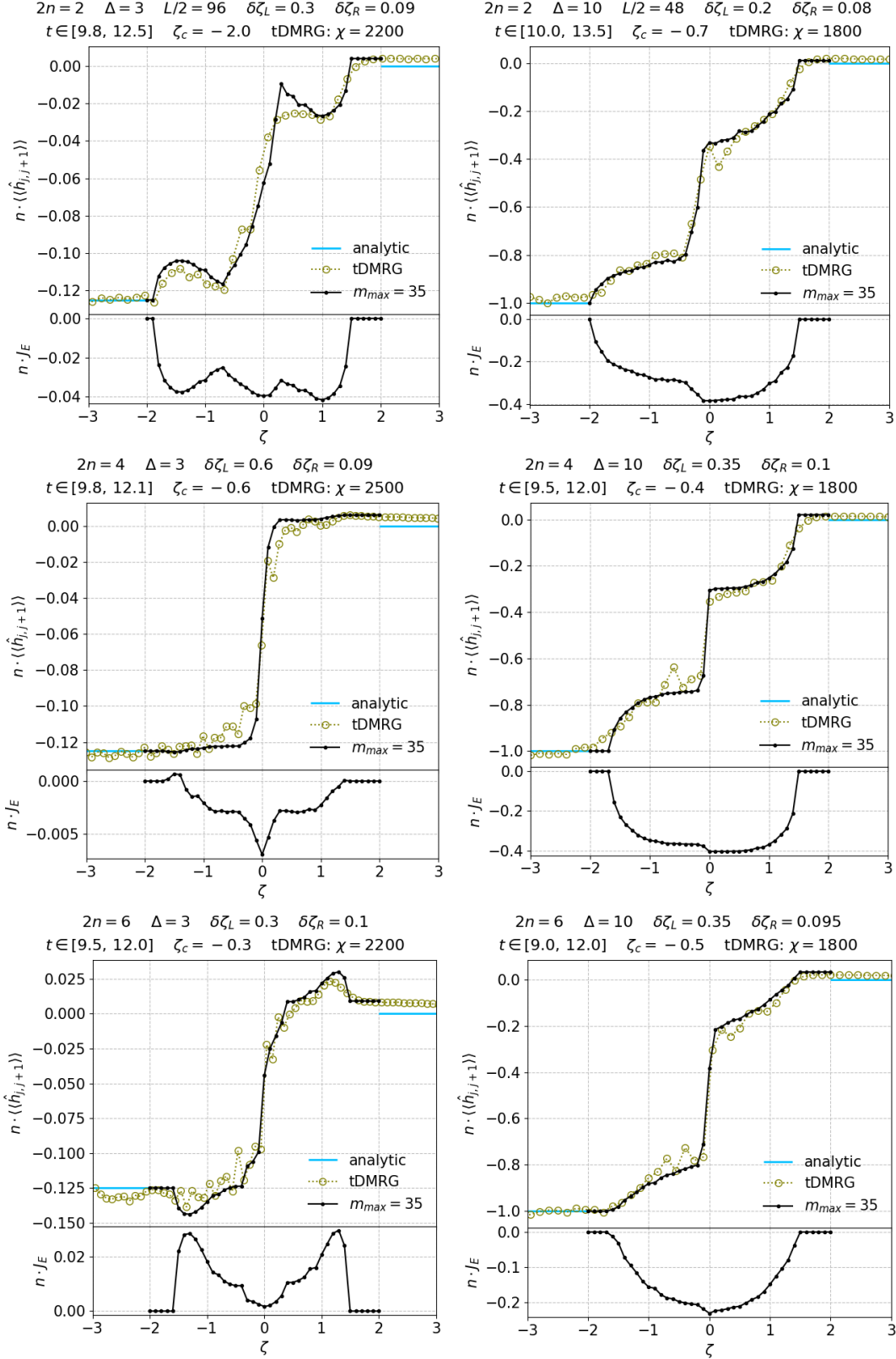


FIGURE 10.1: Bipartitioning protocol from the $2n$ -rainbow state (with $2n \in \{2, 4, 6\}$) and the infinite-temperature thermal state. In all panels, the upper subplot shows (green empty circles) the ζ -averaged energy density $\langle \hat{h} \rangle(\zeta)$ compared to (black solid lines) the GHD prediction. Light-blue horizontal lines are the analytical values of boundary energy density. The lower subplot shows the corresponding GHD current. Resolving the nearly specular profile for $(\Delta, 2n) = (3, 2)$ required increasing the number of Gaussian quadrature points to 200–250, thereby reducing numerical artefacts. Both the energy and current profiles are multiplied by n , to remove the trivial effect of the $\propto 1/n$ vertical shrinking and enable an easier comparison of the profiles.

- on the left we have to find the value computed in (1.10).

Thus, this provides a test for the reliability of our GHD and tDMRG predictions, in particular, typically:

- In the GHD solution, we observe a quite fast convergence in m_{max} for the rainbow side (we recall that here we are considering small rainbow sizes $2n$). By contrast, in the infinite-temperature thermal state for $m_{max} \approx 30$ long strings still cause significant deviations in the energy density profile, which reproduce the behaviour in Fig.???. This is due to the fact that, as observed in §2, infinite-temperature root densities decay as a power law for increasing string length, while for generic states they usually decay much faster, say exponentially.
- In our tDMRG averaged data, if we keep too many low- t points we observe spikes and oscillations which are highly sensible to slight variations of t_{min} . On the other side, if we choose a high t_{max} , too far from the tDMRG validity regime, we are warned by deviations from the boundary analytical value.
- Moreover, our tDMRG simulations are performed on a finite lattice with $L/2 = 48$. However, when $\Delta = 3$ and $2n = 2$ a strong boundary effect arose, so that to observe the correct constant boundary energy density (which assumes an infinite lattice), we had to double the length up to $L/2 = 96$.

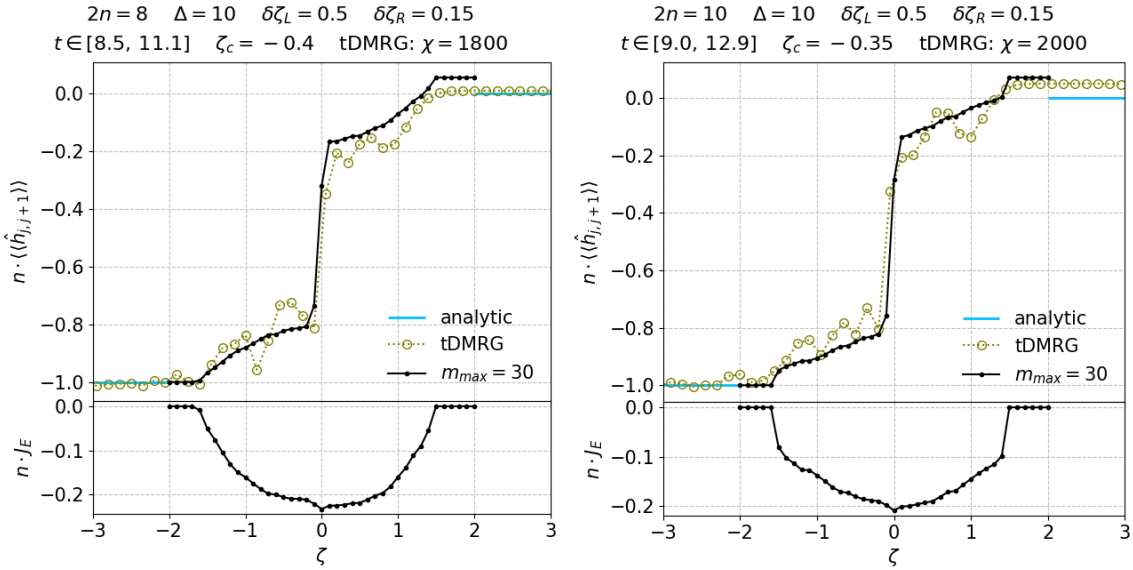


FIGURE 10.2: Bipartitioning protocol from the $2n$ -rainbow state (with $2n \in \{8, 10\}$) and the infinite-temperature thermal state. Here we only sampled $\Delta = 10$. In all panels, the upper subplot shows (green empty circles) the ζ -averaged energy density $\langle \hat{h} \rangle(\zeta)$ compared to (black solid lines) the GHD prediction. Light-blue horizontal lines are the analytical values of boundary energy density. The lower subplot shows the corresponding GHD current. Both the energy and current profiles are multiplied by n , to remove the trivial effect of the $\propto 1/n$ vertical shrinking and enable an easier comparison of the profiles.

Rainbow-Size Limitations: The focus of our interest would have been the $2n$ -rainbow state as $n \rightarrow +\infty$. However, for the following two reasons (the first dealing with tDMRG and the second with the string-charge duality) we only considered bipartitioning protocols up to $2n = 10$:

- As already discussed, in order to produce collapse plots our "bins" $2\delta\zeta$ have to guarantee an efficient average of microscopic inhomogeneities, e.g. at least $t\delta\zeta \approx n$. In principle, this is possible for every size $2n$ given that sufficiently long times are accessible.

However, $t \approx 10\text{--}15$ already requires a maximal bond dimension $\chi_{max} \approx 2000$, hence we are not able to reach later times. Moreover, the accessible time-interval for fixed $\chi_{max} \approx 2000$ rapidly shortens as n increases, since the maximal bond dimension at $t = 0$ is 2^n . Assuming $\delta\zeta \approx 0.1\text{--}0.3$, then $2n = 10$ is very close to be our limit for meaningful collapse plots.

- Our recursive construction of the $\hat{\mathbb{T}}_s^{(n)}$ operator (7.34), encoding the information about the $2n$ –rainbow state, involves an increasing number of products and sums of $\hat{\mathbb{L}}_s$ –elements as n grows, requiring increasing numerical precision to avoid indeterminate ratios or the growth of numerical errors. This problem worsens when considering higher-spin $\hat{\mathbb{T}}_s$ operators, where matrices are bigger and the q –deformed $\{\hat{s}^z, \hat{s}^\pm\}$ given by (7.12) become more complicated.

Indeed, $\hat{\mathbb{T}}_s$ operators and the elements of $\hat{\mathbb{L}}_s$ act on a pair of ancillary spaces, i.e. have dimension $(2s+1)^2 \times (2s+1)^2$, while the explicit final sum of (7.34) counts 4^n terms, each involving the product of $2n$ of these matrices. Simultaneously, $\{\hat{s}^z, \hat{s}^\pm\}$ are given by sums of $2n$ operators and their coefficients $\sqrt{[2s-k]_q[k+1]_q}$ blow up as $e^{\eta s}$. Moreover, while for small rainbows strings of increasing length rapidly become irrelevant, this is no longer true as $2n \rightarrow +\infty$, since we proved that the $2n$ –rainbow state LSS approaches the infinite-temperature root densities for growing n .

Finally, since the discrepancy between zero and the GHD thermal energy density at fixed m_{max} is independent of n while the relevant energy window shrinks as $1/n$, the corresponding relative error grows $\propto n$.

Interpretation of Results and Conclusions: Despite the limitations discussed above, which prevent us from exploring initial states with large rainbow sizes, some meaningful observations can still be drawn looking at Fig.10.1 and Fig.10.2, regarding the convergence of the $2n$ –rainbow state LSS to that of infinite temperature as $2n \rightarrow +\infty$.

First, consistently with the analytical boundary values of the energy density, *the vertical range over which fluctuations can develop shrinks proportionally to $1/n$* . This suggests that the profile becomes increasingly flat, and ultimately constant, in the limit $n \rightarrow +\infty$.

However, this effect was expected and not primarily interesting for our purposes. Thus, in Fig.10.1 and Fig.10.2 we multiplied both the energy and current profiles to remove it and focus on an easier comparison of the profiles.

Indeed, we recall that our main question is whether the increasing similarity between the $2n$ –rainbow state and the infinite-temperature thermal state causes ballistic transport to be suppressed sufficiently rapidly that density profiles approach a *step function*, besides the $1/n$ rescaling. In §5 we displayed that this is not the case of the free XX model ($\Delta = 0$): the energy density profile undergoes a smooth transition upon varying ζ between the two boundary values.

Here, in the genuinely interacting XXZ model, Fig.10.1 and Fig.10.2 reveal that nontrivial ballistic transport is present for very small n , but progressively vanishes as n increases.

We emphasize that *this "avoided-mixing" is interesting given the very different nature of the two initial halves of the bipartitioning protocol*, namely the pure $2n$ –rainbow state, with its highly structured microscopic entanglement pattern, and the mixed, completely uncorrelated and strikingly simple infinite-temperature thermal state.

Our last interesting observation deals with the presence of *angular (i.e. non-analytic) points in GHD predictions*. We will discuss their origin and interpretation later in §10.3, here we just mention that tDMRG is not able to reproduce them mainly due to the finite χ_{max} and to the finite-time and finite-size effects, as proposed in [1].

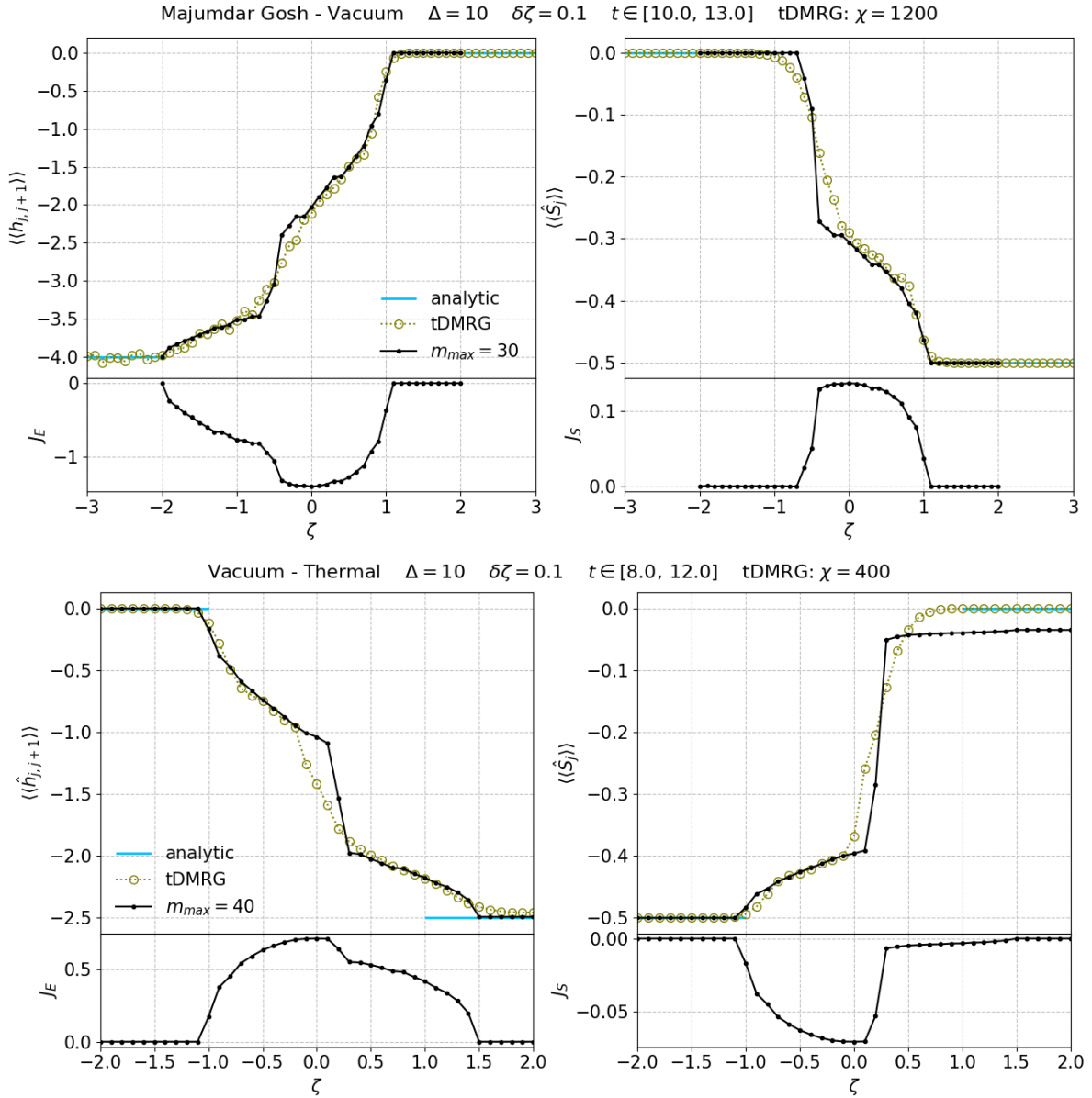


FIGURE 10.3: ABOVE: Energy density (left) and local magnetization (right) time evolution in the bipartitioning protocol between the Majumdar-Ghosh state and the vacuum $|\downarrow\downarrow\downarrow \dots\rangle$.

BELOW: bipartitioning protocol between the vacuum and the infinite-temperature state.

In both panels, the upper subplot shows (green empty circles) the ζ -averaged observable density (energy or magnetization) compared to (black solid lines) the corresponding GHD prediction. The lower subplots shows the corresponding GHD currents (black dashed lines). Light-blue horizontal lines indicate the analytical values of boundary energy density.

For this figure only, the energy spectrum is shifted downward by $\Delta/4$, so that the vacuum energy density is zero, while that of the infinite-temperature state is $-\Delta/4$.

10.2 ADDITIONAL TEST: EXPANSION INTO VACUUM

As a test of the reliability of our tDMRG procedure (complementary to §8.4) for our tDMRG procedure, we considered the following two bipartitioning protocols.

The corresponding results are plotted in Fig.10.3:

- *MG - Vacuum:* We already defined the MG state in §8.4. As vacuum state we mean instead the fully polarized $|\downarrow\downarrow\downarrow \dots\rangle$ state, which has the minimum magnetization, or equivalently no Bethe excitation. The energy density and current profile of this bipartitioning protocol has been studied in [1] as well and our results turn out in agreement with that article.
- *Vacuum - Thermal:* Bipartitioning protocol between the vacuum state $|\downarrow\downarrow\downarrow \dots\rangle$ and the infinite-temperature thermal state.

Together with the comparison to [1], we can also test our results by comparing the boundary energy density with their analytic values.

We already know the exact result for the thermal state and the vacuum. The MG state energy density can be computed straightforwardly in the following way.

- Consider without loss of generality three sites $(j, j+1, j+2)$ s.t. in the initial state $(j, j+1)$ are in a triplet Bell pair $|1, 0\rangle$, while $(j+1, j+2)$ are unentangled.
- Clearly $\langle h_{j+1, j+2} \rangle = 0$.
- In order to compute $\langle h_{j, j+1} \rangle$ it is convenient to employ that

$$\begin{aligned} \langle 0, 0 | \vec{S}_j \vec{S}_{j+1} | 1, 0 \rangle &= \frac{(|\vec{S}_j \vec{S}_{j+1}|^2 - |\vec{S}_j|^2 - |\vec{S}_{j+1}|^2)}{2} = \frac{1}{2} \left(-2 \cdot \frac{3}{4} \right) = -\frac{3}{4} \\ \langle 0, 0 | \hat{S}_j^z \hat{S}_{j+1}^z | 0, 0 \rangle &= \frac{1}{2} \left(2 \cdot \langle \uparrow\downarrow | \hat{S}_j^z \hat{S}_{j+1}^z | \uparrow\downarrow \rangle \right) = -\frac{1}{4} \\ \langle h_{j, j+1} \rangle &= -\frac{3}{4} - \frac{1}{4}(\Delta - 1) = -\frac{2 + \Delta}{4} \end{aligned}$$

- Finally, by averaging over these two results:

$$\lim_{\zeta \rightarrow -\infty} \langle \langle h_{\zeta t, \zeta t+1} \rangle \rangle = -\frac{2 + \Delta}{8}$$

- If, in order to have zero energy for the vacuum state we shift downward the energy spectrum by $\Delta/4$, the energy density becomes

$$\lim_{\zeta \rightarrow -\infty} \langle \langle h_{\zeta t, \zeta t+1} \rangle \rangle = -\frac{2 + 3\Delta}{8}$$

In particular, this is -4 when $\Delta = 10$, in agreement with both our GHD and tDMRG results in Fig.10.3.

Interpretation of Fig.10.3: A first clarification deals with the fact that *the convergence of the GHD predictions to the tDMRG results (or the analytically-known values) is much slower for the local magnetization with respect to energy density.*

The reason is that the energy ϵ_m of an m -string (2.30) is almost independent of m . Hence, since longer string are less abundant, they give only little energetic contribution for increasing m . By contrast, the magnetization of an m -string is linear in m .

Shifting to observations of more physical interest, as for the 2–rainbow state, here we expect (and find) the presence of ballistic transport. However, the main difference with the above framework is that previously both the Hamiltonian and the initial state were invariant under global spin flip. The LQSS inherits the same symmetry, preventing any fluctuation of the GHD magnetization profile around zero or the development of ballistic spin transport, as already discussed in §7.8. We did not display the corresponding plots since any observed nontrivial GHD profile must be due to numerical errors or effects coming from the finiteness of m_{max} . By contrast, now the two halves exhibit different local magnetization and nontrivial particle transport is expected.

10.3 NON-ANALYTIC POINTS AND LIGHT-CONES

As we mentioned, apart from the sharp jump in $\zeta = 0$, Fig.10.1, Fig.10.2, and Fig.10.3 display additional *angular points at $\zeta \neq 0$ in the GHD predictions*.

An explanation of this phenomenon was proposed in [102].

The main idea is to start from (4.16). We recall that to obtain this relation for the LQSS we had to assume that $\zeta = v_{n,\zeta}(\lambda)$ has a unique solution $\forall \lambda$.

Consider observables computed on the right side of the system, i.e. on some ray $\zeta = x/t > 0$ (all the following discussion can be easily recast for $\zeta < 0$ and we will simply convert the final result). Let us define the set $V_m(\zeta) := \{v_{m,\zeta}(\lambda) : \lambda \in [-\pi/2, \pi/2]\}$. The above observables receive contributions from m –strings only if $\zeta \leq \max[V_m(\zeta)]$, i.e. right-moving m –string ballistic carriers spreading from the junction have been able to reach x within t . If we define $\bar{\zeta}_m(\lambda)$ as the lightcone of such λ –rapidity carriers, i.e. as the (unique) solution of

$$\bar{\zeta}_m(\lambda) = v_{m,\bar{\zeta}_m(\lambda)}(\lambda)$$

then the condition $\zeta \leq \max[V_m(\zeta)]$ is equivalent to

$$\zeta \leq \zeta_m^+ := \max\{\bar{\zeta}_m(\lambda) : \lambda \in [-\pi/2, \pi/2]\} \quad [\zeta > 0]$$

where the right side is easily interpreted as the highest possible velocity of an m –string ballistic carrier spreading from the junction.

For observables computed on the left half of the system the analogous conclusion is

$$\zeta \geq \zeta_m^- := \min\{\bar{\zeta}_m(\lambda) : \lambda \in [-\pi/2, \pi/2]\} \quad [\zeta < 0]$$

In [102] $\zeta_m^\pm$ are called *m –th positive/negative light cone*, respectively, corresponding to the outermost right/left ray receiving nontrivial contributions from bound states of m quasiparticles.

When ζ approaches one of these lightcones, a non-analytic point (with a typical square root behaviour) turns out in the profile of local observables:

$$\langle \hat{O} \rangle(\zeta) \approx f(\zeta) + A \cdot \Theta[\pm(\zeta_m^\pm - \zeta)] \cdot \sqrt{\pm(\zeta_m^\pm - \zeta)}$$

Chapter 11

BEYOND RELAXATION: ENTANGLEMENT DYNAMICS

In §5 we provided an exact analysis of the entanglement dynamics in bipartitioning protocols in the noninteracting XX model, with the two halves of the system prepared in $2n$ -rainbow states with different n .

However, in the same chapter we mentioned that the quasiparticle picture has been extended to describe the entanglement dynamics in genuinely interacting integrable models, under suitable assumptions. Thus, in this chapter we consider the entanglement dynamics in the interacting XXZ model as well.

We recall that in §9 we exposed how the tDMRG naturally returns the entanglement dynamics when the initial state is pure. Instead, if the initial state is half-purified (e.g. pure $\hat{\rho}_L$ and thermal $\hat{\rho}_R$), it only returns the time evolution of the von Neumann entropy of the left subsystem. Since the system is not globally in a pure state, the von Neumann entropy is not a measure of the entanglement between the two parts of the chain.

Precisely, we first consider the situation in which the left part of the chain is prepared in an *integrable state* [103]. Indeed, the entanglement dynamics arising from these initial states is known to be described within the original entangled-pair quasiparticle picture. We conjecture that the von Neumann entropy is the sum of the thermal entropy carried by one-particle excitations generated in the initially-thermal subsystem and the entanglement entropy due to entangled pairs generated in the initially-pure subsystem. This is supported by tDMRG simulations.

We also consider quenches in which the left part is prepared in n -rainbow states with $n > 2$. Interestingly, for generic $n > 2$ the state is not integrable. In the XX model, our conjecture holds by replacing pairs with entangled multiplets. In the XXZ model there is no known generalization describing initial states breaking the pair-structure and nonintegrable initial states generally display a different entanglement dynamics. However, in our particular setting, we show that the standard pair-structure quasiparticle picture is surprisingly accurate.

11.1 CONJECTURE FOR THE DYNAMICS OF $S_N(\hat{\rho}_L)$ FROM HYBRID PURE-THERMAL INITIAL STATES

In Fig. 11.1 and 11.2 we display our tDMRG simulations for the time evolution of the von Neumann entropy of $\hat{\rho}_L$ in the XX and XXZ model, respectively. We recall that it does not have a particularly direct physical interpretation. Mathematically, it quantifies the degree of mixedness of $\hat{\rho}_L$. Numerically, it gives an idea of the increasing computational resources required to study the dynamics.

However, from the viewpoint of the tDMRG evolution of our purified initial state, it is exactly the bipartite entanglement. Thus, *the observed linear scaling of this quantity naturally raises the*

question whether this framework could be traced back to some quasiparticle picture.

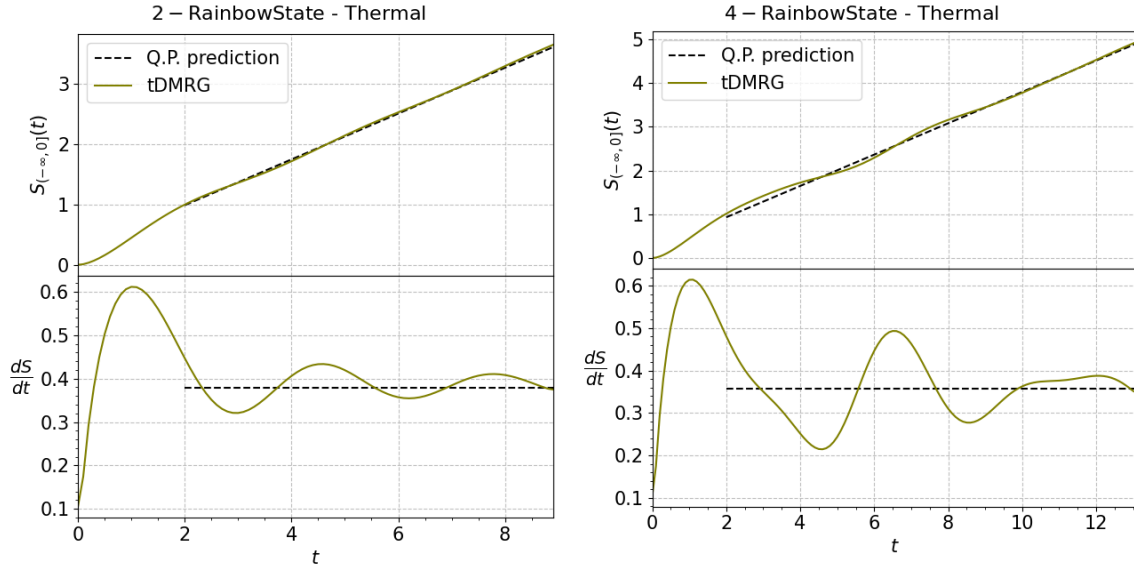


FIGURE 11.1: Bipartitioning protocols between the 2/4–rainbow state and the infinite-temperature state in the XX model. Comparison between tDMRG data and the quasiparticle-picture prediction based on our conjecture.

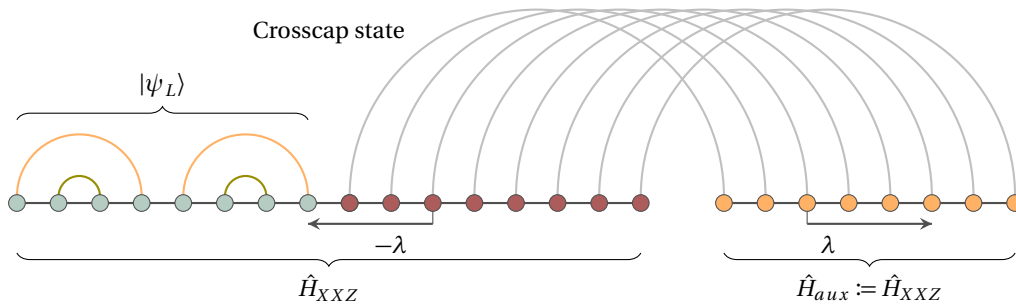
Since we observe a linear growth of $S_N(\hat{\rho}_L)$, we wonder whether it can be traced back to some underlying quasiparticle picture. The main issue is that, as clear in the XX model, any thermal state has no entangled modes, thus it is incompatible with a pair-structure quasiparticle picture. However, it has thermodynamic entropy (which is zero in $|\psi_L\rangle$), carried by single-particle excitations, and, as discussed in §9, $S_N(\hat{\rho}_L)$ counts both the spreading of quantum correlations and the transfer of thermodynamic entropy. Thus, we expect that *the left-subsystem contributes to $S_N(\hat{\rho}_L)$ by its entanglement entropy, due to shared multiplets, while the right side by its thermal entropy, carried by single-particle excitations*. Here we focus on the infinite-temperature state, but this applies to finite-temperature states as well.

Apart from the above motivation, we can also justify this conjecture in the half-purified perspective, where no thermodynamic entropy is involved. Indeed, the fundamental idea is that this dynamics does not depend on how we define the purification.

In principle, the half-purified system has a macroscopically inhomogeneous Hamiltonian: ancillary sites are decoupled from the physical system and their Hamiltonian $\hat{H}_{aux}$ is arbitrary, e.g. $\hat{H}_{aux} = 0$ in our tDMRG. While bipartite initial states have been widely studied, to date the same does not hold for inhomogeneous Hamiltonians [98]. However, in our setting inhomogeneity is apparent since it is simply due to the presence of decoupled sites.

Instead of labelling staggered physical/ancillary sites, we can rather put all physical sites together on the left and all the decoupled auxiliary sites on the right. The Hamiltonian is now homogeneous around the junction. Then, a possible choice for the purification is given by the *crosscap state* [33]: every site j in $\mathcal{H}_R$ (which counts $L/2$ sites) is in a Bell pair with the site $j + L/2$, falling within $\mathcal{H}_R$. In the XXZ model, the entanglement spreading from the crosscap state (which is an *integrable state*) is known to be described within a generalized quasiparticle picture in which the two quasiparticles of an entangled pair are generated separately in these two macroscopically-distant sites. The main difference in our setting is that, of these two antipodal sites, one evolves with $\hat{H}_{XXZ}$ and the other with $\hat{H}_{aux}$. Anyway, we can arbitrarily fix $\hat{H}_{aux} = \hat{H}_{XXZ}$. Our conjecture is thus equivalent to: *the right-site quasiparticles spread in a decoupled ancillary system, they*

cannot affect the entanglement dynamics in our bipartition.



This picture consistently predicts one-particle carriers for the right subsystem instead of a pair structure. Their entanglement content in the crosscap state is the same as the thermodynamic entropy of the infinite temperature state and, given that both the left and right contributions are entanglement entropies, there is no doubt that they can be simply summed. Moreover, since the purification does not affect the computation of observables on the physical space, the GHD predictions for the LQSSs still holds, consistently with all formulas for the entanglement dynamics involving these root densities (see next section). Given this additional justification, for practical applications and generalizations it is more convenient and versatile to consider the original conjecture with the sum of entanglement and thermodynamic entropy.

Some Explicit Results in the XX Model: Let us first consider the XX model. In §5 we exactly computed the entanglement contribution of an half of the system prepared in the 2– or 4–rainbow state. For the thermal right side, $s_R(k) = \log(2)$ since $\langle \hat{n}_k \rangle = 1/2$. The corresponding prediction is in agreement with the tDMRG simulation, as displayed in Fig.11.1.

$$\begin{aligned} \frac{S_R(t)}{t} &= \int_0^\pi \frac{dk}{2\pi} \sin(k) \log(2) \approx 0.221 \\ \frac{S(t)}{t} &\approx \begin{cases} 0.378 & \text{2–rainbow state - thermal} \\ 0.358 & \text{4–rainbow state - thermal} \end{cases} \end{aligned} \quad (11.1)$$

Importance of Testing the Conjecture in the XXZ Model: However, a fundamental remark is that, despite in thermal states there are no correlated modes with different momenta, *in the XX free model we would have obtained the same prediction by simply assuming the thermal side to generate entangled pairs*, as in the ordinary quasiparticle picture. Indeed, in free systems excitation trajectories are straight lines: only one of the two entangled particles in a given pair can cross the junction and one can forget about the pair structure. As a consequence, as pointed out in [2, 3], in these models, even when the initial state is pure, the entanglement production rate coincides with the rate at which the two halves exchange thermodynamic entropy. *In order for our conjecture to return truly different predictions with respect to the ordinary pair-structure quasiparticle picture, we have to turn to the genuinely interacting XXZ model.*

Pair Structure and Integrable States: The generalization to interacting systems requires $|\psi_L\rangle$ to be consistent with the pair-structure behind the quasiparticle picture in the XXZ model.

Indeed, the legitimacy of the pair-structure assumption is both state and model dependent. In free models, it can be directly checked from the form factor. In interacting integrable models, the origin of entangled opposite-rapidity pairs is more elusive. For instance, initial states annihilated by all odd local Bethe-ansatz charges (which are parity-odd) typically have nonzero overlap only with Bethe eigenstates containing pairs of opposite rapidities. This fails for example in the XX model [92], but holds in the generic- Δ XXZ model. States satisfying this annihilation requirement are called *integrable initial states* [103]. Although this is not a rigorous validity proof, general

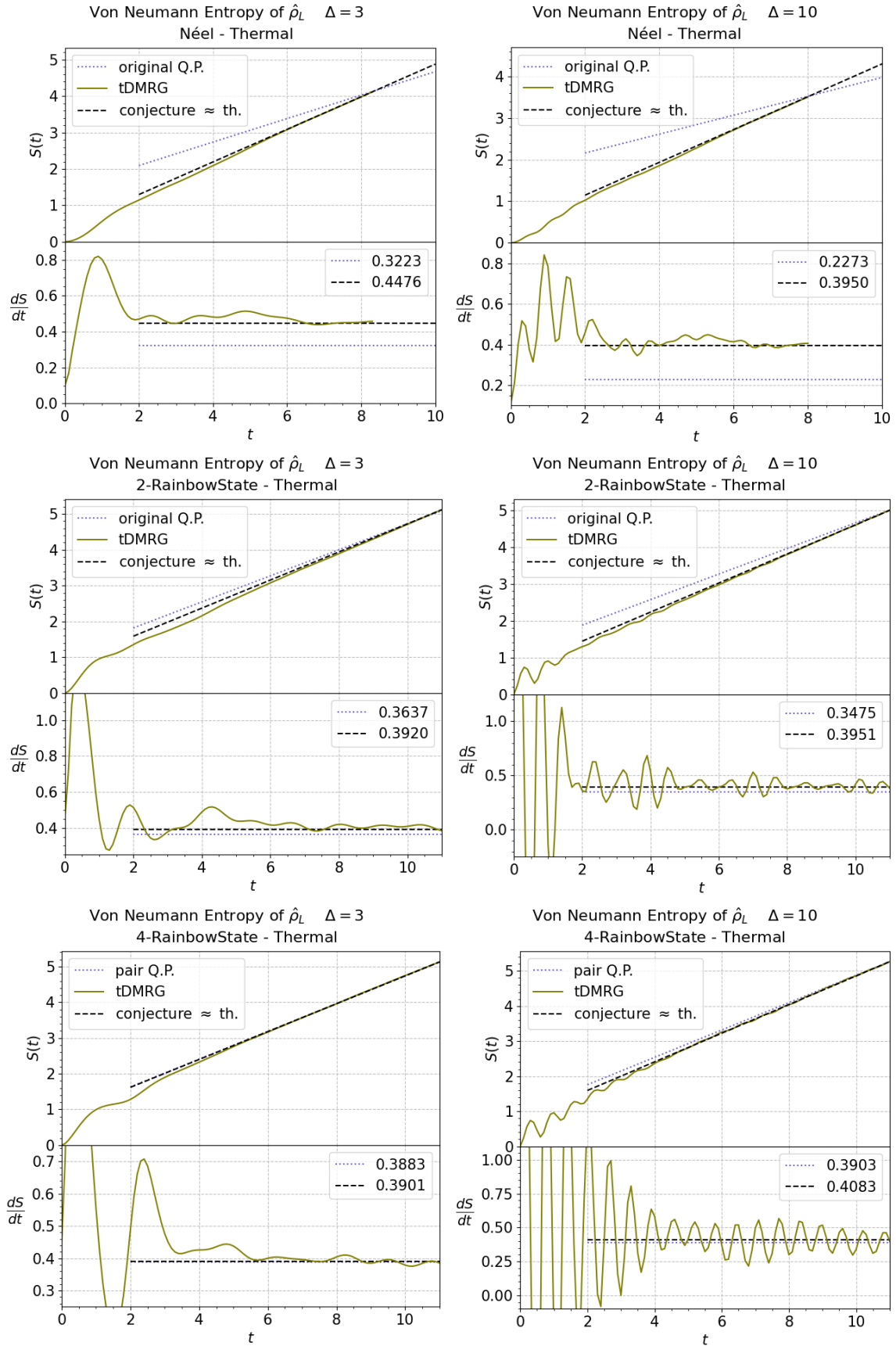


FIGURE 11.2: Green solid lines display the linear growth of the non Neumann entropy of the left subsystem after the bipartitioning protocol between $|\psi_L\rangle$ (Néel or $2n$ -rainbow state) and the infinite-temperature thermal state. Black dashed lines are the prediction of our conjecture (11.9). It turns out indistinguishable from the rate of exchange of thermodynamic entropy. Violet dotted lines are the original quasiparticle picture prediction (11.2).

integrable initial states are expected to be well described by the pair-structure quasiparticle picture. Conversely, this does not exclude states outside this class for which the opposite-rapidity quasiparticle picture nevertheless provides a valid description.

All 2-site periodic states, for example the Néel state $|\Psi\rangle = |\uparrow\downarrow\rangle^{L/2}$ and the 2–rainbow state, are integrable. Instead, $2n$ –rainbow states for $n > 1$ are not integrable. This is e.g. proven by explicitly checking that they are not annihilated by the third local charge, whose density is $\hat{q}_j^{(3)} \propto i[\hat{h}_{j,j+1}, \hat{h}_{j+1,j+2}]$ [66]. However, we test the prediction on these states, as well.

11.2 AN OVERVIEW ON THE QUASIPARTICLE PICTURE FOR BIPARTITIONING PROTOCOLS IN THE XXZ MODEL

Even in ordinary bipartitioning protocols from pure states, the main difficulty with the quasiparticle picture in interacting models is that, as mentioned in §4, quasiparticles generated in the bulk can experience velocity variations while passing through different ζ –rays. However, for $\mathbb{A} = (-\infty, 0]$ a strikingly simple expression for the entanglement dynamics is recovered [2, 3], only depending on the dressed velocities and root densities evaluated at $\zeta = 0$:

$$\frac{S^{Q.P.}(t)}{t} = \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \text{sgn}(\lambda) v_{m,\zeta=0}(\lambda) s_m^{(\zeta=0)}(\lambda) d\lambda \quad (11.2)$$

where we assumed $\text{sgn}(\lambda) = \text{sgn}[v_{m,\zeta=\pm\infty}(\lambda)]$ and $s_m^{(\zeta=0)}(\lambda)$ is the Yang-Yang entropy content (2.32) of λ –rapidity m –strings in the $\zeta = 0$ LQSS. The factor $\text{sgn}(\lambda)$ accounts for the possibility that interactions are sufficiently strong for the quasiparticles forming an entangled pair to end up in the same half of the system as $t \rightarrow +\infty$. Indeed, if both the quasiparticles of a pair cross the junction, the asymptotic velocity of one and only one of them has opposite sign with respect to the initial emission velocity in the bulk. Thus, this factor makes the total contribution of such a pair vanish. We expect this formula to fail within our half-pure half-thermal setting. Indeed, according to our conjecture the entropy contribution of the right subsystems is carried by single particles (not shared pairs): every particle crossing the junction does contribute and the above cancellation must not be performed. In agreement with this argument, (11.2) does not match the slopes in Fig. 11.2.

To get a correct expression of the slope of the von Neumann entropy of $\hat{\rho}_L$ we can follow the steps of [2], but assuming the entropy contributions to be as described above.

We will not replicate all the calculations: the reader is referred to the original article for further details. Here we simply summarize the main notations and results, pointing out the differences connected to our conjecture. Our final formula will also be easily interpretable without this more formal introduction.

Some Notations, Quasiparticle Motion and Conservation of Yang-Yang Entropy: First, let $X_{m,\lambda}(x_0, t)$, as a function of t , be the position of an m –string with rapidity λ with initial condition (x_0, t_0) . According to the usual semiclassical picture, it is the solution of

$$\frac{\partial X_{m,\lambda}(x_0, t)}{\partial t} = v_{m,\zeta_{m,\lambda}(x_0, t)}(\lambda) \quad \text{where} \quad \zeta_{m,\lambda}(x_0, t) := \frac{X_{m,\lambda}(x_0, t) - x_0}{t} \quad \text{and} \quad X_{m,\lambda}(x_0, t_0) = x_0$$

Eventually we will take the limit $t/t_0 \rightarrow 0$. It is convenient to rewrite the above differential equation in terms of the ray-variable ζ , yielding

$$\frac{\partial \zeta_{m,\lambda}(x_0, t)}{\partial t} = \frac{v_{m,\zeta_{m,\lambda}(x_0, t)}(\lambda) - \zeta_{m,\lambda}(x_0, t)}{t} \quad \zeta_{m,\lambda}(x_0, t_0) = \frac{x_0}{t_0} \quad (11.3)$$

As in the derivation of (4.16), we assume that $\forall (\lambda, m)$ the equation $\zeta = v_{m,\zeta}(\lambda)$ has a unique solution, which we label as $\bar{\zeta}_m(\lambda)$.

In particular, (λ, m) –particles spreading from the junction along $\bar{\zeta}_m(\lambda)$, always keep along this same ζ –ray. This observation is useful since the spacetime trajectories of (λ, m) –particles with different initial conditions cannot cross.

$$\begin{aligned} x_0/t_0 < \bar{\zeta}_m(\lambda) &\leftrightarrow \zeta_{m,\lambda}(x_0, t) < \bar{\zeta}_m(\lambda) \quad \forall t \\ x_0/t_0 > \bar{\zeta}_m(\lambda) &\leftrightarrow \zeta_{m,\lambda}(x_0, t) > \bar{\zeta}_m(\lambda) \quad \forall t \end{aligned}$$

Then, the above differential equation (11.3) can be integrated from t_0 to t as

$$\int_{x_0/t_0}^{\zeta_{m,\lambda}(x_0, t)} \frac{d\zeta}{v_{m,\zeta}(\lambda) - \zeta} = \log \frac{t}{t_0}$$

As $t \rightarrow +\infty$, the left-hand-side has to diverge logarithmically. This requires

$$\lim_{t \rightarrow +\infty} \zeta_{m,\lambda}(x_0, t) = \bar{\zeta}_m(\lambda) \quad \forall x_0$$

More properly, the $t \rightarrow +\infty$ limit is meant as $x_0/t \rightarrow 0$. Independently on the initial position, all trajectories with the same (m, λ) become asymptotically linear, parallel to $\bar{\zeta}_m(\lambda)$, and we additionally recall approaching this behaviour that they never cross each other.

With a slight abuse of notation, henceforth we change variable $x_0 \rightarrow \zeta_0 := x_0/t$ and omit the dependence on t (assumed fixed):

$$\zeta_{m,\lambda}(\zeta_0) := \zeta_{m,\lambda}(\zeta_0 t, t) = \frac{X_{m,\lambda}(\zeta_0 t, t)}{t}$$

Finally, it is useful to define the inverse function $\zeta_0^{[m,\lambda]}(\zeta)$ of $\zeta_{m,\lambda}(\zeta_0)$ yielding the initial condition ζ_0 of an (m, λ) –particle which is on the ray ζ at time t . It satisfies [2]

$$\zeta_0^{[m,\lambda]}(\zeta_{m,\lambda}(\zeta_0)) = \zeta_0 \quad \partial_\zeta \zeta_0^{[m,\lambda]}(\zeta) = \frac{\zeta_0^{[m,\lambda]}(\zeta)}{\zeta - v_{m,\zeta}(\lambda)} \quad (11.4)$$

As we previously discussed, the Yang-Yang entropy density $S_{YY}(\zeta)/L$ of the LQSS can be expressed in terms of the contributions $s_{m,\zeta}(\lambda)$ (2.32) of m –strings with rapidity λ . Despite not being a proper quasilocal conserved charge, the Yang-Yang entropy satisfies a continuity equation

$$\zeta \partial_\zeta s_{m,\zeta}(\lambda) = \partial_\zeta [v_{m,\zeta}(\lambda) s_{m,\zeta}(\lambda)] \quad (11.5)$$

$$\text{with the boundary condition} \quad s_{m,\pm\infty}(\lambda) = s_m^{R/L}(\lambda)$$

Thus it is by definition a conserved quantity and this enable assuming that it is transported by ballistic carriers. In particular, since quasiparticle trajectories do not cross and the interval $[0, \zeta_0]$ is mapped at time t onto $[\bar{\zeta}_m(\lambda), \zeta_{m,\lambda}(\zeta_0)]$

$$\int_{\bar{\zeta}_m(\lambda)}^{\zeta_{m,\lambda}(\zeta_0)} s_{m,\zeta}(\lambda) d\zeta = \int_0^{\zeta_0} s_{m,\zeta}^{(0)}(\lambda) d\zeta = \zeta_0 s_{m,\zeta_0}^{(0)}(\lambda)$$

$$\text{where} \quad s_{m,\zeta_0}^{(0)}(\lambda) := s_m^R(\lambda) \cdot \theta[\zeta_0] + s_m^L(\lambda) \cdot \theta[-\zeta_0]$$

Simultaneously, exploiting (11.5) to explicitly integrate the left-hand side yields

$$s_{m,\zeta}(\lambda) = \left(\frac{\zeta_0^{[m,\lambda]}(\zeta)}{\zeta - v_{m,\zeta}(\lambda)} \right) s_{m,\zeta_0^{[m,\lambda]}(\zeta)}^{(0)}(\lambda) \quad (11.6)$$

Before proceeding, here we recall a relation derived in [2], useful in performing integrals over $d\zeta_0$ involving $s_{m,\zeta_0}^{(0)}(\lambda)$.

$$d\zeta_0 = \frac{\partial \zeta_0^{[m,\lambda]}(\zeta)}{\partial \zeta} d\zeta = \frac{\zeta_0^{[m,\lambda]}(\zeta)}{\zeta - v_{m,\zeta}(\lambda)} d\zeta$$

where we employed (11.4). Therefore, given (11.6),

$$s_m^{(0)}(\lambda, \zeta_0) d\zeta_0 = s_{m,\zeta}(\lambda) d\zeta$$

Explicit Expression of the Von Neumann Entropy Growth from Half-Thermal Initial States:

According to our assumptions, the von Neumann entropy of the left (initially pure) subsystem is given by the sum of two contributions $S_{L/R}$ of different nature. First, the quasiparticle-picture left-hand-site contribution is

$$\begin{aligned} \frac{S_L^{QP}(t)}{t} &= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} d\lambda \int_{-\infty}^{+\infty} d\zeta_0 s_m^{(0)}(\lambda, \zeta_0) \cdot \theta[\zeta_{m,\lambda}(\zeta_0)] \cdot \theta[-\zeta_{m,-\lambda}(\zeta_0)] \cdot \theta[-\zeta_0] = \\ &= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} d\lambda \int_{-\infty}^{+\infty} d\zeta s_{m,\zeta}(\lambda) \cdot \theta[\zeta] \cdot \theta[-\zeta_{m,-\lambda}(\zeta_0^{[m,\lambda]}(\zeta))] \cdot \theta[-\zeta_0^{[m,\lambda]}(\zeta)] \end{aligned}$$

where the presence of $\theta[-\zeta_0]$, which selects the quasiparticles generated on the left subsystem, is the only difference with the expression considered in [2]. Thus, all the steps of the original article still apply, reducing to an expression only depending on velocities and root densities at $\zeta = 0$. The effect of $\theta[-\zeta_0]$ is that, unlike (11.2), only quasiparticles crossing the junction from left to right ($v_{m,\zeta=0}(\lambda) > 0$) are counted.

$$\frac{S_L^{QP}(t)}{t} = \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \theta[v_{m,\zeta=0}(\lambda)] \cdot \text{sgn}(\lambda) v_{m,\zeta=0}(\lambda) s_m^{(\zeta=0)}(\lambda) d\lambda \quad (11.7)$$

The right-subsystem contribution, instead, coincides with the right contribution to the usual total rate of exchange of thermodynamic entropy, which we write below for completeness since it represents an upper limit for the slope of the von Neumann entropy:

$$\begin{aligned} \frac{S^{th}(t)}{t} &:= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} d\lambda \int_{-\infty}^{+\infty} d\zeta_0 s_m^{(0)}(\lambda, \zeta_0) \cdot \{ \theta[\zeta_{m,\lambda}(\zeta_0)] \cdot \theta[-\zeta_0] + \theta[-\zeta_{m,\lambda}(\zeta_0)] \cdot \theta[\zeta_0] \} = \\ &= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} d\lambda \int_{-\infty}^{+\infty} d\zeta s_{m,\zeta}(\lambda) \cdot \{ \theta[\zeta] \cdot \theta[-\zeta_0^{[m,\lambda]}(\zeta)] + \theta[-\zeta] \cdot \theta[\zeta_0^{[m,\lambda]}(\zeta)] \} = \\ &= \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} |v_{m,\zeta=0}(\lambda)| s_m^{(\zeta=0)}(\lambda) d\lambda \end{aligned}$$

The $\text{sgn}(\lambda)$ factor is disappeared, since all quasiparticles crossing the junction contribute. Again, turning to our framework, we only need the term with $\theta[\zeta_0]$, yielding a factor $\theta[-v_{m,\zeta=0}(\lambda)]$.

$$\frac{S_R^{th}(t)}{t} = \sum_{m=1}^{+\infty} \int_{-\pi/2}^{\pi/2} \theta[-v_{m,\zeta=0}(\lambda)] \cdot |v_{m,\zeta=0}(\lambda)| s_m^{(\zeta=0)}(\lambda) d\lambda \quad (11.8)$$

Finally, given (11.7) and (11.8), we expect the von Neumann entropy slope for the left (initially pure) subsystem to be given by

$$\frac{S^{Q.P.}(t)}{t} \leq \frac{S^{conj}(t)}{t} = \frac{S_L^{QP}(t) + S_R^{th}(t)}{t} \leq \frac{S^{th}(t)}{t} \quad (11.9)$$

In free models these three values coincide.

11.3 DISCUSSION OF RESULTS IN THE XXZ MODEL

Explicit evaluations of the above expression for the Néel state and $2n$ -rainbow states with $n \in \{1, 2\}$ are:

	Néel $\Delta = 3$	Néel $\Delta = 10$	2-R $\Delta = 3$	2-R $\Delta = 10$	4-R $\Delta = 3$	4-R $\Delta = 10$
$S_L^{QP/th}/t$	0.10610	0.01925	0.15865	0.15328	0.19406	0.19985
S_R^{QP}/t	0.21619	0.20810	0.20502	0.19426	0.19421	0.19043
S_R^{th}/t	0.34154	0.37576	0.23336	0.24185	0.19609	0.20841
S^{QP}	0.32229	0.22735	0.36366	0.34754	0.38827	0.39028
$S^{conj} \approx S^{th}$	0.44764	0.39502	0.39201	0.39513	0.39015	0.40826

The comparison with the direct tDMRG numerical simulation is shown in Fig.11.2.

In computing the analytical predictions, since the infinite-temperature $\{\theta_m\}$ decay much slower than that of the $2n$ –rainbow states (power-law vs. exponentially), we approximated the latter as zero for $m > 35$ and solved Eqs.(2.42), (4.16), and (4.17) using $m_{max} = 50$. Indeed, for $m \geq 35$ the θ –ratios of the rainbow states were generally several orders of magnitude smaller than those of the infinite-temperature thermal state. By comparing with the values obtained using $m_{max} = 49$, we observed deviations of order $10^{-8} - 10^{-5}$, thus the truncation has almost negligible effects for our purposes. Thus, we do not display the uncertainties in Table ?? and we simply truncate values at order 10^{-5} .

In all cases, up to our numerical precision, we found $S_L^{QP}/t \approx S_L^{th}/t$, implying that the difference between our conjecture and the rate of exchange of thermodynamic entropy is not resolvable. Instead, $S_R^{QP}(t)$ and $S_R^{th}(t)$ display a macroscopic difference, allowing us to distinguish between the prediction of our conjecture and the original pair-structure quasiparticle picture.

The fact that the difference between the thermal and pair-structure predictions is more resolvable in the right-side contribution has a straightforward interpretation. Indeed, the dressing due to interactions is known to have stronger effect on the trajectories of long strings, while it is unlikely that a pair of entangled short strings ends up with both quasiparticles in the same subsystem. For most pure initial states, the LSS root densities decay exponentially, thus we expect conserved quantities to receive the almost total contribution from short strings. Instead, the infinite-temperature root densities display a power-law decay and longer strings are expected to give a higher relative contribution.

This interpretation is in agreement with the fact that, in the very paper [5], the authors study bipartitioning protocols from a variety of pure initial state, but conclude that in all cases the difference $|S^{th} - S^{QP}|$ is generally negligible or very difficult to resolve, at the typical timescale accessible to tDMRG. Instead, at least in the first four plots in Fig.11.2, this difference is clearly distinguishable even at short times.

$2n$ –rainbow states with $n > 1$: As previously discussed, $2n$ –rainbow states with $n > 1$ are *not* integrable states. Moreover, in the XX model their entanglement dynamics requires introducing entangled pairs of $2n$ species of quasiparticles (but, we recall, the integrability condition itself has deeply different implications for the Bethe overlaps in the $\Delta = 0$ and $\Delta \neq 0$ cases, thus the multiplet structure in the XX model may be likely to collapse in the XXZ model, as well). A further test in this direction may be to measure the tripartite mutual information [25] in this setting, which is sensible to the presence of multiplets.

Anyway, our tDMRG simulations in Fig.11.2 display a surprisingly good agreement with our prediction based on the pair-structure quasiparticle picture.

CONCLUSIONS

In this last section we recall the aim of the present work and our main original results.

In this work we proved that, in the gapped XXZ model and the XX model, the LSS of our family of increasingly entangled $2n$ –rainbow states approaches the infinite-temperature thermal state. This was our extension of a conjecture [99] proposed for "long random states" in the XXX model. For this purpose, we computed the form factor (5.7) of our states in the XX model and then the ABA quasilocal charges in the XXZ model, for both the $2n$ –rainbow states and the $T \rightarrow +\infty$ state. In particular, we proved that infinite-temperature charges can be numerically computed using (8.6), similarly to the usual case of periodic product states. We also gave a simple recursive expression (7.34) for the $\hat{T}$ operator involved in their computation on the $2n$ –rainbow states.

Then, we considered bipartitioning protocols between the $2n$ –rainbow states and the infinite-temperature thermal state. Unlike the XX model, in the XXZ model we observed a strong suppression of ballistic transport, with the development of a sharp vertical jump at the junction in the energy-density profile. In performing tDMRG simulations, we defined a half-purification of our hybrid pure|thermal initial state.

Finally, we focused on the entanglement dynamics. In the XX model it can be described within a known generalization of the original quasiparticle picture, involving entangled multiplets of $2n$ species of quasiparticles, with correlations given by (5.12).

Turning to the XXZ model and our pure|mixed initial state, the concept of entanglement dynamics becomes more controversial and it is correct only when referring to the half-purified system. The bipartite entanglement computed on the half-purified state coincides with the von Neumann entropy $S_N(\hat{\rho}_L)$ of the reduced density matrix of that half of the system which was initially pure. We conjectured that the initially-pure subsystem contributes to $S_N(\hat{\rho}_L)$ with entangled pairs, while the initially-thermal subsystem with single particles carrying thermodynamic entropy, as expressed by (11.9). We also depicted a complementary description in the half-purified setting justifying the sum of these two contributions of different nature.

Unexpectedly, despite the fact that $2n$ –rainbow states are not *integrable initial states* [103] for $n > 1$, and that in the XX model they generate entangled multiplets of $2n$ quasiparticles, their entanglement dynamics seems to be well captured by the usual quasiparticle picture.

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